Real Analysis

— *EYNTKA* Series —

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Contents

0.0.1 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7⋆; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Introduction

The Riemann integral is not wrong, but it is fragile in a way that a first course in analysis cannot quite bring itself to admit. It computes $\int_0^1 x^2\,dx$ perfectly well. What it does badly is pass to a limit. If $f_n \to f$ pointwise with every f_n Riemann integrable, f need not be: enumerate the rationals in $[0,1]$ and let f_n be the indicator of the first n of them, and the limit is the indicator of $\mathbb{Q} \cap [0,1]$, a function with no Riemann integral at all. The trouble is not the pathology of the example. It is that the theorems one wants, the ones that let lim pass through $\int$, come loaded with uniform-convergence hypotheses that the analysis of the late nineteenth century could rarely verify. Fourier had written down series for functions nobody could integrate, Dirichlet and Riemann had asked when those series converge, and the answers kept stalling on the same obstruction: the class of integrable functions was too small and, worse, not closed under the operations being performed on it.

Lebesgue's reply, in 1902, was to stop slicing the domain and start slicing the range. Rather than partition $[a,b]$ and ask how much f varies over each piece, partition the values of f and ask *how large is the set where f lies between c and $c+\varepsilon$.* That question is only as good as one's ability to assign a size to a genuinely arbitrary set, and so the integral turns out to rest on a prior and more basic object: a countably additive measure, defined on as large a family of sets as consistency permits. The exchange is a good one. The three convergence theorems (monotone, Fatou, dominated) ask almost nothing of the reader, and L^1 becomes complete. The price is stated at the very start of Chapter 1: no such measure can be defined on *all* subsets of $\mathbb{R}^n$ while remaining translation-invariant and normalizing the cube, and a Vitali set is the proof.

The integrable functions, the modes of convergence, and the duality pairings built here are what *Functional Analysis* [13] reorganizes into Banach spaces, what *Harmonic Analysis* [14] runs the Fourier transform over, and what *Probability* [7] reads, with no change of mathematics whatsoever and only a change of vocabulary, as a probability space. A reader who has met a σ-algebra as "the events" and a measure as "the probability" has already met the objects of this book. What is added here is that they are constructed rather than posited.

Goal

The order of the chapters is the order in which the objects have to exist. Chapter 1 builds the measure. Carathéodory's construction turns an outer measure, which is easy to write down and badly behaved, into a countably additive measure on a σ-algebra, which is what the rest of the book needs. The same construction reappears later for Lebesgue-Stieltjes, Hausdorff, product and Haar measures, so it is worth watching once in full.

Chapter 2 builds the integral on top of it, in the three stages the construction forces: simple functions, then nonnegative measurable functions by supremum, then general ones by splitting into positive and negative parts. The convergence theorems arrive here because this is the first point at which they can be stated, and with them the modes of convergence and the uniform integrability that decides when domination can be weakened. Product measures and Fubini-Tonelli close the chapter, since an integral over a product is the first thing one wants that a single measure space cannot express.

Chapter 3 asks how two measures compare. Allowing a measure to take negative values gives the Hahn and Jordan decompositions, and comparing a measure against a reference measure gives absolute continuity and the Radon-Nikodym derivative. The chapter then shows this derivative is a derivative: the Lebesgue differentiation theorem and the theory of functions of bounded variation recover the fundamental theorem of calculus in the form the Lebesgue integral supports.

The last three chapters each add structure to the underlying space and ask what the theory gains. Chapter 4 adds a topology: on a locally compact Hausdorff space the Riesz-Markov theorem identifies Radon measures with positive linear functionals on $C_c(X)$, which converts the problem of building a measure into the problem of building an average. Chapter 5 adds a metric and a scale parameter, producing a family of measures that assign a dimension rather than a volume, with the Kakeya problem as the payoff. Chapter 6 adds a group, and spends the functional built in Chapter 4 to produce a translation-invariant measure on any locally compact group.

What the reader will be able to do

By the end of this book, the reader should be able to:

1. Construct a measure from scratch (exhibit a premeasure on an algebra, push it through Carathéodory's criterion, and say exactly when the extension is unique), and recognize that construction when it recurs downstream under another name: Lebesgue, Lebesgue-Stieltjes, Hausdorff, product, Haar.

2. Justify interchanging a limit and an integral without hesitating over which theorem applies, including the cases where the natural hypothesis is uniform integrability rather than domination.

3. Differentiate one measure against another: decide when $\nu \ll \mu$, produce the Radon-Nikodym derivative, and connect it to the classical fundamental theorem of calculus through functions of bounded variation.

4. Move between a measure and the functional it integrates against, in both directions, on a locally compact Hausdorff space, which is what makes Haar measure available to the representation theory downstream.

5. Read a research-level statement about the size of a set (Hausdorff dimension, a Besicovitch set, the Kakeya conjecture) and locate what is known, what is open, and which of the tools in this book the argument uses.

Omitted Material

Four omissions here are deliberate, and none of them is a gap in the theory. Each is a boundary drawn so that a neighbouring volume can own the subject properly.

- **The L^p spaces as Banach spaces.** The notation $\|f\|_p$ appears throughout, and density statements are proved in it, but completeness, duality, and the Riesz-Fischer theorem are not here. They need the machinery of normed linear spaces, which is a subject and not a section, and they are developed in *Functional Analysis* [13]. The seam is deliberate: a reader who arrives from Folland or Royden expecting an L^p chapter should go there for it.
- **The Fourier transform, and all of it.** Fourier inversion, Plancherel, the Schwartz space and tempered distributions, and the Hausdorff-Young inequality are the subject of *Harmonic Analysis* [14]. This book constructs the measures those theorems integrate against and stops there. The one place the boundary is visibly strained is the Kakeya chapter, whose maximal-function conjecture is Fourier-analytic in motivation, and the statement is given here because Hausdorff dimension is, with the analytic machinery cited.
- **Ergodic theory.** Measure-preserving transformations, the Birkhoff and von Neumann theorems, mixing and entropy are the natural sequel to a measure space with a group acting on it, which Chapter 6 constructs. They are reserved for *Ergodic Theory* [3], because they are a dynamical subject rather than a measure-theoretic one: the interest is in the orbit, not the measure.
- **Rectifiability, currents, and the geometric theory.** Hausdorff measure appears here as a way of assigning a dimension, and the Kakeya chapter is where that pays off. What it does not become is geometric measure theory proper: Frostman's lemma, Riesz energies, projection and rectifiability theorems, the Besicovitch-Federer structure theorem. That layer sits directly on this one and is developed in *Geometric Measure Theory* [4]. For the classical account, Mattila's *Geometry of Sets and Measures in Euclidean Spaces* and, at the far end, Federer's *Geometric Measure Theory*.

1

Measures

As a start, we might want to define a reasonable definition of what it means to measure any subset of $\mathbb{R}^n$. Such a function would be of the form $\mu : P(\mathbb{R}^n) \to [0, \infty]$. If μ were to exist, we would want our common geometric intuitions to hold, that is:

1. If $E_1, E_2, ..., E_n, ...$ is a countable collection of pair-wise disjoint subsets, then
$$\mu\left(\bigcup_i E_i\right) = \sum_i \mu(E_i)$$
2. If E and F are sets such that there is a function that only translates, rotates, and reflects E to get to F (i.e. a rigid motion), then $\mu(E) = \mu(F)$
3. If C is a unit cube in $\mathbb{R}^n$, then $\mu(C) = 1$

However, these three axioms are inconsistent with one another! We have actually shown this by showing if μ is defined on $P(\mathbb{R}^n)$ (even when $n = 1$), then we can construct *Vitali sets* that will have to be both 0 and ∞ in measure – a contradiction.

Theorem 1.0.1: Vitali Sets Are Unmeasurable

Vitali sets are unmeasurable

***Proof*:**
Take $[0, 1]$. Partition this set as follows:

$$x \sim y \iff x - y \in \mathbb{Q}$$

In a sense, this is the set of cosets of $\mathbb{R}/\mathbb{Q}$ restricted to $[0, 1]$; we will rely on this intuition to label the equivalence classes. Since $\mathbb{R}/\mathbb{Q}$ is the set of irrational numbers modulo rational numbers, let N_q be an equivalence class of contains x that is equivalent to the irrational number q. Notice that there are uncountably many such equivalence relations as there are uncountably many irrational that do not satisfy the equation $x - \frac{p}{q} = 0$.

Now, pick one $p \in N_q$ for each equivalence class, and let N be the collection of all these p's, so N contains one point from each equivalence class (notice that the axiom of choice must be used to create a non-empty N). We'll show that N is unmeasurable.

First, for each rational number $r \in \mathbb{Q} \cap [0,1)$, define:

$$N + r = \{x + r \mid x \in N \cap [0, 1-r)\} \cup \{x + r \mid x \in N \cap (1-r, 1)\}$$

Then clearly, $N + r \subseteq [0,1)$ for each $r \in \mathbb{Q} \cap [0,1)$, and furthermore, for each $x \in [0,1)$ there exists an r such that $x \in N + r$. To show this, let's say $y \in N$ is the element that was picked from the equivalence class of of $[x]$. Then by our choice of y, $x \in N + r$ since $r = x - y$ if $x \geq y$, or $r = x - y + 1$ if $x < y$. Next, notice that the $N + r$ also forms a partition: if $r \neq s$ and $(N + r) \cap (N + s) \neq \emptyset$ then $x - r$ (or $x - r + 1$) would equal $x - s$ (or $x - s + 1$), but they belong to distinct equivalence classes, for example:

$$0 = y - y = x - r - (x - s) = s - r \qquad \Rightarrow \qquad r = s \Rightarrow\Leftarrow$$

and so $(N + r) \cap (N + s) = \emptyset$.

We have now reached the point where we can declare our contradiction. If μ satisfies the three wanted properties, by (1) and (2):

$$\mu(N) = \mu(N \cap [0, 1-r)) + \mu(N \cap (1-r, 1)) = \mu(N + r)$$

for any $r \in \mathbb{Q} \cap [0,1)$. Since $\mathbb{Q} \cap [0,1)$ is countable and $[0,1)$ is the disjoint union of the $N + r$'s:

$$1 = \mu([0,1)) = \sum_{r \in \mathbb{Q} \cap [0,1)} \mu(N + r)$$

where $1 = \mu([0,1))$ comes from (3). Now, we have two possibilities:

1. Either $\mu(N) = 0$, in which case each $\mu(N + r) = 0$ but then $0 = 1$ – a contradiction
2. Either $\mu(N) \neq 0$ (say k, so $\mu(N + r) = k$, but then $0 = \infty$ – a contradiction

In other words, no mater what $\mu(N)$ is, we will run into a contradiction!

As you can verify, this construction required that we take advantage of the countability condition (the 1st property). You might ask if weakening it to only finitely many will fix the issue, but sort of crazily it does not! In 1924, Banach and Tarski proved the following paradox now known as the *Banach Tarski Paradox*:

> Let U and V be open subsets of $\mathbb{R}^n$, $n \geq 3$. Then there exists a $k \in \mathbb{N}$ and subsets $E_1, E_2, ..., E_k$, $F_1, F_2, ..., F_k$ such that
>
> 1. all E_j's are disjoint and their union is U
> 2. all F_j's are disjoint and their union is V

3. There exists a rigid motion from E_j to F_j for $j = 1, ..., k$

This is kind of insane: this means that we can take a pea-size ball, and make into the size of the earth! This means that the notion of "geometry" is not actually a consistent concept in general set theory!

Fortunately for us, The sets E_j and F_j are very strange sets; most sets that are in fact measurable. To be more precise, it is consistent to say that the axiom of choice fails and that all sets are Lebesgue measurable (that is, a measure on $\mathbb{R}^n$ that is translation and will be generated by the fact that $\mu([a, b]) = b - a$). For the purposes of this book, that means that any set we construct without the axiom of choice (and just doing normal ZF without any sneaky logical cheating) will be measurable. For more details, see

https://math.stackexchange.com/questions/1847052/most-functions-are-measurable

1.1 Sigma-Algebras and Properties

Since the axiom of choice is the source of the problem of unmeasurability in the previous problems, we start by defining a space on which we avoid unmeasurable sets, namely, we want to limit ourselves to *countable* choices and constructions. The definition should be pretty general, as it should accommodate measures many in more general spaces (finite, continuous, $\mathbb{R}^\infty$, probability spaces, function spaces, and so forth). It should also be "arithmetically expandable", as in we should be able to measure smaller parts of the space (perhaps in more convenient orientations) and add them all up. Breaking up a problem into smaller pieces that could be put together is a fundamental way the core principle of an algebra, hence we define the notion of (analytic) *algebra*[1]:

Definition 1.1.1: Algebra and σ-Algebra

Let X be an arbitrary set. Then let $\mathcal{A} \subseteq P(X)$ be a non-empty collection of subsets such that

1. for any finite set of sets $A_1, A_2, ..., A_n \in \mathcal{A}$, $\bigcup_i A_i \in \mathcal{A}$
2. For any $A \in \mathcal{A}$, $A^c \in \mathcal{A}$.

Then $\mathcal{A}$ is called an *algebra*. If it is closed under countable union, we call it a *σ-algebra* (or *σ-field*).

The σ-algebra will be our natural space in which we will show in the next section we can define a "measure function" without the previous contradiction.

I like to think of the finite closure as a more general way of saying that a space is closed under the operation $\cup$. Closure under compliment insures the "opposite" property of what A represents must always be within our space. Adding the fact that it's closed under countable union brings us further way from the realm of algebra, and hence it's a σ-algebra. The term sigma is very common to mean countable, for example, a space σ-compactness if it is the countable union of compact subspaces.

[1] Note to be confused with an algebra from [12]

Definition 1.1.2: Measurable Space

Let X be a set and $\mathcal{M} \subseteq P(X)$ be a σ-algebra. Then $(X, \mathcal{M})$ is called a *measurable space* and a set in $\mathcal{M}$ is called a *measurable set*

Proposition 1.1.3: Properties of σ-algebra

Some immediate properties of a σ-algebra:

1. It is closed under countable intersection, since $\bigcap_i X_i = \left(\bigcup_i X\right)^c$, and by difference, since $X \setminus Y = X \cap Y^c$.
2. $\emptyset$ and $X \in \mathcal{A}$, since for any $E \in \mathcal{A}$, $E \cap E^c = \emptyset \in \mathcal{A}$ and $E \cup E^c = X \in \mathcal{A}$
3. Any countable collection of subsets $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ can be turned into a countable disjoint union, where
$$F_k = E_k \setminus \left(\bigcup_{i=1}^{k-1} E_i\right) = E_k \cap \left(\bigcup_{i=1}^{k-1} E_i\right)^c$$
and by what we've just shown, every $F_k \in \mathcal{A}$, and by construction $\cup_i E_i = \cup_i F_i$. We will use this technique often very soon (note that not all sets are nonempty, that proof requires more set-theory manipulation).
4. Given any family of σ-algebras on X, their intersection is also a σ-algebra. Because of this, given $Y \subseteq X$, there is always a unique smallest σ-algebra $\mathcal{M}(Y)$ that contains Y (there is at least always one, $P(Y)$, and so this concept is well-defined). $\mathcal{M}(Y)$ is called the σ-algebra generated by Y.

A lemma we'll point out right now to simplify future proofs is the following:

Lemma 1.1.4

If $Y \subseteq \mathcal{M}(X)$, then $\mathcal{M}(Y) \subseteq \mathcal{M}(X)$

***Proof*:**
Since $\mathcal{M}(X)$ is a σ-algebra that contains Y, then it will contain all countable unions and compliment of Y, and hence it will contain $\mathcal{M}(Y)$ (i.e. proposition 1.1.3[4]).

Example 1.1.5: σ-Algebras

1. If X is any set, then $\{\emptyset, X\}$ and $P(X)$ are trivially σ-algebras. Once we define measurable function, these will be the initial and final object in measurable spaces.
2. If X is any set, then $P(X)$ is also a σ-algebra. Usually, we will use proposition 1.1.3[5] and other tools to construct a smaller σ-algebra to not work with $P(X)$ since it's too big for most cases.
3. If X is an uncountable set, define
$$\mathcal{A} = \{E \subseteq X \mid E \text{ is countable or } E^c \text{ is countable}\}$$

this set is called the σ-algebra of countable and co-countable sets. It is useful when we want to make sure that the countable union of sets is still countable (or it's compliment is) [a]

[a] Sneakily, we use the axiom of choice for countably many choices to prove this is true

Another common σ-algebra generated on a set X is related to a topology on X, that is, if $\mathcal{T}(X)$ is a topology, of X, we will define the following σ-algebra:

Definition 1.1.6: Borel σ-Algebra

Let X be a set and $\mathcal{T}(X)$ be a topology on X. Define the *Borel σ-algebra* to be

$$\mathcal{B}_X := \mathcal{M}(\{U \subseteq X \mid U \text{ is an open set contained in } X\}$$

The elements of $\mathcal{B}_X$ are called *Borel sets*

The set $\mathcal{B}_X$ will include the open sets, closed sets, countable union and intersection of closed sets, and the countable union of countable intersections of open/closed sets, et cetera.

Since the countable intersection of open sets and countable union of closed sets are not necessarily open or closed sets respectively[2], but are members in $\mathcal{B}_X$, we will give them names: A countable intersection of open sets is called a G_δ set (G for *Gebeit* in German), and a countable union of closed sets is called a F_σ set (F for *fermé* in french). The σ is for *Summe* (sum in German) since it's a union and the δ is for *Durchschnitt* (intersection in German) since it's an intersection. We can continue this pattern and define $G_{\delta\sigma}$ to be the countable union of countable intersection of open sets, $F_{\sigma\delta}$ to be the countable intersection of countable union, and so and so on and so forth for any length of interchanging σ's and δ's. In general, this "chain" of unions and intersections does not need to terminate, and a $F_{\sigma\delta}$ (as a simple example for a chain of length 2, $\mathbb{R} \setminus \mathbb{Q}$ is a $F_{\sigma\delta}$ set, but not a F_σ set; use Baire's Categories theorem). There will be cases in which it will terminate (in fact, the majority of measure's we'll work with will have this terminate after at just F_σ or G_δ "up to a zero set", see theorem 1.4.5, and example ref:HERE for a measure that does not terminate)

One of the most common Borel sets we will be working with is $\mathcal{B}_\mathbb{R}$ with the usual euclidean topology on $\mathbb{R}$. It is useful to have an of different generating sets for $\mathcal{B}_\mathbb{R}$:

Proposition 1.1.7: Generators For $\mathcal{B}_X$

Let $\mathbb{R}$ be the real numbers and $\mathcal{B}_\mathbb{R}$ be the Borel σ-aglebra on $\mathbb{R}$ given the standard euclidean topology $\mathcal{T}(\mathbb{R})$. Then the following generate $\mathcal{B}_\mathbb{R}$:

1. The set of open intervals
2. the set of closed intervals
3. the set of half-open intervals
4. the set of open rays
5. the set of closed rays

[2] it is possible that the countable intersection/union of open/closed sets are open/closed, and so the word "necessarily" is required

***Proof*:**
These proofs are quite easy to prove, and should be done as an exercise. A fact that you might need to recall is that every open set in $\mathbb{R}$ can be written as a countable union of open intervals (hint: $\mathbb{R}$ has a countable basis).

Products and σ-Algebras

Recall that the product between two sets is defined as $A \times B := \{(a, b) \mid a \in A, b \in B\}$, or more generally $\{X_\alpha\}_{\alpha \in A}$ for a collection of non-empty sets X_α[3], with $X = \prod_{\alpha \in A} X_{\alpha},$. As we know from Category Theory, the product comes equipped with set functions π_α that satisfy the universal property of products (or more generally, this is a limit over the discrete category). Though we have yet to give a category where σ-algebras are the objects (we need to define what the morphisms are[4]), we can already construct our set $\prod_\alpha X_\alpha$ that will be the product of the X_α in the category of measurable spaces.

Definition 1.1.8: Product σ-Algebra

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of non-empty sets and $X = \prod_{\alpha \in A} X_\alpha$. Let

$$\bigotimes_{\alpha \in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in M_\alpha,\ \alpha \in A\})$$

to be the *product σ-Algebra* on $\{M_\alpha\}_{\alpha \in A}$, i.e., it is the σ-algebra generated by all of the pre-images of the sets in M_α.

It is not difficult to check that the product σ-algebra is indeed a σ-algebra. If $|A|$ is countable we usually write $\bigotimes_{i=1}^\infty M_i$, and if $|A|$ is finite, we can write $\bigotimes_{i=1}^n M_i$ or $M_1 \bigotimes M_2 \bigotimes \cdots \bigotimes M_n$. Notice that since we allow for countable unions, it is not so easy to represent $\bigotimes_{\alpha \in A} M_\alpha$ as the product of the elements of M_α, (just like for Groups in the category of **Grp**), and hence why we do not use the $\prod$ or $\bigoplus$ notation. On the other hand, the fact that only a countable union of elements is permitted will still limit the product in the number of non-trivial components, hence justifying the $\bigotimes$ notation instead of $\prod$. In fact, it is the same as $\bigoplus$, but instead of “all but finitely many”, we have “all but countably many”:

Proposition 1.1.9: Countable Product σ-algebra

Let $M_n = \mathcal{M}(\mathcal{E}_n)$ be a σ-algebra for $n \in \mathbb{N}$ where $\mathcal{E}_n \subseteq P(X_n)$. If $X_n \in \mathcal{E}_n$, then

$$\bigotimes_{n=1}^\infty M_n = \mathcal{M}\left(\left\{\prod_{i=1}^n E_i \,\middle|\, E_i \in M_i\right\}\right)$$

[3] If they were empty, then the product would be empty
[4] this is done in chapter 2

***Proof*:**
The $\supseteq$ direction is easy (Take $\bigcap_{i=1}^{n} \pi_i(E_i)$ for appropriate values). For $\subseteq$, Since $X_i \in M_i$, we have that every element is inside the right hand side. Make sure you see why $X_n \in \mathcal{E}_n$ makes a difference. By lemma 1.1.4, the result follows.

We can further simplify our representation of $\bigotimes_{\alpha \in S} M_\alpha$ given a generating set for each M_α (such a set always exists, namely M_α itself is always a generating set)

Proposition 1.1.10: Generating set for Product σ-Algebra

Let $\{M_\alpha\}_{\alpha \in A}$ be a non-empty collection of σ-algebras, and let $\mathcal{E}_\alpha \subseteq P(X_\alpha)$ be a generating set for M_α. If $\mathcal{F} = \cup_{\alpha \in A} \mathcal{E}_\alpha$, then:

$$\bigotimes_{\alpha \in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha,\ \alpha \in A\}) = \mathcal{M}(\mathcal{F})$$

Furthermore, if $|A|$ is countable and $X_\alpha \in \mathcal{E}_\alpha$ for each $\alpha \in A$, then

$$\bigotimes_{\alpha \in A} M_\alpha = \mathcal{M}\left(\left\{ \prod_{\alpha \in A} E_\alpha \;\middle|\; E_\alpha \in \mathcal{E}_\alpha,\ \alpha \in A \right\}\right)$$

***Proof*:**
Dealing with the first case, the $\supseteq$ is clear since $E_\alpha \in \mathcal{E}_\alpha$ implies $E_\alpha \in M_\alpha$, so $E_\alpha \in \bigotimes_{\alpha \in A} M_\alpha$. For the $\subseteq$ direction, If $E_\alpha \notin \mathcal{E}_\alpha$, then since $\mathcal{M}(\mathcal{E}_\alpha) = M_\alpha$, some countable unions and intersections of elements of $\mathcal{E}_\alpha$ equal E_α, proving the $\subseteq$ direction.

The countable case follows proposition 1.1.9

Borel Spaces and Products

A topological space of much study in Analysis is metric spaces. As a reminder, if $\{X_\alpha\}_{i=1}^{n}$ is a collection of metric spaces, then $X = \prod_{i=1}^{n} X_i$ has an induced product metric structure. As was shown in [9, Metric Product Space] for the max metric, and as the pointwise bounds $d_{\max} \leq d_2 \leq d_1 \leq 2\, d_{\max}$ extend to the other two, the following metrics on a product space all define the same product topology (for notational simplicity, let's work with two metric spaces X and Y and consider $X \times Y$):

1. $d((x_1, x_2), (y_1, y_2)) = \max\{d_X(x_1, y_1), d_Y(x_2, y_2)\}$
2. $d((x_1, x_2), (y_1, y_2)) = \sqrt{d_X(x_1, y_1)^2 + d_Y(x_2, y_2)^2}$
3. $d((x_1, x_2), (y_1, y_2)) = d_X(x_1, y_1) + d_Y(x_2, y_2)$

We can ask about establishing Borel sets $\mathcal{B}_X$ on $X = \prod_{i=1}^{n} X_i$ with X having the product measure. It is tempting to say that this naturally splits into $\bigotimes \mathcal{B}_{X_i}$, however, this is not always the case! This is mainly due to the fact that not all topological spaces have a countable basis, and since we're allowing countable intersection of open sets, this will actually create more sets than the if we take the product of the Borel spaces:

Proposition 1.1.11: Borel sets on product Metric Space

Let $X_1, ..., X_n$ be metric spaces, and $X = \prod_{i=1}^n X_i$ be the product metric space. Then

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$$

If the spaces X_i are separable (i.e. has a countable dense subset, and since it's a metric space it means there is a countable basis), then

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} = \mathcal{B}_X$$

***Proof*:**
The proof of the $\subseteq$ direction rather simple. By proposition 1.1.10, $\bigotimes_{i=1}^n \mathcal{B}_{X_i}$ is generated by $\mathcal{E} = \left\{\pi_i^{-1}(U_j) \;\middle|\; U_j \text{ is open in } X_j,\ 1 \leq j \leq n\right\}$. Since the sets of $\mathcal{E}$ are open in X, by lemma 1.1.4, $\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$.
For the converse, since each X_j is separable, each have a countable dense subset. Since X_j is a metric space, we can form a countable basis with open balls with rational radii around each points in the countable dense subsets.

As a consequence, $\bigotimes_{i=1}^n \mathcal{B}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R}^n}$, since $\mathbb{R}$ has a countable dense subset.

Elementary Sets

This is a technical result needed for later (maybe introduce it as exercises?)

Definition 1.1.12: Elementary Family

Let X be a set and $\mathcal{E} \subseteq P(X)$. Then $\mathcal{E}$ is called an *elementary family* if

1. $\emptyset \in \mathcal{E}$
2. If $E, F \in \mathcal{E}$ then $E \cap F \in \mathcal{E}$
3. If $E \in \mathcal{E}$, then E^c is a finite disjoint union of elements of $\mathcal{E}$

Proposition 1.1.13: Elementary Families and Algebra

Let $\mathcal{E}$ be an elementary family. Then the collection of finite disjoint unions of element of $\mathcal{E}$ forms an algebra

***Proof*:**
(TBD) Let $U, V \in \mathcal{E}$, and consider $V^c = \coprod_{j=1}^{J} F_j$ (for disjoint $F_j \in \mathcal{E}$). Then

$$U \setminus V = U \cap V^c = \coprod_{j=1}^{J} (U \cap F_j)$$

notice that each $U \cap F_j \in \mathcal{E}$. So

$$U \cup V = (U \setminus V) \cup V = \coprod_{j=1}^{J} (U \cap F_j) \coprod V$$

where $\coprod_{j=1}^{J} (U \cap F_j) \in A$
Continuing inductively, can check Folland for proof, or just do it yourself.

(also, remember that separable means countable dense subset) Once the course starts

1. There is an interseting example ith logical statements!!

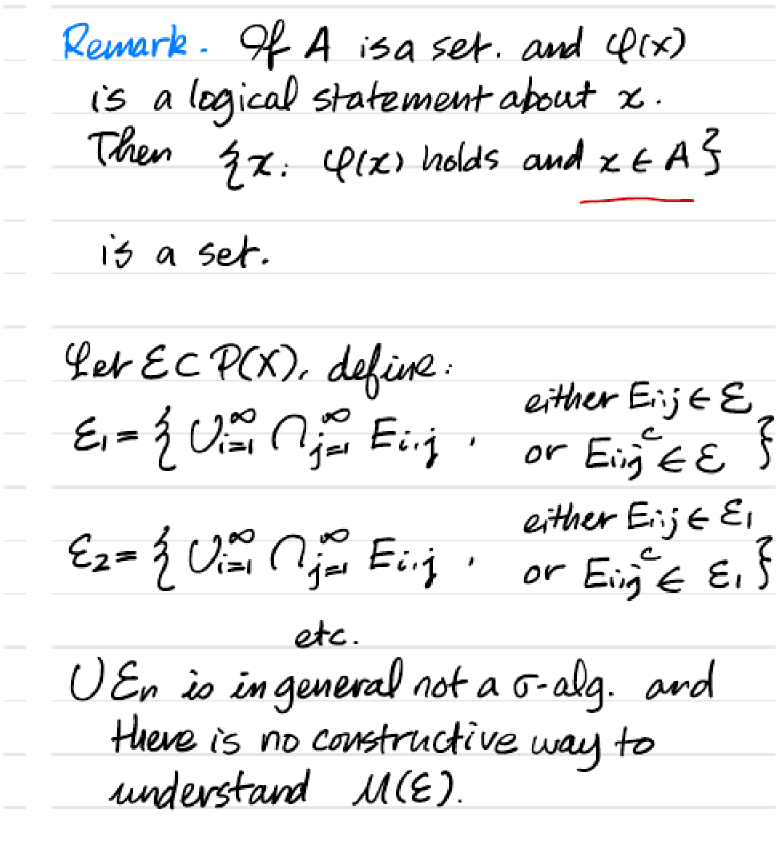


Figure 1.1: fascinating!

1.2 Measure

We can finally define the function which will give us an idea of the "size" of sets. The key idea behind the idea of notion is that the "bigger" a set is, the "larger" the the area. In other words, if μ is a measure on X, then (abusing notation a bit):

$$\mu(A \subseteq B) \ \Leftrightarrow \ \mu(A) \leq \mu(B) \tag{1.1}$$

In otherwords, a measure is an order-homomorphism from $(\mathcal{M}, \subseteq)$ to $([0, \infty], \leq)$. This also means that we want the codomain of a measure to be *totally ordered*, and since we want limits to work, it also have to be *complete*. Furthermore, if we have a countable number of disjoint areas, it should possible to measure them all separately and or all together at once and get the same measure (a sort of "invariance of space"). In fact, if we just assume $\mu(\emptyset) = 0$ (i.e. "nothing" should have no area of measure), then using this "invariance of space" notion, we can deduce equation (1.1), as we'll show in the next proposition, and hence we would want the codomain of a measure to be $[0, \infty]$ (more generally, a minimal complete totally ordered set embeds into $[0, \infty]$, see ref:HERE, showing this is a natural choice). We thus define a measure on these intuitions:

Definition 1.2.1: Measure

Let X be a set along with a σ-algebra $\mathcal{M}$. A function $\mu : \mathcal{M} \to [0, \infty]$ is called a *measure* on $\mathcal{M}$ (or $(X, \mathcal{M})$ or even X if the measure is clear from context) if

1. $\mu(\emptyset) = 0$
2. *(Countably Additivity)* If $\{E_i\}_{i=1}^{\infty}$ is a sequence of disjoint sets, then $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$

Given that $(X, \mathcal{M})$ is a measurable space, along with a μ we can say the space has a measure:

Definition 1.2.2: Measure Space

Let $(X, \mathcal{M})$ be a measurable space. Then if μ is a measure on X, then $(X, \mathcal{M}, \mu)$ is a *measure space*

If the 2nd condition of a measure was limited to a finite sequence (or more generally, all but countably many of the sets in the sequence or nonempty), then we would say that it respects finite additivity. If μ respects finite but not countable additivity, μ is called a *finite additive measure*.

Most measure's that we will work with have some finiteness condition associated with it:

Definition 1.2.3: Finiteness Condition on μ

Let $(X, \mathcal{M}, \mu)$ be a measure space.

1. If $m(X) < \infty$, then μ is called a *finite measure*
2. if there exists sets $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ such that $\bigcup_i^{\infty} E_i = X$ and $\mu(E_i) < \infty$ for all E_i, then μ is called a *σ-finite measure*. More generally, if $E = \bigcup_{i=1}^{\infty} E_i$ for some sequence of E_i, then E is called σ-finite on μ (some say that E is said to be σ-finite on μ)
3. if for for all $E \in \mathcal{M}$ such that $\mu(E) = \infty$, there exists a $D \subseteq E$, $D \in \mathcal{M}$ such that $\mu(D) < \infty$, then μ is called a *semi-finite measure*

If $\mu(X) < \infty$, then this implies for all $S \in \mathcal{M}$, $\mu(S) < \infty$ (as can quickly be checked). Naturally, all finite measure's are σ-finite, and all σ-finite measure are semi-finite, but the converses are not true (as we'll explore in the following examples). Note too that most measure's we will be working over will be σ-finite (think of $\mathbb{R}$ with the intuitive notion of length of an interval).

Example 1.2.4: Measures

1. Let X be nonempty, $\mathcal{M} = P(X)$, and $f : X \to [0, \infty]$. Then we can define

$$\mu_f(E) = \sum_{e \in E} f(e)$$

where the sum can be infinite (the domain of f is the positive part of the extended real numbers). Notice that f is semifinite if and only if $f(x) < \infty$ for all $x \in X$, and μ is σ-finite if and only if μ is semi-finite and $\{x \mid f(x) > 0\}$ if countable.

If $f(x) = 1$ for all $x \in X$, then μ_f is called the *counting measure*. On the other hand, if for some $x_0 \in X$, $f(x_0) = 1$ and $f(x) = 0$ if $x \neq x_0$, then μ_f is called a *point mass* or *Dirac measure* at x_0

2. Let X be an uncountable set an $\mathcal{M}$ be the σ-algebra of countable or co-countable sets (recall example 1.1.5). Define
$$\mu(E) = \begin{cases} 0 & E \text{ is countable} \\ 1 & E \text{ is co-countable} \end{cases}$$
Then μ is a measure!

3. An example of an finite additive measure that is not a measure is the following: let X be some infinite set and $\mathcal{M} = P(X)$. Define μ to be
$$\mu(E) = \begin{cases} 0 & E \text{ is finite} \\ \infty & E \text{ is infinite} \end{cases}$$
Then μ is an finite additive measure, but not a measure

Proposition 1.2.5: Properties of Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then

1. *(Monotonicity)* Let $E, F \in \mathcal{M}$ and $E \subseteq F$. Then $\mu(E) \leq \mu(F)$
2. *(Subadditivity)* if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$, then $\mu\left(\cup_i \mu(E_i)\right) \leq \sum_i \mu(E_i)$
3. *(Continuity from Below)* if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$, and $E_1 \subseteq E_2 \subseteq \cdots$, then
$$\mu\left(\cup_i E_i\right) = \lim_{i \to \infty} \mu(E_i)$$
4. *(Continuity from Above)*: if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ and $E_1 \supseteq E_2 \supseteq$ *and* $\mu(E_1) < \infty$. Then
$$\mu\left(\cap_i E_i\right) = \lim_{i \to \infty} \mu(E_i)$$

***Proof*:**
here. Do them as exercises! I also like this solution by Hytham Farah for continuity from above:

$$\begin{aligned}
\mu\Big(\bigcup_{n\geq 1} F_n\Big) &= \mu\Big(\bigcup_{n\geq 1} E_n\Big) \\
&= \mu\Big(\big(\bigcap_{n\geq 1} E_n^c\big)^c\Big) \\
&= \mu(X) - \mu\Big(\bigcap_{n\geq 1} E_n^c\Big) \\
&= \mu(X) - \lim_{n\to\infty} \mu(E_n^c) \\
&= \mu(X) - \lim_{n\to\infty} \big(\mu(X) - \mu(E_n)\big) \\
&= \lim_{n\to\infty} \mu(E_n) \\
&= \sum_{i=1}^{\infty} F_n.
\end{aligned}$$

Notice that for (d), we can instead limit it to $\mu(E_i) < \infty$ for some i, and the proof remains the same. We still require that some i exists such that $\mu(E_i) < \infty$, for it's possible that all $\mu(E_i)$ is infinite, but $\mu(\cap_i E_i) < \infty$, for example take the measurable space $(\mathbb{N}, P(\mathbb{N}))$ with the counting measure. Then the sets

$$E_i = \{n \in \mathbb{N} \mid n \geq i\}$$

then every E_i is infinite but their intersection is empty and thus has measure 0 in the counting measure.

zero measure sets

Frequently, there are sets in our measure that have a "trivial size" in the current measure. This is akin to when we integrate, the border has no effect on the value of the integral. Such sets are said to be zero sets or null set:

Definition 1.2.6: Null Set

Let $(X, \mathcal{M}, \mu)$ be a measure space and $E \in \mathcal{M}$. Then if $\mu(E) = 0$, E is said to be a *null set* or *zero set*.

By countably subadditivity, the countable union of null sets is a null set, hence null sets are "invariant" under countable union. Due to this, we can work with measurable sets in $\mathcal{M}$ *up to null sets*. We shall often state in proofs that a statement is true *almost everywhere* (often abbreviated to a.e.) if it is true on the sets except for the null sets.

If we take $E \in \mathcal{M}$ such that $\mu(E) = 0$, then by monotonicity, if $F \subseteq E$ the $\mu(F) = 0$, *provided that* $F \in \mathcal{M}$. This is not always the case:

Example 1.2.7: Not Complete Space

Partition $[0, 1]$ into 10 (really any number) of subintervals, $\left[\frac{i}{n}, \frac{i+1}{n}\right]$ for $0 \leq i \leq 9$, and let each of their measures be 0. Take $\mathcal{M}$ to be the σ-algebra defined on this set. Then is clearly an uncountable number of sets that do not have a measure defined on them that are subset of zero-measure sets.

However, adding all of these F's that are subsets of null sets does not break our measure. A measure who's domain includes all subsets of null sets is a *complete measure* (i.e. the σ-algebra is closed under subsets of null sets). Proving that this is still a measure is useful to eliminate future "trivial obstructions" to our proofs:

Theorem 1.2.8: Completing a Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} \mid \mu(N) = 0\}$ and

$$\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subseteq N \text{ for some } N \in \mathcal{N}\}$$

Then $\overline{\mathcal{M}}$ is also a σ-algebra, and there is a *unique* extension $\overline{\mu}$ of μ that is a complete measure on $\overline{\mathcal{M}}$ that restricts to the measure of μ on $\mathcal{M}$

***Proof*:**

First of all, since $\mathcal{M}$ and $\mathcal{N}$ are both closed under countable union, so $\overline{\mathcal{M}}$ is closed under countable union. To show it's closed under compliments, consider some $E \cup F \in \overline{\mathcal{M}}$ ($E \in \mathcal{M}$, $F \subseteq N \in \mathcal{N}$). Without loss of generality, assume $E \cap N = \emptyset$ (if not, take $F \setminus E$ and $N \setminus E$ and continue the proof, then this will have he same value as our original $E \cap N$). Then

$$E \cup F = (E \cup N) \cap (N^c \cup F)$$

So we get

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F)$$

Since $E \cup N \in \mathcal{M}$, $(E \cup N)^c \in \mathcal{M}$. Since $N \setminus F \in \mathcal{N}$, Since it's closed under union, $(E \cup F)^c \in \overline{\mathcal{M}}$. Thus $\overline{\mathcal{M}}$ is a σ-algebra.

Since $\overline{\mathcal{M}}$ is a σ-algebra, we can try to define a measure, and indeed

$$\overline{\mu}(E \cup F) = \mu(E)$$

is a well-defined measure. If

$$E_1 \cup F_1 = E_2 \cup F_2$$

Then $E_1 \subseteq E_2 \cup F_2$, and so by monotonicity:

$$\mu(E_1) \leq \mu(E_2 \cup F_2) \leq (E_2 \cup N_2) = \mu(E_2) + \mu(N_2) = \mu(E_2)$$

Similarly, $\mu(E_1) \geq \mu(E_2)$. Thus $\overline{\mu}(E_1 \cup F_1) = \overline{\mu}(E_2 \cup F_2)$, and so $\overline{\mu}$ is well-defined. Since $\overline{\mu}$ has all subsets of zero-sets in its domain, it is a complete measure.

Furthermore, if $\overline{\nu}$ was another complete measure on $\mathcal{M}$, then $\overline{\nu} = \overline{\mu}$, (since if it were different, then some subset of a zero-set must have non-zero measure, a contradiction to monotonicity) showing that $\overline{\mu}$ is unique.

Definition 1.2.9: Completion of Measure

Let $(X, \mathcal{M}, \mu)$ be a measurable space. Then $\overline{\mathcal{M}}$ and $\overline{\mu}$ is called the *completion of* $\mathcal{M}$ *with respect to* μ and the *completion of* μ respectively

1.3 Outer Measure

We now move on to constructing a particular type of measure which is draws inspiration from how we measured area's when we calculated Riemann Integrals. In the abstract, when we tried defining a Jordan measure (or content[5]), we approximated the inner area of a shape E in $\mathbb{R}^n$ by the limit of sums of rectangles contained in E, and the outer-area of E in $\mathbb{R}^n$ by the limit of the sums of rectangles containing E. The two numbers are called the inner and outer area of E. If the two matched, we would call the result the area of E. This intuition can be generalized to the countable case, except we can replace rectangles with some other "basic sets" that we might want that form an algebra. Furthermore, the intuition of having the "inner area" and "outer-area' is actually a bit restricting to work with in the general measure case, and so that condition will be replaced with a different measureability condition we will soon introduce.

For the sake of generality, we will start by definition a more general notion of the measure called the outer-measure, and then build-up on how form the outer-measure, we can get a measure that satisfies this generalization of Riemann integration:

Definition 1.3.1: Outer Measure

A function $\mu^* : P(X) \to [0, \infty]$ is called an *outer measure* if

1. $\mu^*(0) = \emptyset$
2. if $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$
3. $\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (even if the sets are disjoint!)[a]

[a]We will return to see when equality holds in definition 1.3.3

Note that we have not yet proven that an outer-measure is a measure; in particular notice that the domain is the powerset. The name "outer-measure" gets its name from the following general construction:

Proposition 1.3.2: Outer Measure Construction

Let $\mathcal{E} \subseteq P(X)$ where $\emptyset, X \in \mathcal{E}$, and let $\rho : \mathcal{E} \to [0, 1]$ where $\rho(\emptyset) = 0$. For any $A \subseteq X$, define

$$\mu^*(A) = \inf\left\{ \sum_{1}^{\infty} \rho(E_j) \;\middle|\; E_j \in \mathcal{E} \text{ and } A \subseteq \bigcup_{1}^{\infty} E_j \right\}$$

Then μ^* is an outer measure

***Proof*:**
First, this measure is well-defined since for any $A \subseteq X$, If we take $E_i = X$, then $A \subseteq \cup_i E_i$. Next, we verify the 3 properties

1. Clearly, $\mu^*(\emptyset) = 0$ by taking $E_i = \emptyset$ for all i

[5]since "Jordan measures" only allow finite additivity, some don't like to use the word measure and opt for the word content. In this book, we would say *finite additive measure* as has already been stated earlier

2. If $A \subseteq B$, then let $\mathcal{A}$ and $\mathcal{B}$ sets that correspond to $\mu^*(A)$ and $\mu^*(B)$. Then since all of the elements of $\mathcal{A} \supseteq \mathcal{B}$, by the property of the infimum $\inf \mathcal{A} \leq \inf \mathcal{B}$, which transferring notations gives us $\mu^*(A) \leq \mu^*(B)$

3. Let $\{E_i\}_{i=1}^{\infty} \subseteq P(X)$ be a collection of (not necessarily disjoint) subsets. Let $\epsilon > 0$. Consider $\mu^*(E_k)$ for some $E_k \in \{E_i\}_{i=1}^{\infty}$. By the property of the infimum, there must exist some $\{F_j^k\}_{j=1}^{\infty}$ such that $E_i \subseteq \cup_j F_j^k$ where $\sum_{j=1}^{\infty} \rho(F_j^k) \leq \mu^*(E_k) + \epsilon 2^{-k}$.

   Now, let $E = \cup_i E_i$. By construction, $E \subseteq \cup_{i,k=1}^{\infty} F_j^k$, and

   $$\sum_{j,k} \rho(F_j^k) \leq \mu^*(A) + \epsilon$$

   Since ϵ can be arbitrarily small, this completes the proof.

As mentioned before, an outer-measure is not necessarily a measure. What we will do to allow this to be a measure is single out sets that have the following property

Definition 1.3.3: μ^*-measureable

Let $E \subseteq X$ be a set and μ^* be an outer measure on X. Then E is called μ^**-measurable* if

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

In other word, if E can naturally "split" a set S into a part that's "inside" E or "outside" E. The set $S \subseteq X$ is called the *test set*.

By definition of the outer measure, since $S = (S \cap E) \cup (S \cap E^c)$:

$$\mu^*(S) \leq \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

Therefore, what is more important is to check the $\geq$ direction. Furthermore, if $\mu^*(A) = \infty$. Then $\geq$ holds trivially, so to show a set is μ^* measurable, it is equivalent to show that for all finite measure sets A, $\mu^*(S) \geq \mu(S \cap E) + \mu(S \cap E^c)$.

As mentioned in the definition, this gives us a sense of being able to measure the "inside" and "outside" of a set. If $E \subseteq A$, that is the μ^*-measurable set belong in a better set, then $\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \setminus E)$ which can be thought of as saying that we are measuring S from the "inside" of E and the "outisde" of E, and getting the same result. In fact, if $\mu^*(X) < \infty$, then we can define $\mu_*(E) = \mu^*(X) - \mu(E^c)$, and have that E is μ^*-measurable if and only if $\mu_*(E) = \mu^*(E)$ aligning with our intuition for inside equalling outside (see exercise ref:HERE). We do not do this method since it requires some finiteness condition on μ. We can conversely define μ_* in terms of supremums and need that E contains the collection of $\cup_i A_i$, and that the elements of $\{A_i\}_{i=1}^{\infty}$ must be disjoint (see exercise ref:HERE).

The big thing that is in the way of saying that an outer measure is a measure is that its domain is $P(X)$, which can cause problems when X is uncountable. However, all sets that satisfy μ^*-measureability turn out form a σ-algebra and make μ^* a complete measure!

Theorem 1.3.4: Carathéodory's Theorem

Let μ^* be an outer measure on X, and let $\mathcal{M}$ be the collection of μ^*-measurable sets. Then $\mathcal{M}$ forms a σ-algebra, and the restriction of μ^* to $\mathcal{M}$ is a complete measure

Notice that in the process of defining this measure, we started with the powerset, and then simply choose all sets that restrict (which we will show forms a σ-algebra). In this way proving a function is an outer-measure is a little less finicky since we don't need to be so careful in defining over which σ-algebra we want to define it over: the σ-algebra comes with μ^*

***Proof*:**
Let $\mathcal{M}$ be the collection of μ^*-measurable sets. Clearly, $\emptyset \in \mathcal{M}$ since for all $S \subseteq X$:

$$\mu^*(S \cap \emptyset) + \mu^*(S \cap X) = 0 + \mu^*(S)$$

implying μ^*-measureability. Next, $\mathcal{M}$ is closed under compliments, since if $E \in \mathcal{M}$, then

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \mu^*(S \cap E^c) + \mu^*(S \cap E)$$

which also shows that $X \in \mathcal{M}$. Finally, we need to show that $\mathcal{M}$ is closed under countable unions. We'll start by showing finite unions, which then means it suffices to show the union of two sets is closed (due to induction). Let $A, B \in \mathcal{M}$. As we remarked, it suffices to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. Since A and B are μ^* measurable, we have

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) && E \text{ is the test set} \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) \\ &\quad + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) && E \cap A \text{ and } E \cap A^c \text{ are the test sets} \end{aligned}$$

Then notice we can write

$$A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$$

Thus, using subadditivity, we can re-write 3 of the terms in the previous expression as:

$$\mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) \geq \mu^*(E \cap A \cup B)$$

Since $E \cap A^c \cap B^c = E \cap (A \cup B)^c$, we get:

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

showing that $\mathcal{M}$ is closed under finite union. Since $\mathcal{M}$ is closed under compliment and contains $\emptyset$, $\mathcal{M}$ is an algebra. Even better, If we let A and B be disjoint, and let $A \cup B$ be the test set, then since A is μ^* measurable:

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B)$$

showing that $\mu^*|_{\mathcal{M}}$ is a finitely additive measure.

Next, we must show that $\mathcal{M}$ is a σ-algebra, which as a consequence will show that $\mu^*|_{\mathcal{M}}$ is a measure. Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$. To simplify our task, re-construct $\{E_i\}_{i=1}^{\infty}$ to be the set of disjoint sets $\{D_i\}_{i=1}^{\infty}$. Let $B_n = \bigcup_{i=1}^{n} D_i$ and $B = \bigcup_{i=1}^{\infty} D_i$. Our goal is to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c)$. We will start by considering $\mu^*(E \cap B_n)$:

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap B_n \cap D_n) + \mu^*(E \cap B_n \cap D_n^c) && E \cap B_n \text{ is the test set} \\ &= \mu^*(E \cap D_n) + \mu^*(E \cap B_{n-1}) \end{aligned}$$

Notice that on the right hand side we have diminished to the case of $E \cap B_{n-1}$ from $E \cap B_n$. Using induction, we can see that the expression simplifies to

$$\mu^*(E \cap B_n) = \sum_{i=1}^{n} \mu^*(E \cap D_i)$$

Continuing, since B_n is measurable

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \overset{!}{\geq} \mu^*(E \cap B_n) + \mu^*(E \cap B^c) = \sum_{i=1}^{n} \mu^*(E \cap D_i) + \mu^*(E \cap B_n^c)$$

where $\overset{!}{\geq}$ comes from monotonicity after switching B_n^c to B^c. Since this equality holds true for all n, we get:

$$\begin{aligned} \mu^*(E) &\geq \sum_{i=1}^{\infty} \mu^*(E \cap D_i) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(\bigcup_i E \cap D_i\right) + \mu^*(E \cap B^c) && \text{subadditivity} \\ &= \mu^*(E \cap B) + \mu^*(E \cap B^c) && \text{by construction of } E \cap B \\ &\geq \mu^*(E) && \text{subadditivity} \end{aligned}$$

Since we got $\mu^*(E) \geq \mu^*(E)$, we see that in fact all inequalities are equalities, getting us that

$$\mu^*(E) = \mu^*(E \cap B) + \mu^*(E \cap B^c)$$

showing us that $B \in \mathcal{M}$. Furthermore, replacing $E = B$ in the previous proof shows us that

$$\mu^*(B) = \sum_{i=1}^{\infty} \mu^*(D_i)$$

giving us that μ^* is a measure on $\mathcal{M}$.

Finally, we must show that $\mathcal{M}$ is complete, that is, if $E \in \mathcal{M}$ such that $\mu^*(E) = 0$, then all subsets of E are in $\mathcal{M}$. Notice that since μ^* is defiend on all of $P(X)$, it is equivalent to show that if $\mu^*(A) = 0$, then $A \in \mathcal{M}$. Then for any test set E, by monotonicity:

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) = 0 + \mu^*(E \cap A^c) \leq \mu^*(E)$$

meaning equality holds, showing that $A \in \mathcal{M}$. Thus, $\mu^*|_{\mathcal{M}}$ is a complete measure, as we sought to show.

Carathéodory's Theorem is quite useful being able to state the existence of a measure given an outer-measure, but it's not good to construct a measure (for example, can you tell me a non-trivial element in $\mathcal{M}$ without directly appealing to the definition?). The following concept is used to more systematically construct the set $\mathcal{M}$ and the measure μ induced by μ^*:

Definition 1.3.5: Premeasure

Let $(X, \mathcal{A})$ be an algebra. Then μ_0 is called a *premeasure* if

1. $\mu_0(\emptyset) = 0$
2. **"Respects" countable union**: If $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ is a collection of disjoint such that $\cup_i A_i \in \mathcal{A}$, then $\mu_0(\cup_i A_i) = \sum_i \mu_0(A_i)$

Note that the premeasure is automatically finitely additive since we can let all but finitely many of the terms in (2) be $\emptyset$ (and hence are a stronger condition than finitely additive measure). However, it is not yet a measure since it's domain is not necessarily closed under countable union. However, if it is closed for countable union for a particular collection, that collection must satisfy the "countable" condition. Interestingly, by the definition of an algebra, this implies that the countable union can be represented as a finite union and intersection of elements from $\mathcal{A}$.

As before, we can define semifinite and σ-finite in a similar way. Using μ_0, we can define the following outer-measure:

$$\mu^* : P(X) \to [0, \infty] \quad \mu^*(E) = \inf \left\{ \sum_i^{\infty} \mu_0(A_i) \;\middle|\; A_i \in \mathcal{A},\ E \subseteq \cup_i A_i \right\}$$

Proposition 1.3.6: Premeasure To Measure

Let μ_0 be a premeasure on $\mathcal{A}$, and let μ^* be defined as above. Then

1. $\mu^*|_{\mathcal{A}} = \mu_0$
2. Every set in $\mathcal{A}$ is μ^*-measurable

Essentially, μ^* is an extension of μ_0 and the associated restriction of μ^* to a measure will contain $\mathcal{A}$

***Proof*:**

1. Let's show $\leq$ and $\geq$ for any $E \in \mathcal{A}$. The $\leq$ direction is obvious since $\mu^*(E) \leq \mu_0(E)$ since $E \subseteq \bigcup_{i=1}^{\infty} A_i$ where $A_1 = E$ and $A_i = \emptyset$ for $i > 1$, and so it is part of the set over which we take the infimum, and so the result *has* to be smaller.

   For the $\geq$ direction, let's say $E \subseteq \bigcup_{i=1}^{\infty} A_i$ for any $\{A_i\}_{i=1}^{\infty} \in \mathcal{A}$. Let $B_n = E \cap \left(A_n \setminus \cup_1^{n-1} A_i\right)$. Then the collection $\{B_i\}_{i=1}^{\infty}$ is a disjoint collection of members of $\mathcal{A}$ whose union is E. Thus, by the property of premeasure

   $$\mu_0(E) = \sum_1^{\infty} \mu_0(B_i) \overset{!}{\leq} \sum_1^{\infty} \mu_0(A_i)$$

   where the $\overset{!}{\leq}$ inequality comes from the fact that $\mu_0(B_i) \leq \mu_0(A_i)$. Since this is true for any arbitrary collection $\{A_i\}_{i=1}^{\infty}$, it follows that

$$\mu_0(E) \leq \mu^*(E)$$

Thus establishing equality

2. We need to show that for any $E \subseteq X$ and $A \in \mathcal{A}$, $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$. As mentioned earlier, it suffice to prove $\geq$ where $\mu^*(E) < \infty$.

   Let $\epsilon > 0$. By the property of the infimum, there exists a subset $\{B_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \cup_i B_i$ such that $\sum_i^{\infty} \mu_0(B_i) \leq \mu^*(E) + \epsilon$. By part (1), μ^* is [countably] additive on $\mathcal{A}$ (since it equals the pre-measure)

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{i=1}^{\infty} \mu_0(B_i \cap A) + \sum_{i=1}^{\infty} (B_i \cap A^c) && \text{additivity} \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) && \text{monotinicity} \end{aligned}$$

   and since ϵ was arbitrary, A must be μ^*-measurable, completing the proof

We can in fact be more precise with the measures that extend a premeasure:

Theorem 1.3.7: Premeasure Extensions

Let $\mathcal{A} \subseteq P(X)$ be an algebra, μ_0 a premeasure on $\mathcal{A}$, and $\mathcal{M} = \mathcal{M}(\mathcal{A})$ (i.e. $\mathcal{M}$ is the σ-algebra generated by $\mathcal{A}$). Let μ be the extension of μ_0 from proposition 1.3.6. Then if ν is another measure on $\mathcal{M}$ that extends μ_0, then

$$\nu(E) \leq \mu(E) \qquad \forall E \in \mathcal{M}$$

with equality when $\mu(E) < \infty$ (i.e., it possible that μ is infinite while ν is finite).
If μ_0 is σ-finite, then μ is the unique extension of μ_0 to a measure on $\mathcal{M}$

In other words, we can extend a pre-measure to a measure by knowing what it does on the algebra (which in most cases will be a collection of simpler sets, like intervals or rectangles), and then what happens on the rest of the measurable sets we use the infimum definition. If we define another extension of μ_0, then that measure will be *finer* than the μ given by μ^*, i.e, μ is the *coarsest* extension. If μ_0 is σ-finite, then this will in fact be the only extension of μ_0

***Proof*:**
Let's say ν is an extension of μ_0 so that $(X, \mathcal{M}(\mathcal{A}), \nu)$ is a measure space. To show that $\nu(E) \leq \mu(E)$, let's say $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \cup_i A_i$. Then since ν is a measure, it is sub-additive on non-disjoint sets, and so

$$\nu(E) \leq \sum_i \nu(A_i)$$

Since ν extends μ_0, by proposition 1.3.6 $\nu|_{\mathcal{A}} = \mu_0$, so

$$\nu(E) \leq \sum_i \nu(A_i) = \sum_i \mu_0(A_i)$$

Since $\{A_i\}_{i=1}^{\infty}$ was an arbitrary choice, this works for any collection, and so

$$\nu(E) \leq \mu(E) \quad [= \mu^*(E)]$$

We show that $\nu(E) = \mu(E)$ if $\mu(E) < \infty$. Let $A = \cup_i A_i$ for some collection $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$. Then

$$\nu(A) = \lim_{n\to\infty} \nu\left(\bigcup_{i=1}^{n}(A_i)\right) = \lim_{n\to\infty} \mu\left(\bigcup_{i=1}^{n}(A_i)\right) = \mu(A)$$

If μ is not σ-finite, then there can be more than one extension:

Example 1.3.8: 2 extensions of pre-measure

1. Let $\mathcal{A}$ be the collection of finite unions of the sets of the form $(a, b] \cap \mathbb{Q}$ with $-\infty \leq a \leq b \leq \infty$. Then $\mathcal{A}$ is an algebra on $\mathbb{Q}$

   ***Proof*:**
   Recall proposition 1.1.13 where we can construct an algebra using an elementary family. Thus, we will show that the collection $\mathcal{A}$ satisfies the 3 properties of an elementary family:

   (a) To get the empty-set in $\mathcal{A}$, we will understand the notion of "finite union" to also include "no union". Otherwise, since $\mathbb{Q}$ is dense in $\mathbb{R}$, all intervals $(a, b]$ intersected with $\mathbb{Q}$ is nonempty (by MAT157 knowledge). Since $a < b$, the singleton interval is not admissible, and so we will have to assume this "no union" property

   (b) Let $E, F \in \mathcal{A}$. We want to show that $E \cap F \in \mathcal{A}$. By definition, $E = (a, b] \cap \mathbb{Q}$ and $F = (c, d] \cap \mathbb{Q}$. Then by some basic set-theory properties, we know that the intersection of two left-open/right-closed intervals is again a left-open/right-closed interval or $\emptyset$. If their intersection is $\emptyset$, then we're done. If not, let $(e, f] = (a, b] \cap (c, d]$. Then $E \cap F = (e, f] \cap \mathbb{Q}$. Thus, $E \cap F \in \mathcal{A}$

   (c) Let $E \in \mathcal{A}$, so that $E = (a, b] \cap \mathbb{Q}$. Let $i \in \mathbb{N}$ $(0 \notin \mathbb{N})$ and:

   $$E_i = (a - i, a - (i + 1)] \cap \mathbb{Q} \quad F_i = (b + (i - 1), b + i] \cap \mathbb{Q}$$

   then each $E_i, F_i \in \mathcal{A}$, $E_i \cap E_j = F_i \cap F_j = \emptyset$ if $i \neq j$ and $E_i \cap F_j = \emptyset$ for all $i, j \in \mathbb{N}$. Let $E = \{E_i\}_{i=1}^{\infty}$ and $F = \{F_i\}_{i=1}^{\infty}$ and let $G = E \cup F$. Then

   $$\bigcup_{g\in G} G = E^c$$

   since G is a countable union of disjoint elements of $\mathcal{A}$, this completes the proof.

   Thus, by proposition 1.7, $\mathcal{A}$ is an algebra.

2. The σ-algebra generated by $\mathcal{A}$ is $P(\mathbb{Q})$

***Proof*:**
Let $x \in \mathbb{Q}$ Notice that

$$\{x\} = \bigcap_{k=1}^{\infty} \left(x - \frac{1}{k}, x\right] \cap \mathbb{Q}$$

hence, every singleton is inside the σ-algebra of $\mathbb{Q}$. Since every subset of $\mathbb{Q}$ is countable, every subset is the countable union of singletons. Since every singleton is inside the σ-algebra, we have that the σ-algebra is $P(\mathbb{Q})$.

3. Define μ_0 on $\mathcal{A}$ by $\mu_0(\emptyset) = 0$ and $\mu_0(A) = \infty$ for $A \neq \emptyset$. Then μ_0 is a premeasure on $\mathcal{A}$, and there is more than one measure on $P(\mathbb{Q})$ whose restriction to $\mathcal{A}$ is μ_0

***Proof*:**
μ_0 is trivially a pre-measure. By definition, $\mu_0(\emptyset) = 0$. For any nonempty $A \in \mathcal{A}$, we have $\mu_0(A) = \infty$. When checking countable additivity, unless all the sets are empty, we get that both sides are infinite.

Thus, we can define μ to be the μ^* induced by μ_0 restricted to all measurable sets. By construction, all sets are measurable. If a sets is non-empty, then it will always be covered by a nonempty set, which has measure ∞. Thus, If $E \in P(X)$ for any nonempty test set $S \subseteq \mathbb{Q}$

$$\infty = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \infty$$

and if $S = \emptyset$

$$0 = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = 0$$

Thus, $\mathcal{M}_\mu = P(\mathbb{Q})$.

We can define another extension of μ_0 using the counting measure, in particular, let μ_C be the counting measure on $P(\mathbb{Q})$. Then since every $A \in \mathcal{A}$ has countable cardinality

$$\mu_C(A) = \infty$$

Thus, $\mu_C|_{\mathcal{A}} = \mu_0$. However

$$\mu_C(\{1\}) = 1 \neq \infty = \mu(\{1\})$$

Thus, we have defined two separate extensions of μ_0.

1.4 Borel Measure on the Real Line

The full title of this section should really be "Borel Measures on the Real line defined through Distributions", but that title seems to cumbersome. Nevertheless, this is exactly what we will be doing in this section. We will extensively concentrating on these types of measures, since they are, in a sense, the focus of "real analysis". From a high-level perspective, the fact that we are studying "real analysis" justifies why we will take so much time understanding this measure. More concretely, these measure are pervasive in analysis in general: they are central to defining integration, to the study of Fourier series, to studying L^p spaces, to linking differentiation to integration, to generalizing the notation of integration of groups (in particular, it will make sense to integrate locally compact topological groups), and so much more. Another way of looking at it is that the euclidean topology is central to the study of many fields of mathematics and physics, and is the basis for our notion of manifolds (which are locally euclidean). Therefore, getting a sense of how to define notions of "size" in $\mathbb{R}$ (and then generalize to $\mathbb{R}^n$, $\mathbb{R}^\infty$, or manifolds) gives us a solid first step in gaining an intuition on measures.

One way to define a measure in $\mathbb{R}$ is by using our intuition of the fact that the length of (a, b) is $b - a$. This is a good intuition and will be the basis to the generalization of this idea we'll do in order to be able to define a larger class of measures based on this idea which will also have more uses outside of $\mathbb{R}$. Here is how we will motivate the construction of these measures: let μ be a finite measure on Borel sets (a measure define on Borel sets is called the *Borel Measure*). Define:

$$F(x) = \mu((\infty, x])$$

F is sometimes called the *distribution function* of μ (this vocabulary will come back often in probability). Intuitively, it shows the "accumulated size" of the space. Clearly, F is an increasing function, and is right-continuous since

$$(\infty, x] = \bigcap_k (\infty, x + 1/k]$$

or any decreasing function, $x_n \searrow x$ (i.e. limits are preserved from the right). Furthermore, if we take $a < b$, then $(-\infty, b] = (-\infty, a] \cup (a, b]$, thus

$$\mu((a, b]) = F(b) - F(a)$$

Notice how similar this is to the usual result from Riemann integration! In the following, instead of defining F in terms of μ, we will define μ from an increasing right-continuous function F by showing how F define's a pre-measure and using Carathéodory. The special case of $F(x) = x$ will give us our usual notion of length in $\mathbb{R}$.

Throughout this section, we will use sets of the form $(a, b]$ where $a < b$, along with the edge cases of (a, ∞) and $\emptyset$. Let all intervals of this type be called h-intervals (h for half open). The union or compliment of two h-intervals are h-intervals (or two disjoint h-intervals) and that h-intervals are closed under finite operations, and so h-intervals from a algebra $\mathcal{A}$. By proposition 1.1.7, the σ-algebra by the set $\mathcal{A}$ is $\mathcal{B}_\mathbb{R}$. Using this, we'll first show that we can define a premeasure μ based off of an F:

Proposition 1.4.1: Distribution and Premeasure

Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right continuous. If $(a_i, b_i]$ is a set of disjoint h-intervals. If we define μ_0 to be:

$$\mu_0 \left(\bigcup_i^\infty (a_i, b_i] \right) = \sum_{i=1}^\infty F(b_i) - F(a_i)$$

and $\mu_0(\emptyset) = 0$, then μ_0 is a premeasure on the algebra $\mathcal{A}$

***Proof*:**
We will first show that μ_0 is well-defined, then we simply must show that $\mu\left(\bigcup_i (a_i, b_i]\right) = \sum_i \mu_0((a_i, b_i])$.

To show it's well-defined, we must show that for any finite representation of $(a, b]$, $\mu_0((a, b]) = F(b) - F(a)$. To that end, let $\{(a_i, b_i]\}_{i=1}^n$ be disjoint where $(a, b] = \coprod i = 1^n [a_i, b_i)$. Since the co-domain is commutative, we can re-arrange the order that we are taking the union (i.e., re-index the set) so that

$$a < a_1 < b_1 = a_2 < b_2 = a_3 < \cdots < b_n = b$$

Then the resulting μ_0 is simply an alternating series:

$$\sum_{j=1}^{n} F(b_j) - F(a_j) = F(b) - F(a)$$

showing that it is independent of representation of the intervals. Thus, if we have any disjoint collection $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^n$ such that $\cup_i I_i = \cup_j J_j$, since we can always form alternative series:

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j)$$

and so μ_0 is well-defined. Since it is well-defined, then we also get that μ_0 is finitely additive (essentially by definition), and since $\mu_0(\emptyset) = 0$, it is at least a finite measure.

Next, we need to show that it is a premeasure. Since $\mu_0(\emptyset) = 0$, it remains to check that if $\{A_i\}_{i=1}^\infty \subseteq \mathcal{A}$ where $\cup_i^\infty A_i \in \mathcal{A}$, then

$$\mu_0\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty \mu_0(A_i)$$

Since $\cup_i^\infty I_i = I \in \mathcal{A}$, there exists (at least one) subsets $\{A_i\}_{i=1}^n \subseteq \mathcal{A}$ such that $\cup_i^\infty I_i = \cup_i^n A_i$. Therefore the elements from I can be partitioned so that the union of each partition represents a single h-interval. By finite additivity of μ_0, it suffices to consider what happens at one of these partitions. So, let J be the union of one of the families so that $(a, b]J = \cup_{j=1}^\infty I_{i_j}$. We will show that $\mu_0(\cup_j^\infty J_j) = \sum_j^\infty \mu_0(J_j)$ by showing $\leq$ and $\geq$. For the $\geq$ direction, notice that

$$\mu_0(J) = \mu_0\left(\bigcup_j^n J\right) + \mu_0\left(J \setminus \bigcup_j^n J\right) \geq \mu_0\left(\bigcup_j^n J\right) = \sum_j^n \mu_0(J)$$

Thus, as $n \to \infty$ we have $\mu_0(J) \geq \sum_j^\infty \mu_0(J_j)$.

For the $\leq$ direction, we will take advantage of the right-continuity of F.

(Might not be the case TBD p.34 Folland) First, if either b is infinite, the result holds automatically. If a is infinite, then b would have to also be infinite, and so we have the zero-intervals. So, let a and b be finite.

Let $\epsilon > 0$. Then since F is right-continuous, there exists a δ such that $F(a+\delta) - F(a) < \epsilon$ or similarly $-F(a) < -F(a+\delta) + \epsilon$. Similarly, for the same epsilon, there exists a δ_j such that $F(b_j + \delta_j) - F(b_j) < \frac{\epsilon}{2^n}$. Notice that $(a_j, b_j] \subseteq (a_j, b_j + \delta_j)$, so the open intervals cover $[a, b]$, and since $[a, b]$ is compact, there exists a finite sub-cover! If $\{(a_j, b_j + \delta_j)\}_{i=1}^N$ is the finite subcover of $[a, b]$, if we discard all sets that are contained in a bigger set in this family, then we get that

$$b_j + \delta_j \in (a_{j+1}, \beta_{j+1} + \delta_{j+1}) \quad \text{for } j = 1, \ldots N-1$$

With this, we get the following sequence of inequalities:

$$\begin{aligned}
\mu_0(J) &= F(b) - F(a) \\
&\leq F(b) - F(a+\delta) + \epsilon \\
&\leq F(b_N + \delta_N) - F(a_1) + \epsilon && \text{extrema of cover} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(a_{j+1}) - F(a_j))\epsilon && \text{fancy 0} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(b_{j+1} + \delta_j) - F(a_j))\epsilon && \text{expanding} \\
&< \sum_{j=1}^{N} [f(b_j - f(a_j) + \epsilon 2^{-n}] + \epsilon \\
&< \sum_{j=1}^{\infty} \mu_0(J_j) + 2\epsilon
\end{aligned}$$

and since ϵ was arbitrary, the result follows.

This result let's us prove that *any* increasing right-continuous function defines a measure!

Theorem 1.4.2: Distributions define Measure

If $F : \mathbb{R} \to \mathbb{R}$ is an increasing right-continuous function, there is a unique Borel measure μ_F on $\mathbb{R}$ such that $\mu_F((a,b]) = F(b) - F(a)$ for all $a, b \in \mathbb{R}$. This function is unique up to the following condition: if G is another such function, then $F - G$ is constant.
Conversely, if μ is a Borel measure that is bounded on all finite Borel sets, we can define

$$F(x) = \begin{cases} \mu((0,x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((-x,0]) & \text{if } x < 0 \end{cases}$$

Then F is an increasing right-continuous function

***Proof*:**
First, by the previous proposition, F induces a premeasure on $\mathcal{A}$. If $F - G$ is constant, then the sums will cancel out in $F(b) - F(a)$, and so $\mu_F = \mu_G$. Similarly, if the two are equal, F and G can only differ up to a constant and that these measures are σ-finite (recall that $\mathbb{R} = \bigcup_{-\infty}^{\infty}(-i, i+1])$.

For the converse, note that if μ is monotonic, then so is F. Next, above/bellow continuity of μ will imply right-continuity of F for $x \geq 0$ and $x < 0$. Therefore, by construction, $\mu = \mu_F$ on $\mathcal{A}$, and so by proposition 1.4.1, $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$ by uniqueness in theorem 1.3.7.

Note that F need not be left-continuous. Something with the point-density measure that proves the counter-case.

We have been using intervals of the form $(a, b]$ and right-continuous function. The same theory can be developed with $[a, b)$ intervals and left-continuous functions. If we restrict μ to being a finite Borel measure, then we get $\mu = \mu_F$ where $F(x) = \mu((-\infty, x])$ (i.e. it differs up to a constant of the μ_F when μ is a Borel measure). Next, by the construction of the measure from the pre-measure, the resulting measure is in fact a complete measure. Even better, taking completion of μ_F and then applying Carathéodory is the same as just taking completion of μ after applying Carathéodory to μ_F, and so taking the completion on μ_0 or μ_F does not make a difference! The complete measure is also usually denoted by μ_F and called the *Lebesgue-Stieltjes measure* of F.

As a consequence of the Lebesgue-Stieltjes measure μ being complete, if we have a measure on a Borel space (which we usually call a Borel measure), then then then the set of measurable sets $\mathcal{M}^*$ might be larger than the number of Borel sets (in particular in that it may contain some zero sets that are not Borel sets).

The Lebesgue-Stieltjes measure has a couple of nice properties worth exploring. Let μ be a complete Lebesgue-Stieltjes measure on $\mathbb{R}$ associated to F, and let $\mathcal{M}_\mu$ be the associated μ^*-measurable sets. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\left\{\sum_1^\infty F(b_i) - F(a_i) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i]\right\} = \inf\left\{\sum_1^\infty \mu((a_i, b_i]) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i]\right\}$$

Notice that the contribution of the right end points of every h intervals is clearly negligable and so we can replace it with open intervals

Lemma 1.4.3: h-Intervals to Open Intervals

Let μ be the complete Lebesgue-Stieltjes measure on $\mathbb{R}$ associated to F. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\left\{\sum_1^\infty \mu((a_i, b_i)) \;\middle|\; E \subseteq \cup_i^\infty (a_i, b_i)\right\}$$

***Proof*:**
This is the same epsilon trick you have seen before, right it down here as an exercise later

In fact, we can write down $\mu(E)$ in even simple terms

Theorem 1.4.4: Lebesgue-Stieltjes in Terms of Open and Compact sets

Let μ be the complete Lebesgue-Stieltjes measure on $\mathbb{R}$ associated to $\mathbb{F}$. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf\{\mu(U) \mid U \supseteq E \text{ and } U \text{ is open}\}$$

$$\mu(E) = \sup\{\mu(K) \mid K \subseteq E \text{ and } U \text{ is compact}\}$$

***Proof*:**
By lemma 1.4.3, for all $\epsilon > 0$, there exists open intervals (a_i, b_i) such that $E \subseteq \cup_i^\infty (a_i, b_i)$ and $\mu(E) \leq \sum_i^\infty (a_i, b_i) + \epsilon$. If we let $U = \cup_i (a_i, b_i)$, we have that $\mu(U) \leq \mu(E) + \epsilon$. Conversely, since $E \subseteq U$, $\mu(U) \geq \mu(E)$, and so the result for the first statements follows.

For the 2nd equivalence, let's first suppose E is bounded. If E is closed, then it is compact and so E is contained in the set that we are taking the sup over, and so equality is immediate. If E was open, then we can choose an $\epsilon > 0$ such that $\overline{E} \setminus E \subseteq U$ and $\mu(U) \leq \mu(\overline{E} \setminus E) + \epsilon$. Set $K = \overline{E} \setminus U$. Then K is compact, and so

$$\begin{aligned}
\mu(K) &= \mu(E) - \mu(E \cap U) \\
&= \mu(E) - [\mu(U) - \mu(U \setminus E)] \\
&\geq \mu(E) - [\mu(U) - \mu(\overline{E} \setminus E)] \\
&= \mu(E) - \mu(U) + \mu(\overline{E} \setminus E) \\
&\geq \mu(E) - \epsilon
\end{aligned}$$

showing that any compact set arbitrarily approximates K.

If E is unbounded, let $E_i = E \cap (i, i+1]$ so that $E = \cup_i E_i >$ Then by what we've just shown, for all $\epsilon > 0$, there exists a $K_j \subseteq E_j$ such that $\mu(K_j) \leq \mu(E_j) - \frac{\epsilon}{2^n}$. Take $H_n = \cup_{-n}^{n} K_i$. Then H_n is compact, $H_n \subseteq E$, and

$$\mu(H_n) \geq \mu(\cup_{-n}^{n} E_i) - \epsilon$$

Since this equality holds when $n \to \infty$, the result follows

This gives us a strikingly easy representation of representing every measurable subset of $\mathbb{R}$!

Theorem 1.4.5: $E \subseteq \mathbb{R}$ Representation

Let $E \subseteq \mathbb{R}$. Then the following are equivalent

1. $E \in \mathcal{M}_\mu$
2. $E = V \setminus N_1$ where V is a G_δ set and $\mu(N_1) = 0$
3. $E = W \cup N_2$ where W is a H_σ set and $\mu(N_2) = 0$

***Proof*:**
Clearly, (2) and (3) imply (1). For (1) implies (2) or (3), I would highly recommend you try and solve this in the $\mu(E) < \infty$ case to jog your memory.

For the $\mu(E) = \infty$, TBD (exercise 25 p.37 Folland)

This means that all borel sets, or more generally sets in $\mathcal{M}_\mu$ (also contain all unions of borel sets with measure zero sets). Note that this theorem and proof would be much harrier if we did not assume the measure was complete. '

I will leave this section with another way of interpreting finite measurable sets that Folland left as an exercise (we'll use the notation of $A \triangle B = (A \setminus B) \cup (B \setminus A)$)

Proposition 1.4.6: Diminishing Difference

Let $E \in \mathcal{M}_\mu$ and $\mu(E) < \infty$. Then for every $\epsilon > 0$, there exists a set A which is the finite union of open intervals and

$$\mu(E \triangle A) < \epsilon$$

1.4.1 Lebesgue Measure

We will study the case where the measure μ is defined through the distribution $F(x) = x$, i.e., when F represents our usual intuition of length! This is arguably the most important measure on $\mathbb{R}$, and most certainly the most used. For that reason, it has some special symbology: we will denote the measure by m instead of μ and the set of measurable sets by $\mathcal{L}$. If we restrict m to $\mathcal{B}_{\mathbb{R}}$, we will also write it as m.

The first properties worth establishing is to see that measurable sets in $\mathbb{R}$ inherit some of our intuitions how a "shape" should act under translation and dilation, that it's invariant under translation (just like we assumed for the construction of Vitali sets!) and it scales linearly with dilation:

Proposition 1.4.7: Scaling and Dilation

Let $(X, \mathcal{L}, m)$ be a Lebesgue measure space. Define

$$E + s := \{e + s \mid e \in E\} \qquad rE := \{re \mid e \in E\}$$

Then $E + s, rE \in \mathcal{L}$ for all $r, s \in \mathbb{R}$, and:

$$m(E + s) = m(E) \qquad m(rE) = |r|m(E)$$

***Proof*:**
We essentially notice that open intervals are invariant under translation and operate in the form described in the proposition, then apply theorem 1.3.7 (which is a really cool application of this theorem!!)

For any $E \in \mathcal{B}_{\mathbb{R}}$ define $m_s(E) = m(E + s)$ and $m^r(E) = M(rE)$. Then m_s and m^r clearly agree with m and $|r|m$ on finite unions of intervals, and so by theorem 1.3.7 they agree on all of $\mathcal{B}_{\mathbb{R}}$. As a consequence, if $m(E) = 0$, then $m(E + s) = m(rE) = 0$. Thus, for all sets in $\mathcal{L}$ (i.e. all unions of borel sets and Lebesgue zero-sets) must be preserved and linearly scaled (respectively) under translation and dilation:

$$m(E + s) = m(E) \qquad m(rE) = |r|m(E)$$

as we sought to show

Since we are working over $\mathcal{M}(\mathcal{B}_{\mathbb{R}})$, the next interesting questions we may ask about m is how it interacts with the topological sets on $\mathbb{R}$. For starters, it is clear (or should be immediately proven) that any singleton has measure 0. Therefore, by countable additivity, every countable set has measure zero; in particular $m(\mathbb{Q}) = 0$. Notice that the rational are dense which is a "topologically large" concept, however it is measure-theoretically small. In fact, we can even have a collection of *open dense* sets, and still be measure-theoretically small. Take some enumeration $\{r_i\}_{i=1}^{\infty}$ of the rational numbers in $[0, 1]$ and let I be the indexing set for this enumeration $i \in I$. Let I_n be the interval of size $\frac{\epsilon}{2^n}$ which is centered at r_n. Then set $U = \bigcup_n I_n$. By construction, U must be dense and open in $(0, 1)$, however

$$\mu(U) \leq \sum_{i=1}^{\infty} \mu(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon$$

and hence can be as measure-theoretically small as we want. From this we can also find the opposite, where we have something topologically small but measure-theoretically big, by taking $K = [0, 1] - U$. By monotonicity

$$\mu(K) \geq 1 - \epsilon$$

and yet K is nowhere dense.

There is at least one intuition that can remain: a nonempty *open* set will have non-zero measure. For more on how open/closed sets interact with the measure, see the exercises.

Note that the converse that a non-zero measure set will have an open intervals, is not actually true! For that, we can explore the concept of *cantor sets*

Cantor Set

(was gonna look through: (commented so that this complies

Definition 1.4.8: Classical Cantor Set

Let $I = [0, 1]$. From this, we'll construct the cantor set iteratively

1. Start by removing the middle third $I_1 = I - \left(\frac{1}{3}, \frac{2}{3}\right)$
2. Given I_i, remove the open middle third of each connected compoennt

Let $C = \cap_{n=0}^{\infty} I_n$ be the Cantor set. In particular, it is the *standard middle thirds Cantor set.* We'll usually refer to this set as "the" Cantor set.

Notice that C is measurable by continuity from above since $C_1 \subseteq C_2 \subseteq \cdots \subseteq C_n \subseteq$.

Proposition 1.4.9: Properties Of Cantor Sets

Give C the standard subspace metric of $[0, 1]$. Then C is:

1. nonempty and uncountable
2. compact
3. nowhere dense $(\text{int}(\overline{A}) = \emptyset$ for all $A \subseteq C$
4. $m(C) = 0$
5. totally disconnected

***Proof*:**
Using some topological properties, since C is the intersection of compact spaces, it is compact.

Next, the cantor set is nonempty since $0 \in C_k$ for all k, and so $0 \in C$. To show C is uncountable, Notice that we can represent every element of the uncountable set $[0, 1]$ in it's base 3 expansion, namely for appropriate $a_k \in \{0, 1, 2\}$:

$$x = \sum_{k=1}^{\infty} a_k 3^{-k}$$

Then take:

$$f : [0,1] \to C \qquad x = \sum_{k=1}^{\infty} a_k 3^{-k} \mapsto$$

To show total disconnectedness, let's take the same I as before. I is closed in $\mathbb{R}$, and thus in C^n. Note that I^c is the union of finite closed sets, and thus is also closed. Thus, I is clopen in C^n. Thus, $C \cap I$ is also a clopen neigborhood of x in C. Since x was arbitrary, C is totally disconnected.

Finally uncountablenss come from C being compact and complete, which makes C uncountable (exercise).

This construction does not rely on the size of the middle-third; in fact, we can make it converge to any value we want. To see this, first notice that To do this, we first prove a lemma:

Lemma 1.4.10

Suppose $\{\alpha_i\}_{i=1}^{\infty} \subseteq (0,1)$.

1. $\prod_1^{\infty}(1-\alpha_i) > 0$ if and only if $\sum_1^{\infty} \alpha_i < \infty$ (Compare $\sum_1^{\infty} \log(1-\alpha_i)$ to $\sum_1^{\infty} \alpha_i$)
2. Given $\beta \in (0,1)$, there always exists a sequence $\{\alpha_i\}_{i=1}^{\infty}$ such that $\prod_1^{\infty}(1-\alpha_i) = \beta$

***Proof*:**
First, we see that we can convert $\prod_i^{\infty}(1-\alpha_i)$ to $\sum_i^{\infty} \log(1-\alpha_j)$ by considering

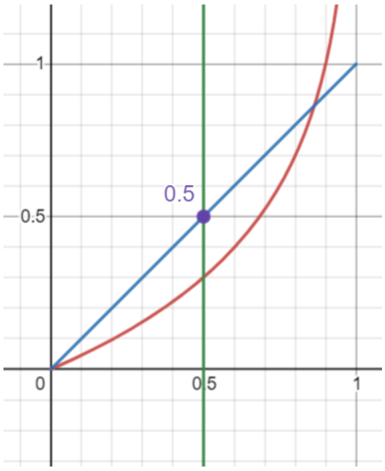
$$\begin{aligned}
\log\left(\prod_i^{\infty}(1-\alpha_i)\right) &= \log\left(\lim_{n\to\infty}\prod_i^{n}(1-\alpha_i)\right) \\
&\overset{!}{=} \lim_{n\to\infty} \log\left(\prod_i^{n}(1-\alpha_i)\right) \\
&= \lim_{n\to\infty} \sum_i^{n} \log(1-\alpha_i) \\
&= \sum_{i=1}^{\infty} \log(1-\alpha_i)
\end{aligned}$$

where $\overset{!}{=}$ comes from continuity of log for $1-\alpha_i > 0$. Notice that it's impossible that $1-\alpha_i \leq 0$ form some α_i terms since $\alpha_i \in (0,1)$, and so this proof works. Thus, we get the equivalent condition of

$$\sum_i^{\infty} \log(1-\alpha_i) > -\infty$$

With this setup, we proceed with the proof

($\Leftarrow$) Let $\sum_i^{\infty} \alpha_i < \infty$. Then we know that $\alpha_i \to 0$. Thus, pick an N such that for all $i > N$, $\alpha_i < 0.5$. Then notice that $\log(1-\alpha_i) < \alpha_i$. This is easier to see with a visual aid:



In particular, for all $0 < x < 1/2$, we have $|\log(1-x)| - x > 0$, showing that x is greater than $\log(1-x)$ for all $x \in (0, 1/2)$.

Then from here, this is a simple application of the Weiestrass M-test. Let $K = \sum_i^N \log(1-\alpha_i)$ to eliminate the irregular part. Then since $|\log(1-\alpha_i)| < \alpha_i$, for all $i > N$ and $\sum_i^\infty \alpha_i$ converges, by the Weiestrass M-test the $\sum_{i=N+1}^\infty \log(1-\alpha_i)$ converges. Let's say it converges to L. Then

$$\sum_{i=1}^{\infty} \log(1-\alpha_i) = K + L > -\infty$$

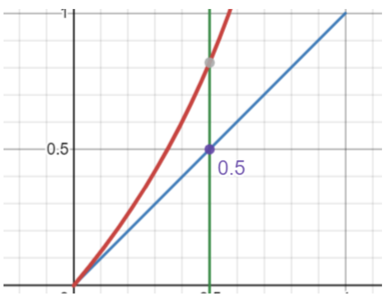
as we sought to show.

($\Rightarrow$) let $\prod(1-\alpha_i) > 0$ so equivalently $\sum_i^\infty \log(1-\alpha_i) > -\infty$ so $\sum_i^\infty \log(1-\alpha_i) = K$ for appropriate $K \in \mathbb{R}$. Then since the series converges, we have that

$$eK = e\sum_i^{\infty} \log(1-\alpha_i) = \sum_i^{\infty} e\log(1-\alpha_i)$$

i.e., we can scale the terms by e. We will show why this scaling is important in a moment; before we do we must setup a little more.

Since the series converges, it must be that $e\log(1-\alpha_i) \to 0$. So pick N such that for all $i > N$, $e\log(1-\alpha_i) < 1/2$. Then notice that $\alpha_i < e\log(1-\alpha_i)$. This is easier to see with a visual aid:



Or, to carry out some of the computations, if you raise $e\log(1-\alpha_i) = \log(1-\alpha_i)e$ to the power of e, you get

$$(e^{\log(1-\alpha_i)})^e = (1-\alpha_i)^e \geq \alpha_i^e \quad \forall \alpha_i \in (0, 1/2]$$

and so the inequality holds. Then the proof continues as for the $\Leftarrow$ direction by setting up an appropriate constant term to bound the irregularities, then using the Weiestrass M test bound every α_i by $\log(1-\alpha_i)$, and since $\sum_i^\infty \log(1-\alpha_i)$ converges, we get that $\sum_{i=N}^\infty \alpha_i$ also converge, and since it will be the sum of a constant plus the convergent value, $\sum_{i=1}^\infty \alpha_i$ converges, as we sought to show.

Now, let β be given. Let $x_i = \prod_{i=1}^n (1-\alpha_i)$. We'll construct a sequence where

$$x_1 = \beta + \frac{1-\beta}{2} \qquad x_2 = \beta + \frac{1-\beta}{4} \qquad \cdots \qquad x_n = \beta + \frac{1-\beta}{2^n} \cdots$$

to accomplish this, let

$$\alpha_i = 1 - \frac{\beta + \frac{1-\beta}{2^i}}{\beta + \frac{1-\beta}{2^{i-1}}} \tag{1.2}$$

This construction comes from solving the following equation:

$$\left(\beta + \frac{1-\beta}{2}\right) y_2 = \beta + \frac{1-\beta}{4} \quad \Rightarrow \quad y_2 = \frac{\beta + \frac{1-\beta}{4}}{\beta + \frac{1-\beta}{2}}$$

so $\alpha_2 = 1 - y_2$, and clearly $\alpha_2 \in (0, 1)$ since the denominator is bigger than the numerator in the above calculations. Generalizing this pattern, we get the that α_i would be of the from presented in equation (1.2) and $\alpha_i \in (0, 1)$. Thus, we get that

$$\lim_{n\to\infty} \prod_i^n (1-\alpha_i) = \lim_{n\to\infty} \beta + \frac{1-\beta}{2^n} = \beta$$

as we sought to show

Now, let's say we are starting with $[0, 1]$ but on each iteration eliminates intervals that sum to α_i. Let K_i be the ith step in this itteration. Notice that:

$$m(K_{i+1}) = (1-\alpha_i)m(K_i)$$

Then by what we've just shown, we can make $K = \cap_i^\infty K_i$ be any value up to and including 0 and 1.

Proposition 1.4.11: All Cantor Set Homeomorphism

Let C_n be a cantor set where $m(C_n) = n$. Then $C_n \cong C_0$

***Proof*:**
section 9 of Pugh – will get back to it

You can also prove uncountability directly by making a string represent which side of the split interval you go to.

Next, you might've seen many uncountable sets which are dense in $\mathbb{R}$. Here we have an absolutely facsinating result: C is *nowhere dense in* $[0, 1]$.

Definition 1.4.12: Dense, Somewhere Dense, Nowhere Dense

If $S \subseteq M$ and $\overline{S} = M$, then S is dense in M. The set S is somewhere dense if there exists an open nonempty set $U \subseteq M$ such that $\overline{S \cap U} \supset U$. If S is not somewhere dense, then it is *nowhere dense*

Proposition 1.4.13: C Nowhere Dense

The Cantor set contains no interval and is nowhere dense in $\mathbb{R}$

***Proof*:**
To prove it doesn't contain an interval, assume it does, then chose an n large enough, that is $(\frac{1}{3})^n < b - a$, so that it will not be in the interval.

For nowhere dense, assume ti's somewhere then, then $\overline{C \cap U} \supset U \supset (a, b)$ – a contradiction.

We know present how Cantor sets are almost initial objects in compact metric space

Theorem 1.4.14: Cantor Surjection Theorem

Given a compact nonempty metric space M. there is a continuous surjection of C onto M

***Proof*:**
exercise

Finally, we present how all cantor sets are homeomorphic.

Theorem 1.4.15: Moore-Kline Theorem

We say M is a Cantor space if it is compact, nonempty, perfect, and totally disconnected (like the cantor set). Every cantor space M is homeomorphic to the standard middle-thirds Cantor set C.
In other words, those properties *define* Cantor sets.

***Proof*:**
Pugh. p.112

Corollary 1.4.16: Cantor And Fat Cantor Set

The fact Cantor set is homeomorphic to the standard Cantor set

***Proof*:**
Using 1.4.15, the result is immediate.

This is intersting, since it shows that meausre is *not* a topological poperty! This should makes sense with $(0,1)$ and $\mathbb{R}$ being homeomorphic with the same measures, but different, but this is an even more extreme example.

Corollary 1.4.17: Product Of Cantor Sets

A cantor set is homeomorhic to its own cartesian product; $C \equiv C \times C$.

***Proof*:**

A fact that Folland added that I thought was cool is that since every subset of the cantor set is Lebesgue Measurable (since the cantor set has measure zero and the Lebesgue measure is complete), we have that

$$|\mathcal{L}| = |P(\mathbb{R})| > \aleph_1 \quad \text{but} \quad |\mathcal{B}_{\mathbb{R}}| = \aleph_1$$

so there are a *tone* of zero-measure sets, however, we can essentially ignore them because they are zero measure sets!!

For those who are Categorically inclined, Cantor sets are almost like the initial objects of the category of compact metric spaces. They are not initial because there does not exist a unique uniformally continuous function from the cantor set to any compact metric space, but there always will exist at least 1 such function.

Exercise 1.4.1

1. If $E \in \mathcal{L}$ and $m(E) > 0$, for any $\alpha < 1$, there is an open interval I such that $m(E \cap I) > \alpha m(I)$

2. If $E \in \mathcal{L}$, and $m(E) > 0$, the set $E - E = \{x - y \mid x, y \in E\}$ contains an interval centered at 0 (If I is as in the previous exercise with $\alpha > \frac{3}{4}$, then $E - E$ contains $(-\frac{1}{2}m(I), \frac{1}{2}m(I))$).

3. Let E be a Lebesgue measurable set (so $E \in \mathcal{L} = \overline{\mathcal{M}}(\mathcal{B}_{\mathbb{R}})$)

    1. If $E \subseteq N$ where N is a non-measurable set described in section 1.1, then $m(E) = 0$
    2. if $m(E) > 0$, then E contains a non-measurable set (it suffices to assume $E \subseteq [0,1]$. In notation of section 1.1 $E = \cup_{r \in R} E \cap N_r$

2

Integration

In this chapter, we now work with function's between measure spaces that preserve measure structure! Measurable functions feel like the types of functions we "usually" work with, the same way measurable sets are the sets we "usually" work with. Being continuous can be a bit strong (ex. being a monotone function doesn't imply continuous), but measurability rectifies that by in a sense expanding what are the possible "pre-images" we allow to take (ex. Monotone functions *are* always measurable, as we will demonstrate).

After defining measurable functions, we will focus on real and complex functions (i.e. functions with codomain $\mathbb{R}$ or $\mathbb{C}$). In the real case, I like to think of real functions as given every point in the domain X a value or a weight, in contrast to a measure which is a real function which simply assigns a constant value at each point (this will become more clear as we introduce the definitions). For the complex case, each point is given a weight and a *phase* or *state*[1]. In terms of computation, we simply split the problem into the real and imaginary component. Then, we will define how to "measure" a real (resp. complex) measurable function. We then focus on measurable functions which have "finite measure", and will call them *(Lebesgue) integrable functions*; integrable functions will allow us to have some regularity in how we can combine measurable functions without running into some problems with infinities (as we'll soon show). One result of this is that we can think of the integral as being a "measure" for a function, and since integrable function will have "finite measure", it will make it more meaningful to see which functions are "close" (in the appropriate sense of the word close) to a certain function.

A word must be said about Lebesgue integrability vs. Riemann integrability. The key difference is the allowed measurable sets. For Riemann integrability, all our measurable sets are *rectangles*: every partition of the area of integration can be seen as a sum of indicator functions over pair-wise disjoint

[1]you can think of every input having a natural magnitude that periodically shifts. The phase $e^{i\theta}$ tracks this. So if $f(a) = 10$ and $f(b) = 10e^{-i\pi} = -10$, then a is at the beginning of its phase with magnitude 10, while b is half way through its phase which shifted it's magnitude to -10. Addition adds the interference (ex. $(2+i)+(-2-2i) = -i$) while

rectangles. On the other hand, Lebesgue integration is over *measurable* sets. The strategy shall shift so that instead of partitioning the domain based off of its geometry, we shall take partition it using $f^{-1}([-k,k])$ (or really any measurable covering of $\mathbb{R}$, and appropriately modified for $\mathbb{C}$) where we shall show the pre-image of a measurable set is measurable if f is measurable. This shall solve many headache the Riemann integral imposes (ex. any rectangle of $\chi_{\mathbb{Q}}$ on $[0,1]$ will have have rational and irrational points, making it unfeasible to measure).

After we have defined integrable functions, we will have to take a moment to compare different types of convergences, since at that stage we will have 4 different ways functions can converge. We will then move on to trying to bring the theory of the product of σ-algebras to measure and integrable functions more generally, allowing us to construct larger measures and integrable functions from smaller components. Importantly, we will show that we can find the a measure $\mu \times \nu$ by separating any problem to only measuring "slices" of the set with respect to μ, then "slices" of the set with respect to ν (this is Fubini's Theorem).

2.1 Measurable Functions

Recall that if $f : X \to Y$ is a function, then $f^{-1} : P(Y) \to P(X)$ is a function where $f^{-1}(E) = \{x \in X \mid f(x) \in E\}$ which preserves unions, intersections, and compliments. Thus, if $\mathcal{N}$ is a σ-algebra on Y, $f^{-1}(\mathcal{N})$ is a σ-algebra on X. If X and Y are measure space, if $f^{-1}(\mathcal{N})$ maps into $\mathcal{M}$, we will call f *measurable*:

Definition 2.1.1: Measurable Function

Let $f : X \to Y$ be a function and $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. Then f is called a $(\mathcal{M}, \mathcal{N})$-*measurable function* (or just measurable) if for all $E \in \mathcal{N}$

$$f^{-1}(E) \in \mathcal{M}$$

Just like in topology, this could be thought of as a surjection of the measurable space on X onto the measurable space on Y. The composition of two measurable functions is measurable (in particular, the composition of an $(\mathcal{M}, \mathcal{N})$-measurable function with $(\mathcal{N}, \mathcal{O})$-measurable function is $(\mathcal{M}, \mathcal{O})$-measurable), and that the identity function $\mathrm{id} : X \to X$ is clearly a measurable function, and so the collection of measurable functions and measurable spaces from a category, usually denoted **Meas**.

Just like lemma 1.1.4 was useful for simplifying proofs, we proof the following:

Lemma 2.1.2: Measurable Function and Generating Sets

Let $\mathcal{N}$ be generated by $\mathcal{E}$. Then $f : X \to Y$ is a $(\mathcal{M}, \mathcal{N})$-measurable function if and only if for all $E \in \mathcal{E}$, $f^{-1}(E) \in \mathcal{M}$

***Proof*:**
The $\Rightarrow$ direction is *a-fortiori* true by definition. The $\Leftarrow$ direction comes from f^{-1} preserving unions and compliments, so $\left\{E \subseteq Y \;\middle|\; f^{-1}(E) \in \mathcal{M}\right\}$ is a σ-algebra that contains $\mathcal{E}$, and so contains $\mathcal{N}$.

From this, there is a very nice corrollary when $\mathcal{M}$ is a collection of Borel sets:

Corollary 2.1.3: Continuous Function $\Rightarrow$ [Borel] Measurable Function

Let $f : X \to Y$ be a continuous function and Y have the borel σ-algebar. Then f is a measurable function.

***Proof*:**
Since f is continuous, the pre-image of an open set (in particular, open intervals) is open, and hence by proposition 1.1.7 and lemma 2.1.2 it is measurable

Naturally, in analysis, the most important measurable functions will be of the form

$$f : \mathbb{R} \to \mathbb{R} \qquad f : \mathbb{R} \to \mathbb{C}$$

This also means that the domain can either have the $\mathcal{B}_{\mathbb{R}}$ σ-algebra or $\mathcal{L}$ σ-algebra.

Example 2.1.4: Measurable Functions
Most functions you have worked with will in fact be measurable

1. As already shown, all continuous functions are measurable
2. All monotone functions are measurable: the pre-image of closed rays will either be open or closed rays, and so by proposition 1.1.7, they are measurable.
3. All indicator functions over measurable sets are measurable (hence, bump functions are essentially measurable)
4. All Riemann integrable functions are measurable (we will show this this in section 2.3.1)
5. We will soon show that the point-wise limit of measurable functions is also measurable, hence almost all examples of functions that shall be studied are measurable.

It is in practice rare to work with non-measurable functions unless a contrived example is used, like a measurable function purposefully built to map to non-measurable sets (like in the case of the composition of two Lebesgue measurable functions)

2.1.5 Convention e will *always* assume the codomain will have $\mathcal{B}_{\mathbb{R}}$ (resp. $\mathcal{B}_{\mathbb{C}}$) σ-algebara. In fact, if we have a function $f : X \to \mathbb{R}$ (resp. $f : X \to \mathbb{C}$), then we will sometimes say f is $\mathcal{M}$-measurable instead of $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable (resp. $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable). If the domain is also Borel, we will say that f is *Borel measurable* (and omit the Borel measure if it's $\mathbb{R}$ or $\mathbb{C}$). Similarly, if the domain is $\mathcal{L}$, we will call f Lebesgue measurable.

Because of the convention of the co-domain being Borel, it is possible that the composition of two Lebesgue measurable functions is *not* Lebesgue measurable: if $f : \mathbb{R} \to \mathbb{R}$, $g : \mathbb{R} \to \mathbb{R}$ are two Lebesgue measurable functions where $E \in \mathcal{B}_{\mathbb{R}}$, $g^{-1}(E) \in \mathcal{L}$ but $g^{-1}(E) \notin \mathcal{B}_{\mathbb{R}}$, it is possible that $f^{-1}(g^{-1}(E)) \notin \mathcal{L}$

Example 2.1.6: Composition Not Lebesgue Measurable
Let $f : [0,1] \to [0,1]$ be the cantor function form section 1.5, and let $g(x) = f(x) + x$.

1. g is a bijection from $[0,1]$ to $[0,2]$ and $h = g^{-1}$ is continuous from $[0,2]$ to $[0,1]$

   ***Proof*:**
   First, since f is increasing (though not injective) and x is strictly increasing, then $f + x$ is injective. Since f and x are both continuous, so is $g = f + x$. Hence the pre-image of a closed set must be closed. If we consider $g^{-1}([0,2]) = D$, then using the fact that g is a monotonic functions, since $g(0) = 0$ and $g(1) = 2$, it must be that $D = [0,1]$, showing that g is indeed surjective, and hence bijective.

   To show that $g^{-1} : [0,2] \to [0,1]$, is continuous, bijective, and the codomain is Hausdorff, then g is homeomorphic (recall that these imply that g is an open map: if $U \subseteq [0,1]$ then $S = [0,1] - U$ is closed, and since g is continuous $g([0,1] - U) = [0,2] - g(U)$ is closed, so $g(U)^c$ is closed, so $g(U)$ is open), and hence g^{-1} is continuous.

2. If C is the cantor set, then $m(g(C)) = 1$

   ***Proof*:**
   We will take advantage of translation invariance of the Lebesgue measure. Let C be the cantor set on $[0,1]$. Then C can be thought of as the interval $[0,1]$ minus all of the open sets that are eliminated during construction:

   $$C = [0,1] - I_1 - I_{21} - I_{22} - \cdots - I_{ik} - \cdots$$

   Let $I = \coprod_{i,k} I_{ik}$ collection of these sets so that $C = [0,1] - I$. Since $m(C) = 0$, $m(I) = 1$. Consider now $g(I)$. Since g is injective and I is the disjoint union of sets, then there exists sets $\{J_j\}_{j=1}^{\infty}$ such that $J_j = g(I_{ik})$. We now examine the measure of the sets J_j.

   By definition, I_{ik} does not contain any points of the cantor set, and so for appropriate $c_{ik} \in C$,

   $$g_{I_{ik}}(x) = f(c_{ik}) + x$$

   that is, g on the interval I_{ik} is of the form $x + f(c_{ik})$ for constant $f(c_{ik})$! Since the Lebesgue measure is translation invariant, we get:

   $$m(g(I_{ik})) = m(g(I_{ik}) - f(c))$$

   and so, since the I_{ik} intervals are all disjoint and g is injective:

   $$g(\coprod I_{ik}) = \coprod g(I_{ik})$$

   and so the measure of $g(I)$ is in fact exactly the measure of I, that is, $m(g(I)) = m(J) = 1$. Thus:

   $$2 = g(m([0,2])) = m(g(C) \coprod J) = m(C) + m(J) = m(g(C)) + 1$$

   and so $m(g(C)) = 1$, as we sought to show.

3. by exercise 29, $g(C)$ contains a Lebesgue non-measurable set A. Let $B = g^{-1}(A)$. Then B is Lebesgue measurable, but not Borel

> ***Proof*:**
> Since $A \subseteq g(C)$, $B \subseteq g^{-1}(g(C)) = C$. Since $m(C)$ is zero, by completeness, $m(A) = 0$, and so A is Lebesgue measurable.
>
> Next, to show that B is not Borel-measurable, For the sake of contradiction let's assume it was. Then since g^{-1} is continuous, it is Borel-measurable, meaning the pre-image (with respect to g^{-1}) of a measurable set is measurable. However, $g(B) = A$, which is not measurable, and hence B is not Borel measurable.

4. There exists a Lebesgue measurable function F and a continuous function G on $\mathbb{R}$ such that $F \circ G$ is not Lebesgue measurable

> ***Proof*:**
> Let F be the indicator function on B, I_B, and $G = g^{-1}$. Then F and G are clearly Lebesgue measurable, However, G^{-1} does not map all Lebesgue measurable sets to Lebesgue measurable sets (in particular, B maps to a Borel set), and so
>
> $$(F \circ G)^{-1}(B) = (G^{-1} \circ F^{-1})(B) = (g(I_B^{-1}(B)) = g(B) = A$$
>
> However, A is not Lebesgue measurable set, and so $F \circ G$ is not Lebesgue measurable.

Just like for Borel sets earlier, we have the following equivalence for measurable functions with codomain $\mathbb{R}$:

Proposition 2.1.7: Borel sets and Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $f : X \to \mathbb{R}$. Then the following are equivalent:

1. f is $\mathcal{M}$-measurable
2. $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
3. $f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
4. $f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
5. $f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$

> ***Proof*:**
> This follows from proposition 1.1.7 and lemma 2.1.2.

Similarly, we will soon be encountering co-domains which are product spaces (ex. $\mathbb{R}^n$ or $\mathbb{C}^n$). In particular, let's say we had some set X, and some family of measurable spaces $\{(Y_\alpha, \mathcal{M}_\alpha)\}_{\alpha \in A}$, and $f : X \to Y_\alpha$ is a map for each $\alpha \in A$. Then we can impose a unique smallest σ-algebra on X that makes each f_α a measurable function, in particular, $\mathcal{M}_X = \left\{ f_\alpha^{-1}(E_\alpha) \;\middle|\; E_\alpha \in \mathcal{M}_\alpha,\ \alpha \in A \right\}$ [2]. This σ-algebra on X is called the *σ-algebra generated by* $\{f_\alpha\}_{\alpha \in A}$.

[2] A similar strategy is used in the constrution of *weak* topologies. If you're more categorical, this constructions makes X satisfy the universal property of products in the category of **Meas**

Proposition 2.1.8: Product of Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $(Y_\alpha, \mathcal{N}_\alpha)$ be a collection of measurable spaces, so $Y = \prod_{\alpha \in A} Y_\alpha$, $\mathcal{N} = \bigotimes_{\alpha \in A} \mathcal{N}_\alpha$, and $\pi_\alpha : Y \to Y_\alpha$ is the coordinate map. Then $f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$-measurable if and only if $f_\alpha = \pi_\alpha \circ f$ is $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable.

***Proof*:**
Let's say f is $(\mathcal{M}, \mathcal{N})$-measurable. Then since the composition of two measurable functions is measurable, so are each f_α is measurable (since π_α is measurable by definition of $\bigotimes$).

Conversely, if every f_α is $(\mathcal{M}, \mathcal{N}_\alpha)$-measurable, then for each $E_\alpha \in \mathcal{N}_\alpha$, $f^{-1}(\pi_\alpha^{-1}(E_\alpha)) = f^{-1}(E_\alpha) \in \mathcal{M}$, and since these sets generate $\mathcal{M}$, we have that f is $(\mathcal{M}, \mathcal{N})$-measurable by proposition 2.1.7

This proposition also let's us define what it means to take the "product" of two functions. As a consequence, we can give an easy criterion for when complex functions are measurable:

Corollary 2.1.9: Complex Function Measureability

Let $f : X \to \mathbb{C}$ be a complex function. Then f is $\mathcal{M}$-measurable if and only if $\mathrm{Re} f$ and $\mathrm{Im} f$ are both measurable.

***Proof*:**
This is simply taking advantage of the topology of $\mathbb{C}$, namely

$$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} \stackrel{!}{=} \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$$

where the $\stackrel{!}{=}$ equality comes from the fact that $\mathbb{R}$ is separable (see proposition 1.1.11). And so proposition 2.1.8 with the fact that Re and Im can be considered projections completes the proof.

Sometimes, we want to work with the extended real line (or in general some compactification of a topological space). If $\overline{\mathbb{R}} = [-\infty, \infty]$, Then let

$$\mathcal{B}_{\overline{\mathbb{R}}} = \left\{ E \subseteq \overline{\mathbb{R}} \;\middle|\; E \cap \mathbb{R} \in \mathcal{B}_{\mathbb{R}} \right\}$$

Notice that is we put the metric ρ on $\overline{\mathbb{R}}$ to be $\rho(x, y) = |\tan^{-1}(x) - \tan^{-1}(y)|$, then $\mathcal{B}_{\overline{\mathbb{R}}}$ is indeed the usual definition of a Borel σ-algebra (exercise). It should be verified that $\mathcal{B}_{\overline{\mathbb{R}}}$ can be generated by open or closed rays, and so we will say $f : X \to \overline{\mathbb{R}}$ is $\mathcal{M}$-measurable if it is $(\mathcal{M}, \mathcal{B}_{\overline{\mathbb{R}}})$-measurable.

Next, if the codomain has some notion of $+$ and $\cdot$, we establish that the basic "$\mathbb{C}$-algebra operations" work on measurable functions:

Proposition 2.1.10: Arithmetics on Measurable Functions

Let $f : X \to \mathbb{C}$ and $g : X \to \mathbb{C}$ be $\mathcal{M}$-measurable functions. Then $f + g$, fg, and zf for $z \in \mathbb{C}$ are $\mathcal{M}$-measurable functions

Note that we have limited ourselves to $\mathbb{C}$. This still works with $\overline{\mathbb{R}}$ with the appropriate care of the $\infty - \infty$ and $0 \cdot \infty$ case (exercise 2 in book)

> ***Proof*:**
> The proof for zf being measurable is similar to that of $+$ and $\cdot$, so is left as an exercise.
>
> Since f and g, are measurable, then their "cartesian product" is measurable by proposition 2.1.8
>
> $$F : X \to \mathbb{C} \times \mathbb{C} \qquad F(x) = (f(x), g(x))$$
>
> In particular, F is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathbb{C}})$-measurable. Next, consider $p : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, $p(x, y) = x + y$ and $t : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$, $t(x, y) = xy$ (p for plus, t for times). Then since p and t are continuous, by corollary 2.1.3 p and t are $(\mathcal{B}_{\mathbb{C}\times\mathbb{C}}, \mathcal{B}_{\mathbb{C}})$-measurable. Thus, $p \circ F$ and $t \circ F$ is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$-measurable, and by definition
>
> $$f + g = p \circ F \qquad fg = t \circ F$$
>
> completing the proof

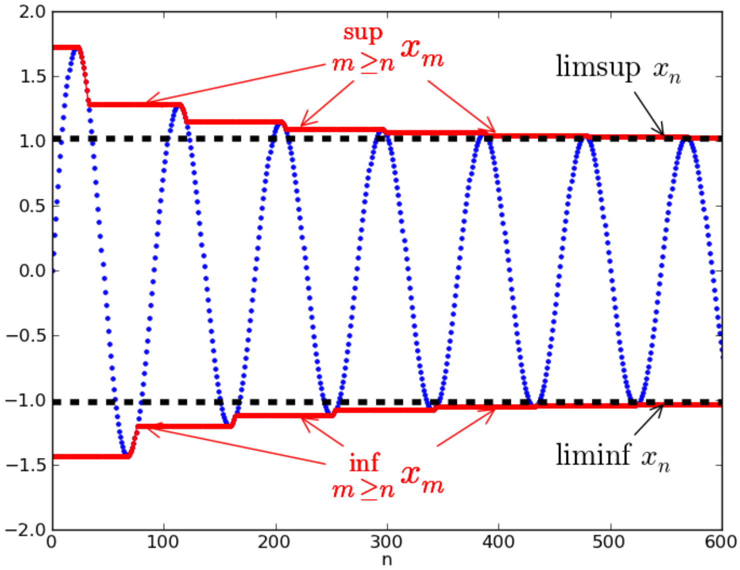
The next operations we'll show are measurable are sup, inf, lim sup and lim inf. This is perhaps one of the most important functions to be measurable since they are at the heart of many constructions in analysis. Recall that

$$\sup_i f_i(x) = \sup \{f_i(x) \mid i \in \mathbb{N}\}$$

and so this expression means we're taking the set $f_i(x)$ for each i and taking the sup of that set. In other words, for every point x we're taking the supremum of the set $\{f_1(x), f_2(x), ...\}$. Symmetrically the same for inf. For lim sup, recall that:

$$\limsup_i f_i(x) = \lim_{i\to\infty} \sup \{f_k(x) \mid k \geq i\}$$

that is, we are constantly shrinking the set over which we are doing the supremum, which in a sense means we're trying to "hug" the value from above as close as possible in the limit. Symmetrically the same for lim inf. If you're visual, I like to keep this image in mind: let's say the blue line is the value of $f_i(x_0)$ for a specified x_0. Then:



Comment here for future reference:

Proposition 2.1.11: Measurable functions and Limit Bounding

Let $f_i : X \to \overline{\mathbb{R}}$ be $\mathcal{M}$-measurable functions for each i and consider $\{f_i\}_{i=1}^{\infty}$. Then

$$g_1(x) = \sup_i f_i(x) \qquad g_2(x) = \inf_i f_i(x)$$

$$g_3(x) = \limsup_{i\to\infty} f_i(x) \qquad g_4(x) = \liminf_{i\to\infty} f_i(x)$$

The codomain $\overline{\mathbb{R}}$ was used so that the sup and inf is still considered defined if sup $= \pm\infty$ or inf $= \pm\infty$.

***Proof*:**
To show $\sup_i f_i$ and $\inf_i f_i$ are measurable, since the codomain domain is $\overline{\mathbb{R}}$, which is equivalent measure-theoretically to $\mathbb{R}$, we will show the pre-image of open rays is the union of measurable sets:

$$\begin{aligned} x \in g_1^{-1}((a,\infty]) &\Leftrightarrow g_1(x) > a \\ &\Leftrightarrow f_i(x) > a && \text{for some } i \\ &\Leftrightarrow x \in f_i^{-1}((a,\infty]) && \text{for some } i \\ &\Leftrightarrow x \in \bigcup_{i=1}^{n} f_i^{-1}((a,\infty]) \end{aligned}$$

and similarly for g_2, except considering $(-\infty, a]$, therefore:

$$g_1^{-1}((a,\infty]) = \bigcup_{i}^{\infty} f_i^{-1}((a,\infty]) \qquad g_2^{-1}(-\infty,a])) = \cup_i^{\infty} f_i^{-1}([-\infty,a))$$

since each f_i is measurable, we have a union of measurable sets, so g_1 and g_2 are measurable by proposition 1.1.11. Next, let

$$h_k(x) = \sup_{i>k} f_i(x) = \sup \{f_k(x) \mid k > i\}$$

then by what we've just established, h_k is measurable for each k. Then by of $\limsup$:

$$g_3 = \inf_k h_k$$

and so g_3 is measurable since we just established the inf of measurable functions is measurable. Similarly for g_4.

as an immediate corrollary:

Corollary 2.1.12: Measurability of Max and Min

Let f, g be measurable functions. Then so is $\max\{f, g\}$ and $\min\{f, g\}$

Corollary 2.1.13: Measurable Preserved under Pointwise limit

Let $\{f_i\}$ be a sequence of measurable function. If $\lim_{i\to\infty} f_i(x)$ exists for every x, and we define $f : X \to \overline{\mathbb{R}}$ to be

$$f(x) = \lim_{i\to\infty} f_i(x)$$

then f is measurable (i.e. point-wise convergence preserves measureability)

Recall that point-wise convergence *does not preserve continuity* (the steeple function's are the usual example). However, measureability is sufficiently general to be preserved under this weaker form of convergence! Naturally, if the sequence of functions $\{f_i\}$ are all $\mathcal{M}$-measurable and uniformally convergent, they are measurable

***Proof*:**
If f exists (i.e., every $f_n(x)$ converges pointwise to $f(x)$), then by proposition 2.1.11, $f = g_3 = g_4$, completing the proof

2.1.14 Remark Note how important it is that *all* points converge! It is possible that a single point doesn't converge (ex. $\lim f_i(0)$ doesn't converge), and so the limit is *not* a measurable function, in fact it doesn't even need to be a function. Take for example the collection of functions $f_k : \mathbb{R} \to \mathbb{R}$ with the properties:

1. f_k is continuous (it can even be smooth)
2. $\int_{\mathbb{R}} f(x)dx = 1$ (In the Riemann sense), or really any constant $c \in \mathbb{R}_{>0}$
3. $\text{supp} f \subseteq \left[\frac{-1}{k}, \frac{1}{k}\right]$

In other words, the collection of f_k are all smooth bump functions with volume 1, but the area in which it has support is shrinking, and so the "bump" of each f_k must be getting higher and higher (to define such functions, consider starting with $f_1 = e^{-\frac{1}{1-x^2}}$ where $x \in (-1, 1)$ and 0 otherwise). Then $f_k \to 0$ at all but 1 point, namely $f_k(0)$ does *not* converge. As a consequecne, this limit is *not* a function!
This example shows the important of somehow "bounding" your area; we will get back to this observation in section 2.2. Some might think that this example is a bit cheeky, it still feels like the function is measurable if we just circumnavigate this problem by making the domain of the function $\mathbb{R} \cup \{\infty\}$, there is the trouble of defining a measure on the 1-point compactification of $\mathbb{R}$ while extending the measure of $\mathbb{R}$[a]. If the reader wonders if there is a way of defining some generlization of a function that will accept the fact that the f_k converge to something that still somehow preserves the fact that $\int f = 1$, then we will show in in chapter [13, Distribution Theory] that this converges to something called a *distribution*[b]

[a] In particular, $\mu(\{\infty\}) = \infty$, as can be shown using continuity form above, and so somehow we have "gained" an infinite amount of area. On the other-hand, if we just ignore this and consider that ∞ is just some point like any other point, then we have "lost" area

[b] The example named here is a distribution called the *delta distribution*, or sometimes *delta function* for reasons that is covered in chapter [13, Distribution Theory]

Now that we proved that these operations preserving measurability, we will introduce some new notation to decompose real and complex functions which will soon become useful when defining integrable functions.

Definition 2.1.15: Real and Complex Decomposition

Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{C}$ be real and complex measurable functions respectively. Then

$$f^+ = \max\{f, 0\} \qquad f^- = \max\{-f, 0\}$$

and

$$\text{sgn}(g(x)) = \begin{cases} \frac{g(x)}{|g(x)|} & g(x) \neq 0 \\ 0 & g(x) = 0 \end{cases} \qquad |g(x)| = |z| = |x + iy| = \sqrt{x^2 + y^2}$$

which gives the decomposition of:

$$f = f^+ - f^-, \; g = (\text{sgn} g)|g|$$

The functions f^+ and f^- are clearly well-defined and measurable. For the complex case, $|\cdot|$ is continuous everywhere and $z \mapsto \operatorname{sgn} z$ is continuous everywhere except 0. It is measurable since if $U \subseteq \mathbb{C}$ is open, then $\operatorname{sgn}^{-1}(U)$ is either V for some open set V, or $V \cup \{0\}$, meaning sgn is Borel measurable. In some textbooks, arg is used instead of sgn.

2.1.1 Standard Representation

Recall that every measurable function is the point-wise limit of a bunch of easy to work with function, in particular, linear combination of characteristic functions! As a reminder, if $\chi_E : X \to \mathbb{R}$ (or $\chi_E : X \to \{0, 1\}$) where $E \subseteq X$, then:

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

Then χ_E is called the characteristic (or indicator) function. It is sometimes also denoted as 1_E or I_E.

Definition 2.1.16: Simple function

Let $\{E_i\} \subseteq P(\mathbb{C})$ be some finite collection of sets. Then the finite linear combination of characteristic functions on E_i multiplied by a complex constant:

$$f = \sum_{i=1}^{n} z_i \chi_{E_i}$$

is called a *simple function*

Example 2.1.17: Simple Functions

1. Show that $|f(X)| < \infty$ then f is in fact a simple function (hint: let $E_i = z_i$ for each $z_i \in f(X)$). In fact, each coefficient can also be distinct (let it be the respective z_i from the range)
2. Show that if f and g are simple functions, so are $f + g$ and fg

Using these, we can now show that any measurable function is simply the limit of simple functions!

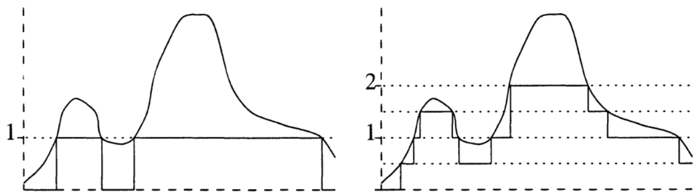
Theorem 2.1.18: Simple Function Approximation

Let $(X, \mathcal{M})$ be a measurable space. Then

1. If $f : X \to [0, \infty]$ is measurable, there exists a sequence of functions $\{\varphi_i\}_{i=1}^{\infty}$ such that
$$0 \leq \varphi_1 \leq \varphi_2 \leq \cdots \leq f$$
where $\lim_{n\to\infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \to f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformally converges)

2. If $f : X \to \mathbb{C}$ is measurable, there exists a sequence of simple functions $\{\varphi_i\}_{i=1}^{\infty}$ such that
$$0 \leq |\varphi_1| \leq |\varphi_2| \leq \cdots \leq |f|$$
where $\lim_{n\to\infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \to f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformally converges)

***Proof*:**
Essentially, you will get characteristic functions to converge to the value like so:



In particular, for $n \in \mathbb{N}_{\geq 0}$ and $0 \leq k \leq 2^{2n} - 1$, let

$$E_n^k = f^{-1}\left(\left(\frac{k}{2^n}, \frac{k+1}{2^n}\right)\right) \qquad F_n = f^{-1}((2^n, \infty])$$

and

$$\varphi_n = \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \chi_{E_n^k} 2^n \chi_{F_n}$$

then we get that $\varphi_n \leq \varphi_{n+1}$ for all n and $0 \leq f - \varphi_n \leq \frac{1}{2^n}$ where $f \leq 2^n$. The rest is left to check.

Generalizing to $\mathbb{C}$, decomposing to $f = g + ih = (g^+ + g^-) = i(h^+ + h^-)$, where the $+$ and $-$ functions are the positive and negative parts of f and g, then we can repeat the process for each of these and get $\psi_n^+, \psi_n^-, \zeta_n^+, \zeta_n^-$ and let $\varphi_n = (\psi_n^+ - \psi_n^-) + i(\zeta_n^+ - \zeta_n^-)$.

This shows that measurable functions are a very natural type of function to work with in analysis: they are all a pointwise-limit (and on a bounded domain a uniform limit) of simple functions. The particular sequence of simple functions will be called the *standard representation of f*. Using this result, we can define a new definition of equality up to zero-measure sets:

Proposition 2.1.19: Measurability up to Zero Set

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be two measure with ν being a complete measure, and let $f : X \to Y$ be $(\mathcal{M}, \mathcal{N})$-measurable. Then:

1. If $f = g$ up to a zero measure set with respect to μ (shorthanded to μ-a.e.), then g is measurable
2. If f_n is measurable for each $n \in \mathbb{N}$ and $f_n \to f$ μ-a.e, then f is measurable

***Proof*:**

1. Let $N = \{x \in X \mid f(x) \neq g(x)\}$. Let $B \in \mathcal{N}$ and consider $g^{-1}(B)$. Then by definition:
$$\begin{aligned} g^{-1}(B) &= \{x \in M \mid g(x) \in B\} \\ &= \{x \in \mathcal{M} \mid g(x) \in B,\ g(x) = f(x)\} \cup \{x \in \mathcal{M} \mid g \in B,\ g(x) \neq f(x)\} \end{aligned}$$
the latter set is a subset of N, and so is measurable by the completeness of f. The former set is of the form:
$$\begin{aligned} \{x \in \mathcal{M} \mid g(x) \in B,\ g(x) = f(x)\} &= \{x \in \mathcal{M} \mid f(x) \in B,\ g(x) = f(x)\} \\ &= f^{-1}(B) \cap \{x \mid f(x) = g(x)\} \\ &= f^{-1}(B) \cap (X - \{x \mid f(x) \neq g(x)\}) \\ &= f^{-1}(B) \cap (X - N) \end{aligned}$$
and since f measurable the pre-image of a measurable set is too, and since N is measuable, $X - N$ is too, so we have that $g^{-1}(B)$ is measurable.
2. Let's say N is the set on which $f_n \not\to f$. Define
$$g_n = f_n \chi_{X-N} + f \chi_N$$
Then notice that $g_n = f_n$ μ-a.e., and so by part 1 are measurable, and by construction $g_n \to f$ everywehre, and so f is measurable!

However, if μ is not complete, it turns out to make little difference since we can complete the measure, produce the result, and restrict back to the original measure without changing the result:

Proposition 2.1.20: Extending To Complete Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space and $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. if f is a $\overline{\mathcal{M}}$-measurable function on X, there is a $\mathcal{M}$-measurable function g such that $f = g$ $\overline{\mu}$-a.e.

***Proof*:**
If f is a simple function, then we see that the completion will at most remove finitely many points when we restrict every χ_E $(E \in \overline{\mathcal{M}})$ to g, and so $f = g$ $\overline{\mu}$-a.e.

Now let f be a $\overline{\mathcal{M}}$-measurable function, and by theorem 2.1.18 choose a sequence of $\overline{\mathcal{M}}$-measurable functions $\{\varphi_i\}_{i=1}^{\infty}$ that converge pointwise to f. The idea is that each $\varphi_i = \psi_i$ $\overline{\mu}$-a.e., that is, there exists a E_i where $\varphi_i(x) \neq \psi_i(x)$ for all $x \in E_i$ but $\overline{\mu}(E_i) = 0$, and the countable union of these will still be measure 0.

In particular Let $E = \bigcup_i^{\infty} E_i$. By definition of completion, there must exist some N such that $\mu(N) = 0$ and $E \subseteq N$. Let $g = \lim_{\chi_{X-N}} \psi_n$. Then g is measurable by corollary 2.1.13, and clearly $f = g$ on N^c, completing the proof.

2.2 Integration of non-negative Function

We are now in a position to define how to integrate functions (integrable functions coming soon)! Throughout this section, let $(X, \mathcal{M}, \mu)$ be a measure space

Definition 2.2.1: L^+

Let
$$L^+ = \{f : X \to [0, \infty] \mid f \text{ is a } \mathcal{M}\text{-measurable function}\}$$

Sometimes, L^+ is written as $L^+(X)$, $L^+(\mu)$, or $L^+(\mu, X)$ to emphasize which space of functions we are dealing with. If it is unambiguous, we use L^+.

Definition 2.2.2: Integral of Simple Function

Let $f = \sum_1^n a_i \chi_{E_i}$ be a simple function. Then define the *integral of f with respect to μ* to be

$$\int f d\mu = \sum_i^n a_i \mu(E_i)$$

In other words, we simply took the measure of the domain of χ_{E_i} for each i. As usual, we take the convention that $0 \cdot \infty = 0$. We will also allow $\int f = \infty$ to be a valid result. If there is no confusion, we will write $\int f$ instead of $\int f d\mu$. It should also be emphasized that it is the measure of the *domain* and the coefficient from the *codomain* that we are taking.

If we want to integrate over just some $A \subseteq X$ where $A \in \mathcal{M}$, then $\varphi|_A$ is also a simple function (simply take $\chi_{E \cap A}$ for each characteristic function). We usually denote this by

$$\int_A f d\mu = \int f \chi_A \, d\mu$$

Because of this notation, we can write $\int f d\mu$ as

$$\int f d\mu = \int_X f d\mu$$

Proposition 2.2.3: properties of Integrals of Simple Functions

Let φ and ψ be simple functions in L^+. Then

1. if $c \geq 0$, $\int c\varphi d\mu = c \int \varphi d\mu$
2. $\int(\varphi + \psi)d\mu = \int \varphi d\mu + \int \psi d\mu$
3. if $\varphi \leq \psi$ then $\int \varphi \leq \int \psi$
4. The map $A \mapsto \int_A 1 d\mu$ is a measure on $\mathcal{M}$ (more generally, $A \mapsto \int_A \varphi d\mu$ is a meaure on $\mathcal{M}$ for a simple function φ)

***Proof*:**
Do these as midterm exercises

With this definition, we can define integrable function more generally:

Definition 2.2.4: Integral of Measurable Function

Let $f \in L^+$. Then define

$$\int f d\mu := \sup \left\{ \int \varphi d\mu \,\middle|\, \varphi \text{ is a simple function, } 0 \leq \varphi \leq f \right\}$$

It's important to see why the integral of a simple function will give the same result as the "simple integral" (i.e. integral over simple functions from earlier), in particular since the set over which we are containing the supremum will contain the simple function in question, and so the maximum will be achieved by itself (and proposition2.2.3?). Furthermore, results 1 and 3 from proposition 2.2.3 holds for the definition of integration just given. The other results, in particular linearity, will be established soon. We can also see how the definition of the integral solidifies the idea that the integral is a sort of "weighted measure": the integral of a simple function is just a sum of measure of sets weighted by some constant, and we take the supremum of all such sets. We will in fact take a closer look at defining measure in terms of integration of functions in chapter 3.

Given the supremum definition of the integral, it could be rather hard to find the integral of a function in L^+. We essentially never check this, and treat the following theorem as the de-facto way of finding the integral:

Theorem 2.2.5: Monotone Convergence Theorem (MCT)

If $\{f_n\}_{i=1}^{\infty} \subseteq L^+$ is a sequence such that $f_i \le f_{i+1}$ for all $i \in \mathbb{N}$ and $f = \lim_{n\to\infty} f_n$ $(= \sup_n f_n)$, then

$$\int f = \lim_{n\to\infty} \int f_n$$

or, since $f = \lim_{n\to\infty} f_n$:

$$\int \lim_{n\to\infty} f = \lim_{n\to\infty} \int f_n$$

i.e. the limit commutes

It might be tempting to then say that $\int$ is "continuous" in the appropriate domain since it commutes with limits. However, the f_i must be *increasging* for the monotone convergence theorem to apply, and hence it is better to say that $\int$ is continuous from bellow. We will later show that the limit need not always commute (and hence it will be innapropriate to say that the integral is continuous on the space of measurable functions).

2.2.6 Remark any times in Analysis, when the limit can commute with something (ex. integral, another limit, functions), then the proof that it commutes is usually given a name (ex. sequential continuity, MCT, DCT, Fatou lemma, Moore-Osgood theorem, etc.) see:

wikipedia: Interchanging Limits

***Proof*:**
We'll show $\le$ and $\ge$. For $\le$, notice that $\{\int f_n\}$ is an increasing sequence, and so converges (possibly to ∞). Furthermore, $f_n \le f$ for all n, and so by proposition 2.2.3 updated for general integral functions, we have:

$$\int f_n \le \int f$$

and so

$$\lim_{n\to\infty} \int f_n \le f$$

Conversely, we'll show that if we "shrink" the left hand side by a factor of $\alpha \in (0,1)$ so that we get $\ge$, then since this will be true for all α, we will get the result we want.

Fix some $\alpha \in (0,1)$, and choose a simple function where $0 \le \varphi \le f$ (such a function exists by theorem 2.1.18). Let

$$E_n = \left\{x \in \overline{\mathbb{R}} \;\middle|\; f_n(x) \ge \alpha\varphi(x)\right\}$$

Then $\{E_n\}_{n=1}^{\infty}$ is an increasing sequence (via inclusion) of measurable sets who's union is X (since $\lim_{n\to\infty} f_n = f$ and the f_n's are increasing). Then

$$\int f_n \ge \int_{E_n} f_n \ge \alpha \int_{E_n} \varphi$$

Since the map $E_n \mapsto \int_{E_n} \varphi$ is measurable by proposition 2.2.3 and the E_n's are continuous from above, we have

$$\lim_{n\to\infty}\int_{E_n}\varphi = \int \varphi$$

and so

$$\lim_{n\to\infty}\int f_n \geq \alpha \int \varphi$$

Since this is true for all $\alpha < 1$, it is true for $\alpha = 1$, and so $\int f_n \geq \int \varphi$. Taking the supremum over all φ, we get that

$$\lim_{n\to\infty}\int f_n \geq \int f$$

And since we showed the $\leq$ direction, we in fact have:

$$\lim_{n\to\infty}\int f_n = \int f$$

as we sought to show

Thus, to find the value of $\int f$, it suffices to find some sequence of simple functions $\{\varphi_n\}$ that converges to f, which always exists since every measurable function has a standard representation (theorem 2.1.18).

An immediate consequence of the Monotone Convergence Theorem is that linearity also applies to integrals:

Proposition 2.2.7: Linearity of Integrals

Let $\{f_i\}_{i=1}^N \subseteq L^+$ where $N \in \mathbb{N} \cup \{\infty\}$ and let $f = \sum_i^N f_i$. Then

$$\int f = \sum_i^N \int f_n$$

***Proof*:**
Let's first consider the case where $n = 2$ so that we have f_1 and f_2. Let $\{\varphi_i\}$ and $\{\psi_i\}$ be a sequence for the standard representation of f_1 and f_2. Then clearly $\{\varphi_i + \psi_i\}$ is a sequence for a standard representation of $f_1 + f_2$. Then by the Monotone Convergence Theorem and linearity

of integrals with simple functions:

$$\begin{aligned}
\int f_1 + f_2 &= \lim_{i\to\infty} \int \varphi_i + \psi_i \\
&\overset{\text{MCT}}{=} \int \lim_{i\to\infty} \varphi_i + \psi_i \\
&= \int \lim_{i\to\infty} \varphi_i + \lim_{i\to\infty} \psi_i \\
&= \int \lim_{i\to\infty} \varphi_i + \int \lim_{i\to\infty} \psi_i \\
&\overset{\text{MCT}}{=} \lim_{i\to\infty} \int \varphi_i + \int \lim_{i\to\infty} \psi_i \\
&= \int f_1 + \int f_2
\end{aligned}$$

By induction, it holds that

$$\int \sum_i^n f_i = \sum_i^n \int f_i$$

Letting $n \to \infty$, we can once again apply the Monotone Convergence Theorem and get

$$\begin{aligned}
\int \sum_i^\infty f_i &= \int \lim_{n\to\infty} \sum_i^n f_i \\
&\overset{\text{MCT}}{=} \lim_{n\to\infty} \int \sum_i^n f_i \\
&= \lim_{n\to\infty} \sum_i^n \int f_i \\
&= \sum_i^\infty \int f_n
\end{aligned}$$

completing the proof

The next result is a generalization from the same result with Riemann integrals, except we replace the condition of "finitely many points" with "null-set amount of points". It follows the trend of integration "forgetting" up to a zero-measure amount of points:

Proposition 2.2.8: Zero Integral

Let $f \in L^+$. Then $\int f = 0$ if and only if $f = 0$ a.e.

***Proof*:**
Starting with simple functions, this is evident in both directions. If φ was our simple function and $\int \varphi = 0$, then $\sum_i^n a_i\mu(E_i) = 0$, so either each $a_i = 0$ or each $\mu(E_i) = 0$ which immediately implies φ is zero a.e. (not necessarily everywehre since it's possible that $E_i \neq \emptyset$ but $\mu(E_i) = 0$).

Conversely, if each $\mu(E_i) = 0$ a.e., then clearly $\int \varphi = 0$.

Moving onto to a measurable function f, the $\Leftarrow$ comes almost immediately. If $f = 0$ a.e., then for every simple function $\varphi \leq f$, then $\varphi = 0$ a.e., which we have established means $\int \varphi = 0$. Then by definition of $\int f$:

$$\int f = \sup_{\varphi \leq f} \int \varphi = 0$$

For the $\Rightarrow$ direction, assume that $\int f = 0$, and for the sake of contradiction let's say $f \neq 0$ a.e. The idea is that we can then construct a simple function φ where $\varphi \leq f$ whose integral is greater than zero, but $\int f = \sup_{\varphi \leq f} \int \varphi$. To that end take:

$$\bigcup_{i}^{\infty} E_n = \bigcup_{i}^{\infty} \left\{ x \;\middle|\; f(x) > \frac{1}{n} \right\} = \{x \mid f(x) > 0\} = E$$

Since $f \neq 0$ a.e., it must be that $\mu(E) > 0$, and so by construction for some n, $\mu(E_n) > 0$. But then

$$f \geq n^{-1} \chi_{E_n} \qquad \Leftrightarrow \qquad \int f \geq n^{-1} \mu(E_n) > 0$$

so the supremum of values in f contain simple functions of nonzero measure, showing that $\int f > 0$, a contradiction to our original assumption.

This theorem also shows that if $f = g$ a.e., then $\int f = \int g$, since $f - g = 0$ a.e., so $\int f - g = 0$, so $\int f = \int g$. Using this, we can upgrade the Monotone Conversely Theorem by needing the convergence to happen up to a measure zero set:

Corollary 2.2.9: MCT a.e.

Let $\{f_n\}_{i=1}^{\infty} \subseteq L^+$ and $f \in L^+$. Then if $f_1 \leq f_2 \leq \cdots \leq f$ and $f_n \to f$ a.e., then

$$\int f = \lim_{n \to \infty} \int f_n$$

***Proof*:**
Let $f_n(x) \uparrow f(x)$ for all $x \in E$ such that $\mu(E^c) = 0$. Then $f - f\chi_E = 0$ a.e. which implies $f_n - f_n \chi_E = 0$ a.e., so by the Monotone Conversely Theorem

$$\int f = \int f \chi_E \overset{\text{MCT}}{=} \lim_{n \to \infty} \int f_n \chi_E = \lim_{n \to \infty} \int f_n$$

as we sought to show.

We next address the fact that the sequence $\{f_i\}$ must be increasing (at least a.e.) for the MCT to work (recall we relied on the sequence being increasing in the construction of one of the direcitons of the inequality). If it were not, then it need not be the case that the limit passes through the integral!

Example 2.2.10: not increasing $\not\Rightarrow$ Limit commutes with $\int$

Let $X = \mathbb{R}$ and μ be the Lebesgue measure. Consider the two following examples:

$$\{n\chi_{(0,1/n)}\}_{n=1}^{\infty}$$

then $n\chi_{(0,1/n)} \to 0$. However, notice that $\int n\chi_{(0,1/n)} = 1$ for each element in the sequence. Thus:

$$0 = \int \lim_{n\to\infty} n\chi_{(0,1/n)} \neq \lim_{n\to\infty} \int n\chi_{(0,1/n)} = 1$$

Even if the function is bounded, the problem remains: take $\chi_{(n,n+1)}$ which also converges pointwise to 0 but the sequence of the integrals of the functions is a sequence of constants 1 and so converges to 1 which is clearly not 0.

This is interesting if we interpret it from a sort of "physics" perspective: somehow, we have "lost our mass" in the process of taking our limit. There is in fact essentially 3 ways we can "lose mass', the first two begin the counter-examples above, the last one being a function that oscillates faster and faster (this last one requires a function that is also defined in the negative, and so we will get back to this)

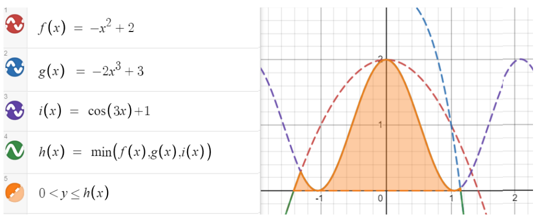
Notice that in both examples, there is some "escaping to infinity" going on. Perhaps if we have some overarching "bound" to the sequence, then we can pass the limit through, and that will indeed be the case. We address this when talking about the Dominated Convergence Theorem in the next section. However, not all hope is lost for the general case; there is weakening to only the lim inf and the equality to $\leq$ which works in the general case:

Theorem 2.2.11: Fatou's Lemma

If $\{f_n\} \subseteq L^+$ is *any* sequence, then

$$\int \liminf_n f_n \leq \liminf_n \int f_n$$

To visualize this, consider:



The integral of the orange region weill be less than or equal to the lim inf of those functions over that region. In this case, the integral would be equal, but in the case where the mass somehow "escapes", we will have an inequality.

***Proof*:**
This comes directly from properties of the $\liminf$. Let $k \geq 1$. Then

$$\inf_{i \geq k} f_i \leq f_j \quad \forall\ j \geq k$$

Then by theorem 2.2.3

$$\int \inf_{i \geq k} f_i \leq \int f_j \quad \forall j \geq k$$

and since this holds for all $j \geq k$, we get:

$$\int \inf_{i \geq k} f_i \leq \inf_{j \geq k} \int f_j$$

Since the $\liminf$ defines an increasing sequence of function, by letting $k \to \infty$ and applying the Monotone Conversely Theorem:

$$\int \liminf f_n = \lim_{k \to \infty} \int \inf_{n \geq k} f_n \leq \liminf \int f_n$$

as we sought to show

Like before, the result holds if we loosen the condition so that $f_n \to f$ a.e.

Corollary 2.2.12: Fatou's Lemma a.e.

let $\{f_n\} \subseteq L^+$ be any sequence and $f \in L^+$. If $f_n \to f$ a.e. Then

$$\int f \leq \liminf_n \int f_n$$

***Proof*:**
Using proposition 2.2.9, make $f_n \to f$ everywhere arguing that it is equivalent to do so, and then the rest is Fatou's Lemma.

Finally, we show that if $\int f < \infty$, we can guarantee that f cannot "explode" too much, and that even better, the set over which the value of f is non-zero is σ-finite, i.e., it is not "pathological"

Proposition 2.2.13

Let $f \in L^+$. If $\int f < \infty$, then $\{x \mid f(x) = \infty\}$ is a null set and $\{x \mid f(x) > 0\}$ is σ-finite

***Proof*:**
was left as an exercise to the reader in the book

2.3 Integration of Real and Complex Functions

We'll again let $(X, \mathcal{M}, \mu)$ be the fixed measure space for this section, except the codomain now will be either $\mathbb{R}$ or $\mathbb{C}$. For the case of $\mathbb{R}$, we can quickly extend our results from last section by splitting f into f^+ and f^- (which are both measurable if f is by corrollary 2.1.12), and if at least one of f^+ or f^- is finite, then

$$\int f = \int f^+ - \int f^-$$

Notice that if they are both infinity, we have $\infty - \infty$ which is in general not well-defined. Thus, we will limit our attention to functions where $\int |f| < \infty$ since $|f| = f^+ + f^-$:

For the case of $\mathbb{C}$, using the triangle inequality we get that:

$$|f| = |\text{Re}f + \text{Im}f| \leq |\text{Re}f| + |\text{Im}f| \leq 2|f|$$

and so both $\text{Re}f$ and $\text{Im}f$ must be finite to get a well-defined result. In that case, we define

$$\int f := \int \text{Re}f + i \int \text{Im}f$$

We give a name for complex functions (the real functions being a special case) that satisfy this finiteness condition:

Definition 2.3.1: Integrable Function

Let $f : X \to \mathbb{C}$ be measurable. We say f is *integrable* if

$$\int |f| < \infty$$

Let L^1 represent the set of all integrable functions.

Like with L^+, we sometimes write $L^1(\mu)$, $L^1(X)$, or $L^1(X, \mu)$. If the codomain is $\mathbb{R}$, then we get that $\int |f^+| < \infty$ and $\int |f^-| < \infty$, as we claimed for our extension of the integral to $\mathbb{R}$. Furthermore, for any $E \in \mathcal{M}$, we will say that f is *integrable on* E if $\int_E |f| < \infty$.

The set of L^1 functions are well-behaved under addition and scalar multiplication in the sense that they form a vector-space:

Proposition 2.3.2: Vector Space on L^1

Let L^1 be the set of integrable functions. Then with $+$ and scalar multiplication, L^1 forms a vector-space (over $\mathbb{R}$ or $\mathbb{C}$ respective to the codomain). Furthermore, $\int$ is a linear functional on L^1 (i.e. $\int : L^1 \to \mathbb{R}$ is a linear functional)

***Proof*:**
the proof that $cf + dg$ is integrable comes from the triangle inequality ($|cf + dg| \leq |c||f| + |d||g| < \infty$). The fact that $cf \in L^1$ for any $c \in \mathbb{R}$ and $f \in L^1$ follows similarly.

$\int$ being a functional, we need to show linearity and $\int cf = c \int f$. For linearity, assume $h = f + g$

is integrable. Then $h^+ - h^- = f^+ - f^- + g^+ - g^-$, or, re-arranging, we get:

$$h^+ + f^- + g^- = h^- + f^+ + g^+$$

These are all functions in L^+, and so by linearity for of integrals for L^1 functions:

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+$$

re-arraning, we get:

$$\begin{aligned}
&= \int f + g \\
&= \int h = \int h^+ - \int h^- = \int f^+ - \int f^- + \int g^+ - \int g^- \\
&= \int f + \int g
\end{aligned}$$

showing linearity. Through a similar means, we can also show $\int cf = c \int f$, completing the proof.

2.3.3 Remark Since it is a vector-space, it has a basis and a dimension. What dimension is this vector-space? Since it is a real vector-space, it is isomorphic to $\mathbb{R}^N$ for appropriate cardinal N (N may be finite, countable, uncountable, etc). If $N \geq \aleph_0$, that means that it is rather difficult to define a measure on the space L^1[a]. Trying to find ways to study "measure" notions infinite dimensional vector-spaces (ex. boundedness, σ-finiteness, etc.) is accomplished by defining various different concepts like *seminorms*, *norms*, or *inner products*, which we shall do in chapter [13, Banach and Fréchet Spaces].

[a] For an illustration of the oddities of infinite dimensional vector-spaces, consider the [hyper]-volume of a unit ball S^n as $n \to \infty$; you should find that it goes to 0!

Next, we would like to establish some properties of integrable functions. One result is a including a useful inequality including absolute values. Another is how the results still work as long as things happen "a.e.". Finally, the set of all "interesting" points (points where $f(x) \neq 0$) are still σ-finite:

Proposition 2.3.4: Properties of Integrable Functions

Let $f, g \in L^1$. Then:

1. $\left|\int f\right| \leq \int |f|$
2. $\{x \mid f(x) \neq 0\}$ is σ-finite
3. $\int_E f = \int_E g$ for all $E \in \mathcal{M}$ if and only if $\int |f - g| = 0$ if and only if $f = g$ a.e.

***Proof*:**

1. The result is essentially immediate in the $\mathbb{R}$ case. In the $\mathbb{C}$ case, let α where $|\alpha| = 1$ be such that it "rotates" the integral so that it's in the real case, and then apply a similar strategy to that of the $\mathbb{R}$ case.

2. Notice that $\{x \mid f(x) \neq 0\} = \{x \mid f^+(x) > 0\} \cup \{x \mid f^-(x) < 0\}$, which we've shown in proposition 2.2.13 to be σ-finite

3. The fact that $\int |f-g| = 0$ if and only if $f = g$ a.e. comes from proposition 2.2.8, (since $|f-g| \in L^+$) so we will prove $\int_E f = \int_E g$ if and only if $\int |f-g| = 0$. Let's say $\int |f-g| = 0$. Then to show that $\int_E f = \int_E g$, it is equivalent to show that
$$\left| \int_E f - \int_E g \right| = 0$$
by the epsilon principle (i.e. $a = b$ if and only if $|a-b| < \epsilon$ for all ϵ). Using proposition 2.3.4, we get that:
$$\left| \int_E f - \int_E g \right| \leq \int \chi_E |f-g| \leq \int |f-g| = 0$$
completing the $\Rightarrow$ direction.
For the other direction, let's argue contra-positively and assume $\int |f-g| \neq 0$ a.e. Let
$$u = \operatorname{Re}(f-g) \qquad v = \operatorname{Im}(f-g)$$
so that u^+, u^-, v^+, v^- are the corresponding functions. Since $f \neq g$ a.e., at least one of these must have a set with positive measure. Without loss of generality, let's say $E = \{x \mid u^+(x) > 0\}$ and $\mu(E) > 0$. Then
$$\operatorname{Re}\left(\int_E f - g \right) = \operatorname{Re}\left(\int_E f - \int_e g \right) = \int_E u^+ + \int_E u^- = \mu(E) + 0 > 0$$
since $u^- = 0$ on E. Thus, the measure will be positive.

As a consequence of this, we usually think of integrable functions *up to a zero set*. If we wanted to, we can limit our scope to of integration to functions f that are measurable on E and E^c has zero-measure to functions where $f = 0$ on E^c. This also means that if we are working with an $\overline{\mathbb{R}}$-value function that is finite a.e., then it is equivalent to treat it as a $\mathbb{R}$-value function.

Proposition 2.3.4 also allows us to re-define L^1 in terms of functions that are equivalent up to a measure zero set. That is, we form an equivalence relation on L^1 if and only if $f = g$ a.e., and label the collection of equivalence classes as L^1. We abuse notation and write "$f \in L^1$" to mean we are selecting a function that is integrable (or an a.e.-defined integrable function). This is because we will usually be working with functions instead of the equivalence classes, and so this is a common abuse of notation.

The reason to introduce this new definition of L^1 is so that we can define a new metric on L^1:

Definition 2.3.5: L^1 metric

Let L^1 be the set of equivalence classes as defined above. Then

$$\rho(f, g) = \int |f - g|$$

is a metric on L^1

Symmetry and triangle inequality come immediately without even needing to use the fact that these are equivalence classes. Where the new construction of L^1 becomes important is to say $\int |f - g| = 0$ if and only if $f = g$. Since this is true a.e., then we need these two to be in an equivalence class. Since we defined a metric, we can define a notion of convergence, in particular, f_n converges to f in L^1 if and only if $\int |f - f_n| \to 0$, meaning $f_n \to f$ almost everywhere (however, L^1 convergence does not imply a.e. point-wise convergence, and a.e. point-wise convergence does not imply L^1 convergence, as we'll show in section 2.4).

We will soon explore more properties of this metric space, in particular, we would want this metric space to be complete so that the limit of a sequence of L^1 function's is L^1 and find some nice dense set of L^1 (so far, we only have the [point-wise] limit of measurable function is measurable, but no such result yet for integrable functions). To do so, we need more tools to analyze swapping limits and integrals. The next result we'll cover allows us to generalize the Monotone Conversely Theorem to the general case of $f_n \to f$ a.e., with the extra condition that each $|f_n|$ is bounded by a single $|g|$ with finite area (in this way, we are getting rid of the "escaping to infinity" cases in example 2.2.10)

Theorem 2.3.6: Dominated Convergence Theorem (DCT)

Let $\{f_n\}_{i=1}^{\infty} \subseteq L^1$ be a sequence of integrable functions such that

1. $f_n \to f$ a.e.
2. there exists a nonnegative $g \in L^1$ such that for all $n \in \mathbb{N}$, $|f_n| \leq g$ a.e.

Then

$$\int f = \lim_{n\to\infty} \int f_n$$

or equivalently:

$$\int \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int f_n$$

***Proof*:**
By proposition 2.1.19 and 2.1.20 , f is measurable (up to a null set, in which case we can appropriately redefine f on that null-set). Since $|f_n| < g$ for all $n \in \mathbb{N}$, $f < g$, and so $f \in L^1$.

Since all L^1 functions are broken down to real parts (complex functions into 2 real parts), it suffices to show the result for real value f_n and f. By assumption, we have $g + f_n \geq 0$ and $g - f_n \geq 0$. Thus, by Fatou's Lemma:

$$\int g + \int f \leq \liminf \int (g + f_n) = \int g + \liminf \int f_n$$

$$\int g - \int f \leq \liminf \int (g - f_n) = \int g - \limsup \int f_n$$

Therefore, $\limsup \int f_n \leq \int f \leq \liminf \int f_n$, and so the two value are actually equal and so

$$\int f = \lim_{n\to\infty} \int f_n$$

as we sought to show

Note that functions can still converge even if the hypothesis of the DCT fails:

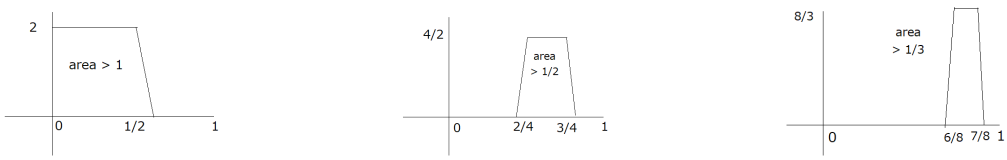
Example 2.3.7: Converges Without DCT

Construct a sequence of functions $\{f_n\}$ such that all the following hold:

1. $f_n : [0, 1] \to \mathbb{R}$ is continuous
2. $f_n \geq 0$
3. $\lim_{n\to\infty} f_n(x) = 0$ for all $x \in [0, 1]$
4. $\lim_{n\to\infty} \int_0^1 f_n(x)\, dm = 0$
5. $\sup_n f_n(x)$ is *not* integrable on $[0, 1]$ with respect to the Lebesgue measure

(that is, just because the hypothesis of the dominated convergence theorem fails does not mean the conclusion does not hold: it is not an if and only if)

Proof. The proof essentially is a formalizing the following images:



(a) f_1 (b) f_2 (c) f_3

Figure 2.1: Visual intuition for question 3

We'll first define the indicator functions, then make these functions continuous. Let

$$g_1(x) = \begin{cases} 2^1/1 & x \in \left[0, \frac{1}{2}\right) \\ 0 & x \in \left[1 - \frac{1}{2}, 1\right] \end{cases} \quad g_2(x) = \begin{cases} 2^2/2 & x \in \left[\frac{1}{2}, \frac{1}{4}\right) \\ 0 & x \in \left[1 - \frac{1}{4}, 1\right] \end{cases} \quad g_3(x) = \begin{cases} 2^3/3 & x \in \left[\frac{3}{4}, \frac{7}{8}\right) \\ 0 & x \in \left[1 - \frac{7}{4}, 1\right] \end{cases}$$

and so on, defining:

$$g_n(x) = \begin{cases} 2^n/n & x \in [2^{n-1} - 2/2^{n-1}, 2^{n-1} - 1/2^{n-1}) \\ 0 & x \in [1 - 2^{n-1} - 1/2^{n-1}, 1] \end{cases}$$

Next, we can make each g_n continuous by simply adding a line to omit the jump:

$$f_1(x) = \begin{cases} 2^1/1 & x \in [0, 1/2) \\ -2^{10\cdot 1}\left(x - \left(\frac{1}{2} - 1/2^{10\cdot 1}\right)\right) & x \in (1/2, 1/2 + 1/2^{10\cdot 1}] \\ 0 & x \in [1 - (1/2 + 1/2^{10\cdot 1}), 1] \end{cases}$$

and defining each f_n similarly. Then it is clear by our construction that $f_n \to 0$: since the trapezoids are moving away, they will never cross our values again once they pass them (this is the opposite of the proof for question 2 where they constantly loop back, since the proof is similar the details were ommited for this proof). Furthermore,

$$\int f_n = 1/n + 1/2^{10n}$$

Thus, $\int f_n \xrightarrow{n\to\infty} 0$, showing that the integral also converges to 0. However,

$$\int \sup_n f_n \geq \int \sup_n g_n = \sum_{k=1}^{\infty} 1/k = \infty$$

showing that the $\sup_n f_n$ is *not* integrable, as we sought to show.

□

Therefore, the DCT is a sufficient, but not necessary condition for convergence. Sufficient and necessary conditions are a bit harder to come by, and we will not need to characterize limits commuting with integrals in these other ways, but for those who are curious:

https://journalofinequalitiesandapplications.springeropen.com/articles/10.1186/s13660-020-02502-w

This theorem will be key in finding the integral of many function constructed by limits, and will be used to prove completeness of L^1 (after defining Cauchy sequences in L^1 in section ref:HERE) A more immediate result is that the Dominated convergence theorem allows to define a form of countable linearity for L^1 functions:

Theorem 2.3.8: Linearity for Integrable Functions

Let $\{f_i\}_{i=1}^{\infty} \subseteq L^1$ be a sequence of functions, and $f = \sum_{i=1}^{\infty} f_i$. If $\sum_{i=1}^{\infty} |f_i| < \infty$, then $\sum_{i=1}^{\infty} f_i$ converges a.e. to a function in L^1, and

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

***Proof*:**
First, since $\sum_{i=1}^{\infty} |f_i| < \infty$, by proposition 2.2.7,

$$\int \sum_{i=1}^{\infty} |f_i| = \sum_{i=1}^{\infty} \int |f_i|$$

Thus, $g = |\sum_{i=1}^{\infty} f_i|$ is a function in L^1. The function g is finite a.e. by proposition 2.3.4[2], so by the dominated convergence theorem, any partial sum of functions will commute, and so taking the limit, we get:

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

as we sought to show

The dominated convergence theorem also gives us a dense set in L^1 to work with, namely the simple functions:

Theorem 2.3.9: Simple Functions Dense in L^1

Let $f \in L^1$. Then for all $\epsilon > 0$, there exists an integrable simple function $\varphi = \sum a_i \chi_{E_i}$ such that

$$\int |f - \varphi| d\mu < \epsilon$$

Furthermore, if μ is a Lebesgue-Stieltjes measure on $\mathbb{R}$, the sets E_i can be taken to be finite union of open intervals, and better, there is a continuous function g that vanishes outside a bounded interval (i.e. $g = 0$ outside the bounded interval) such

$$\int |f - g| d\mu < \epsilon$$

***Proof*:**
We first prove the general statement. By theorem 2.1.18(2) there is a sequence of simple functions $\{\varphi_n\}_{n=1}^{\infty}$ with $0 \leq |\varphi_1| \leq |\varphi_2| \leq \cdots \leq |f|$ and $\varphi_n \to f$ pointwise. Each φ_n is integrable, for $|\varphi_n|$ is a simple function with $0 \leq |\varphi_n| \leq |f|$, so definition 2.2.4 gives $\int |\varphi_n| \leq \int |f| < \infty$. Each $|f - \varphi_n|$ is measurable, being the composition of the measurable $f - \varphi_n$ (proposition 2.1.10) with the continuous modulus (corollary 2.1.3, composed as noted at the start of this chapter). Pointwise we have $|f - \varphi_n| \leq |f| + |\varphi_n| \leq 2|f|$ with $2|f| \in L^1$, and $|f - \varphi_n| \to 0$. The dominated convergence theorem (theorem 2.3.6) applied to the sequence $|f - \varphi_n|$ therefore gives $\int |f - \varphi_n| \, d\mu \to 0$, so $\int |f - \varphi_n| \, d\mu < \epsilon$ for n large enough. Take $\varphi = \varphi_n$ for such an n.

Before turning to the second statement, we record a fact that costs nothing here and will be needed again: every set on which φ is nonzero has finite measure. Writing $\varphi = \sum_{i=1}^{m} a_i \chi_{E_i}$ with distinct nonzero values a_i and disjoint sets $E_i = \varphi^{-1}(a_i)$, definition 2.2.2 computes $|a_i| \, \mu(E_i) = \int_{E_i} |\varphi| \, d\mu \leq \int |f| \, d\mu < \infty$, so $\mu(E_i) < \infty$ for every i.

Now suppose μ is a Lebesgue-Stieltjes measure on $\mathbb{R}$, and run the first part with $\epsilon/3$ in place of ϵ. For measurable sets E and F we have $|\chi_E - \chi_F| = \chi_{E \triangle F}$ pointwise, so

$$\int |\chi_E - \chi_F| \, d\mu = \mu(E \triangle F)$$

and approximating an indicator in the L^1 metric is the same as approximating its set in measure. Since each $\mu(E_i)$ is finite, proposition 1.4.6 provides a finite union of open intervals A_i with $\mu(E_i \triangle A_i) < \epsilon/(6m|a_i|)$, and after merging overlaps the intervals making up each A_i may be taken disjoint. The intervals may moreover be taken bounded. Indeed $\mu(A_i) \leq \mu(E_i) + \mu(E_i \triangle A_i) < \infty$, so if one of them is unbounded, say (c, ∞), then the tails $[N, \infty)$ for $N > c$ decrease to $\emptyset$

and satisfy $\mu([N,\infty)) \leq \mu(A_i) < \infty$, so continuity from above (proposition 1.2.5(4)) gives $\mu([N,\infty)) \to 0$. Replacing (c,∞) by (c,N) for N large enough keeps $\mu(E_i \triangle A_i) < \epsilon/(3m|a_i|)$, and symmetrically for an interval unbounded below. Setting $\psi = \sum_{i=1}^m a_i \chi_{A_i}$, the pointwise triangle inequality together with linearity (proposition 2.2.7) and monotonicity of the integral (noted after definition 2.2.4) gives

$$\int |\varphi - \psi| \, d\mu \leq \sum_{i=1}^m |a_i| \int |\chi_{E_i} - \chi_{A_i}| \, d\mu = \sum_{i=1}^m |a_i| \, \mu(E_i \triangle A_i) < \frac{\epsilon}{3}$$

It remains to trade each indicator of an interval for a continuous function, and by the same bookkeeping it suffices to approximate $\chi_{(a,b)}$ for a single bounded open interval (a,b) to any prescribed accuracy. For $0 < \delta < (b-a)/2$ let g_δ be the continuous trapezoid that equals 0 outside (a,b), equals 1 on $[a+\delta, b-\delta]$, and interpolates linearly on $[a, a+\delta]$ and $[b-\delta, b]$. Then $0 \leq \chi_{(a,b)} - g_\delta \leq \chi_{(a,a+\delta)} + \chi_{(b-\delta,b)}$ pointwise, so

$$\int |\chi_{(a,b)} - g_\delta| \, d\mu \leq \mu((a, a+\delta)) + \mu((b-\delta, b))$$

As $\delta \downarrow 0$ the intervals $(a, a+\delta)$ decrease to $\emptyset$, and they have finite measure because $\mu((a, a+\delta]) = F(a+\delta) - F(a) < \infty$ (theorem 1.4.2), so continuity from above (proposition 1.2.5(4)) sends $\mu((a, a+\delta))$ to 0, and likewise $\mu((b-\delta, b)) \to 0$. Choosing δ small enough on each of the finitely many disjoint intervals making up the A_i, and summing the resulting trapezoids with the weights a_i, produces a continuous function g with $\int |\psi - g| \, d\mu < \epsilon/3$. The function g vanishes outside a bounded interval, since it vanishes outside the finitely many bounded intervals just used. Finally,

$$\int |f - g| \, d\mu \leq \int |f - \varphi| \, d\mu + \int |\varphi - \psi| \, d\mu + \int |\psi - g| \, d\mu < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

completing the proof

Nothing in the first half of this argument is special to the first power: the same increasing simple approximation and the same dominating trick, now raised to the p-th power, approximate any function whose p-th power is integrable. To state this cleanly, for $1 \leq p < \infty$ and a measurable $f : X \to \mathbb{C}$ define the notation

$$\|f\|_p := \left(\int |f|^p \, d\mu \right)^{1/p}$$

and write $f \in L^p(\mu)$ (or just $f \in L^p$) to mean that f is measurable with $\|f\|_p < \infty$, so that $p = 1$ recovers definition 2.3.1. For now this is pure integration shorthand. The quantity $\|\cdot\|_p$ turns out to be a seminorm, but even its triangle inequality (Minkowski's inequality) belongs to the study of L^p as a normed space carried out in [13], and the results below do not use it.

Theorem 2.3.10: Simple Functions Dense in L^p

Let $(X, \mathcal{M}, \mu)$ be a measure space, $1 \leq p < \infty$, and let $f : X \to \mathbb{C}$ be measurable with $\int |f|^p \, d\mu < \infty$. Then for every $\epsilon > 0$ there is a simple function $\varphi = \sum_{i=1}^m a_i \chi_{E_i}$ with $\mu(E_i) < \infty$ for every i such that

$$\|f - \varphi\|_p < \epsilon$$

***Proof*:**
By theorem 2.1.18(2) choose simple functions φ_n with $0 \leq |\varphi_1| \leq |\varphi_2| \leq \cdots \leq |f|$ and $\varphi_n \to f$ pointwise. The function $|f - \varphi_n|^p$ is measurable, being the composition of the measurable $f - \varphi_n$ (proposition 2.1.10) with the continuous $z \mapsto |z|^p$ (corollary 2.1.3, composed as noted at the start of this chapter). Pointwise, $|f - \varphi_n| \leq |f| + |\varphi_n| \leq 2|f|$, and since $t \mapsto t^p$ is increasing on $[0, \infty)$,

$$|f - \varphi_n|^p \leq 2^p |f|^p$$

The right side is integrable by hypothesis, and $|f - \varphi_n|^p \to 0$ pointwise, so the dominated convergence theorem (theorem 2.3.6) gives $\int |f - \varphi_n|^p \, d\mu \to 0$. Fix n with $\int |f - \varphi_n|^p \, d\mu < \epsilon^p$ and set $\varphi = \varphi_n$, so that $\|f - \varphi\|_p < \epsilon$.

For the finiteness claim, write $\varphi = \sum_{i=1}^m a_i \chi_{E_i}$ with distinct nonzero values a_i and disjoint sets $E_i = \varphi^{-1}(a_i)$. Then $|\varphi|^p = \sum_{i=1}^m |a_i|^p \chi_{E_i}$ is simple with $|\varphi|^p \leq |f|^p$, so definition 2.2.2 together with definition 2.2.4 gives

$$\sum_{i=1}^m |a_i|^p \, \mu(E_i) = \int |\varphi|^p \, d\mu \leq \int |f|^p \, d\mu < \infty$$

Since every a_i is nonzero, every $\mu(E_i)$ is finite, completing the proof

The finiteness of the $\mu(E_i)$ is automatic rather than an extra hypothesis: a simple function with $\|\varphi\|_p < \infty$ must assign finite measure to each of its nonzero level sets, so it vanishes outside a set of finite measure. This observation is what will let the approximation be pushed one step further once the underlying space carries a topology. On a locally compact Hausdorff space with a Radon measure, the simple functions here can be traded for continuous functions of compact support (theorem 4.2.4).

For the purposes of Ordinary Differential Equations (ODE's), we want to establish is a quick result to relate differentiable functions and integral of functions:

Theorem 2.3.11: Relating Derivative and Integral

Let $(X, \mathcal{M}, \mu)$ be a measure space, and suppose $f : X \times [a, b] \to \mathbb{C}$ for $-\infty < a < b < \infty$ and that $f(\cdot, t) : X \to \mathbb{C}$ is integrable for every $t \in [a, b]$, so we can define $F(t) = \int_X f(x, t) \, d\mu$.

1. Suppose that there exists a $g \in L^1$ such that $|f(x, t)| \leq g(x)$ for all x, t. Then if $\lim_{t \to t_0} f(x, t) = f(x, t_0)$ for every x, $\lim_{t \to t_0} F(t) = F(t_0)$; that is, if $f(x, \cdot)$ is continuous at every x, so is F.

2. Suppose $\frac{\partial f}{\partial t}$ exists and there is a $g \in L^1(\mu)$ such that $\left|\frac{\partial f}{\partial t}(x, t)\right| \leq g(x)$ for all x, t. Then F is differentiable and
$$F'(t) = \int \frac{\partial f}{\partial t}(x, t) \, d\mu(x)$$

The way I like to think about is that there is a system X that at different times t acts/behaves differently. This difference in how the system acts at any given time t is captured by $f(x, t)$. The function F capture the "total value" of this action/behavior (note that f is non-negative too, so all behavior has a "non-negative" output).

Now, if the behavior varies continuously, then the total output varies continuously, and if the behavior is smooth smooth enough so that it is differentiable (at any slice), then F is also differentiable, and it is in fact even easy to compute the value of F'.

***Proof*:**

1. Since $|f(x,t)|$ is dominated by g, simply apply the Dominated Convergence Theorem:

$$\lim_{t\to t_0} F(t) = \lim_{t\to t_0} \int f(x,t)\, d\mu(x) = \int \lim_{t\to t_0} f(x,t)\, d\mu(x) = \int f(x,t_0)\, d\mu(x) = F(t_0)$$

where $\{t_n\}$ is any sequence converging to t (since any $f(x,t)$ satisfies the bound, no more restrictions are needed on the choice of sequence)

2. Since $\frac{\partial f}{\partial t}$ exists, the limit of the derivative exists, and so define

$$\frac{\partial f}{\partial t} = \lim_{t_n\to t_0} h_n(x) \qquad h_n(x) = \frac{f(x,t_n) - f(x,t_0)}{t_n - t_0}$$

$\{t_n\}$ is any sequence converging to t. Since h_n difference and division of measurable functions, by proposition 2.1.10 it is measurable, and by the Dominated Convergence Theorem $\frac{\partial f}{\partial t}$ is measurable. Furthermore, by the Mean Value Theorem:

$$|h_n(x)| \le \sup_{t\in[a,b]} \left|\frac{\partial f}{\partial t}(x,t)\right| \le g(x)$$

and so by the Dominated Convergence Theorem again:

$$\begin{aligned}
F'(t_0) &= \lim_{t_n\to t} \left(\frac{F(t_n) - F(t_0)}{t_n - t_0}\right) \\
&= \lim_{t_n\to t} \left(\int h_n(x)\, d\mu(x)\right) \\
&\overset{\text{D.C.T.}}{=} \int \lim_{t_n\to t} h_n(x)\, d\mu(x) \\
&= \int \frac{\partial f}{\partial t}\, d\mu(x)
\end{aligned}$$

completing the proof

The final fact we shall present is a continuation of proposition 2.2.13 in the sense that we aim to further establish connections between some subsets of the domain and the resulting integral value in the co-domain, and see that if $\int$ is limited to some finite area of integration E ($\int_E < \infty$), does it actually respect continuity? In fact, it does! Even better, it is in a sense uniformally continuous with respect to the measure of the set that is being integrated over.

Proposition 2.3.12: $\int$ Uniform Continuity

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f \in L^1$. Then for all $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\mu(E) < \delta \ \Rightarrow \ \int_E |f| < \epsilon$$

***Proof*:**

Without loss of generality, let $f \geq 0$. Let $\epsilon > 0$, and choose φ such that $\int f < \int \varphi + \epsilon/2$, i.e. $\int f - \int \varphi < \epsilon/2$. Next, since φ is a simple function, we have that $\sup \varphi = M < \infty$ (being the finite sum of constant terms). Now, set $\delta = \frac{\epsilon}{2M}$ so that $\mu(E) < \delta$.

$$\int_E f - \varphi \leq \int f - \varphi$$

then re-arranging we get:

$$\int_E f \leq \int f - \varphi + \int_E \varphi < \frac{\epsilon}{2} + \frac{M\epsilon}{2M} = \epsilon$$

as we sought to show.

As a consequence of this, if we have a space of integrable functions $\mathcal{F}$ with the sup norm (convergence in the sup norm is equivalent to convergence), then for any $\mu(E) < \infty$, $\int_E$ is in fact *continuous*, that is, if $f_n \rightrightarrows f$, then:

$$\int_E \lim f_n = \lim \int_E f_n$$

let's fix some E where $\int_E 1 = M < \infty$. Take $f_n \rightrightarrows f$ so that $|f(x) - f_n(x)| < \epsilon$ for all $x \in E$ when $n > N$. Then:

$$\begin{aligned}\left|\int_E f(x)dx - \int_E f_n(x)\right| &= \left|\int_E f(x) - f_n(x)\right| \\ &\leq \int_E |f(x) - f_n(x)| \\ &\leq \int_E \epsilon \\ &= \epsilon M\end{aligned}$$

It quickly follows form here that the limit does indeed commute.

We close off this section by proving a useful result allowing for differentiation and integration to commute:

Theorem 2.3.13: Leibniz Integral Rule

Let $f(x,t)$ and its partial derivative $\frac{\partial f}{\partial x}(x,t)$ be continuous functions on a rectangle $R = [c,d] \times [a,b]$. If $x \in (c,d)$, then the function $F(x)$ defined by:

$$F(x) = \int_a^b f(x,t)\,dt$$

is differentiable, and its derivative is given by:

$$\frac{d}{dx}\left(\int_a^b f(x,t)\,dt\right) = \int_a^b \frac{\partial f}{\partial x}(x,t)\,dt$$

Namely, if the variables are independent, differentiation and integration commute.

***Proof*:**
By the definition of the derivative, we have:

$$F'(x) = \lim_{h\to 0} \frac{F(x+h) - F(x)}{h}$$

Substitute the integral definition of F:

$$F'(x) = \lim_{h\to 0} \frac{1}{h}\left(\int_a^b f(x+h,t)\,dt - \int_a^b f(x,t)\,dt\right)$$

Because the integral of a sum/difference is the sum/difference of the integrals, we can combine them into a single integral:

$$F'(x) = \lim_{h\to 0} \int_a^b \frac{f(x+h,t) - f(x,t)}{h}\,dt$$

To prove the theorem, we must show that we can move the limit inside the integral. We do this by applying the Mean Value Theorem.

For a fixed t, consider the function f with respect to its first variable. By the Mean Value Theorem, there exists a number ξ_h strictly between x and $x+h$ such that:

$$\frac{f(x+h,t) - f(x,t)}{h} = \frac{\partial f}{\partial x}(\xi_h, t)$$

Substitute this back into our integral equation:

$$F'(x) = \lim_{h\to 0} \int_a^b \frac{\partial f}{\partial x}(\xi_h, t)\,dt$$

Now, we need to show that as $h \to 0$, the integral of $\frac{\partial f}{\partial x}(\xi_h, t)$ converges to the integral of $\frac{\partial f}{\partial x}(x,t)$.
Consider the absolute difference:

$$\left|\int_a^b \frac{\partial f}{\partial x}(\xi_h, t)\,dt - \int_a^b \frac{\partial f}{\partial x}(x,t)\,dt\right| \le \int_a^b \left|\frac{\partial f}{\partial x}(\xi_h, t) - \frac{\partial f}{\partial x}(x,t)\right| dt$$

Because $\frac{\partial f}{\partial x}$ is continuous on the closed and bounded (compact) rectangle $R = [c, d] \times [a, b]$, it is uniformly continuous on R.

Uniform continuity means that for any $\epsilon > 0$, there exists a $\delta > 0$ such that whenever the distance between two points in R is less than δ, the difference between the function values at those points is less than $\frac{\epsilon}{b-a}$.

Since ξ_h is between x and $x + h$, the distance between (ξ_h, t) and (x, t) is simply $|\xi_h - x|$, which is strictly less than $|h|$.

Therefore, if we choose $|h| < \delta$, we are guaranteed that:

$$\left|\frac{\partial f}{\partial x}(\xi_h, t) - \frac{\partial f}{\partial x}(x, t)\right| < \frac{\epsilon}{b-a}$$

for all $t \in [a, b]$.

Substituting this bound into our absolute difference gives:

$$\int_a^b \left|\frac{\partial f}{\partial x}(\xi_h, t) - \frac{\partial f}{\partial x}(x, t)\right| dt < \int_a^b \frac{\epsilon}{b-a}\, dt = \left(\frac{\epsilon}{b-a}\right)(b-a) = \epsilon$$

Since ϵ can be arbitrarily small, this proves that:

$$\lim_{h\to 0} \int_a^b \frac{\partial f}{\partial x}(\xi_h, t)\, dt = \int_a^b \frac{\partial f}{\partial x}(x, t)\, dt$$

as we sought to show.

2.3.1 Relating Lebesgue and Riemann Integral

Let's consider $f \in L^1(m)$ to be a Lebesgue integrable function on $\mathbb{R}$. As a preliminary to this book, Riemann integration of real-valued functions have been studied. So far, we have done nothing to show that the definitions of the integral will produce the same result. They surely must, since the Riemann integral correctly captures the area under surfaces. In this section, we show how the Lebesgue integral is in fact an extension of the Riemann integral, meaning if we restrict to the set of all Riemann integrable functions (which will all be Lebesgue integral), then the Lebesgue and Riemann integral give the same result.

As a quick reminder of Riemann integration, choosing to follow Darboux' characterization, let $[a, b]$ be a closed interval (or more generally a compact subset of $\mathbb{R}^n$) Then for any finite partition $P = \{t_0, ..., t_n\}$ where $a = t_0 < t_1 < \cdots < t_n = b$, define:

$$U_P(f) = \sum_{i=1}^n M_i(t_i - t_{i-1}) \qquad L_P(f) = \sum_{i=1}^n m_i(t_i - t_{i-1})$$

where M_i and m_i are the supremum and infimum of f on $[t_{i-1}, t_i]$. Then take:

$$\overline{I}_a^b(f) = \inf_P U_P(f) \qquad \underline{I}_a^b = \sup_P L_p(f)$$

over all possible partitions P. If $\overline{I}_a^b(f) = \underline{I}_a^b$, then their common value is denoted as $\int_a^b f(x)\, dx$ and f is called Riemann integrable on $[a, b]$.

We'll now compare this to the Lebesgue integral:

Theorem 2.3.14: Lebesgue-Vitali Theorem

Let f be a bounded real-valued function on $[a, b]$.

1. If f is Riemann integrable, then f is Lebesgue measurable (and hence integrable on $[a, b]$ since f is bounded) and
$$\int_a^b f(x)\,dx = \int_{[a,b]} f\,dm$$
2. f is Riemann integrable if and only if $\{x \in [a, b] \mid f \text{ is discontinuous at } x\}$ has Lebesgue measure 0.

Note that $\chi_{\mathbb{Q}}$ on $[0, 1]$ is discontinuous everywhere, with discontinuities having Lebesgue measure 1, was shown to not be Riemann integrable (this was in fact the example of a non Riemann-integrable function presented in the preliminary chapter). However, it is certainly Lebesgue integrable!

***Proof*:**

1. Let f be Riemann integrable. Let M_i and m_i are defined as the supremum and infimum as defined above, and for each partition P, let:
$$G_P = \sum_{i=1}^{n} M_i \chi_{(t_{i-1},t_i]} \qquad g_P = \sum_{i=1}^{n} m_i \chi_{(t_{i-1},t_i]}$$
so $U_p(f) = \int G_P\,dm$ and $L_p(f) = \int g_P\,dm$. Now, there exists a sequence of partitions P_n such that $\max_i(t_i - t_{i-1})$ tends to zero and P_{n+1} is a refinement of P_n (meaning g_{P_n} increase in value while G_{P_n} decreases) such that $U_{P_n}(f)$ and $L_{P_n}(f)$ converge to $\int_a^b f(x)\,dx$. Let $G = \lim G_{P_n}$ and $g = \lim g_{P_n}$. Since $g \leq f \leq G$, by the dominated convergence theorem:
$$\int G\,dm = \int g\,dm = \int_a^b f(x)\,dx$$
Therefore, $\int G - g = 0$ and so $G = g$ a.e. by proposition 2.2.8, and so $G = f$ a.e.

   Finally, since G is measurable (being the point-wise limit of measurable functions)and m is complete, f is measurable and
$$\int_{[} a, b] f\,dm = \int G\,dm = \int_a^b f(x)\,dx$$
2. This was exercise 23 in the textbook

The big advantage we have with this is that we can now use the computational techniques that were developed for Riemann integrals to compute particular Lebesgue integrals! In most cases in analysis, functions are locally Riemann integrable, and so we might what does the generality give us. The main thing it gives us is *completeness*. So though computing Lebesgue integrals might no be so easy (or if they are not an elementary Riemann integrable function), they are easier to work with in a theoretical framework, and help us define concepts like L^p space (which are important for Fourier Analysis).

(Note that in the infinite case, the Riemann and Lebesgue integral need not match. For example, $\sin(x)/x$ is Riemann integrable and produce a finite value, but it is not Lebesgue integral, producing an infinite value))

(A quick word on the Gamma function)

2.3.2 The Pushforward Measure

So far, whenever we integrated, the measure space was fixed once and for all. However, it is common to have a measurable function $F : X \to Y$ between two measurable spaces where only X carries a measure. Since F^{-1} preserves unions, intersections, and complements (as recalled at the start of chapter 2), we can use F to "transport" the measure from X over to Y: to measure a set $E \subseteq Y$, simply measure the set of points of X that F sends into E. The result is important enough to warrant a name:

Definition 2.3.15: Pushforward Measure

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces, $F : X \to Y$ an $(\mathcal{M}, \mathcal{N})$-measurable function, and μ a measure on $(X, \mathcal{M})$. Then the *pushforward measure* (or *image measure*) $F_*\mu$ on $(Y, \mathcal{N})$ is defined by

$$(F_*\mu)(E) = \mu(F^{-1}(E)) \qquad E \in \mathcal{N}$$

Note that $\mu(F^{-1}(E))$ makes sense precisely because F is measurable: $F^{-1}(E) \in \mathcal{M}$ for every $E \in \mathcal{N}$ (definition 2.1.1). The next theorem justifies calling $F_*\mu$ a measure, and shows that integrating against $F_*\mu$ on Y is the same as integrating the composed function against μ on X. It is our first "change of variables" theorem: the variable of integration is exchanged via the substitution $y = F(x)$, and no compensating factor appears since $F_*\mu$ was built out of μ to begin with.

Theorem 2.3.16: Pushforward Change of Variables

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces, $F : X \to Y$ an $(\mathcal{M}, \mathcal{N})$-measurable function, and μ a measure on $(X, \mathcal{M})$. Then:

1. $F_*\mu$ is a measure on $(Y, \mathcal{N})$
2. If $g : Y \to [0, \infty]$ is $\mathcal{N}$-measurable (i.e. $g \in L^+(F_*\mu)$), then $g \circ F \in L^+(\mu)$ and
$$\int_Y g \, d(F_*\mu) = \int_X g \circ F \, d\mu$$
3. If $g : Y \to \mathbb{C}$ is $\mathcal{N}$-measurable, then $g \in L^1(F_*\mu)$ if and only if $g \circ F \in L^1(\mu)$, in which case the formula in (2) holds as well.

***Proof*:**
For (1), first $(F_*\mu)(\emptyset) = \mu(F^{-1}(\emptyset)) = \mu(\emptyset) = 0$. Next, let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{N}$ be a disjoint collection. The pre-images $F^{-1}(E_i)$ are then also disjoint (a point x lying in two of them would have $F(x)$ lying in two disjoint sets), and since F^{-1} preserves unions, countable additivity of μ gives:

$$(F_*\mu)\left(\bigcup_{i=1}^{\infty} E_i\right) = \mu\left(\bigcup_{i=1}^{\infty} F^{-1}(E_i)\right) = \sum_{i=1}^{\infty} \mu(F^{-1}(E_i)) = \sum_{i=1}^{\infty} (F_*\mu)(E_i)$$

showing that $F_*\mu$ is a measure.

For (2), we follow the standard ascent: characteristic functions first, then simple functions by linearity, then all of L^+ by the Monotone Convergence Theorem. Before starting, note that $g \circ F$ is $\mathcal{M}$-measurable, being the composition of measurable functions (as noted at the start of this chapter), so indeed $g \circ F \in L^+(\mu)$. Now let $g = \chi_E$ for some $E \in \mathcal{N}$. Then $\chi_E \circ F = \chi_{F^{-1}(E)}$, since $\chi_E(F(x)) = 1$ exactly when $F(x) \in E$, that is, when $x \in F^{-1}(E)$. Therefore, by definition 2.2.2:

$$\int_Y \chi_E \, d(F_*\mu) = (F_*\mu)(E) = \mu(F^{-1}(E)) = \int_X \chi_{F^{-1}(E)} \, d\mu = \int_X \chi_E \circ F \, d\mu$$

Next, let $g = \sum_{i=1}^n a_i \chi_{E_i}$ be a simple function in $L^+(F_*\mu)$. Then $g \circ F = \sum_{i=1}^n a_i \chi_{F^{-1}(E_i)}$ is again simple, and applying linearity (proposition 2.2.7) on both sides of the characteristic-function case:

$$\int_Y g \, d(F_*\mu) = \sum_{i=1}^n a_i \, (F_*\mu)(E_i) = \sum_{i=1}^n a_i \, \mu(F^{-1}(E_i)) = \int_X g \circ F \, d\mu$$

Finally, let $g \in L^+(F_*\mu)$ be arbitrary. By theorem 2.1.18, there is a sequence of simple functions $\varphi_1 \leq \varphi_2 \leq \cdots$ converging pointwise to g. Then $\{\varphi_i \circ F\}$ is a sequence of simple functions increasing pointwise to $g \circ F$, since composing with F changes nothing pointwise: at each $x \in X$, $\varphi_i(F(x))$ increases to $g(F(x))$. Applying the Monotone Convergence Theorem (theorem 2.2.5) on both sides of the simple-function case:

$$\int_Y g \, d(F_*\mu) = \lim_{i\to\infty} \int_Y \varphi_i \, d(F_*\mu) = \lim_{i\to\infty} \int_X \varphi_i \circ F \, d\mu = \int_X g \circ F \, d\mu$$

For (3), apply (2) to $|g| \in L^+(F_*\mu)$: since $|g| \circ F = |g \circ F|$,

$$\int_Y |g| \, d(F_*\mu) = \int_X |g \circ F| \, d\mu$$

so the two sides are finite together, i.e. $g \in L^1(F_*\mu)$ if and only if $g \circ F \in L^1(\mu)$ (definition 2.3.1). In that case, decompose $g = (\operatorname{Re} g)^+ - (\operatorname{Re} g)^- + i(\operatorname{Im} g)^+ - i(\operatorname{Im} g)^-$ as in definition 2.1.15. Each of the four functions lies in $L^+(F_*\mu)$ with finite integral (each is bounded by $|g|$), and composition with F respects the decomposition (for instance $(\operatorname{Re} g)^+ \circ F = (\operatorname{Re}(g \circ F))^+$, as both compute $\max(\operatorname{Re} g(F(x)), 0)$ at each x). Applying (2) to each of the four pieces and recombining by linearity (theorem 2.3.8) gives the formula for g, completing the proof.

The theorem says that the integral does not care what space is "underneath" the measure: all that matters is how the measure distributes mass among the sets that the integrand can distinguish, and

$F_*\mu$ distributes mass on Y exactly as μ does on X, seen through F. Note also that no invertibility of F was needed anywhere. The theorem goes to work in section 2.7, where it does the abstract bookkeeping in the proof of the classical change of variables theorem on $\mathbb{R}^n$. There the substitution $y = T(x)$ *does* introduce a compensating factor, namely $|\det D_x T|$, and the whole geometric content of that theorem is in identifying the factor.

2.4 Modes of Convergence

So far, we have accumulated the following forms of convergence of functions:

1. uniform convergence $f_n \rightrightarrows f$
2. point-wise convergence $f_n \to f$
3. point-wise convergence a.e. $f_n \to f$ a.e.
4. L^1 convergence $f_n \to f$ if and only if $\int |f_n - f| \to 0$

The first 3 are weaker than then the next: uniform convergence $\implies$ point-wise convergence $\implies$ a.e. convergence, but the converse is false in each case. Furthermore, L^1 convergence implies none of those 3, and none of those 3 imply L^1 convergence. The examples to keep in mind to compare are:

1. $f_n = n^{-1}\chi_{[0,n]}$
2. $f_n = \chi_{[n,n+1]}$
3. $f_n = n\chi_{[0,1/n]}$
4. $f_1 = \chi_{[0,1]}$, $f_2 = \chi_{[0,1/2]}$, $f_3 = \chi_{[1/2,1]}$, $f_4 = \chi_{[0,1/4]}$, $f_5 = \chi_{[1/4,2/4]}$, and so on

We now introduce another way of thinking about convergence: instead of saying that the $\int f_n$ approaches that of f (i.e., evaluate the integral and look at the limiting behavior of the integral of the function), we will want there to exist an N such that $n \geq N$, the measure of the difference between f_n and f are epsilon away from the function we are converging to f. This is not the same thing as L^1 convergence, since this (1), (3) and (4) all converge to 0 in this type of convergence. Before any further analysis let's define our terms:

Definition 2.4.1: Cauchy in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *cauchy in measure* if:

$$\forall \epsilon > 0,\ \exists N \text{ such that } \forall n, m \geq N,\ \mu(\{x \mid |f_m(x) - f_n(x)| \geq \epsilon\}) < \epsilon$$

Equivalently, we can write the definition as $\forall \epsilon > 0$,

$$\mu(\{x \mid |f_n(x), f_m(x)| \geq \epsilon\}) \to 0 \text{ as } n, m \to \infty$$

Definition 2.4.2: Convergence in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *convergent in measure* or *converge in measure* if:

$$\forall \epsilon > 0, \lim_{n\to\infty} \mu(\{x \mid |f(x) - f_n(x)| \geq \epsilon\}) = 0$$

With these definition, notice that (2) is in fact *not* Cauchy in measure: the difference between the two functions.

Proposition 2.4.3: L^1 convergence Then Measure Convergence

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f_n \to f$ in L^1. Then $f_n \to f$ in measure

***Proof*:**
Let ϵ be given, and define

$$E_{\epsilon,n} = \mu(\{x \mid |f_n(x) - f(x)| \geq \epsilon\})$$

Then:

$$\int |f - f_n| \geq \int_{E_{\epsilon,n}} |f - f_n| \geq \epsilon\mu(E_{\epsilon,n})$$

Thus:

$$\mu(E_{n,\epsilon}) \leq \epsilon^{-1} \int |f - f_n| \to 0$$

showing that it converges in measure.

Note that convergence in measure does not imply convergence in L^1 as we saw in example (1) and (3).

With this established, we can go on to establish why we care about convergence in measure: it gives us another way of finding point-wise convergent functions by establishing a condition weaker than L^1 convergence:

Theorem 2.4.4: Cauchy in Measure Then Pointwise a.e.

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ Cauchy in measure. Then there exists a f such that $f_n \to f$ in measure, and furthermore there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \to f$ a.e.
If $f_n \to g$ in measure, then $f = g$ a.e.

***Proof*:**

We can choose a subsequence $\{g_j\} = \{f_{n_j}\}$ of $\{f_n\}$ such that if $E_j = \{x : |g_j(x) - g_{j+1}(x)| \geq 2^{-j}\}$, then $\mu(E_j) \leq 2^{-j}$. If $F_k = \bigcup_{j=k}^{\infty} E_j$, then $\mu(F_k) \leq \sum_k^{\infty} 2^{-j} = 2^{1-k}$, and if $x \notin F_k$, for $i \geq j \geq k$ we have

$$(2.31) \qquad |g_j(x) - g_i(x)| \leq \sum_{l=j}^{i-1} |g_{l+1}(x) - g_l(x)| \leq \sum_{l=j}^{i-1} 2^{-l} \leq 2^{1-j}.$$

Thus $\{g_j\}$ is pointwise Cauchy on F_k^c. Let $F = \bigcap_1^{\infty} F_k = \limsup E_j$. Then $\mu(F) = 0$, and if we set $f(x) = \lim g_j(x)$ for $x \notin F$ and $f(x) = 0$ for $x \in F$, then f is measurable (see Exercises 3 and 5) and $g_j \to f$ a.e. Also, (2.31) shows that $|g_j(x) - f(x)| \leq 2^{1-j}$ for $x \notin F_k$ and $j \geq k$. Since $\mu(F_k) \to 0$ as $k \to \infty$, it follows that $g_j \to f$ in measure. But then $f_n \to f$ in measure, because

$$\{x : |f_n(x) - f(x)| \geq \epsilon\} \subset \{x : |f_n(x) - g_j(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |g_j(x) - f(x)| \geq \tfrac{1}{2}\epsilon\},$$

and the sets on the right both have small measure when n and j are large. Likewise, if $f_n \to g$ in measure,

$$\{x : |f(x) - g(x)| \geq \epsilon\} \subset \{x : |f(x) - f_n(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |f_n(x) - g(x)| \geq \tfrac{1}{2}\epsilon\}$$

for all n, hence $\mu(\{x : |f(x) - g(x)| \geq \epsilon\}) = 0$ for all ϵ. Letting ϵ tend to zero through some sequence of values, we conclude that $f = g$ a.e.

Corollary 2.4.5: L^1 convergence and Finding Subsequence

Let $f_n \to f$ in L^1. Then there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \to f$ a.e.

***Proof*:**

By proposition 2.4.3, $f_n \to f$ in measure, and by theorem 2.4.4, there exists a subsequence that converges to f a.e.

As we saw, $f_n \to f$ a.e. does not imply $f_n \to f$ in measure, as we saw in example (2). This is due to the "escape to infinity" that is happening, which is why we care about the dominated convergence theorem to help us stop such "sneaky infinity problem". However, if μ was a finite measure, then $f_n \to f$ a.e. does imply $f_n \to f$ in measure. In fact, since the area is bounded, we can state some even stronger convergence conditions:

Theorem 2.4.6: Egorov's Theorem

Let $(X, \mathcal{M}, \mu)$ be a measure space and suppose that $\mu(X) < \infty$. If $\{f_i\}$ are measurable complex-valued functions such that $f_n \to f$ a.e., then for every $\epsilon > 0$, there exists a $E \subseteq X$ such that $\mu(E) < \epsilon$ and $f_n \to f$ uniformly on E^c

Proof:

Without loss of generality we may assume that $f_n \to f$ everywhere on X. For $k, n \in \mathbb{N}$ let

$$E_n(k) = \bigcup_{m=n}^{\infty} \{x : |f_m(x) - f(x)| \geq k^{-1}\}.$$

Then, for fixed k, $E_n(k)$ decreases as n increases, and $\bigcap_{n=1}^{\infty} E_n(k) = \varnothing$, so since $\mu(X) < \infty$ we conclude that $\mu(E_n(k)) \to 0$ as $n \to \infty$. Given $\epsilon > 0$ and $k \in \mathbb{N}$, choose n_k so large that $\mu(E_{n_k}(k)) < \epsilon 2^{-k}$ and let $E = \bigcup_{k=1}^{\infty} E_{n_k}(k)$. Then $\mu(E) < \epsilon$, and we have $|f_n(x) - f(x)| < k^{-1}$ for $n > n_k$ and $x \notin E$. Thus $f_n \to f$ uniformly on E^c.

The type of convergence in this theorem is sometimes called *almost uniform convergence*. Almost uniform convergence implies convergence a.e. and convergence in measure.

2.5 Uniform Integrability and the Vitali Convergence Theorem

Convergence in measure by itself does not move integrals: examples (1) and (3) of section 2.4 converge to 0 in measure while their integrals stay at 1. The dominated convergence theorem closes this gap with an outside hypothesis, a single integrable g lying above the whole sequence, and it is natural to ask what exact control the domination provides. The answer is a condition on the family itself, one that forbids any member from keeping a fixed amount of area on the set where it is large.

Definition 2.5.1: Uniformly Integrable Family

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_\alpha\}_{\alpha \in A} \subseteq L^1$ a family of integrable functions. The family is *uniformly integrable* if

$$\lim_{M \to \infty} \sup_{\alpha \in A} \int_{\{x \mid |f_\alpha(x)| > M\}} |f_\alpha| = 0$$

The integral appearing in the definition is the area of f_α over the set where $|f_\alpha|$ exceeds M, the *tail* of f_α at M. Each individual $f_\alpha \in L^1$ has tails vanishing as $M \to \infty$, as part (1) of the next example shows, so the content of the definition is the uniformity: one M must work for the entire family at once.

Example 2.5.2: Uniformly Integrable Families

1. A single integrable function is uniformly integrable, and hence so is any finite subset of L^1. Let $f \in L^1$, and set $g_n = |f|\chi_{E_n}$ where $E_n = \{x \mid |f(x)| > n\}$. Then $g_n \leq |f|$, and $g_n \to 0$ a.e. because an integrable function is a.e. finite (proposition 2.2.13 applied to $|f|$). By the dominated convergence theorem (theorem 2.3.6),

$$\int_{E_n} |f| = \int g_n \to 0$$

   Since $\{x \mid |f(x)| > M\} \subseteq E_n$ whenever $M \geq n$, the tail at M is bounded by the tail at n, and so the tails vanish along all of $M \to \infty$. For a finite family $\{f_1, \ldots, f_N\}$, the supremum of the tails is the largest of N quantities that each vanish as $M \to \infty$.

2. A dominated family is uniformly integrable. If $g \in L^1$ is nonnegative and $|f_\alpha| \leq g$ a.e. for every $\alpha \in A$, then up to a null set $\{x \mid |f_\alpha(x)| > M\} \subseteq \{x \mid g(x) > M\}$, so

$$\int_{\{x \mid |f_\alpha(x)| > M\}} |f_\alpha| \leq \int_{\{x \mid g(x) > M\}} g$$

   for every α, and the right side vanishes as $M \to \infty$ by part (1) applied to g.

Uniform integrability converts into two estimates: sets of small measure carry uniformly small area, and on a finite measure space the family is bounded in L^1.

Lemma 2.5.3: Uniform Absolute Continuity

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_\alpha\}_{\alpha \in A} \subseteq L^1$ a uniformly integrable family.

1. For every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $E \in \mathcal{M}$ with $\mu(E) < \delta$ and all $\alpha \in A$,

$$\int_E |f_\alpha| < \epsilon$$

2. If $\mu(X) < \infty$, then $\sup_{\alpha \in A} \int |f_\alpha| < \infty$

***Proof*:**
For (1), let $\epsilon > 0$ and choose $M > 0$ such that

$$\sup_{\alpha \in A} \int_{\{x \mid |f_\alpha(x)| > M\}} |f_\alpha| < \frac{\epsilon}{2}$$

and take $\delta = \epsilon/(2M)$. If $\mu(E) < \delta$, then splitting E over where $|f_\alpha|$ is at most M and where it exceeds M,

$$\int_E |f_\alpha| \leq M\mu(E) + \int_{\{x \mid |f_\alpha(x)| > M\}} |f_\alpha| < M\delta + \frac{\epsilon}{2} = \epsilon$$

for every $\alpha \in A$.

For (2), choose $M > 0$ so that the supremum of the tails at M is less than 1. The same splitting, applied to $E = X$, gives

$$\int |f_\alpha| \leq M\mu(X) + \int_{\{x \mid |f_\alpha(x)|>M\}} |f_\alpha| \leq M\mu(X) + 1$$

for every $\alpha \in A$, and the bound is finite because $\mu(X) < \infty$, completing the proof.

Applied to the one-member family $\{f\}$, part (1) says that the integral of an L^1 function is an absolutely continuous quantity: sets of small measure carry small area. The same phenomenon returns in chapter 3 in the language of signed measures, as theorem 3.2.3 and its corollary corollary 3.2.4. Note that part (1) makes no finiteness assumption on μ, only part (2) does. With these estimates, uniform integrability is exactly what separates convergence in measure from convergence in L^1 on a finite measure space:

Theorem 2.5.4: Vitali Convergence Theorem

Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty$, let $\{f_n\} \subseteq L^1$, and let f be a measurable function. Then the following are equivalent:

1. $\{f_n\}$ is uniformly integrable and $f_n \to f$ in measure
2. $f \in L^1$ and $f_n \to f$ in L^1

***Proof*:**

(1) $\Rightarrow$ (2) We first extract an a.e. convergent subsequence. The sequence is Cauchy in measure: for any $\epsilon > 0$,

$$\{x \mid |f_n(x) - f_m(x)| \geq \epsilon\} \subseteq \{x \mid |f_n(x) - f(x)| \geq \epsilon/2\} \cup \{x \mid |f_m(x) - f(x)| \geq \epsilon/2\}$$

since if x lies in neither set on the right, the triangle inequality gives $|f_n(x) - f_m(x)| < \epsilon$, and both sets on the right have measure tending to 0 as $n, m \to \infty$. By theorem 2.4.4, there is a g such that $f_n \to g$ in measure and a subsequence $f_{n_j} \to g$ a.e., and the uniqueness clause of the same theorem gives $g = f$ a.e. Thus $f_{n_j} \to f$ a.e.

Next, $f \in L^1$. Since $|f_{n_j}| \to |f|$ a.e., corollary 2.2.12 gives

$$\int |f| \leq \liminf_j \int |f_{n_j}| \leq \sup_n \int |f_n| < \infty$$

where the supremum is finite by lemma 2.5.3(2).

Now let $\epsilon > 0$, and let $\delta > 0$ be as in lemma 2.5.3(1) for the family $\{f_n\}$: whenever $\mu(E) < \delta$, $\int_E |f_n| < \epsilon$ for every n. The same δ controls f, for if $\mu(E) < \delta$, then $|f_{n_j}|\chi_E \to |f|\chi_E$ a.e., so corollary 2.2.12 again gives

$$\int_E |f| \leq \liminf_j \int_E |f_{n_j}| \leq \epsilon$$

Set $A_n = \{x \mid |f_n(x) - f(x)| \geq \epsilon\}$. Convergence in measure provides an N such that $\mu(A_n) < \delta$ for all $n \geq N$. For such n, using $|f_n - f| < \epsilon$ on A_n^c and $|f_n - f| \leq |f_n| + |f|$

on A_n,

$$\int |f_n - f| = \int_{A_n^c} |f_n - f| + \int_{A_n} |f_n - f| \le \epsilon\mu(X) + \int_{A_n} |f_n| + \int_{A_n} |f| \le \epsilon(\mu(X) + 2)$$

Hence $\limsup_n \int |f_n - f| \le \epsilon(\mu(X)+2)$, and since $\mu(X) < \infty$ and $\epsilon > 0$ was arbitrary, $\int |f_n - f| \to 0$.

(2) $\Rightarrow$ (1) Convergence in measure is proposition 2.4.3. For uniform integrability, first note that

$$\int |f_n| \le \int |f_n - f| + \int |f|$$

and the right side is bounded in n because $\int |f_n - f| \to 0$, so $C = \sup_n \int |f_n|$ is finite. Let $\epsilon > 0$. Choose N such that $\int |f_n - f| < \epsilon/2$ for all $n > N$. The single function f is uniformly integrable by example 2.5.2(1), so lemma 2.5.3(1) applied to the family $\{f\}$ produces a $\delta > 0$ with $\int_E |f| < \epsilon/2$ whenever $\mu(E) < \delta$. For $M > 0$, write $E_n^M = \{x \mid |f_n(x)| > M\}$. Then

$$M\mu(E_n^M) \le \int_{E_n^M} |f_n| \le \int |f_n| \le C$$

so $\mu(E_n^M) \le C/M < \delta$ for every n once $M > C/\delta$. For such M and $n > N$,

$$\int_{E_n^M} |f_n| \le \int_{E_n^M} |f_n - f| + \int_{E_n^M} |f| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

The finite family $\{f_1, \dots, f_N\}$ is uniformly integrable by example 2.5.2(1), so there is an M_0 such that $\int_{E_n^M} |f_n| < \epsilon$ for all $n \le N$ whenever $M \ge M_0$. For $M > \max(M_0, C/\delta)$, then, $\sup_n \int_{E_n^M} |f_n| \le \epsilon$. Since $\epsilon > 0$ was arbitrary, the tails vanish uniformly as $M \to \infty$, completing the proof.

The chain from almost-everywhere convergence deserves its own statement, since almost everywhere convergence is the hypothesis most often available.

Corollary 2.5.5: Vitali Convergence from Almost Everywhere Convergence

Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty$, let $\{f_n\} \subseteq L^1$ be uniformly integrable, and suppose $f_n \to f$ a.e. for a measurable function f. Then $f \in L^1$ and $f_n \to f$ in L^1.

***Proof*:**
Almost-everywhere convergence implies convergence in measure on a finite measure space: for $\epsilon, \eta > 0$, theorem 2.4.6 gives a set E with $\mu(E) < \eta$ off of which the convergence is uniform, so $\{x \mid |f_n(x) - f(x)| \ge \epsilon\} \subseteq E$ for all large n and $\mu(\{x \mid |f_n(x) - f(x)| \ge \epsilon\}) < \eta$. Theorem 2.5.4 then returns $f \in L^1$ and $f_n \to f$ in L^1, completing the proof.

The dominated convergence theorem on a finite measure space is now a special case. If $f_n \to f$ a.e. with $|f_n| \le g$ a.e. for a fixed nonnegative $g \in L^1$, then $\{f_n\}$ is uniformly integrable by example 2.5.2(2), and corollary 2.5.5 returns $f_n \to f$ in L^1, which carries the integrals along because $\left|\int f_n - \int f\right| \le \int |f_n - f|$. No dominating function is required, only the uniform integrability it implies.

The Vitali convergence theorem is the sharp form of this circle of ideas in two senses. First, uniform integrability is necessary for L^1 convergence, which is the direction (2) $\Rightarrow$ (1), while domination is not. Second, uniform integrability is strictly weaker than domination: take disjoint intervals $J_n \subseteq [0,1]$ with $m(J_n) = (n^2 \log n)^{-1}$ for $n \geq 2$ (the lengths sum to less than 1) and $f_n = n\chi_{J_n}$. The tail of f_n at M is $(n \log n)^{-1}$ when $n > M$ and 0 otherwise, so the family is uniformly integrable, yet any g with $g \geq f_n$ a.e. for all n satisfies $g \geq \sum_{n\geq 2} n\chi_{J_n}$ a.e., and since the J_n are disjoint, $\int g \geq \sum_{n\geq 2}(n \log n)^{-1} = \infty$ by the integral test.

Finally, the finiteness of μ cannot be dropped from the statement. On $(\mathbb{R}, m)$, the sequence $f_n = n^{-1}\chi_{[0,n]}$ of example (1) of section 2.4 converges to 0 uniformly, hence in measure (for $n > \epsilon^{-1}$ the set $\{x \mid |f_n(x)| \geq \epsilon\}$ is empty), and is uniformly integrable because $|f_n| \leq 1$ makes every tail past $M = 1$ vanish, yet $\int f_n = 1$ for every n. Only the implication (1) $\Rightarrow$ (2) used the finiteness of μ, as inspection of the proof shows. The general theorem on an infinite measure space replaces the finiteness of μ with a tightness hypothesis on the family, one that rules out exactly this kind of "escape to infinity".

2.6 Product Measure

As we defined in definition 1.1.8, If $\mathcal{M}$ and $\mathcal{N}$ are σ-algebras, then we can define a new σ-algebra $\mathcal{M} \otimes \mathcal{N}$. If $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are measure spaces, we will show that we can define a new measure on $\mathcal{M} \otimes \mathcal{N}$ using μ and ν. We will use the tools we have done in constructing measure to construct this measure, and show that that there is a natural measure $\mu \times \nu$ with respect to the outer-measure given our space is σ-finite (this is just the uniqueness clause of theorem 1.3.7).

First, we'll give a name to sets in $\mathcal{M} \otimes \mathcal{N}$ that will be easier to measure (these will be our generating sets)

Definition 2.6.1: [Measurable] Rectangle

Let $A \in \mathcal{M}$ and $B \in \mathcal{N}$. Then $A \times B \in \mathcal{M} \otimes \mathcal{N}$ is called a *measurable rectangle* or just *rectangle* for short

Notice that if $A \times B$ and $C \times D$ are rectangles, they form an elementary family:

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D) \qquad (A \times B)^c = (X \times B^c) \cup (A^c \times B)$$

and so by proposition 1.1.13, the collection of finite disjoint unions of rectangles forms an algebra $\mathcal{A}$, and they generate $\mathcal{M} \otimes \mathcal{N}$.

Furthermore, not all sets in $\mathcal{M} \otimes \mathcal{N}$ are measurable rectangles: if we take $(\mathbb{R}, \mathcal{L}(\mathbb{R}), m)$ and consider $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$, then the set of points $\Delta = \{(x,x) \mid x \in \mathbb{R}\}$ is in $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$ (being the countable intersection of ever smaller squares on the line), however it is very far from being the product of two sets in $\mathcal{L}(\mathbb{R})$.

We'll now define how to integrate simple functions on $X \times Y$ with the associated measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$: Let $A \times B$ be a rectangle that is the (finite or countable) disjoint union of rectangle $A_i \times B_i$. Then notice that:

$$\chi_{A\times B}(x,y) = \chi_A(x)\chi_B(y)$$

and so:

$$\chi_{A\times B}(x,y) = \sum_i \chi_{A_i\times B_i}(x,y) = \sum_i \chi_{A_i}(x)\chi_{B_i}(y)$$

If we integrate with respect to x, treating $\chi_{B_i}(y)$ as some constant, then by theorem 2.2.7

$$\int \chi_A(x)\chi_B(y)d\mu = \mu(A)\chi_B(y)d\mu = \mu(A)\chi_B(y) = \sum_i \mu(A_i)\chi_{B_i}(y)$$

It might be confusing which variable is associated to which measure, in which case it is common to write:

$$\int \chi_A(x)\chi_B(y)d\mu(x)$$

to emphasize the function with the x variable is the one being measured by μ. Integrating with respect to y yields:

$$\int \mu(A)\chi_B(y)d\nu = \mu(A)\nu(B) = \sum_i \mu(A_i)\nu(B_i)$$

Thus, we can define $\pi : \mathcal{A} \to \mathbb{R}$ where if $E \in \mathcal{A}$ is a finite disjoint union of rectangles $A_i \times B_i$, then:

$$\pi(E) = \sum_i \mu(A_i)\nu(B_i)$$

with the usual convention of $0 \times \infty = 0$ (i.e., if one of the dimensions is "0", then we are making the value 0). Notice that π is independent of representation, since any finite disjoint union of rectangles have a common refinement. Thus π is a pre-measure, and so by theorm 1.3.7, π generates an outer measure on $X \times Y$ whose restriction to $\mathcal{M} \otimes \mathcal{N}$ is a measure where restricted to $\mathcal{A}$ is π (or conversely, whose measure extends π). Such a measure is given a name:

Definition 2.6.2: Product Measure

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces, and define π as in the previous construction. Then the induced outer-measure is called the *product measure* and is usually denoted $\mu \times \nu$.

Notice further that if μ and ν are σ-finite, so $\mu(X) = \sum_i^\infty \mu(A_i)$ and $\nu(Y) = \sum_i^\infty \nu(B_i)$, then $(\mu \times \nu)(X \times Y) = \sum_{i,j} \mu(A_i)\nu(B_j)$, and since all terms are finite, $\mu \times \nu$ is also σ-finite, in which case by the same theorem as invoked in defining $\mu \times \nu$, the product measure is the unique measure such that $\mu \times \nu(A \times B) = \mu(A)\nu(B)$.

Induction extends this construction to any finite product of measures. Since later sections lean on this extension (most importantly for the n-dimensional Lebesgue measure), it deserves a proper record:

Corollary 2.6.3: n-fold Product Measure

Let $(X_i, \mathcal{M}_i, \mu_i)$ be measure spaces for $i = 1, ..., n$. Then there exists a measure $\mu_1 \times \cdots \times \mu_n$ on $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$ such that for all $A_i \in \mathcal{M}_i$:

$$(\mu_1 \times \cdots \times \mu_n)(A_1 \times \cdots \times A_n) = \prod_{i=1}^{n} \mu_i(A_i)$$

Furthermore, if every μ_i is σ-finite, then $\mu_1 \times \cdots \times \mu_n$ is σ-finite and is the *unique* measure on $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$ with this property.

***Proof*:**
We induct on n. There is nothing to prove for $n = 1$, and $n = 2$ is the construction we just carried out: definition 2.6.2 produces $\mu_1 \times \mu_2$, its value on a rectangle is $\pi(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$ since the extension of theorem 1.3.7 agrees with π on $\mathcal{A}$, and the paragraph above settled σ-finiteness and uniqueness.

For the inductive step, suppose the result holds for $n-1$ factors, and write $\lambda = \mu_1 \times \cdots \times \mu_{n-1}$ for the resulting measure on $\mathcal{M}' = \mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_{n-1}$, a σ-algebra on $X' = X_1 \times \cdots \times X_{n-1}$. The $n = 2$ case applied to $(X', \mathcal{M}', \lambda)$ and $(X_n, \mathcal{M}_n, \mu_n)$ produces the measure $\lambda \times \mu_n$ on $\mathcal{M}' \otimes \mathcal{M}_n$. Identifying $X' \times X_n$ with $X_1 \times \cdots \times X_n$ in the natural way, we claim that

$$\mathcal{M}' \otimes \mathcal{M}_n = \mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$$

so that we may simply define $\mu_1 \times \cdots \times \mu_n := \lambda \times \mu_n$.

To see the claim, first note both sides are generated by rectangles: by definition 1.1.8, $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$ is generated by the pre-images $\pi_i^{-1}(A_i) = X_1 \times \cdots \times A_i \times \cdots \times X_n$, each of which is a rectangle, while conversely every rectangle satisfies $A_1 \times \cdots \times A_n = \bigcap_{i=1}^n \pi_i^{-1}(A_i)$; applying lemma 1.1.4 in both directions, $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$ is generated by the collection of rectangles, and the same reasoning shows $\mathcal{M}' \otimes \mathcal{M}_n$ is generated by the two-factor rectangles $B \times A_n$ ($B \in \mathcal{M}'$, $A_n \in \mathcal{M}_n$). Now, every n-factor rectangle is a two-factor rectangle, since $A_1 \times \cdots \times A_n = (A_1 \times \cdots \times A_{n-1}) \times A_n$ with $A_1 \times \cdots \times A_{n-1} \in \mathcal{M}'$, and so $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n \subseteq \mathcal{M}' \otimes \mathcal{M}_n$ by lemma 1.1.4. Conversely, fix $A_n \in \mathcal{M}_n$ and consider

$$\mathcal{D} = \{B \subseteq X' \mid B \times A_n \in \mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n\}$$

This collection contains X' and every $(n-1)$-factor rectangle (the product with A_n being an n-factor rectangle), and it is closed under countable unions (as $(\bigcup_i B_i) \times A_n = \bigcup_i (B_i \times A_n)$) and under complements (as $B^c \times A_n = (X' \times A_n) \setminus (B \times A_n)$, a difference of two members of the σ-algebra). Thus $\mathcal{D}$ is a σ-algebra containing the rectangles that generate $\mathcal{M}'$, so $\mathcal{M}' \subseteq \mathcal{D}$: every two-factor rectangle lies in $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$, and so $\mathcal{M}' \otimes \mathcal{M}_n \subseteq \mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$, again by lemma 1.1.4. This proves the claim.

The rectangle formula is now immediate: for $A_i \in \mathcal{M}_i$,

$$(\lambda \times \mu_n)\big((A_1 \times \cdots \times A_{n-1}) \times A_n\big) = \lambda(A_1 \times \cdots \times A_{n-1})\, \mu_n(A_n) = \prod_{i=1}^{n} \mu_i(A_i)$$

using the $n = 2$ rectangle formula and then the inductive hypothesis.

Finally, suppose every μ_i is σ-finite, say $X_i = \bigcup_{k=1}^{\infty} A_i^k$ with $\mu_i(A_i^k) < \infty$, where we may take each sequence to be increasing (replace A_i^k with $A_i^1 \cup \cdots \cup A_i^k$). Then $X_1 \times \cdots \times X_n = \bigcup_{k=1}^{\infty} A_1^k \times \cdots \times A_n^k$, and each of these rectangles has measure $\prod_{i=1}^{n} \mu_i(A_i^k) < \infty$, so $\mu_1 \times \cdots \times \mu_n$ is σ-finite. For uniqueness, let ρ be any measure on $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$ satisfying the rectangle formula. The rectangles form an elementary family exactly as in the two-factor case, so by proposition 1.1.13 their finite disjoint unions form an algebra $\mathcal{A}_n$, which generates $\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$ (it contains the rectangles and is contained in the σ-algebra). On $\mathcal{A}_n$, both ρ and $\mu_1 \times \cdots \times \mu_n$ are determined by additivity from their (equal) values on rectangles, so they restrict to a common premeasure on $\mathcal{A}_n$, which is σ-finite by the exhausting rectangles above. By the uniqueness clause of theorem 1.3.7, such a premeasure has exactly one extension to a measure on $\mathcal{M}(\mathcal{A}_n) = \mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_n$, and so $\rho = \mu_1 \times \cdots \times \mu_n$, completing the induction and the proof.

In particular, in the σ-finite case, uniqueness settles the associativity of the *measures*: once the grouped σ-algebras are identified with $\bigotimes_1^3 \mathcal{M}_i$ (the mirror of the good-sets identification in the proof, run once per grouping), the differently grouped measures (say $(\mu_1 \times \mu_2) \times \mu_3$ versus $\mu_1 \times (\mu_2 \times \mu_3)$) satisfy the same rectangle formula on the same σ-algebra, so they must agree.

With what we've established so far, we can already measure sets using the outer-measure definition. However, this is very tedious, and ultimately comes down to some measure properties of the respective component measure. There is an easier way to use these component measure to get the resulting measure: notice in our construction of π that we integrated with respect to one variable first, and then the next. We will ostensibly generalizing this trick so that we can measure spaces, or more specifically integrate functions on product spaces, by fixing all variables except one, integrating on it, and then moving the result out (since it's a constant) and integrate the next variable, until we integrate over all variables.

Since we can generalise our results by induction, let's take $n = 2$ and consider the measure space $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$. The following definition captures the idea of fixing everything except one variable:

Definition 2.6.4: n-section

Let $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$ be a product measure space, and consider $E \subseteq X \times Y$. Then for any $x \in X$, $y \in Y$, define the *x-section* E_x and *y-section* E^y to be:

$$E_x = \{y \in Y \mid (x, y) \in E\} \qquad E^y = \{x \in X \mid (x, y) \in E\}$$

If f is a function on $X \times Y$, we define the *x-section* f_x and *y-section* f^y to be:

$$f_x(y) = f(x, y) \qquad f^y(x) = f(x, y)$$

The most simple example of the section of a function would be:

$$(\chi_E)_x = \chi_{E_x} \qquad (\chi_E)^y = \chi_{E^y}$$

Proposition 2.6.5: Measure of Section

1. Let $E \in \mathcal{M} \otimes \mathcal{N}$. Then $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$ (i.e., all sets in a produce σ-algebra can be broken down into x-sections and y-section)
2. If f is $\mathcal{M} \otimes \mathcal{N}$-measurable, then f_x is $\mathcal{N}$-measurable and f^y is $\mathcal{M}$-measurable

***Proof*:**

1. The proof of this fact comes down to taking advantage of the fact that the pre-measure of rectangles generates $\mathcal{M} \otimes \mathcal{N}$. Let $\mathcal{R}$ be the collection of all subsets of $E \subseteq X \times Y$ such that $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$. This clearly contains the empty set, and the set must contain all rectangles, since if $A \neq 0$ and $x \in A$, then $(A \times B)_x = B$ (if $x \notin A$, then $(A \times B)_x = \emptyset$). Now, since
$$\left(\bigcup_i^\infty E_i\right)_x = \bigcup_i^\infty (E_i)_x \qquad (E^c)_x = (E_x)^c$$
by some basic set manipulation, we see that $\mathcal{R}$ is a σ-algebra. Since it contains the generating set of $\mathcal{M} \otimes \mathcal{N}$, we have that $\mathcal{R} \supseteq \mathcal{M} \otimes \mathcal{N}$. Thus, all the sets in $\mathcal{M} \otimes \mathcal{N}$ can be decompose in this way (note that we have not proved equality: we have not proved that if every slice is measurable then E is measurable)
2. By some set-theory manipulation, and part (1):
$$f_x^{-1}(B) = (f^{-1}(B))_x \qquad f^y{-}1(B) = (f^{-1}(B))^y$$
and so f_x and f^y must be $\mathcal{N}$ and $\mathcal{M}$ measurable (respectively)

With this, we are almost ready to start proving we can integrate $A \times B \in \mathcal{M} \otimes \mathcal{N}$ with respect to 1 measure at a time. We first need to prove a technical result for the next theorem which gives (again) another way of constructing σ-algebra. This one is particularly useful, since it will let us prove certain sets are σ-algebras by using the MCT and DCT:

Definition 2.6.6: Monotone Class

Let X be a set and $P(X)$ the powerset of X. Then the subset $\mathcal{C} \subseteq P(X)$ is called a *monotone class* if

1. it is cosed under countable unions of increasing sets, that is, if $E_1 \subseteq E_2 \subseteq \cdots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\cup_i E_i \in \mathcal{C}$
2. it is cosed under intersection unions of decreasing sets, that is, if $E_1 \supseteq E_2 \supseteq \cdots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\cap_i E_i \in \mathcal{C}$

Every σ-algebra is a monotone class (we've even used this property in a couple of proofs). Furthermore, the intersection of any family of monotone class is again a monotone class, and so we can define the unique smallest monotone class that contains the set $\mathcal{E} \subseteq P(X)$, where $\mathcal{E}$ is called the monotone class generated set, and the monotone class is called the monotone class generated by $\mathcal{E}$, and is dentoed $\mathcal{C}(\mathcal{E})$

Lemma 2.6.7: Monotone Class Lemma

Let X be a set and $\mathcal{A} \subseteq X$ be an algebra on some subsets of X. Then the monotone class $\mathcal{C}$ generated by $\mathcal{A}$ is equal to the σ-algebra $\mathcal{M}$ generated by $\mathcal{A}$, that is:

$$\mathcal{C}(\mathcal{A}) = \mathcal{M}(\mathcal{A})$$

***Proof*:**
Since $\mathcal{M}$ is itself a monotone class, we automatically have $\mathcal{E} \subseteq \mathcal{M}$, so we'll prove $\mathcal{E} \supseteq \mathcal{M}$.

This is just a lot of set-theory manipulation, I'll leave it for another time

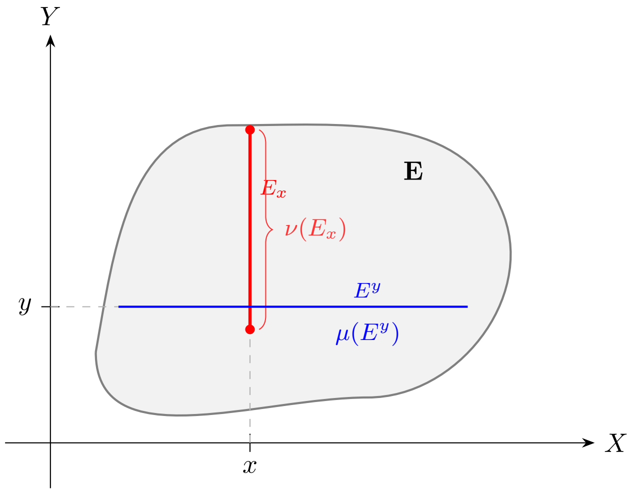
We now are able to start proving the main results:

Theorem 2.6.8: Fubini-Tonelli for Characteristic Functions

Let $(X, \mathcal{M}, \mu)$ and $(N, \mathcal{N}, \nu)$ be measure spaces with μ, ν being σ-finite. Then if $E \in \mathcal{M} \otimes \mathcal{N}$, the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable on X and Y respectively, and

$$(\mu \times \nu)(E) = \int \nu(E_x) d\mu = \int \mu(E^y) d\nu$$

If you are visual, you can think of $\mu(E_x)$ as measuring the area under the slice (inside the appropriate measure so that it doesn't have measure 0), and see that are saying there exists a function that maps x to every slice of the graph:



In particular, it maps x the measure of that slice, so we get a new function which at every slice captures the effect of the X component. After that, we integrate the function to get the effect from the Y component.

Furthermore, it will turn out that we can measure the slices with respect to X, or with respect to Y, and get the same result, hence the second statement of this theorem! After proving the theorem, we'll go over why the assumption in the statement of the theorem are important.

***Proof*:**
We'll first prove it for when μ and ν are finite, and then generalize for when they are σ-finite. The way we'll do it is by starting with a dense subset of $\mathcal{M} \otimes \mathcal{N}$ for which we know the conclusion holds, and then generalize the result using the MCT and DCT. In particular, we will be taking advantage that rectangles already satisfy the property of the theorem, and since rectangles are a generating set for $\mathcal{M} \otimes \mathcal{N}$, we will be able to extend this result any measurable set.

Let $\mathcal{C}$ be the subset of $\mathcal{M} \otimes \mathcal{N}$ for which the result of the theorem is true, that is, the two maps are measurable and we can integrate one way or another. Then if $A \in \mathcal{M}$ and $B \in \mathcal{N}$, we have that $A \times B \in \mathcal{C}$. To see this, first compute the values $\nu(E_x)$ and $\mu(E^y)$:

$$\nu(E_x) = \chi_A(x)\nu(B) \qquad \mu(E^y) = \mu(A)\chi_B(y)$$

Then the functions is measurable since it's a slice (theorem 2.6.5), and:

$$\int \nu(E_x)d\mu = \int \chi_A(x)\nu(B)d\mu = \mu(A)\nu(B) = \int \mu(A)\chi_B(y)d\nu = \int \mu(E^y)d\nu$$

Showing the two integrals are equivalent, and since $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$, all equality hold, and so $E = A \times B \in \mathcal{C}$. By additivity of the integral, it follows that all finite disjoint unions of rectangles are also in $\mathcal{C}$. Thus, by lemma 2.6.7, it suffices to show that $\mathcal{C}$ is a monotone class, for then it is a σ-algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, meaning $\mathcal{C} \supseteq \mathcal{M} \otimes \mathcal{N}$, and so all measurable sets can be broken down in this way.

First, let $\{E_n\}_{n=1}^\infty \subseteq \mathcal{C}$ be an increasing subset and let $E = \bigcup_n^\infty E_n$. We want to show that $E \in \mathcal{C}$. Since $E_n \in \mathcal{C}$ for all $n \in \mathbb{N}$, the sequence of y-section functions f_n, $f_n(y) = \mu((E_n)^y)$ are measurable and increase pointwise to $f(y) = \mu(E^y)$. Since the limit of measurable functions is measurable (corollary 2.1.13), f is measurable. Thus, we can find the value of $\int \mu(E^y)d\nu$ by applying the Monotone Convergence Theorem:

$$\begin{aligned}
\int \mu(E^y)d\nu &= \int \lim_{n\to\infty} \mu((E_n)^y)d\nu \\
&= \lim_{n\to\infty} \int \mu((E_n)^y)d\nu && \text{MCT} \\
&= \lim_{n\to\infty} (\mu \times \nu)(E_n) && E_n \in \mathcal{C} \\
&= (\mu \times \nu)(E) && \cup_i E_i = E
\end{aligned}$$

Doing the same trick with respect to x, we can see that:

$$\int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

which combined together shows us that:

$$\int \mu(E^y)d\nu = \int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

and so $E \in \mathcal{C}$.

Next, let $\{E_n\}_{n=1}^{\infty} \subseteq \mathcal{C}$ be a decreasing subset and let $\bigcap_n E_n = E$. Then since $E_1 \in \mathcal{C}$, the function f_1, $f_1(y) = \mu((E_1)^y)$ is measurable. In fact, it's in L^1, since:

$$\mu((E_1)^y) \leq \mu(X) < \infty \qquad \nu(Y) < \infty$$

and so the resulting integral must have finite measure. Furthermore, f_1 will dominate the sequence of functions f_n, and so we can apply the Dominated Convergence Theorem to do the same limit trick we have just done and conclude that $E \in \mathcal{C}$. Therefore, $\mathcal{C}$ is a monotone class, hence a σ-algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, and the result is true for all measurable sets.

Now, suppose that μ and ν are σ-finite. Then we can write $X \times Y$ to be the union of an increasing sequence of rectangles $\{X_i \times Y_i\}$ of finite measure. Thus, for any $E \in \mathcal{M} \otimes \mathcal{N}$, take $E \cap (X_i \times Y_i)$, then this set now has finite measure, and so what we have just proved applies giving us:

$$(\mu \times \nu(E \cap (X_i \times Y_i)) = \int \chi_{X_i}(x)\nu(E_x \cap Y_i)d\mu = \int \mu(E^y \cap X_i)\chi_{Y_i}(y)d\nu$$

and so, a final application of the Monotone Convergence Theorem gives us our desired result

The hypothesis that $\mu \times \nu$ is σ-finite is necessary:

Example 2.6.9: Necessity of σ-finite

To compute the iterated integrals, we start by finding the x and y section. Using the definition of the diagonal we get that: $(\chi_D)_x = \chi_{\{x\}}$ and $(\chi_D)_y = \chi_{\{y\}}$. Thus, we get:

$$\int \left[\int \chi_{\{y\}} d\mu(x)\right] d\nu(y) = \int 0\nu(y) = 0$$

and

$$\int \left[\int \chi_{\{x\}} d\nu(y)\right] d\mu(x) = \int 1 d\mu(x) = 1$$

showing that the iterated iterated iterated integrals are not equal. For the integral $\int \chi_D \, d(\mu \times \nu)$, since χ_D is a simple function, we have shown in class that the integral is the measure of D with respect to the product measure $\mu \times \nu$ (namely, since χ_D will be in the supremum set over which we are computing the integral), which is:

$$\int \chi_D d(\mu \times \nu) = \mu \times \nu(D) = \inf \left\{ \sum_i^{\infty} \mu(A_i)\nu(B_i) \;\middle|\; A_i \times B_i \in \mathcal{A},\ D \subseteq \bigcup_i A_i \times B_i \right\}$$

where $\mathcal{A}$ is the collection of all rectangles of $\mathcal{M} \otimes \mathcal{N}$. We'll show $\mu \times \nu(D) = \infty$.

Take any countable cover of D by rectangle $\mathcal{C}$ (so $\{A_i \times B_i\}_{i=1}^{\infty} = \mathcal{C}$ for appropriate $A_i \times B_i$). Without loss of generality, we can make these covers disjoint; we can refine the rectangles. If we can show that there must be a non-zero measure set A_i and an infinite measure set B_i such that $A_i \times B_i \in \mathcal{C}$ then we are done. For the sake of contradiction, let's say no such set exists in any countable cover $\mathcal{C}$. We'll show that $\mathcal{C}$ doesn't cover D.

Split $\mathcal{C}$ into $F \subseteq \mathcal{C}$ and $I \subseteq \mathcal{C}$ where F is the collection of all sets where A_i has ***finite*** non-zero measure (the sets have finite measure since if $A_i \subseteq X$, $\mu(A_i) \leq \mu(X) = 1$), and I is the set where

B_i has ***infinite*** measure. By our contradiction assumption, these set's are disjoint, since the first component of the sets in I must have zero measure.

The set F can only cover countably many points of D: Since each B_i has finite measure, any set $A_i \times B_i \in F$ can cover at most finitely many points of D (given appropriate A_i), and so $\bigcup_{A_i \times B_i \in F} A_i \times B_i$ can at most cover countably many points of D (since the 2nd component of D is covered by elements from B_i given appropriate A_i). Therefore, an uncountable number of points are not being covered by F.

The set I misses at least uncountably many points. Since each component A_i has zero-measure and the collection of x components of D (i.e. the set $[0,1]$) has measure one, the union of the sets A_i cannot cover all of the x-components (given appropriate B_i components) or else $0 = \mu(\bigcup_i A_i) \geq \mu([0,1]) = 1$, a contradiction. If the B_i component's don't line up correctly, then this set cover's even fewer points of D. Therefore, I can at most cover a measure zero amount of x-components points from D. Since any set two sets containing each other with a different measure (ex. $\alpha \subseteq \beta$, $0 = \mu(\alpha) \leq (\beta) = 1$) must have a difference with at least uncountable cardinality[a] ($|B \setminus A| \geq \aleph_1$), the set I must miss at least uncountably many points.

However, the cover F can only cover at most countably many points. Therefore, the cover $F \cup I = \mathcal{C}$ cannot cover all of D, a contradiction to the assumption that $\mathcal{C}$ is a cover for D. Hence, there is at least one set $A_i \times B_i$ with $\mu(A_i) > 0$ and $\mu(B_i) = \infty$ for any cover of D, meaning all covers of D must have a set with infinite measure, and so $\infty \leq \mu \times \nu(D)$, and hence

$$\int \chi_D \, d(\mu \times \nu) = \infty$$

and so the two iterated integrals, as well as the integral of χ_D has unequal measures, as we sought to show.

[a] If two sets containing each other had differed by countably many point, then they would have the same measure

Notice that the functions within the integral of the previous theorem are in fact the values of characteristic functions! Hence by the linearity of the integral, we have the result for simple functions. Thus, with the statement proven for simple functions on $\mathcal{M} \otimes \mathcal{N}$, we move on to proving it for integrable functions in general:

Theorem 2.6.10: Fubini-Tonelli Theorem

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ-finite measures. Then:

1. (Tonelli) If $f \in L^+(X \times Y)$, then the functions $g(x) = \int f_x d\nu$ and $h(y) = \int f^y d\mu$ are in $L^+(X)$ and $L^+(Y)$ respectively, and

$$\int \left[\int f(x,y) d\nu(y)\right] d\mu(x) = \int f(x,y) d(\mu \times \nu) = \int \left[\int f(x,y) d\mu(x)\right] \nu(y)$$

2. (Fubini) If $f \in L^1(\mu \times \nu)$, then $f_x \in L^1(\nu)$ for a.e. $x \in X$ and $f^y \in L^1(\mu)$ for a.e. $y \in Y$. Furthermore, the a.e. defined functions $g(x) = \int f_x d\nu$ and $h(y) = \int f^y d\mu$ are in $L^1(\mu)$ and $L^1(\nu)$ respectively, and

$$\int \left[\int f(x,y) d\nu(y)\right] d\mu(x) = \int f(x,y) d(\mu \times \nu) = \int \left[\int f(x,y) d\mu(x)\right] d\nu(y)$$

The resulting integral is called the *iterated integral*

***Proof*:**

1. This essentially comes down to using MCT and the standard representation of measurable functions. By theorem 2.6.8, we have the simple functions satisfy the theorem. To extend it to integrable functions, take $f \in L^+(X \times Y)$, and let $\{f_n\}$ be some simple representation for it, that is, a sequence of simple functions converging upwards to f. From $\{f_n\}$, we have a corresponding $\{g_n\}$ and $\{h_n\}$ converging upwards to g and h: $g_n = \int (f_n)_x$ and $h_n = \int (f_n)^y$. By theorem 2.6.8, these functions are measurable, and so h and g are measurable. With this, we can put together all our information to form the following chain of equalities:

$$\begin{aligned}
\int \left[\int f_x \, d\nu\right] d\mu = \int g d\mu = \int \lim g_n \, d\mu & \\
= \lim \int g_n \, d\mu & \\
= \lim \int f_n \, d(\mu \times \nu) & \qquad \text{theorem } 2.6.8 \\
= \int f \, d(\mu \times \nu) & \qquad \text{MCT} \\
= \lim \int f_n \, d(\mu \times \nu) & \\
= \lim \int h_n \, d\nu & \qquad \text{theorem } 2.6.8 \\
\int \left[\int f^y \, d\mu\right] d\nu = \int h d\nu = \int \lim h_n \, d\nu &
\end{aligned}$$

Thus establishing the equality of Tonelli's Theorem. Furthermore, since $\int f \, d(\mu \times \nu) < \infty$, then $g < \infty$ a.e. and $h < \infty$ a.e., so $\int f_x < \infty$ and $\int f^y < \infty$, and therefore $f_x \in L^1(\nu)$ and $f^y \in L^1(\mu)$, which some of the preliminary results for Fubini's Theorem.

2. Let $f \in L^1(\mu \times \nu)$, so $f^+ \in L^+(\mu \times \nu)$ and $f^- \in L^+(\mu \times \nu)$ (or Ref and sgn$f$). Thus, the result applies to each component function, and so by linearity applies to f

Often, the parenthesis are dropped, and we combine the integrals into a double integral:

$$\int \left[\int f(x,y)\, d\mu(x) \right] d\nu(y) = \int \int f(x,y)\, d\mu(x)\, d\nu(y) = \int \int f\, d\mu\, d\nu$$

Also, you might wonder if instead of assuming $f \in L^+(X \times Y)$ or $f \in L^1(X \times Y)$, is it possible to instead assume that each $f_x \in L^+(\nu), f^y \in L^+(\mu)$ (L^1 respectively) and that the iterated integrals $\int \int f\, d\mu\, d\nu$ and $\int \int f\, d\nu\, d\mu$ existed, then $f \in L^+(X \times Y)$ (L^1 respectively), that is, is the converse true? The answer is actually no:

Example 2.6.11: Converse not True

Let $X = Y = \mathbb{N}$, $\mathcal{M} = \mathcal{N} = P(\mathbb{N})$, $\mu = \nu$ be the counting measures. Define:

$$f(m,n) = \begin{cases} 1 & m = n \\ -1 & m = n+1 \\ 0 & \text{otherwise} \end{cases}$$

Then $\int |f| d(\mu \times \nu) = \infty$, and $\int \int f d\mu d\nu$, $\int \int f d\nu d\mu$ exist and are unequal.

This is not quite a converse because the two iterated integrals are unequal, and I don't know of an example where they are equal, but $\int f d(\mu \times \nu)$ is not equal to it (i.e. infinity).

One practical application of Fubini-Tonelli's Theorem is to solve integrals by integrating over the simpler variable first. Usually, one first uses Tonelli's theorem to evaluate the integral $\int f\, d(\mu \times \nu)$ as an iterated integral $\int \int f\, d\mu\, d\nu$ and show that the result is finite, at which point Fubini's Theorem can be invoked to get $\int \int f\, d\mu\, d\nu = \int \int f\, d\nu\, d\mu$.

To close off the discussion on product measures, notice that if μ and ν are complete measure, $\mu \times \nu$ is almost never a complete measure. Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces, take $A \in \mathcal{M}$ is a non-empty set with $\mu(A) = 0$, and suppose $\mathcal{N} \neq P(Y)$ (For example, choose $(\mathbb{R}, \mathcal{L}, m)$ for both measure spaces). If $E \in P(Y) \setminus \mathcal{N}$, then $A \times E \notin \mathcal{M} \times \mathcal{N}$ since not all slices of $A \times E$ (in particular, the slice E) are in the respective measure spaces (proposition 2.6.5). However, $A \times E \subseteq A \times Y$, and $(\mu \times \nu)(A \times Y) = 0$ (since the measure is finite, any out-measure is equal to the usual outer-measure on these sets, which is $\mu(A)\nu(Y) = 0 \cdot n = 0$, since we take on the convention that $0 \cdot \infty = 0$). Thus, $\mu \times \nu$ is not complete. If we want, we can apply theorem 1.2.8 to complete the measure, however we can no longer directly apply Fubini-Tonelli's Theorem. (Put Why Here!). We thus have to slightly amend Fubini-Tonelli's theorem for the completion:

Theorem 2.6.12: Fubini-Tonelli's Theorem for Complete Measures

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be complete σ-finite measures and let $(X \times Y, \mathcal{L}, \lambda)$ be the completion of $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$. If f is $\mathcal{L}$-measurable is either:

1. non-negative, that is $f \geq 0$
2. $f \in L^1(\lambda)$

Then f_x is $\mathcal{N}$-measurable for a.e. x and f^y is $\mathcal{M}$-measurable for a.e. y, and if (2) applies as well, then f_x and f^y are integrable for a.e. x and y. Furthermore, $x \mapsto \int f_x \, d\nu$ and $y \mapsto \int f^y \, d\mu$ are measurable (and in the case of (2), also integrable) and

$$\int f \, d\lambda = \int \int f(x, y) \, d\mu(x) \, d\nu(y) = \int \int f(x, y) \, d\nu(y) \, d\mu(x)$$

***Proof*:**
We'll do the following:

1. If $E \in \mathcal{M} \otimes \mathcal{N}$ and $\mu \times \nu(E) = 0$, then $\nu(E_x) = \mu(E^y) = 0$ for a.e. x and y
2. If f is $\mathcal{L}$-measurable and $f = 0$ λ-a.e., then f_x and f^y are integrable for a.e. x and y, and $\int f_x d\nu = \int f^y d\mu = 0$ for a.e. x and y (here, the completeness of μ and ν is needed)

lemma 1:

Proof. Note that $\chi_E \in L^+(X \times Y)$ and $(X, \mathcal{M}, \mu)$ $(Y, \mathcal{N}, \nu)$ are σ-finite, and so Tonelli's Theorem applies, meaning

$$\int \chi_E d(\mu \times \nu) = \int \left[\int (\chi_E)^y d\mu(x) \right] d\nu(y) = \int \left[\int (\chi_E)_x d\nu(y) \right] d\mu(x)$$

First, since χ_E is an indicator function, we see that $(\chi_E)_x = \chi_{E_x}$, since $\chi_{E_x}(y) = \chi_E(x, y)$. Similarly, $(\chi_E)^y = \chi_{E^y}$. Therefore, we have:

$$\begin{aligned}
0 &= \int \chi_E d(\mu \times \nu) \\
&= \int \left[\int (\chi_E)^y d\mu(x) \right] d\nu(y) \\
&= \int \left[\int (\chi_E)_x d\nu(y) \right] d\mu(x)
\end{aligned}$$

showing that the function $x \mapsto \int \chi_{E_x} d\nu$ and $y \mapsto \int \chi_{E^y} d\mu$ is 0 a.e., which implies $\nu(E_x) = 0$ and $\mu(E^y) = 0$ a.e. □

lemma 2: Multiple parts of this proof take advantage of the fact that if μ is a measure and $\mu(A) = 0$, then $\int_A f d\mu = 0$ (since all simple functions φ approximating f with the property of $\varphi \leq f$ will have their integral be of the form $\sum_i^n c_i \chi_{A_i} = \sum_i^n c_i \cdot 0 = 0$ for $A_i \subseteq A$, and so the supremum of the integral of such simple functions must also be 0)

Proof. Let f be $\mathcal{L}$-measurable and $f = 0$ λ-a.e. We want to show that f_x and f^y is integrable for a.e. x and y and $\int f_x d\nu = \int f^y d\mu = 0$ for a.e. x and y. What we'll do is take advantage of Fubini-Tonelli and theorem 2.12 (which states that there exists a $\mathcal{M} \otimes \mathcal{N}$-measurable function g such that $f = g$ λ-a.e.)

First, since $f = g$ λ-a.e., $f - g = 0$ λ-a.e., meaning $(f-g)_x = f_x - g_x = 0$ for a.e. x (assuming completeness) and $f^y - g^y = 0$ for a.e. y (assuming completeness), and so $f_x = g_x$ and $f^y = g^y$ for a.e. x and y. This will be useful to us later.

Next, Consider $\int f d\lambda$. We want to take advantage of Fubini's theorem, which is true for g, and so we must re-write this expression in such a way that we get it equal to g. Let's say $S = \{(x,y) \in X \times Y \mid f(x,y) \neq g(x,y)\}$. We know that $\lambda(S) = 0$, and so:

$$0 = \int f\, d\lambda = \int_S f\, d\lambda + \int_{S^c} f\, d\lambda = \int_{S^c} f\, d\lambda \tag{2.1}$$

Since $S^c \in \mathcal{L}$, we have that $S^c = A \coprod N$ where $A \in \mathcal{M} \otimes \mathcal{N}$ and $N \subseteq B$ where $\mu \times \nu(B) = 0$. Thus, we get:

$$\int_{S^c} f d\lambda = \int_A f d\lambda + \int_N f d\lambda = \int_A f d\lambda$$

since $f = g$ on S^c, in particular $f = g$ on $A \in \mathcal{M} \otimes \mathcal{N}$, and λ is the extension of $\mu \times \nu$, we can re-write this as:

$$\int_A f d\lambda = \int_A g d(\mu \times \nu)$$

and so Fubini-Tonelli applies, giving us the iterated integrals are equal to 0 (carrying the 0 from equation (2.1)):

$$\int \left[\int_{A_x} g_x d\nu(y) \right] d\mu(x) = 0$$

and similarly for the y-section. Since $f = g$ on S^c, by what we've shown earlier, $f_x = g_x$ on A_x, (and similarly $f^y = g^y$ on A^y) and so:

$$\int_{A_x} f_x d\nu(y) = 0 \qquad \text{for a.e. } x$$

and similarly for the y-sections, which brings us close to our goal (it is possible that $A_x \subsetneq X$ or $A^y \subsetneq Y$). We now have to add back in points which we eliminated since the integral was zero on S and N:

$$\int_S f d\lambda = 0 \qquad \int_N f d\lambda = 0$$

Since they are measure zero sets in $\mathcal{L}$, there exists a $C, D \in \mathcal{M} \otimes \mathcal{N}$ such that $\mu \times \nu(C) = \mu \times \nu(D) = 0$ and $S \subseteq C$, $N \subseteq D$. Let $E = C \cup D$ so that $S \cup N \subseteq E$ and $\mu \times \nu(E) = 0$. Thus, we have:

$$\int_E f d\lambda = 0$$

since $\mu \times \nu(E) = 0$ and $E \in \mathcal{M} \otimes \mathcal{N}$, by the first lemma, $\mu(E^y) = \nu(E_x) = 0$ for a.e. x and y. But then, for a.e. x and y, $\int_{E_x} f_x d\nu(y) = \int_{E^y} f^y d\mu(x) = 0$. Since $X \times Y = S^c \cup S = A \cup E$ (by how we've broken the set down), then $X = A_x \cup E_x$ and $Y = A^y \cup E^y$, and so

$$0 \le \int f_x d\nu(y) \le \int_{A_x} f_x d\nu(y) + \int_{E_x} f_x d\nu(y) = 0$$

for a.e. x, implying $\int f_x d\nu(y) = 0$ for a.e. x (and similarly for y). Hence, they are integrable for a.e. x and y, completing the proof. □

With this, we can prove Fubini-Tonelli's Theorem for complete measures. First, let's build-up some result and notation. Let $f \in L^+(\lambda)$ (resp. $f \in L^1(\lambda)$). Then by theorem 2.12, there exists a g such that $f = g$ λ-a.e., meaning that $f - g = 0$ λ-a.e. Thus, by the 2nd lemma, $\int (f-g)_x = 0$ and $\int (f-g)^y = 0$ are integrable for a.e. x and y. Therefore, $\int f_x - g_x = 0$ and $\int f^y - g^y = 0$ for a.e. x and y, and so $\int f_x = \int g_x$ and $\int f^y = \int g^y$ for a.e. x and y.

We can now start concluding the proof. Since μ and ν are complete and $f_x = g_x$ ($f^y = g^y$) for a.e. x and y, by proposition 2.11, f_x is $\mathcal{N}$-measurable and f^y is $\mathcal{M}$-measurable for a.e. x and y, completing the 1st assertion of the theorem. In the $f \in L^1(\lambda)$ case, the proof we've just done in the previous paragraph shows that each f_x and f^y are integrable for a.e. x and y.

Next, since $G_x : x \mapsto \int g_x d\nu$ and $G^y : y \mapsto \int g^y d\mu$ is measurable (resp. integrable), by theorem 2.11, $F_x : x \mapsto \int f_x d\nu$ and $F^y : y \mapsto \int f^y d\mu$ is measurable (resp. integrable). Finally, since $F_x = G_x$ for a.e. x and $F^y = G^y$ for a.e. y, their integrals match, and since g is $\mathcal{M} \otimes \mathcal{N}$ measurable, Tonelli's (resp. Fubini's) theorem applies, and so by one more application of theorem 2.11 :

$$\int f \, d\lambda = \int \int f(x,y) \, d\mu(x) \, d\nu(y) = \int \int f(x,y) \, d\nu(y) \, d\mu(x)$$

Completing the proof

2.7 The n-dimensional Lebesgue Integral

In this section, we focus on $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ with the completed product measure $m \times m \times \cdots \times m = m^n$ on $\mathcal{L} \otimes \mathcal{L} \otimes \cdots \otimes \mathcal{L} = \mathcal{L}^n$. The domain $\mathcal{L}^n$ of m^n is called the set of *Lebesgue measurable sets in* $\mathbb{R}^n$: $(\mathbb{R}^n, \mathcal{L}^n, m^n)$ (sometimes, we will restrict m^n to have domain $(\mathcal{B}_{\mathbb{R}})^n = \mathcal{B}_{\mathbb{R}^n}$). If it is unambiguous, we will usually write $(\mathbb{R}^n, \mathcal{L}^n, m)$, dropping the power on the m, and for the $n = 1$ case, we will often write $\int f(x) \, dx := \int f \, dm$.

The following 3 theorems are simply generalizations of previous properties of the Lebesgue measure. In the following, let $E = \prod_{i=1}^n E_i$ where $E_i \subseteq \mathbb{R}$. We will call each E_i the *sides* of E

Theorem 2.7.1: Simplifying outer-measure

Suppose $E \in \mathcal{L}^n$

1. $m(E) = \inf \{m(U) \mid U \supseteq E,\ U \text{ is open}\} = \sup \{m(K) \mid K \subseteq E,\ K \text{ is compact}\}$
2. $E = A_1 \cup N_1 = A_2 \setminus N_2$ where $A_1 \in F_\sigma$, $A_2 \in G_\delta$, and $m(N_1) = m(N_2) = 0$
3. If $m(E) < \infty$, for all $\epsilon > 0$, there is a finite disjoint collection of rectangles $\{R_i\}_{i=1}^n$ whose sides are intervals such that $m(E \Delta \cup_i^n R_i) < \epsilon$

***Proof*:**
p. 70 in folland

Theorem 2.7.2: Completeness of $L^1(m)$

Let $f \in L^1(m)$ and let $\epsilon > 0$. Then there exists a simple function $\varphi = \sum_{i=1}^n a_i \chi_R$ where each R_i is a product of intervals such that

$$\int |f - \varphi| < \epsilon$$

and there is a continuous function that vanishes outside a bounded set such that

$$\int |f - g| < \epsilon$$

***Proof*:**
Just like in theorem 2.3.9, approximate f by a simple function, then apply proposition 2.7.1(3) to approximate φ appropriately. Finally, approximate φ by continuous functions by applying the natural generalization of the argument in theorem 2.3.9

Theorem 2.7.3: Translation Invariance in $\mathbb{R}^n$

Let $(\mathbb{R}^n, \mathcal{L}^n, m)$ be a measure space. Then m is translation invariant, that is, for any $a \in \mathbb{R}^n$, if we define $\tau_a : \mathbb{R}^n \to \mathbb{R}^n$, $\tau_a(x) = a + x$, then

1. If $E \in \mathcal{L}^n$, then $\tau_a(E) \in \mathcal{L}^n$ and $m(E) = m(\tau_a(E))$
2. If $f : \mathbb{R}^n \to \mathbb{C}$ is Lebesgue measurable, then so is $f \circ \tau_a$. Furthermore, if either $f \geq 0$ or $f \in L^1(m)$, then $\int (f \circ \tau_a)\, dm = \int f\, dm$

Notice that we did not mention scaling. This is because scaling is not abstracted to linear transformations which requires more work to show how the scaling of a linear transformation affects the value, and so we will prove that later in this section.

***Proof*:**
Folland p.71

(here, some on cube approximation and Jordan content)

I'll for now simply state the effect of linear transformations and come back to this:

Theorem 2.7.4: Linear Transformation on Integrability

Suppose $T \in GL_n(\mathbb{R})$. Then:

1. If f is a Lebesgue measurable function on $\mathbb{R}^n$, so is $f \circ T$. If $f \geq 0$ or $f \in L^1(m)$, then:
$$\int f\, dm = |\det(T)| \int f \circ T\, dm$$
2. If $E \in \mathcal{L}^n$, then so is $T(E)$, and $m(T(E)) = |\det(T)| m(E)$

***Proof*:**
here

With linear transformations understood, the general statement on how diffeomorphisms affect the integral is in reach. The idea is one of the payoffs of differentiability: near each point x, a C^1 map T is well approximated by its derivative D_xT, which is a linear map, and theorem 2.7.4 tells us exactly how a linear map scales volume, namely by the absolute value of its determinant. A differentiable map should therefore scale volume *locally* at x by the factor $|\det D_xT|$, and adding up (integrating!) these local contributions should recover the measure of the image. The change of variables theorem makes this heuristic precise, and contains the familiar u-substitution from single-variable calculus as its $n = 1$ case.

Let us fix the terms. Let $\Omega \subseteq \mathbb{R}^n$ be open. A map $T = (T_1, ..., T_n) : \Omega \to \mathbb{R}^n$ is C^1 if every partial derivative $\partial T_i/\partial x_j$ exists and is continuous on Ω, and D_xT denotes the linear map given by the matrix $(\partial T_i/\partial x_j(x))_{i,j}$ of partial derivatives at x (if T is itself linear, then $D_xT = T$ for every x). The map T is a C^1 *diffeomorphism* onto its image if $T(\Omega)$ is open, $T : \Omega \to T(\Omega)$ is a bijection, and both T and T^{-1} are C^1. Differentiating $T^{-1} \circ T = \mathrm{id}_\Omega$ with the chain rule [6] shows that D_xT is then invertible at every $x \in \Omega$, with

$$(D_xT)^{-1} = D_{T(x)}(T^{-1})$$

(In fact, by the inverse function theorem [6], it would have been enough to ask that T be injective and C^1 with D_xT invertible everywhere: the openness of $T(\Omega)$ and the smoothness of T^{-1} then come for free. That refinement is not needed here.) Note that in particular T is a homeomorphism onto $T(\Omega)$, so it sends open sets to open sets and compact sets to compact sets, and for *any* $E \subseteq \Omega$, $T(E) = (T^{-1})^{-1}(E)$: images under T are pre-images under the continuous map T^{-1}, which is what will make measurability manageable.

Before the theorem comes a small geometric tool making precise the idea that open sets in $\mathbb{R}^n$ are built out of cubes the same way open sets in $\mathbb{R}$ are built out of intervals. By a *dyadic cube of generation* k (for $k \in \mathbb{Z}_{\geq 0}$) is meant a set of the form

$$Q = \prod_{i=1}^{n} \left[\frac{m_i}{2^k}, \frac{m_i + 1}{2^k}\right) \qquad m_1, ..., m_n \in \mathbb{Z}$$

For fixed k, the dyadic cubes of generation k partition $\mathbb{R}^n$, and each is contained in exactly one dyadic cube of generation $k - 1$, called its *parent*. Consequently, two dyadic cubes (of any generations) are either disjoint or nested. By corollary 2.6.3, $m(Q) = 2^{-kn}$ (the sides are intervals of length 2^{-k}), and the boundary ∂Q is null: it is covered by $2n$ rectangles each having one side a single point, and a rectangle with finite sides one of which has length 0 has measure 0 by the same corollary. In particular $m(\overline{Q}) = m(Q)$, and integrals over Q and $\overline{Q}$ agree, since the integrands differ only on the null set ∂Q (proposition 2.2.8).

Lemma 2.7.5: Dyadic Decomposition of Open Sets

Let $U \subseteq \mathbb{R}^n$ be open and non-empty. Then U is a countable *disjoint* union of dyadic cubes, each of whose closures is contained in U.

***Proof*:**
Call a dyadic cube Q *admissible* if $\overline{Q} \subseteq U$, and let $\mathcal{S}$ consist of every admissible dyadic cube that is either of generation 0 or whose parent is not admissible. There are countably many dyadic cubes in total, so $\mathcal{S}$ is countable.

First, $\mathcal{S}$ covers U: let $x \in U$, and for each k let $Q_k(x)$ be the unique dyadic cube of generation k containing x. Every point of $\overline{Q_k(x)}$ differs from x by at most 2^{-k} in each coordinate, and since U is open, it contains all points differing from x by less than some $\delta > 0$ in each coordinate, so $Q_k(x)$ is admissible once $2^{-k} < \delta$. Taking the smallest such k, either $k = 0$ or the parent $Q_{k-1}(x)$ is not admissible, so $Q_k(x) \in \mathcal{S}$, and $x \in Q_k(x)$.

Second, the members of $\mathcal{S}$ are pairwise disjoint: suppose two distinct members meet. Dyadic cubes are disjoint or nested, so one strictly contains the other, say $Q \subsetneq Q'$, meaning Q' is an ancestor of Q. Then Q is not of generation 0, and its parent P satisfies $P \subseteq Q'$ (the ancestors of Q are nested), so $\overline{P} \subseteq \overline{Q'} \subseteq U$, i.e. the parent of Q is admissible, contradicting $Q \in \mathcal{S}$. Thus $U = \coprod_{Q \in \mathcal{S}} Q$, as we sought to show.

The theorem can now be stated and proved. The formulation follows [11, Theorem 2.47].

Theorem 2.7.6: Change of Variables for C^1 Diffeomorphisms

Let $\Omega \subseteq \mathbb{R}^n$ be open and let $T : \Omega \to T(\Omega)$ be a C^1 diffeomorphism.

1. If f is a Lebesgue measurable function on $T(\Omega)$, then $f \circ T$ is Lebesgue measurable on Ω. If $f \geq 0$ or $f \in L^1(T(\Omega), m)$, then

$$\int_{T(\Omega)} f\, dm = \int_{\Omega} (f \circ T)(x)\, |\det D_x T|\, dm(x)$$

   Moreover, $f \in L^1(T(\Omega), m)$ if and only if $(f \circ T)|\det D_x T| \in L^1(\Omega, m)$.

2. If $E \subseteq \Omega$ is Lebesgue measurable, then $T(E)$ is Lebesgue measurable, and

$$m(T(E)) = \int_E |\det D_x T|\, dm(x)$$

***Proof*:**
Here, a function on the open set $T(\Omega)$ (or a subset $E \subseteq \Omega$) is called Lebesgue measurable if it is measurable with respect to the σ-algebra of Lebesgue measurable subsets of $T(\Omega)$ (resp. Ω), and integrals over subsets are as in definition 2.2.2 and the notation following it.

The proof is long, so let us map it out. Steps 1-3 establish, by hand, the inequality $m(T(Q)) \leq \int_Q |\det D_x T|\, dm$ for cubes. Step 4 upgrades it to all measurable sets (giving the measurability claims along the way), and Step 5 finishes the measurability bookkeeping of part (1). Everything up to that point applies to an arbitrary C^1 diffeomorphism, in particular to T^{-1}, and Step 6 exploits this symmetry to reverse the inequality, proving part (2). Steps 7-8 then convert the measure identity of part (2) into the integral formula of part (1), with the pushforward measure (theorem 2.3.16) doing the abstract work.

Throughout, we use the norm $\|x\| = \max_i |x_i|$ on $\mathbb{R}^n$, so that the closed ball of radius h about a

is precisely the closed cube of side length $2h$ centered at a. Throughout this proof, "cube" means one with sides parallel to the axes. For an $n \times n$ matrix $A = (A_{ij})$ we set $\|A\| = \max_i \sum_j |A_{ij}|$, which is the operator norm for $\|\cdot\|$: the estimate $\|Ax\| = \max_i |\sum_j A_{ij}x_j| \leq \|A\|\,\|x\|$ is direct, and it is an equality for a suitable vector of signs $x \in \{\pm 1\}^n$. Being an operator norm, it is submultiplicative: $\|ABx\| \leq \|A\|\|B\|\|x\|$ for all x, so $\|AB\| \leq \|A\|\|B\|$. Finally, note that $x \mapsto D_xT$ is continuous (its entries are, by the definition of C^1), hence so is $x \mapsto |\det D_xT|$, the determinant being a polynomial in the entries. This function is moreover strictly positive, D_xT being invertible.

Step 1: a cube estimate for C^1 maps. Let $G : \Omega \to \mathbb{R}^n$ be any C^1 map, and let $Q \subseteq \Omega$ be a closed cube with center a and side length $2h$. We claim

$$m(G(Q)) \leq \left(\sup_{y\in Q} \|D_yG\|\right)^n m(Q)$$

Write $s = \sup_{y\in Q} \|D_yG\|$, finite since $y \mapsto \|D_yG\|$ is continuous on the compact set Q. Fix $x \in Q$ and a component index i. The segment $\gamma(t) = a + t(x-a)$, $t \in [0,1]$, stays in Q (cubes are convex), and $t \mapsto G_i(\gamma(t))$ is differentiable with derivative $\sum_j \partial G_i/\partial x_j(\gamma(t))\,(x_j - a_j)$ by the chain rule [6]. By the mean value theorem [2] there is $y_i = \gamma(t_i) \in Q$ with

$$G_i(x) - G_i(a) = \sum_{j=1}^{n} \frac{\partial G_i}{\partial x_j}(y_i)\,(x_j - a_j)$$

and hence $|G_i(x) - G_i(a)| \leq \left(\sum_j |\partial G_i/\partial x_j(y_i)|\right) \max_j |x_j - a_j| \leq \|D_{y_i}G\|\,h \leq sh$. As i was arbitrary, $\|G(x) - G(a)\| \leq sh$, that is, $G(Q)$ is contained in the closed cube of side length $2sh$ centered at $G(a)$. Since $G(Q)$ is compact (hence measurable) and the measure of a cube is the n-th power of its side length (corollary 2.6.3), monotonicity gives $m(G(Q)) \leq (2sh)^n = s^n(2h)^n = s^n m(Q)$, proving the claim.

Step 2: twisting by a linear map. Keep Q as in Step 1, and now let $G = T$ and fix any $S \in GL_n(\mathbb{R})$. The composition $S^{-1} \circ T$ is C^1 on Ω with $D_y(S^{-1} \circ T) = S^{-1}D_yT$ (chain rule again, the derivative of the linear map S^{-1} being itself). Applying theorem 2.7.4(2) to the linear map S and the set $S^{-1}(T(Q)) = (S^{-1} \circ T)(Q)$, and then Step 1 to the C^1 map $S^{-1} \circ T$:

$$m(T(Q)) = |\det S|\, m\big((S^{-1} \circ T)(Q)\big) \leq |\det S| \left(\sup_{y\in Q} \|S^{-1}D_yT\|\right)^n m(Q)$$

Step 3: the exact estimate on a cube. We claim that for every closed cube $Q \subseteq \Omega$,

$$m(T(Q)) \leq \int_Q |\det D_xT|\, dm(x)$$

Fix $\epsilon > 0$. Since $z \mapsto D_zT$ is continuous and each D_zT is invertible, $z \mapsto (D_zT)^{-1}$ is continuous as well (by Cramer's rule, its entries are polynomials in the entries of D_zT divided by the non-vanishing continuous function $\det D_zT$), so $M = \sup_{z\in Q} \|(D_zT)^{-1}\|$ is finite. For $y, z \in Q$, submultiplicativity gives

$$\|(D_zT)^{-1}D_yT\| = \|I + (D_zT)^{-1}(D_yT - D_zT)\| \leq 1 + M\,\|D_yT - D_zT\|$$

Also, $|\det D_xT|$ attains a strictly positive minimum c on the compact set Q. Now, $y \mapsto D_yT$ and $y \mapsto |\det D_yT|$ are uniformly continuous on Q, so there is $\delta > 0$ such that whenever $y, z \in Q$

with $\|y - z\| \leq \delta$:

$$\|(D_zT)^{-1}D_yT\| \leq 1+\epsilon \qquad \text{and} \qquad |\det D_zT| \leq |\det D_yT| + \epsilon c \leq (1+\epsilon)|\det D_yT|$$

Bisecting the sides of Q enough times, divide Q into finitely many closed subcubes $Q_1, ..., Q_N$ with disjoint interiors and side lengths at most δ, and let x_j be the center of Q_j. Step 2 with $S = D_{x_j}T$ on the cube Q_j gives

$$m(T(Q_j)) \leq |\det D_{x_j}T| \left(\sup_{y\in Q_j} \|(D_{x_j}T)^{-1}D_yT\| \right)^n m(Q_j) \leq (1+\epsilon)^n \, |\det D_{x_j}T| \, m(Q_j)$$

since $\|y - x_j\| \leq \delta$ for $y \in Q_j$. Moreover, integrating the pointwise bound $|\det D_{x_j}T| \leq (1+\epsilon)|\det D_xT|$ (valid for $x \in Q_j$) over Q_j:

$$|\det D_{x_j}T| \, m(Q_j) \leq (1+\epsilon) \int_{Q_j} |\det D_xT| \, dm(x)$$

Combining, and using that the Q_j cover Q while $\sum_j \chi_{Q_j} = \chi_Q$ except on the shared boundary faces, which are null (so the integrals agree, as in the discussion before lemma 2.7.5, with proposition 2.2.7 converting the sum of integrals into the integral of the sum):

$$m(T(Q)) \leq \sum_{j=1}^{N} m(T(Q_j)) \leq (1+\epsilon)^{n+1} \sum_{j=1}^{N} \int_{Q_j} |\det D_xT| \, dm(x) = (1+\epsilon)^{n+1} \int_Q |\det D_xT| \, dm(x)$$

As $\epsilon > 0$ was arbitrary, the claim follows.

Step 4: the inequality for all measurable sets. We claim that for every Lebesgue measurable $E \subseteq \Omega$, the set $T(E)$ is Lebesgue measurable and

$$m(T(E)) \leq \int_E |\det D_xT| \, dm(x)$$

We proceed in stages. Note first that for *Borel* $E \subseteq \Omega$ there is no measurability problem at all: $T(E) = (T^{-1})^{-1}(E)$ is a pre-image of a Borel set under the continuous map T^{-1}, hence Borel (corollary 2.1.3).

(a) *Open sets.* Let $U \subseteq \Omega$ be open. By lemma 2.7.5, $U = \coprod_j Q_j$ is a countable disjoint union of dyadic cubes with $\overline{Q_j} \subseteq U \subseteq \Omega$. Since T is injective, $T(U) = \coprod_j T(Q_j)$ is a disjoint union. Monotonicity ($T(Q_j) \subseteq T(\overline{Q_j})$), Step 3 applied to the closed cube $\overline{Q_j}$, and the fact that integrals over Q_j and $\overline{Q_j}$ agree together give $m(T(Q_j)) \leq \int_{Q_j} |\det D_xT| \, dm$, so by countable additivity:

$$m(T(U)) = \sum_j m(T(Q_j)) \leq \sum_j \int_{Q_j} |\det D_xT| \, dm(x) = \int_U |\det D_xT| \, dm(x)$$

the last equality again by proposition 2.2.7, since $\sum_j \chi_{Q_j} = \chi_U$ exactly.

(b) *Borel sets inside a compact set.* Let E be Borel with $E \subseteq K$ for some compact $K \subseteq \Omega$. Using the convention that the distance to the empty set is ∞, let $d = \inf\{\|x - w\| \mid x \in K, \ w \notin \Omega\} > 0$ (positive since K is compact and $\mathbb{R}^n \setminus \Omega$ is closed and disjoint from it), and choose a bounded open set W with $K \subseteq W$ and $\overline{W} \subseteq \Omega$: say $W = \{x \mid \|x - p\| < d/2 \text{ for some } p \in K\}$ intersected with a large open ball containing K. Then $\overline{W}$ is compact, so $C = \sup_{\overline{W}} |\det D_x T| < \infty$. Since $m(E) \le m(K) < \infty$, outer regularity (theorem 2.7.1) provides open sets $V_j \supseteq E$ with $m(V_j) \le m(E) + 1/j$. Set $U_j = W \cap V_j$, an open subset of Ω containing E. By (a):

$$m(T(E)) \le m(T(U_j)) \le \int_{U_j} |\det D_x T|\, dm \le \int_E |\det D_x T|\, dm + C\, m(U_j \setminus E) \le \int_E |\det D_x T|\, dm + \frac{C}{j}$$

using $U_j \setminus E \subseteq \overline{W}$ for the bound by C, and $m(U_j \setminus E) \le m(V_j) - m(E) \le 1/j$ (valid as $m(E) < \infty$). Letting $j \to \infty$ proves the claim for such E.

(c) *All Borel sets.* The sets $K_i = \{x \in \Omega \mid \|x\| \le i \text{ and } \operatorname{dist}(x, \mathbb{R}^n \setminus \Omega) \ge 1/i\}$ (with the same empty-set convention, and distances in $\|\cdot\|$) are compact, increase with i, and exhaust Ω. For Borel $E \subseteq \Omega$, the sets $E \cap K_i$ increase to E, so $T(E \cap K_i)$ increase to $T(E)$, and by continuity from below (proposition 1.2.5), part (b), and the Monotone Convergence Theorem (theorem 2.2.5) applied to $\chi_{E \cap K_i} |\det D_x T| \nearrow \chi_E |\det D_x T|$:

$$m(T(E)) = \lim_{i \to \infty} m(T(E \cap K_i)) \le \lim_{i \to \infty} \int_{E \cap K_i} |\det D_x T|\, dm = \int_E |\det D_x T|\, dm$$

(d) *Null sets, and all Lebesgue sets.* Let $N \subseteq \Omega$ with $m(N) = 0$. By outer regularity there are open $V_j \supseteq N$ with $m(V_j) \le 1/j$, so $B = \Omega \cap \bigcap_j V_j$ is a Borel set with $N \subseteq B \subseteq \Omega$ and $m(B) = 0$. By (c), $m(T(B)) \le \int_B |\det D_x T|\, dm = 0$, the integrand $\chi_B |\det D_x T|$ being zero a.e. (proposition 2.2.8). Since $T(N) \subseteq T(B)$ and m is complete (theorem 1.3.4), $T(N)$ is Lebesgue measurable with $m(T(N)) = 0$: *a C^1 diffeomorphism sends null sets to null sets.* Now let $E \subseteq \Omega$ be Lebesgue measurable. By theorem 2.7.1(2), $E = A \cup N$ with A an F_σ (in particular Borel) set and $m(N) = 0$. Both are subsets of $E \subseteq \Omega$. Then $T(E) = T(A) \cup T(N)$ is the union of a Borel set and a null set, hence Lebesgue measurable, and

$$m(T(E)) \le m(T(A)) + m(T(N)) \le \int_A |\det D_x T|\, dm + 0 \le \int_E |\det D_x T|\, dm$$

completing Step 4. We emphasize that Steps 1-4 used nothing about T beyond its being a C^1 diffeomorphism, so they apply equally to $T^{-1} : T(\Omega) \to \Omega$.

Step 5: measurability of $f \circ T$. Let f be Lebesgue measurable on $T(\Omega)$ and B a Borel subset of the codomain. Then $(f \circ T)^{-1}(B) = T^{-1}(f^{-1}(B))$, and $f^{-1}(B)$ is a Lebesgue measurable subset of $T(\Omega)$, so as in Step 4(d) we may write $f^{-1}(B) = A \cup N$ with A Borel, $m(N) = 0$, and both inside $T(\Omega)$. Now $T^{-1}(A)$ is Borel (T is continuous, so pre-images of Borel sets are Borel), while $T^{-1}(N)$ (the *image* of the null set N under the C^1 diffeomorphism T^{-1}) is null and measurable by Step 4(d) applied to T^{-1}. Hence $(f \circ T)^{-1}(B)$ is Lebesgue measurable, and $f \circ T$ is a Lebesgue measurable function.

Step 6: the reverse inequality, and part (2). We claim that for every Lebesgue measurable $E \subseteq \Omega$:

$$\int_E |\det D_x T|\, dm(x) \le m(T(E))$$

First suppose $E \subseteq K$ for a compact $K \subseteq \Omega$, and fix $\epsilon > 0$. As in Step 4(b), choose $r > 0$ so small that the compact set $K' = \{x \mid \|x - p\| \le r \text{ for some } p \in K\}$ is contained in Ω. On K', the function $|\det D_x T|$ is uniformly continuous with strictly positive minimum, so exactly as in Step 3 there is $\delta \in (0, r]$ such that

$$|\det D_z T| \le (1 + \epsilon)\, |\det D_y T| \qquad \text{whenever } y, z \in K' \text{ with } \|y - z\| \le \delta$$

Fix a generation k with $2^{-k} \le \delta$, and let $Q_1, ..., Q_N$ be the dyadic cubes of generation k that meet K (finitely many, as K is bounded). Each Q_j lies in K': a point of Q_j differs from a point of K by at most $2^{-k} \le r$ in each coordinate. The sets $E_j = E \cap Q_j$ are disjoint, Lebesgue measurable, and their union is E. Fix any $x_j \in Q_j$. Since any two points of Q_j are within δ of each other, integrating the pointwise bound $|\det D_x T| \le (1+\epsilon)|\det D_{x_j} T|$ over E_j gives

$$\int_{E_j} |\det D_x T| \, dm(x) \le (1+\epsilon) \, |\det D_{x_j} T| \, m(E_j)$$

Next, $E_j = T^{-1}(T(E_j))$, so Step 4 applied to the C^1 diffeomorphism T^{-1} and the measurable set $T(E_j) \subseteq T(\Omega)$ gives

$$m(E_j) \le \int_{T(E_j)} |\det D_y(T^{-1})| \, dm(y)$$

For $y \in T(E_j)$ we have $T^{-1}(y) \in E_j \subseteq Q_j$, so $\|T^{-1}(y) - x_j\| \le 2^{-k} \le \delta$, and therefore, using $(D_y(T^{-1})) = (D_{T^{-1}(y)}T)^{-1}$ together with the multiplicativity of the determinant $(\det(A)\det(A^{-1}) = 1)$:

$$|\det D_y(T^{-1})| = |\det D_{T^{-1}(y)} T|^{-1} \le (1+\epsilon) \, |\det D_{x_j} T|^{-1}$$

the inequality being the δ-bound applied to the points $z = x_j$ and $y' = T^{-1}(y)$ of K'. Integrating this over $T(E_j)$:

$$m(E_j) \le (1+\epsilon) \, |\det D_{x_j} T|^{-1} \, m(T(E_j))$$

All quantities here are finite: $m(E_j) \le m(Q_j) < \infty$, the number $|\det D_{x_j} T|$ lies in $(0, \infty)$, and $m(T(E_j)) \le \int_{Q_j} |\det D_x T| \, dm < \infty$ by Step 4. Chaining the two displayed estimates:

$$\int_{E_j} |\det D_x T| \, dm(x) \le (1+\epsilon)^2 \, m(T(E_j))$$

Now sum over j. On the left, $\sum_j \chi_{E_j} |\det D_x T| = \chi_E |\det D_x T|$ exactly, so proposition 2.2.7 gives $\int_E |\det D_x T| \, dm$. On the right, injectivity of T makes the sets $T(E_j)$ disjoint with union $T(E)$, so countable additivity gives $m(T(E))$. Hence

$$\int_E |\det D_x T| \, dm(x) \le (1+\epsilon)^2 \, m(T(E))$$

and $\epsilon \to 0$ proves the claim for E inside a compact set. For general measurable $E \subseteq \Omega$, exhaust with the compact sets K_i of Step 4(c): $\int_{E \cap K_i} |\det D_x T| \, dm \nearrow \int_E |\det D_x T| \, dm$ by the Monotone Convergence Theorem (theorem 2.2.5), while $m(T(E \cap K_i)) \nearrow m(T(E))$ by continuity from below (proposition 1.2.5). Together with Step 4, this proves part (2):

$$m(T(E)) = \int_E |\det D_x T| \, dm(x) \qquad \text{for all Lebesgue measurable } E \subseteq \Omega$$

Step 7: the integral formula on L^+, via the pushforward. Consider the map $T^{-1} : T(\Omega) \to \Omega$, read as a function between the measurable spaces of Lebesgue measurable subsets of $T(\Omega)$ and of Ω. It is measurable in the sense of definition 2.1.1: for measurable $E \subseteq \Omega$, the pre-image $(T^{-1})^{-1}(E) = T(E)$ is measurable by Step 4. Let

$$\lambda = (T^{-1})_* \big(\, m|_{T(\Omega)} \big)$$

be the pushforward of (the restriction to $T(\Omega)$ of) Lebesgue measure under T^{-1}. By theorem 2.3.16(1), λ is a measure on the Lebesgue measurable subsets of Ω, and unwinding the definition, $\lambda(E) = m(T(E))$, which part (2) rewrites as

$$\lambda(E) = \int_E |\det D_x T|\, dm(x)$$

We claim that for every $h \in L^+(\Omega)$:

$$\int_\Omega h\, d\lambda = \int_\Omega h(x)\, |\det D_x T|\, dm(x)$$

The ascent is the same as in the proof of theorem 2.3.16, and we compress it for that reason: for $h = \chi_E$ the claim *is* the display above. For simple h it follows by linearity (proposition 2.2.7) applied on both sides. For general $h \in L^+$, choose simple $\varphi_i \nearrow h$ (theorem 2.1.18), note $\varphi_i|\det D_x T| \nearrow h|\det D_x T|$ pointwise, and apply the Monotone Convergence Theorem (theorem 2.2.5) to both sides of the simple case.

Now let $f \in L^+(T(\Omega))$. By Step 5, $f \circ T \in L^+(\Omega)$. Then theorem 2.3.16(2), applied with $F = T^{-1}$, $\mu = m|_{T(\Omega)}$, and $g = f \circ T \in L^+(\lambda)$, reads

$$\int_\Omega (f \circ T)\, d\lambda = \int_{T(\Omega)} (f \circ T) \circ T^{-1}\, dm = \int_{T(\Omega)} f\, dm$$

since $(f \circ T) \circ T^{-1} = f$. Combining with the claim at $h = f \circ T$:

$$\int_{T(\Omega)} f\, dm = \int_\Omega (f \circ T)(x)\, |\det D_x T|\, dm(x)$$

which is part (1) for $f \geq 0$.

Step 8: the L^1 case. Let $f : T(\Omega) \to \mathbb{C}$ be Lebesgue measurable, and apply Step 7 to $|f| \in L^+(T(\Omega))$: since $|f| \circ T = |f \circ T|$ and $|\det D_x T| \geq 0$,

$$\int_{T(\Omega)} |f|\, dm = \int_\Omega \big|(f \circ T)(x)\, |\det D_x T|\big|\, dm(x)$$

so $f \in L^1(T(\Omega), m)$ if and only if $(f \circ T)|\det D_x T| \in L^1(\Omega, m)$, which is the "moreover". Suppose they are finite, and decompose $f = (\mathrm{Re} f)^+ - (\mathrm{Re} f)^- + i(\mathrm{Im} f)^+ - i(\mathrm{Im} f)^-$ as in definition 2.1.15. Each piece lies in $L^+(T(\Omega))$ with integral at most $\int |f|\, dm < \infty$, and the decomposition is respected by composing with T and multiplying by the non-negative factor $|\det D_x T|$: for instance $\big((\mathrm{Re} f)^+ \circ T\big)|\det D_x T| = \big(\mathrm{Re}\big((f \circ T)|\det D_x T|\big)\big)^+$. Applying Step 7 to the four pieces and recombining by linearity (theorem 2.3.8) yields the formula for f, completing the proof.

Notice how the labour was divided. All the geometry went into part (2): cubes, the mean value theorem, and the linear change of variables theorem 2.7.4 produced the measure identity $m(T(E)) = \int_E |\det D_x T|\, dm$, with the derivative playing its advertised role of best local linear approximation. The passage from measures of sets to integrals of functions, by contrast, contained no geometry at all: it was the pushforward theorem (theorem 2.3.16) plus one monotone-convergence ascent. In the case $n = 1$, with $\Omega = (a, b)$ and T a C^1 bijection onto (c, d) with non-vanishing derivative, the formula reads $\int_c^d f(y)\,dy = \int_a^b f(T(x))\,|T'(x)|\,dx$, which is the u-substitution rule of elementary calculus, now valid for every integrable f, not merely the continuous ones, and with the absolute value explaining what happens when the substitution reverses orientation.

3

Signed Measure and Differentiation

In this chapter, we will take a close look at how to define measure's with respect to integrals. In particular, given the appropriate integrable function f, we will want to define a measure μ such that

$$\mu(A) = \int_A f$$

After defining some terms in section 3.1, we will focus in section 3.2 and 3.3 on how to find a function f given a measure μ so that $\mu(A) = \int_A f$. The last 2 sections then focus on the special case where the function f that we found can actually be the *derivative*. In the simple case where $X = \mathbb{R}$, then we can define:

$$F(x) = \int_{(-\infty,x]} f dx = \int_{-\infty}^{x} f dx$$

and ask ourselves what conditions does f or F requires so that $f = F'$ (or, $f = F'$ a.e.). This line of reasoning will ultimately lead to a generalisation of the Fundamental Theorem of Calculus, which will tell us when can we recover the function f given we know $\int_A f$ for all A.

3.1 Signed Measures

Definition 3.1.1: Signed Measure

Let $(X, \mathcal{M})$ be a measurable space. A *signed measure* on $(X, \mathcal{M})$ is a function: $\nu : \mathcal{M} \to [-\infty, \infty]$ such that:

1. $\nu(\emptyset) = 0$
2. ν has at most one value of $\pm\infty$
3. If $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ is a disjoint collection, then

$$\nu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \nu(E_i)$$

   where if $\nu\left(\bigcup_{i=1}^{\infty} E_i\right) < \infty$, then $\sum_{i=1}^{\infty} \nu(E_i)$ converges absolutely.

Example 3.1.2: Signed Measures

Let $(X, \mathcal{M})$ be a measurable space.

1. Every measure on $\mathcal{M}$ is a signed measure. To distinguish between the two, we will often call measures *positive measures*.
2. Let μ_1 and μ_2 be two measure on $\mathcal{M}$ with at least one being finite. Then $\nu = \mu_1 - \mu_2$ is a signed measure.
3. If f is a $\mathcal{M}$-measurable function, then if at least one of $\int f^+$ or $\int f^-$ is finite,

$$\nu(E) = \int_E f$$

   is a signed measure. In this case, we call f the *extended μ-integrable function* (in contrast to $f \in L^1$ where both terms need to be finite)

The fact that it converges absolutely is to make sure that the order in which we sum the elements does not matter: If it did not converge absolutely, then it is possible to make the sum approach any value we want. Since $\mathcal{M}$ doesn't necessarily have an inheret order (in fact it almost never does), then having this dependence on the order of summing the elements is inconsistent with our notion of area (this is also why we only allow one of the direction to be infinite). It might seem that we must thus be careful on how we define our signed measure to make sure this condition holds, however we will soon show that every signed measrue can be decomposed into 2 measures. If the measure of all sets with positive measure and negative measure are both infinite ($\pm\infty$), then we have contradicted the definition of a signed measure. As just mentioned, all signed measures can be decomposed into two positive measures. We will spend this section proving that fact. In the next section, we will show that under appropriate conditions and choices of signed measure ν and positive meaure μ, we can find a measruable function f such that $\nu(E) = \int_E f d\mu$. This will essentially be the generalization of the derivative!

We start by building up to decomposing a signed measure into two positive meaures:

Proposition 3.1.3: Continuity from Above/Bellow for Signed Meaures

Let ν be a signed measure on $(X, \mathcal{M})$.

1. If $\{E_i\}$ is an increasing sequence in $\mathcal{M}$, then $\nu\left(\bigcup_i E_i\right) = \lim_{i\to\infty} \nu(E_i)$
2. If $\{E_i\}$ is an decreasing sequence in $\mathcal{M}$ where $\nu(E_1)$ is finite, then $\nu\left(\bigcap_i E_i\right) = \lim_{i\to\infty} \nu(E_i)$

***Proof*:**
Very similar to proposition 1.2.5

For the next proposition, we introduce a quick new definition: a set $P \in \mathcal{M}$ is called *positive* (resp. *negative*) if for all $S \subseteq P$ such that $S \in \mathcal{M}$, then $\nu(S) \geq 0$ (resp. $\nu(S) \leq 0$)

Lemma 3.1.4

Any countable union of positive sets is a positive set

***Proof*:**
Use proposition 3.1.3

Theorem 3.1.5: Hahn Decomposition Theorem

If ν is a signed measure on $(X, \mathcal{M})$, there exists a positive set P and a negative set N for ν $P \coprod N = X$ and $P \cap N = \emptyset$. If P' and N' is another such pair, then $P \Delta P$ and $N \Delta N'$ is a null set for ν

***Proof*:**
Let ν be a signed measure, so only ∞ or $-\infty$ can be achieved. If ∞ is achieved, take $-\nu$ so that ∞ is not achieved. Now, let m be the supremum of $\nu(E)$ where E ranges over all positive sets. Thus, there is a sequence of positive sets $\{P_i\}$ such that $\nu(P_i) \to m$. Let $P = \bigcup_i P_i$. Then by the previous lemma and proposition, P is a positive set and $\nu(P) = m$. Since ν does not achieve ∞, $m < \infty$. I claim that $N = X \setminus P$ is negative (thus completing the first assertion of the proof).

First, N cannot contain non-null positive sets. If $E \subseteq N$ was a positive set (i.e. $m(E) > 0$), then $E \cap P = \emptyset$ and $E \cup P$ is a positive set, so $m(E \cup P) = m(E) + m(P) > m$, which is a contradiction to m being the supremum.

Now, let's say there exists a $A \subseteq N$ such that $\nu(A) > 0$. We will reach a contradiction in using the following fact: since A cannot be positive, then there must exist a $B \subseteq A$ such that $\nu(B) < 0$. Then let $C = A \setminus B$ so that $\nu(C) > \nu(A)$ and $C \subseteq A$.

With this, we proceed in forming a contradiction: Define the following sequence $\{A_i\}$ and $\{n_i\}$ as follows:

1. Let n_1 be the smallest positive integer such that there exists a $A_1 \subseteq N$ where $\nu(A_1) > n_1^{-1}$ (this set exists since there is a $A \subseteq N$ where $\nu(A) > 0$
2. Let n_k be the smallest positive integer such that there exists a $A_k \subseteq A_{k-1}$ such that $\nu(A_{k-1}) > \nu(A_k) + n_k^{-1}$ (since A_{k-1} is positive, we use the earlier remark fact to find such a set)

From this sequence of positive decreasing sets, $A_1 \supseteq A_2 \supseteq \cdots$, let $A = \bigcap_i A_i$. Then by the fact that ν does not attain ∞ and the previous proposition:

$$\infty > \nu(A) = \lim \nu(A_i) > \sum_{i=1}^{\infty} n_i^{-1}$$

since the series converges, it must be that $n_i \to \infty$. Finally, once again take $B \subseteq A$ such that $\nu(A) > \nu(B) + n^{-1}$ for some positive integer n. Since $n_i \to \infty$, there exists an i sufficiently large so that $n < n_i$ and $B \subseteq A_i$, meaning we can replace n_i by n in the above construction. However, by construction, we picked the *minimum* integer every-time, and so this contradicts minimality! Therefore, it cannot be that there exists a set $A \subseteq N$ such that $\nu(A) > 0$, and so N is indeed negative.

If P' and N' where two other sets satisfying the decomposition condition, then we have that $P \setminus P' \subseteq P$ (since it's a subset) and $P \setminus P' \subseteq N'$ (since it is P minus all the positive set P') , so it is both positive and negative, which can only be possible if it is a null-set (similarly for $N \setminus N'$)

Using the Hahn decomposition, we can define a positive measure on P and N. Before proceeding to that theorem, we quickly give a name to the relation between the two measures that will be defined:

Definition 3.1.6: Mutually Singular Signed Measures

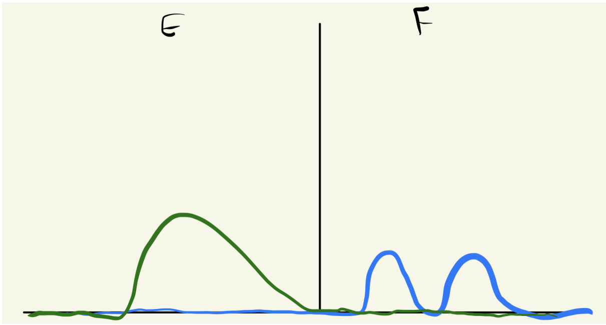
Let ν and μ be be two signed measure on $(X, \mathcal{M})$. Then they are called *mutually singular* or μ *is singular with respect to* ν if there exists a $E, F \in \mathcal{M}$ such that $E \cup F = X$, $E \cap F = \emptyset$, and:

1. $\mu(E) = 0$
2. $\nu(F) = 0$

and we write mutually singular signed measures as:

$$\mu \perp \nu$$

In a sense, this means that these two sets are are meaningful on disjoint sets, in that sense they are "perpendicular" to each other. I like to keep an image like this in mind



Notice that it is possible that μ have zero values on F, and ν have zero values on E. They key is that they *must* have zero values on their respective sets.

Mutually singular measures can be applied more generally than in the context of decomposing a signed measure. For example, of δ_0 is the dirac-delta measure at 0 (i.e. $\delta(\{0\}) = 1$ and zero everywehre else), then $\delta_0 \perp \lambda$.

Theorem 3.1.7: Jordan Decomposition Theorem

If ν is a signed measure, there exists unique positive measure ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$

***Proof*:**
By theorem 3.1.5, let $X = P \cup N$ be the Hahn decomposition of X with respect to ν. Define $\nu^+(E) = \nu(E \cap P)$ and $\nu^-(E) = -\nu(E \cap N)$. Then by definition, $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$.

If there was another decomposition $\nu = \mu^+ - \mu^-$ where $\mu^+ \perp \mu^-$, then we have $E, F \in \mathcal{M}$ such that $E \coprod F = X$ and $E \cap F = \emptyset$, $\mu^+(F) = \mu^-(E) = 0$, so E, F is another Hahn decomposition for ν, and so $P \Delta E$ is null. Thus, for any $A \in \mathcal{M}$

$$\mu^+(A) = \mu^+(A \cap E) = \nu(A \cap E) = \nu(A \cap P) = \nu^+(A)$$

thus, $\mu^+ = \nu^+$, and similarly $\mu^- = \nu^-$, completing the proof.

Definition 3.1.8: Jordan Decomposition

Let ν be a signed measure on $(X, \mathcal{M})$. THen ν^+ and ν^- are caleld the *positive* and *negative variation* of ν, and $\nu = \nu^+ - \nu^-$ is called the *Jordan decomposition of* ν. The *total variation of* ν, denoted $|\nu|$ is defined as $|\nu| = \nu^+ + \nu^-$.

As an exercise, you should verify that $E \in \mathcal{M}$ is ν-null if and only if $|\nu|(E) = 0$, and $\nu \perp \mu$ if and only if $|\nu| \perp \mu$ if and only if $\nu^+ \perp \mu$ and $\nu^- \perp \mu$. With this definition, we will say that ν is finite (resp. σ-finite) if $|\nu|$ is finite (resp. σ-finite)

Finally, we can also get back to showing that any signed measure is of the form $\nu(E) = \int_E d\mu$ for appropriate μ, namely $\mu = |\nu|$ and $f = \chi_P - \chi_N$ for appropriate P and N from a Hahn decomposition. From this, we can define the integration with respect to a signed measure ν as follows:

Definition 3.1.9: Integration w.r.t. Signed Measures

Let ν be a signed measure, $\nu^+ - \nu^-$ be it's Hahn decomposition, and $f : X \to \mathbb{C}$ be a real function. Then we will say that f is *integrable with respect to the signed measure* ν if:

$$f \in L^1(\nu^+) \cap L^1(\nu^-) = L^1(\nu)$$

and define it's integral to be:

$$\int f d\nu = \int f d\nu^+ - \int f d\nu^- \qquad (f \in L^1(\nu))$$

3.2 Lebesgue-Radon-Nikodym Theorem

In this section, we show under what circumstances we can decompose a signed measure μ into $\nu + \rho$ where $\nu(A) = \int_A f$ and ρ is the "left-over" (also known as the singular part). We first define the "opposite" notion of mutually singular signed measures. The motivation for the vocabulary will come in section ref:HERE

Definition 3.2.1: Absolutely Continuous w.r.t. μ

Let ν be a signed measure on $(X, \mathcal{M})$ and μ is a positive measure[a] on $(X, \mathcal{M})$. We say that ν is *absolutely continuous with respect to* μ when if $\mu(E) = 0$ (given $E \in \mathcal{M}$) then $\nu(E) = 0$, and we write is as:

$$\nu \ll \mu$$

[a]this can be extended to a signed measure by taking $\nu \ll \mu$ if and only if $\nu \ll |\mu|$, but we don't need this for this document

I like to remember the direction of the arrow by remembering $\mu(E) = 0 \Rightarrow \nu(E) = 0$, so the arrow is pointing towards ν, so we write the arrow's pointing towards ν: $\nu \ll \mu$. The ν is absolutely continuous with respect to μ, and so it's ν "w.r.t." μ: ν is on the left, the μ on the right. If μ is any measure and 0 is the zero measure, it is always the case that $\mu \ll \mu$ and $0 \ll \mu$. The most important example of an absolutely continuous measures is given by meaures defined via integration:

Definition 3.2.2: Measures Defined By Densities

Let μ be a measure on x and let $f \in L^1(X)$. Then:

$$\nu(E) = \int_E f d\mu$$

is a measure with respect to f.

Clearly $\nu \ll \mu$, since if $\mu(E) = 0$, then $\int_E f d\mu = 0 = \nu(E)$.

It should be verified that $\nu \ll \mu$ if and only if $|\nu| \ll \mu$ if and only if $\nu^+ \ll \mu$ and $\mu^- \ll \mu$. Notice that absolute continuity is sort of the opposite of modular singularity in the sense that if $\nu \perp \mu$ and $\nu \ll \mu$, then it must be that $\nu = 0$. Note that $\mu \perp \nu$ does not imply $\mu(E) = 0 \Rightarrow \nu(E^c) = 0$: take

ν to be the Lebesgue signed measure on $\mathbb{R}$, so that the positive and negative sets are $[0,\infty)$ and $(-\infty,0)$ and experiment.

The following theorem should give you a taste on why the term "continuity" comes around when ν is finite. In particular, it should give you a feeling of uniformally continuous. Later on, we'll show that absolute continuity is stronger than uniform continuity, and so this feeling should make sense:

Theorem 3.2.3: Epsilon-Delta Absolute Continuity

Let ν be a finite signed measure on $(X,\mathcal{M})$ and μ a positive measure on $(X,\mathcal{M})$. Then $\nu \ll \mu$ if and only if for every $\epsilon > 0$, there exists a $\delta > 0$ such that $\mu(E) < \delta \Rightarrow |\nu(E)| < \epsilon$.

***Proof*:**
Since $\nu \ll \mu$ if and only if $|\nu| \ll \mu$, and moreover $|\nu(E)| \leq |\nu|(E)$, it suffices to show it in the case where $\nu = |\nu|$ is positive.

- ($\Leftarrow$) Let the ϵ-δ definition hold so that for any $\epsilon > 0$, there exists a $\delta > 0$ such that for all $E \in \mathcal{M}$ that satisfy $\mu(E) < \delta$, then $\nu(E) < \epsilon$. If we pick all E such that $\mu(E) = 0$, then for any ϵ arbitrarily small, any δ will work, and so $\nu(E) < \epsilon$ for all ϵ, and so it must be that $\nu(E) = 0$
- ($\Rightarrow$) assume the contra-positive; that the ϵ-δ condition is not satisfied, meaning there exists a $\epsilon > 0$ such that for all $\delta > 0$, there is a E_δ such that $\mu(E_\delta) < \delta \Rightarrow \nu(E_\delta) \geq \epsilon$. In particular, for every $n \in \mathbb{N}$, $\mu(E_n) < 2^{-n} \Rightarrow \nu(E_n) \geq \epsilon$. let $F_k = \bigcup_{n=k}^{\infty} E_n$ and $F = \bigcap_k F_k$. Then

  $$\mu(F_k) < \sum_{k=n}^{\infty} \frac{1}{2^n} = 2^{1-k}$$

  and since $F \subseteq F_k$ for all F_k, it must be that $\mu(F) = 0$. However, by assumpiton $\nu(F_k) \geq \epsilon$ for all k. Since ν is finite, $\nu(F_1)$ is finite, so continuity from above implies

  $$\nu(F) = \lim \nu(F_k) \geq \epsilon$$

  and so $\nu \not\ll \mu$

An easy way to find a ν that is absolutey continuous with respect to μ is if we have a measure μ and a extended μ-integrable function f and define $\nu(E) = \int_E f\, d\mu$. If $f \in L^1$, then ν is finite, in which case we have the following corrollary:

Corollary 3.2.4: Absolute Continuity And L^1

If $f \in L^1(\mu)$, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\forall E \in \mathcal{M},\ \mu(E) < \delta \ \Rightarrow \ \left|\int_E f\right| < \epsilon$$

***Proof*:**
apply the previous theorem on Ref and Imf

The definition of ν with respect to $\int_E f d\mu$ is common enough that we define the following notation for it:

Definition 3.2.5: Density Notation

$$d\nu = f d\mu$$

This should give you the feeling of change of variables, and indeed the two concepts will be related. I like to remember this notation by thinking that if we add an integral with respect to E, $\int_E$ to both sides, and imagine there is the constant function in-front of the $d\nu$ on the left hand side, we get back our previous notation of $\nu(E) = \int_E f d\mu$.

We are now in a position to fully classify signed measures given positive measures. The way we will generlize this is to start by remember that an extended μ integrable function f has one part that allow's for an infinite value. Thus, we can think of the domain of f as being split between a signed measure and a positive measure. To avoid pathologies, we will work over σ-finite measures (as usual). We first prove some technical lemma before continuing to the main result:

Lemma 3.2.6

Let μ and ν be finite measures on $(X, \mathcal{M})$. Then either $\mu \perp \nu$, or there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\nu \geq \epsilon\mu$ on E (that is, ν and μ have domain $M_E = \{E \cap F \mid F \in \mathcal{M}\}$, and E is a positive set for $\nu - \epsilon\mu$)

***Proof*:**
Let $X = P_n \coprod N_n$ be the Hahn decomposition for $\nu - n^{-1}\mu$, and let $P = \bigcup_n^\infty P_n$ and $N = \bigcap_n^\infty N_n = P^c$. Then N is a negative set on $\nu - n^{-1}\mu$ for all n a.e., and

$$0 \leq \nu(N) \leq n^{-1}\mu(N)$$

for all n, so $\nu(N) = 0$. If $\mu(P) = 0$, then $\nu \perp \mu$. If $\mu(P) > 0$, then $\mu(P_n) > 0$ for some n, and P_n is a positive set for $\nu - n^{-1}\mu$

Lemma 3.2.7: Sum and Absolute Continuity

Suppose $\{\nu_i\}$ is a sequence of positive meaures. If $\nu_i \perp \mu$ for all i, then $\sum_i^\infty \nu_i \perp \mu$ and if $\nu_i \ll \mu$ for all i then $\sum_i^\infty \nu_i \ll \mu$

***Proof*:**
Let $\nu_i \perp \mu$ for all i, so that $X = A_i \coprod B_i$ where $\nu_i(A_i) = 0$ and $\mu(B_i) = 0$ for all i. Since $\mu(B_i) = 0$ for all i, then $\mu\left(\bigcup_i B_i\right) = 0$. Similarly, $\sum_i^n \nu_i\left(\bigcap_i A_i\right) = 0$ since $\cap_i A_i \subseteq A_i$ for each i. By some basic set theory

$$X = \left(\bigcap_i A_i\right) \coprod \left(\bigcup_i B_i\right) \qquad \left(\bigcap_i A_i\right) \cap \left(\bigcup_i B_i\right) = 0$$

since each x is in either A_i or B_i and $A_i \coprod B_i = X$ for each i. Therefore, we have:

$$\sum_i \nu_i \perp \mu$$

For the second statement, let's say $\nu_i \ll \mu$ for all i. This implies that:

$$\begin{aligned} \mu(E) = 0 &\Rightarrow \nu_1(E) = 0 \\ &\Rightarrow \nu_2(E) = 0 \\ &\vdots \\ &\Rightarrow \nu_n(E) = 0 \\ &\vdots \end{aligned}$$

meaning if $\mu(E) = 0$, then each $\nu_i(E) = 0$. But then $\sum_i \nu_i(E) = 0$, and so

$$\sum_i^\infty \nu_i \ll \mu$$

as we sought to show.

Theorem 3.2.8: Lebesgue-Radon-Nikodym Theorem

Let ν be a signed σ-finite measure on $(X, \mathcal{M})$ and μ be a positive σ-finite measure on $(X, \mathcal{M})$. Then, there exists unique σ-finite measure λ and ρ on $(X, \mathcal{M})$ such that

$$\lambda \perp \mu, \qquad \rho \ll \mu \quad \text{and} \quad \nu = \lambda + \rho$$

Furthermore, there is an extended μ-integrable function f such that $d\rho = f d\mu$, and any two such functions are equal μ-a.e. and so we get:

$$\nu(A) = \int_A f d\mu + \lambda(A)$$

In other words, λ and ρ "split" the measure μ based on where μ is zero or not , and together they can be combined to make ν. Notice that there is no prior relation between the signed measure ν and the positive measure μ. This means that we can replace μ with μ', and find $\lambda' \perp \mu'$ and $\rho' \ll \mu'$, and still get $\nu = \lambda' + \rho'$, and so this decomposition of a signed measure with respect to a measure μ heavily depends on μ. This decomposition might make more sense if we conider the special case where $\nu \ll \mu$, in which case we ar left with $\nu = \rho$. The last statement of the theorem tell's us that:

$$d\rho = f d\mu \qquad \text{which is shorthand for} \qquad \rho(E) = \int_E f d\mu$$

Notice that this looks like the change of varaibles we were talking about earlier. Thus, in this case, this theorem is ultimately about a "change of variables" into the μ measure!

Finally, notice that this "change of variables" show's that the measure ν is sort of the "integral" of the measure of f, and so in a sense f is thus the "derivative" of ν since you "get back" ν by integrating. In other words, this should start giving you Fundamental Theorem of Calculus where $d/dx(\int f)) = f$. However, be weary that the function f is not [always] obtained by taking the limit of some function F, i.e. does does not always exist a function F such that $f(a) = F'(a) = \lim_{x \to a} \frac{F(x+a)-F(a)}{x-a}$. However, in the next section, we will see that in some special cases we *can* define a function F (which we obtain through ν) where $F' = f$. Furthermore, this does not imply that we can compute real measures by defining a distribution F like in the case of Lebesgue measures (which is the other result of the FTC). Thus, the function f can be thought of as a sort of generalization of the notion of derivative.

***Proof*:**
We will break down this proof into 3 cases: when ν and μ are finite positive measures, when ν and μ are σ-finite positive meaures, and finally when ν is a signed measure.

For the first case, let:

$$\mathcal{F} = \left\{ f : X \to [0, \infty] \;\middle|\; \int_E f \, d\mu \leq \nu(E) = \int_E 1 d\nu \text{ for all } E \in \mathcal{M} \right\}$$

Remember that we can define a measure through the integral, so what we are asking for is if we can find all possible functions which can define us a measure that will be smaller than ν. We will end up taking the largest such function f, and labelling $d\rho = f d\mu$, and $d\lambda = d\nu - d\rho$. We can already see that $\rho \ll \mu$. What's left to show is that such an f exists, that $\lambda \perp \mu$ and that the result is unique.

Notice also how the construction can make you think of change of variables. Let's say for a moment that $\int_E f d\mu = \int_E 1 d\nu$. Then it is as though f is correcting the value so that the values are equal. This is actually what we are looking for, so we look for all f such that $\int_E f d\mu \leq \int_E 1 d\nu$ and take the supremum.

First, $\mathcal{F}$ is nonempty since $0 \in \mathcal{F}$. Thus, let $a = \sup \left\{ \int f \, d\mu \;\middle|\; f \in \mathcal{F} \right\}$ (notice that $a \leq \nu(X) < \infty$ since ν is finite), and choose a sequence $\{f_n\} \subseteq \mathcal{F}$ such that $\int f_n d\mu \to a$ and let $f = \sup_n f_n$. We would like to show that $a = \int f$. To that end, notice that if $f, g \in \mathcal{F}$, then $\max(f, g) \in \mathcal{F}$: if $A = \{x \mid f(x) > g(x)\}$, for any $E \in \mathcal{M}$:

$$\int_E \max(f, g) d\mu = \int_{E \cap A} f d\mu + \int_{E \setminus A} g d\mu \leq \nu(E \cap A) + \nu(E \setminus A) = \nu(E)$$

Let $g_n = \max(f_1, f_2, ..., f_n)$. Then $g_n \in \mathcal{F}$, g_n increases pointwise to f, and $\int g_n \, d\mu \geq \int f_n \, d\mu$ by monotonicity. Thus, since a is the supremum, it must be that $\lim \int g_n \, d\mu = a$, and so by the MCT

$$a = \lim \int g_n d\mu = \int \lim g_n d\mu = \int f$$

and so $f \in \mathcal{F}$ (i.e. the supremum is "achieved") and $\int f \, d\mu = a$. In particular, $f < \infty$ a.e., so we may take f to be real-valued everywhere. Let $d\rho = f d\mu$ (we will soon show this $d\rho$ satisfies the conditions of the theorem).

We know define the measure λ which is singular with respect to μ. Let $d\lambda = d\nu - f d\mu$ (i.e. $\lambda(E) = \nu(E) - \int_E f d\mu$). Then $d\lambda$ is positive, since $f \in \mathcal{F}$, and is singular with respect to

μ. It it weren't, then by lemma 3.2.6, there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\lambda \geq \epsilon\mu$ on E. Then $\epsilon\chi_E d\mu \leq d\lambda = d\nu = f d\mu$, or rearranging $(f + \epsilon\chi_E)d\mu \leq d\nu$. Since $f \in \mathcal{F}$ and the $\epsilon\chi_E$ only effects the value on E, we just showed that $f + \epsilon\chi_E \in \mathcal{F}$. However, $\int f + \epsilon\chi_E d\mu = a + \epsilon\mu(E) > a$, contradicting the maximality of a. Hence it must be that $\lambda \perp \mu$.

Therfore, we have that $d\nu = f d\mu + d\lambda$ where $\lambda \perp \mu$. Recalling we set $d\rho = f d\mu$, we get that $d\nu = d\rho + d\lambda$, and by definition of ρ we have $\rho \ll \mu$ (see definition 3.2.2). Converting into our more usual notation, we see that $\nu(E) = \rho(E) + \lambda(E)$ for all E, and so $\nu = \rho + \lambda$ with $\lambda \perp \mu$ and $\rho \ll \mu$. It remains to show that this decomposition is unique. To that end, let's say $d\nu = d\lambda' + f' d\mu$. Then $d\lambda - d\lambda' = (f - f')d\mu$. Then (show that $d\lambda - d\lambda' \perp \mu$) and $(f' - f)d\mu \ll d\mu$, and since both sides must be equal, this forces that $d\lambda - d\lambda' = (f' - f)d\mu = 0$. Thus, $\lambda = \lambda'$, and by proposition 2.3.4, $f = f'$ μ-a.e. Thus, if ν and μ are of finite measure, the theorem is done.

Now let ν and μ be σ-finite measures so that X is a countable union of μ-finite sets and a countable union of ν-finite sets. By refining the two by taking intersections, we get a new collection $\{A_i\} \subseteq \mathcal{M}$ $\mu(A_i)$ and $\nu(A_i)$ is finite for all i and $X = \bigcup_i^\infty A_i$. We can define

$$\mu_i(E) = \mu(E \cap A_i) \qquad \nu_i(E) = \nu(E \cap A_i)$$

Then by what we've just shown, we have the decomposition:

$$d\nu_i = d\lambda_i - f_i d\mu$$

where $\lambda_i \perp \mu_i$. Now $\mu_i(A_i^c) = \nu_i(A_i^c) = 0$, so $\int_{A_i^c} f d\mu = 0$ and

$$\lambda_i(A_i^c) = \nu_i(A_i^c) - \int_{A_i^c} f d\mu = 0$$

for our construciton, let f_i be 0 on A_i^c. Now, let $\lambda = \sum_{i=1}^\infty \lambda_i$ and $f = \sum_{i=1}^\infty f_i$. Then by lemma 3.2.7:

$$d\nu = d\lambda - f d\mu$$

and $d\lambda$ and $f d\mu$ are σ-finite, as desired.

Finally, taking the case of ν being a signed measure, using the Jordan decomposition, apply the preceeding argument to ν^+ and $\nu-$, compelting the proof

Since the decomposition exists, it is given a name:

Definition 3.2.9: Lebesgue Decomposition

Let ν be a signed σ-finite measure and μ a positive σ-finite measure. Then the decomposition

$$\nu = \lambda + \rho \qquad \lambda \perp \mu \quad \rho \ll \mu$$

is called the *Lebesgue decomposition of ν with respect to μ.*

if $\nu \ll \mu$, then the Lebesgue-Radon-Nikodym theorem tell's us that $d\nu = f d\mu$. If the extra condition that $\nu \ll \mu$ is added, the theorem is often called *Radon-Nikodym's Theorem*, and f is given a special name:

Definition 3.2.10: Radon-Nikodym Derivative

Let $\nu \ll \mu$ and f be the function obtain from the Lebesgue Decomposition such that $d\nu = f d\mu$. Then f is called the *Radon-Nikodym derivative of ν with respect to μ*. We often denote f by:

$$\frac{d\nu}{d\mu} := f$$

As a consequence of this notation, we get the following representation of $\nu(E) = \rho(E)$ that looks like a fraction cancellation:

$$d\nu = \frac{d\nu}{d\mu} d\nu$$

Strictly speaking, $\frac{d\nu}{d\mu}$ should really be the class of function's that are equal to f μ-a.e., however just like with L^1, we often abuse notation and pick some function in this class when we use this notation). Most of the time, we will be working with the case of $\nu \ll \mu$. In Distribution theory, we will be studying the case where $\nu \not\ll \mu$.

Writing f in this form is very fitting, as in f respects the same properties as the derivative! Once we reach the special case of the measures being Lebesgue measures and the functions being differentiable functions, then f will in fact be the derivative of f.

Without yet doing concrete computations of f, if we take the Radon-Nikodym of $\nu_1 + \nu_2$ where $\nu_1 + \nu_1 \ll \mu$ then we have

$$\frac{d(\nu_1 + \nu_2)}{d\mu} = \frac{d\nu_1}{d\mu} + \frac{d\nu_1}{d\mu}$$

to see this, let $d(\nu_1 + \nu_2)/d\mu = f$. Since $\nu_1 + \nu_2 \ll \mu$, $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$. Then taking $d\nu_1 = f_1 d\mu$ and $d\nu_2 = f_2 d\mu$, then:

$$\begin{aligned}
&= \int \frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} d\mu \\
&= \int \frac{d\nu_1}{d\mu} d\mu + \int \frac{d\nu_2}{d\mu} d\mu \\
&= \int 1 d\nu_1 + \int 1 d\nu_2 \\
&= \nu_1(E) + \nu_2(E) \\
&= (\nu_1 + \nu_2)(E) \\
&= \int 1 d(\nu_1 + \nu_2) \\
&= \int_E \frac{d(\nu_1 + \nu_2)}{d\mu} d\mu
\end{aligned}$$

and since the two integrals are equal over every measurable set, we get:

$$\frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} = \frac{d(\nu_1 + \nu_2)}{d\mu}$$

which is the same property of differentiable functions. In particular, notice that similarly we have the following chain-rule property:

Proposition 3.2.11: Chain Rule For Measures

Let ν be a signed σ-finite measure on $(X, \mathcal{M})$, μ and λ positive σ-finite measures on $(X, \mathcal{M})$ such that $\nu \ll \mu$ and $\mu \ll \lambda$. Then:

1. If $g \in L^1(\nu)$, then $g(d\nu/d\mu) \in L^1(\mu)$ and

$$\int g d\nu = \int g(d\nu/d\mu) d\mu$$

2. Absolute continuity, $\ll$, is transitive: $\nu \ll \lambda$ and

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda}$$

***Proof*:**

1. This proof uses (what now should be) the standard measure-theory way to generalize results. First, we can consider ν^+ and ν^- seperately, so without loss of generality let's say $\nu \geq 0$. If we let $g = \chi_E$, then:

$$\int g \frac{d\nu}{d\mu} d\mu = \int \int \chi_E f d\mu = \int_E f d\mu$$

which is true by definition of f. Thus, it is true for any simple functions since the integral and the construction is linear, and so it is also true for non-negative function's by the MCT, and so it is true for functions in $L^1(\nu)$ by linearity again.

2. If we take ν to be μ and μ to be λ and set $g = \chi_E \frac{d\nu}{d\mu}$, we get:

$$\nu(E) = \int_E \frac{d\nu}{d\mu} d\mu = \int_E \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda} d\lambda$$

for all $E \in \mathcal{M}$, including X, and so by proposition 2.3.4

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu}\frac{d\mu}{d\lambda} \qquad \lambda\text{-a.e.}$$

as we sought to show

As a consequence, if $\mu \ll \lambda$ and $\lambda \ll \mu$, then $(d\lambda/d\mu)(d\mu/d\lambda) = 1$ λ-a.e. and μ-a.e. Note that all of this is for the special case of $\nu \ll \mu$. If this is not the case (say $\nu \ll \mu$) we get a different notion of "$d\nu/d\mu$ (the example of $\delta_0 \perp \lambda$ has a "Radon-Nikodym derivative" of the dirac delta function).

Note that σ-finiteness is necessary:

Example 3.2.12: Withiout σ-Finiteness

Let $X = [0,1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$, m the Lebesgue measure and μ the counting measure on $\mathcal{M}$

1. $m \ll \mu$ but $dm \neq f d\mu$ for any f

Proof. First, if $\mu(E) = 0$, then $|E| = 0$, which we have shown that this implies $m(E) = 0$, and so $m \ll \mu$.

However, there does not exist an f such that $m(E) = \int_E f d\mu$. To see this, consider any $p \in [0, 1]$. Then

$$0 = m(\{p\}) = \int_{\{p\}} f d\mu = \int_{[0,1]} \chi_{\{p\}} f d\mu \overset{!}{=} f(p)\mu(\{p\}) = f(p)$$

where $\overset{!}{=}$ comes from the fact that $\chi_{\{p\}} f$ is in fact a simple funciton (being of the form $\sum_i c_i \chi_{E_i}$ with $i = 1$, $E_i = \{p\}$ and $c_i = f(p)$), and so matches the definition of the integral for simple functions (which we have shown in class is the value of the usual integl). Hence, we have that $f(p) = 0$ for all $p \in [0, 1]$. However, this implies that $f \equiv 0$. This leads to a contardiction, since:

$$1 = m([0, 1]) = \int_{[0,1]} f d\mu = \int_{[0,1]} 0 d\mu = 0$$

but $1 \neq 0$ $\square$

Next, show that μ has no Lebesgue decomposition with respect to m

Proof. For the sake of contradiction, let $\mu = \lambda + \rho$ where $\lambda \perp m$ and $\rho \ll m$. Then for any $c \in [0, 1]$

$$\mu(\{c\}) = \lambda(\{c\}) = 1$$

since $m(\{c\}) = 0$ and $\rho \ll m$, $\rho(\{c\}) = 0$. But then, λ is never null on non-zero sets by monotonicity, implying that m must *always* be null. However, this is not true since m is the Lebesgue measure – a contradiction. Thus, μ does not have such a decomposition with respect to m. $\square$

Before we move on to our last result of the section, I want to take a moment to talk about generalizations of the Radon-Nikodym theorem. In the theorem, we assumed that the measures must be σ-finite. However, in contrast to previous example where without σ-finiteness the theorem fails badly, we seem to have these assumption because the generalization requires more build-up. In particular, to fix the lack of control of σ-finiteness, we would have to define the notion of being *truly continuous*. If μ is σ-finite, being absolutely continuous and truly continuous are equivalent. The statement of the Radon-Nikodym can be reformulated in terms of truly continuous measures, and I found a stack-exchange post on this:

https://mathoverflow.net/questions/258936/radon-nikodym-theorem-for-non-sigma-finite-measures

Finally, we conclude with this final proposition we'll use later on:

Proposition 3.2.13

Let $\mu_1, ..., \mu_n$ be measures on $(X, \mathcal{M})$. Then there exists a measure μ (namely $\sum_i \mu_i$) such that $\mu_i \ll \mu$

***Proof*:**
exercise

3.3 Complex Measure

We know show how we can generalize the results when the codomain is $\mathbb{C}$. The reason to define complex measure is generalize the notion of the Radon-Nikodym derivative to the complex case so that we may define complex results. Most of the work we have done before translates one-to-one for the complex measures, with the exception of defining the absolute variation which we will spend most of this section covering.

Definition 3.3.1: Complex Measure

Let $(X, \mathcal{M})$ be a measure space. A function $\nu : \mathcal{M} \to \mathbb{C}$ is called a *complex measure* if

1. $\nu(\emptyset) = 0$
2. If $\{E_i\} \subseteq \mathcal{M}$ is a pair-wise disjoint sequence, then if $\sum_i^\infty \nu(E_i)$ converges absolutely, then
$$\nu\left(\bigcup_i^\infty E_i\right) = \sum_i^\infty \nu(E_i)$$

Just like for signed measures, the absolute convergence is so that the order in which we measure the sets does not matter.

Notice that the sequence $X, \emptyset, \emptyset, ...$ is a valid disjoint sequence, and so $\sum_i \nu(X_i) = \nu(X)$ must converge absolutely, and so ν must be a finite measure. Thus, a measure is a complex measure if it is finite. Another way of saying it is that if μ is a positive measure and $f \in L^1(\mu)$ (recall f is a complex function), then $f d\mu$ is also a complex measure. As is usually done with complex valued function, we define the signed measures $\nu_r(E) = \pi_r(\nu(E))$ and $\nu_i(E) = \frac{\pi_i(\nu(E))}{i}$ to take on the real and complex values of ν, so ν_i and ν_r are *finite* signed measures and $\nu = \nu_r + i\nu_i$. As a consequence, the range of ν is a bounded subset of $\mathbb{C}$.

Many of the previous concepts we've seen naturally generalize to the complex case:

1. We define $L^1(\nu)$ to be $L^1(\nu_r) \cap L^1(\nu_i)$
2. For $f \in L^1(\nu)$, set $\int f d\nu := \int f d\nu_r + i \int f d\nu_i$
3. If ν and μ are complex measures, then we that $\mu \perp \nu$ if $\nu_a \perp \nu_b$ for all $a, b \in \{r, i\}$.
4. If λ is a positive measure, we say $\nu \ll \lambda$ if $\nu_r \ll \lambda$ and $\nu_i \ll \lambda$
5. The Lebesgue-Radon-Nikodym Theroem generlizes to complex measures nicely; you would apply theorem 3.2.8 to the real and imaginary part seperately. To make sure there is no ambiguity in understanding the generalization:

   Let ν be a complex measure and μ a σ-finite measure on $(X, \mathcal{M})$. Then there exists a complex measure λ and a $f \in L^1(\mu)$ such that
$$d\nu = d\lambda + f d\mu \qquad \lambda \perp \mu$$

Furthermore, if there existed another λ' such that $\lambda' \perp \mu$ and $d\nu = d\lambda' + f'd\mu$, then $\lambda = \lambda'$ μ-a.e. and $f = f'$ μ-a.e.
If $\nu \ll \mu$, then we denote f a $\frac{d\nu}{d\mu}$

We next want to generalize the notion of *total variation* to the complex case. In the signed measure case, this was an easy task since $\mathbb{R}$ has an inherent order. In the complex case, this is not so straight forward: Given that a surface $S \subseteq \mathbb{C}$ is the image of some complex measure $\nu(E) = S$, what does it mean to capture the "total-variation" on that surface? There is more than one sensible answer to this question, namely since there is no "natural order" to $\mathbb{C}$, and so it would be nice to have succint way to capture it's *total variation*. There is in fact a nice and concise way of defining this which also generalizes the previous way of looking at total variation that captures the idea in 1 formula, and so we will build-up to it.

First, let ν be a complex measure and let $\mu = |\nu_r| + |\nu_i|$. Next, notice that $\nu \ll \mu$, since if $\mu(E) = 0$, then it must be that both $|\nu_r|$ and $|\nu_i|$ are zero, and so $\nu_r(E) = \nu_i(E) = 0$, and so $\nu(E) = 0$. Therefore, the Lebesgue decomposition of ν with respect to μ gives us a f such that $d\nu = f d\mu$. As an example of such an f, let's say that ν was simply a finite signed measure (which is a special case of a complex measure), so $\mu = |\nu|$. Then $f = (\chi_P - \chi_N)$ and:

$$d\nu = (\chi_P - \chi_N)d\mu = (\chi_P - \chi_N)d|\nu|$$

Which is the Jordan decomposition theorem we have shown earlier! Now, notice that $|\chi_P - \chi_N| = 1$ so that we can write:

$$d|\nu| = |\chi_P - \chi_N|d\mu \quad \text{which is shorthand for} \quad |\nu|(E) = \int_E |\chi_P - \chi_N|d\mu$$

If we re-write the last equation as $d|\nu| = |f|d\mu$, then we have that the measure $|\nu|$ is defiend by the absolute value of the measure defined by the derivative with respect to $|\nu|$. If we come back to the case of ν being a general complex measure, then we would get $\mu = |\nu_r| + |\nu_i|$, giving us a less obvious, but certainly existent decomposition $d|\nu| = |f|d\mu$ (i.e. $|\nu|(E) = \int_E |f|d\mu$). Thus, it seems that this property somehow "captures" the idea of total variation. We chose a particular μ for both decomposition, however it turns out that this property we have stated is independent of choice of μ up to an appropriate zero-measure set, and so we will define total-variation with respect to the following property:

Definition 3.3.2: Total Variation And Complex Differentiation

Let ν be a complex measure. Then $|\nu|$ is defined such that when $d\nu = f d\mu$ where μ is a positive measure, then $d|\nu| = |f|d\mu$

What we have shown with the μ we saw earlier is that $d\nu$ is always decomposeable into $d\nu = f d\mu$. We must now show that $|\nu|$ is in-fact independent of choice of μ (i.e. any choice will get the same result up zero an appropriate zero-measure set) given μ has the property of the definition. Let's say $d\nu = f_1 d\mu_1 = f_2 d\mu_2$ satisfy the total-variation property. Let $\rho = \mu_1 + \mu_2$. By proposition 3.2.11, we have that:

$$f_1 \frac{d\mu_1}{d\rho} d\rho = d\nu = f_2 \frac{d\mu_2}{d\rho} d\rho$$

Thus $f_1 \frac{d\mu_1}{d\rho} = f_2 \frac{d\mu_2}{d\rho}$ ρ-a.e. Since $\frac{d\mu_1}{d\rho}$ and $\frac{d\mu_2}{d\rho}$ are both non-negative, we have:

$$|f_1| \frac{d\mu_1}{d\rho} d\rho = \left| f_1 \frac{d\mu_1}{d\rho} d\rho \right| = \left| f_2 \frac{d\mu_2}{d\rho} d\rho \right| = |f_2| \frac{d\mu_2}{d\rho} d\rho \quad \rho\text{-a.e.}$$

Thus:

$$|f_1|d\mu_1 = |f_1|\frac{d\mu_1}{d\rho}d\rho = |f_2|\frac{d\mu_2}{d\rho}d\rho = |f_2|d\mu_2$$

and so, ν is independent of choice of μ. We next give some useful properties of total variation:

Proposition 3.3.3: Properties Of Total Variation

Let $(X, \mathcal{M}, \nu)$ be a complex measure space.

1. $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$
2. $\nu \ll |\nu|$ and $\frac{d\nu}{d|\nu|}$ has absolute value 1 $|\nu|$-a.e.
3. $L^1(\nu) = L^1(|\nu|)$ and if $f \in L^1(\nu)$, then $|\int f d\nu| \leq \int |f| d|\nu|$

***Proof*:**
Pick f such that $d\nu = f d\mu$ (as we've shown exists when defining $|\nu|$). Then:

$$|\nu(E)| = \left|\int_E f d\mu\right| \leq \int_E |f| d\mu = |\nu|(E)$$

showing the first claim, and also showing that $\nu \ll |\nu|$. For the next claim, let $g = \frac{d\nu}{d|\nu|}$ so

$$f d\mu = d\nu = g d|\nu|$$

so $g|f| = f$ μ-a.e. and so $|\nu|$-a.e. Since $|f| > 0$ $|\nu|$-a.e. as well, it must be that $g = 1$ $|\nu|$-a.e.

Next, (left as an exercise by Folland)

The next property we introduce shows that it interacts well with the triangle inequality

Proposition 3.3.4: Triangle Inequality Of Total Variation

Let ν_1 and ν_2 be complex measures on $(X, \mathcal{M})$. Then

$$|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$$

***Proof*:**
By proposition 3.2.13, we can write $\nu_1 = f_1 d\mu$ and $\nu_2 = f_2 d\mu$ for the same μ. Then:

$$d|\nu_1 + \nu_2| = |f_1 + f_2| d\mu \leq |f_1| d\mu + |f_2| d\mu = d|\nu_1| + d|\nu_2|$$

as we sought to show

Finally, though this characterization of total variation is a good definition, it can be hard to work with. The following gives an equivalent definition of total variation that is easier computationally:

Proposition 3.3.5: Equivalence Of Total Variation

Let ν be a complex measure on $(X, \mathcal{M})$. If $E \in \mathcal{M}$, define

1. $\mu_1(E) = \sup\left\{\sum_{i=1}^n |\nu(E_i)| \mid n \in \mathbb{N},\ E_1, ..., E_n \text{ disjiont},\ E = \coprod_{i=1}^n E_i\right\}$
2. $\mu_2(E) = \sup\left\{\sum_{i=1}^\infty |\nu(E_i)| \mid E_1, ... \text{ disjiont},\ E = \coprod_{i=1}^\infty E_i\right\}$
3. $\mu_3(E) = \sup\left\{\left|\int_E f d\nu\right| \mid |f| \leq 1\right\}$

Then $\mu_1 = \mu_2 = \mu_3 = |\nu|$

***Proof*:**
Throughout this proof, let:

$$S_1^E = \left\{\sum_{i=1}^n |\nu(E_i)| \,\middle|\, n \in \mathbb{N},\ E_1, ..., E_n \text{ disjiont},\ E = \coprod_{i=1}^n E_i\right\}$$

and

$$S_2^E = \left\{\sum_{i=1}^\infty |\nu(E_i)| \,\middle|\, E_1, ... \text{ disjiont},\ E = \coprod_{i=1}^\infty E_i\right\}$$

($\mu_1 \leq \mu_2$) Notice that $S_1^E \subseteq S_2^E$ (since for any finite disjoint sequence, we can extend it to an infinite disjoint sequence by letting $E_i = \emptyset$ for the rest of the terms), we get $\sup S_1^E \leq \sup S_2^E$, and since this is true for all E, $\mu_1 \leq \mu_2$

($\mu_2 \leq \mu_3$) In the process we will use the fact that there exists an f such that:

$$d\nu = f d|\nu|$$

By proposition 3.3.3, $\left|\frac{d\nu}{d|\nu|}\right| = |f| = 1$ $|\nu|$-a,e, and so $|f| \leq 1$ $|\nu|$-a.e. For the $|\nu|$-zero-measure points that are not less than or equal to 1, we can simply choose them to be equal to 0. This also implies that $|\overline{f}| \leq 1$. With this information we get that:

$$\overline{f} d\nu = \overline{f} f d|\nu| = |f|^2 d|\nu| = 1 d|\nu| = d|\nu| \quad |\nu|\text{-a.e.}$$

and so:

$$\begin{aligned}\sum_i^\infty |\nu(E_i)| &\leq \sum_i^\infty |\nu|(E_i)\\ &= \int_E d|\nu|\\ &= \int_E \overline{f} d\nu\\ &\leq \left|\int_E \overline{f} d\nu\right|\end{aligned}$$

and since $|f| \leq 1$, this is an element of $\left\{\left|\int_E f d\nu\right| \mid |f| \leq 1\right\}$, and since this was true for arbitrary E, we get: $\mu_2 \leq \mu_3$

($\mu_3 = |\nu|$) We'll first show $\mu_3 \leq |\nu|$. By the definition of the supremum, for all $\epsilon > 0$, there exists an f where $|f| \leq 1$ such that

$$\begin{aligned}
\mu_3 &\leq \left|\int_E f d\nu\right| + \epsilon \\
&\leq \int_E |f| d|\nu| + \epsilon && \text{prop. 3.13[c]} \\
&\leq \int_E d|\nu| + \epsilon && |f| \leq 1 \\
&\leq |\nu|(E) + \epsilon
\end{aligned}$$

and as $\epsilon \to 0$, we have that $\mu_3 \leq |\nu|(E)$. Going the other way, we can again define $\frac{d\nu}{d|\nu|}$, which by proposition 3.13 its absolute value is 1 $|\nu|$-a.e. Let $\overline{f} = \frac{d\nu}{d|\nu|}$ so that $1 = |f|^2 = f\overline{f}$ $|\nu|$-a.e. If f ever has value greater than 1, we can simply redefine them to be 0 since those value have measure 0. Then:

$$\begin{aligned}
|\nu|(E) &= \int_E d|\nu| \\
&= \int_E 1 d|\nu| \\
&= \int_E |f|^2 d|\nu| \\
&= \int_E |f| d|\nu| && \text{since } |f| = 1 \ |\nu|\text{-a.e.} \\
&= \int_E f\overline{f} d|\nu| \\
&= \int_E f \frac{d\nu}{d|\nu|} d|\nu| \\
&= \int_E f d\nu && \text{prop. 3.9} \\
&\leq \left|\int_E f d\nu\right|
\end{aligned}$$

and since $|f| \leq 1$,

$$\left|\int_E f d\nu\right| \in \left\{ \left|\int_E f d\nu\right| \;\middle|\; |f| \leq 1 \right\}$$

and since this is true for arbitrary E:

$$|\nu| \leq \mu_3$$

and so, combining the two inequalities, we get $\mu_3 = |\nu|$

($\mu_3 \leq \mu_1$) We will do as the hint suggets and use simple functions to approximate f, in particular we will use the standard representation from theorem 2.10, let's label the increasing sequence $\{\varphi_k\}_{k=1}^{\infty}$. To take advantage of these functions, first break up ν into positive meaures, $\nu_r^+, \nu_i^+, \nu_r^-, \nu_i^-$ so that:

$$\left|\int_E f d\nu\right| = \left|\int_E f d\nu_r^+ - \int_E d\nu_r^- + i\left(\int_E f d\nu_i^+ - \int_E f d\nu_i^-\right)\right|$$

Then by the definition of integral, for all $\epsilon/4 > 0$, there must exists an $n \in \mathbb{N}$ such that

$$\begin{aligned}
&= \left|\int_E f d\nu_r^+ - \int_E f d\nu_r^- + i\left(\int_E f d\nu_i^+ - \int_e f d\nu_i^-\right)\right| \\
&\leq \left|\int_E \varphi_n d\nu_r^+ + \frac{\epsilon}{4} - \int_E \varphi_n d\nu_r^- - \frac{\epsilon}{4} + i\left(\int_E \varphi_n d\nu_i^+ + \frac{\epsilon}{4} - \int_e \varphi_n d\nu_i^- - \frac{\epsilon}{4}\right)\right| \\
&\leq \left|\int_E \varphi_n d\nu_r^+ - \int_E \varphi_n d\nu_r^- + i\left(\int_E \varphi_n d\nu_i^+ - \int_E \varphi_n d\nu_i^-\right)\right| + \epsilon \\
&= \left|\int_E \varphi_n d\nu\right| + \epsilon
\end{aligned}$$

Then by the definition of a simple function, $\varphi_n = \sum_{i=1}^n c_i \chi_{F_i}$ for appropriate F_i and c_i. Note that $c_i \leq 1$ since $|\varphi_n| \leq |f| \leq 1$. Also, note that $\chi_E \chi_{F_i} = \chi_{E \cap F_i}$. Then by definition of an integral:

$$\left|\int_E \varphi_n d\nu\right| + \epsilon = \left|\sum_{i=1}^n c_i \nu(F_i \cap E)\right| + \epsilon \leq \sum_{i=1}^n |c_i| \left|\nu(F_i \cap E)\right| + \epsilon$$

This is almost the result we need. The problem is that the collection $\{F_i \cap E\}$ is not guaranteed to cover E since there might be portions that are yet to be "precise" enough to bring it into the approximation. Since we only need to get bigger, we can solve this by adding $\bigcap_{i=1}^n F_i^c \cap E$, which is disjoint of every $F_i \cap E$, and so:

$$\begin{aligned}
\sum_{i=1}^n |c_i| \left|\nu(F_i \cap E)\right| + \epsilon &\leq \sum_{i=1}^n |c_i| \left|\nu(F_i \cap E)\right| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon \\
&\leq \sum_{i=1}^n \left|\nu(F_i \cap E)\right| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon
\end{aligned}$$

and so, combining all the inqualities, we get:

$$\left|\int_E f d\nu\right| \leq \sum_{i=1}^n \left|\nu(F_i \cap E)\right| + \left|\nu\left(\bigcap_i^n F_i^c \cap E\right)\right| + \epsilon$$

and so, by the definition of the supremum:

$$\left|\int_E f d\nu\right| \leq \sup S_1^E$$

and since this is true for arbitrary $\left|\int_E f d\nu\right|$:

$$\mu_3 \leq \mu_1$$

and so, we've established $\mu_1 \leq \mu_2 \leq \mu_3 [= |\nu|] \leq \mu_1$, and so they are all equal, as we sought to show

3.4 Differentiation of Euclidean Space

We now analyze the special case where $\mu = m$ is the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$. Since we are working over the Borel sets of $\mathbb{R}^n$, we have a metric available to us, and so we can define the pointwise derivative of ν with respect to m at x:

Definition 3.4.1: Pointwise Derivative

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, m)$ be our measure space, ν a signed or complex measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$, and $x \in \mathbb{R}^n$. Then if $B_r(x)$ is a ball of radius r around x, then if the following limit exists:

$$\lim_{r\to 0} \frac{\nu(B_r(x))}{m(B_r(x))} = F(x)$$

then we say that $F(x)$ is the *pointwise derivative of ν with respect to m at x*

We can replace balls of radius x with sets that shrink suitably quickly in some "regular" way. This will be advantageous to us later, since it will turn out we can make these "nicely shrinking" sets to be $(x, x+h)$ (or the appropriate generalisation in the $\mathbb{R}^n$ case) which will give us

$$\lim_{r\to 0} \frac{\nu(x+h)}{h}$$

notice that if we define ν in terms of a distribution as we've done for the outer measure in section 1.3, say G is our distribution, then our numerator becomes $G(x+h) - G(x)$, so in this case the definition *exactly* matches be the definition of the derivative in one dimension:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \overset{\text{F.t.C.}}{=} \frac{1}{h}\int_x^{x+h} f'(x)dx$$

We will get back to this important case after some more examination of the general case.

If $\nu \ll m$ so that $d\nu = f dm$, then $\nu(B_r(x))/m(B_r(x)) = \frac{1}{m(B_r(x))}\int_{B_r(x)} f d\mu$ is the average value of f on $B_r(x)$ (recall the 1-dimensional intuition of $\frac{1}{b-a}\int_a^b f(x)dx$). Since $r \to 0$, we can hope that $F = f$ m-a.e. (since the averaging is getting more precise with the smaller range), telling us that we can recover a density F with information about f. This will indeed be the case if $\nu(B_r(x))$ is finite for all r, x; this will be what we explore in this section. Essentially, from the point of view of f, what we're showing is that the derivative of the integral of f is f, i.e., the Fundamental Theorem of Calculus (FTC). For the rest of this section, the term integrable and a.e. is with respect to the Lebesgue measure m.

We start with an important technical lemma:

Lemma 3.4.2: build-up lemma I

Let $\mathcal{C}$ be a collection of open balls in $\mathbb{R}^n$ and let $U = \bigcup_{B\in\mathcal{C}} B$. Then for any $c \in \mathbb{R}$ such that $c < m(U)$, there exists a fininte disjoint subset of $\mathcal{C}$, $B_1, ..., B_k$, such that

$$\sum_{i=1}^{k} m(B_i) > \frac{c}{3^n}$$

In other words, for any value c less than $m(U)$, we can show that we only need to pick a finite number of balls from $\mathcal{C}$ and still keep the inequality, up to a correction factor of 3^{-n} where n is the dimension of our space, $\mathbb{R}^n$, and 3 comes form making sure the radius of the balls will be big enough to cover what we need.

***Proof*:**
We essentially sneakily get compactness by using theorem 2.7.1[1]. Using this theorem, we know that there exists a compact set $K \subseteq U$ such that $c < \mu(K)$, so there exists a finite subcover of balls from $\mathcal{C}$, say $A = \{A_1, ..., A_m\}$, that cover K. Since balls have finite radius and there are finitely many balls, let B_1 be the ball with largest radius from A. Then let B_2 be the largest ball from A that are disjoint from B_1. Continuing until we have exhausted all possibilites, we get a collection of balls $B = \{B_1, ..., B_k\}$. If there is an A_i that is not in B, then there must be a B_j such that $A_i \cap B_j \neq \emptyset$, and if j is the smallest integer with this property, then by our construction A_i can have radius that is at most that of B_j (or else we would have picked A_i instead of B_j in our construction). Thus, we can take a ball B_j^* whose radius is 3 times that of B_j and have the same center as B_i (i.e., they are concentric) so that $A_i \subseteq B_j^*$. Then by this construction, $K \subseteq \bigcup_j^k B_j^*$, and so

$$c < \mu(K) \leq \sum_j^k m(B_j^*) = 3^n \sum_j^k m(B_j)$$

and so $\frac{c}{c^n} \leq \sum_j^k m(B_j)$, as we sought to show

With this lemma ready to be referenced, we define another important concept that we will use

Definition 3.4.3: Locally Integrable

Let $f : \mathbb{R}^n \to \mathbb{C}$ be a function. Then we say that f is *locally integrable* (with respect to the Lebesgue measure) if:

$$\forall K \in \mathcal{B}_{\mathbb{R}^n}, m(K) < \infty, \int_K |f(x)|dx < \infty$$

and is denoted L^1_{loc}

Definition 3.4.4: Average

Let $f \in L^1_{\text{loc}}$. Then we define the average $A_r f(x)$ on $B_r(x)$ to be

$$A_r f(x) = \frac{1}{B_r(x)} \int_{B_r(x)} f(y)dy$$

This is simply the observation we made in the previous paragraph but made into a definition. We next prove that varying r or x by a bit will only change $A_r f(x)$ by a bit, i.e., $A_r f(x)$ is continuous:

Lemma 3.4.5: Build-Up Lemma II

Let $f \in L^1_{\text{loc}}$. Then $A_r f(x)$ is continuous on both r $(r > 0)$ and x $(x \in \mathbb{R}^n)$

Notice that f certainly need not be continuous, while $1/m(B_r(x))$ is clearly continuous. It is the $\int$ operation that is forcing this result to be continuous.

***Proof*:**
First, taking advantage that we're working over the Lebesgue measure in $\mathbb{R}^n$, so that if $c = B_1(0)$, then $cr^n = B_r(x)$ and $m(\partial B_r(x)) = 0$ (where $\partial B_r(x) = \{y \mid \|x - y\| = r\}$). This will come back soon.

To actually prove continuity, we'll show that $A_{\lim_{r \to r_0}} f(x) = \lim_{r \to r_0} A_r f(x)$ and $A_r f(\lim_{x \to x_0}) = \lim_{x \to x_0} A_r(x)$. To that end, take a sequene $r \to r_0$ and $x \to x_0$, and notice that the $\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)}$ pointwise on $\mathbb{R}^n \setminus \partial B_{r_0}(x_0)$ (imagine $x = x_0$ is fixed and the radius goes to r_0 by oscilating around r_0 so that the pointwise limit doesn't exist for the border). Thus, $\chi_{B_r(x)} \to \chi_{B_{r_0}(x_0)}$ a.e., and so we have that $\chi_{B_{r_0+1}(x_0)}$ dominates the $\chi_{B_r(x)}$ if $r < r_0 + 1/2$ and $|x - x_0| < 1/2$. Thus, by the Dominated Converting Theorem:

$$\begin{aligned}
A_{\lim\limits_{r \to r_0} r} f(\lim_{x_0 \to x} x) &= \int_{B_{\lim\limits_{r \to r_0} r}(\lim\limits_{x \to x_0} x)} f(y)dy \\
&= \int_{B_r(x)} \lim_{r,x} f(y)dy \\
&= \lim_{r,x} \int_{B_r(x)} f(y)dy && \text{DCT}
\end{aligned}$$

and so $\int_{B_r(x)} f(y)dy$ is continuous. Since $B_r(x) = cr^n$ and r^n is a continuous funciton, and the product of two continuous function is continuous, we have that

$$A_r f(x) = \frac{1}{cr^n} \int_{B_r(x)} f(y)dy$$

is also continuous, as we sought to show

Since $A_r f$ is continuous, it is measurable. Next, we can define the largest possible average given a point x:

Definition 3.4.6: Hardy-Littlewood Maximal Function

Let $f \in L^1_{\text{loc}}$. Then the *Hardy-Littlewood Maximal function* is defined as

$$Hf(x) := \sup_{r>0} A_r|f|(x) = \sup_{r>0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y)|dy$$

Note that Hf is measurable since $(Hf)^{-1} = \bigcup_{r>0} (A - r|f|)^{-1}((a, \infty))$ is open for any $a \in \mathbb{R}$ by the previous lemma. What the next theorem shows us is that if we fix some number $\alpha \in \mathbb{R}_{>0}$, and try to take the measure of all points where $Hf(x) > \alpha$ (i.e., we are measuring the area of the largest possible average over an area of points with that larger average), then this value is in fact bounded by $\int |f(x)|$ provided an appropraite constant C/α:

Theorem 3.4.7: The Maximal Theorem

Let $(\mathbb{R}^n, \mathcal{L}, m)$ be a measure space. There exists a constant C such that for all $f \in L^1$ and all $\alpha > 0$

$$m(\{x \mid Hf(x) > \alpha\}) \leq \frac{C}{\alpha} \int |f(x)| dx$$

***Proof*:**
Let $E_\alpha = \{x \mid Hf(x) > \alpha\}$. By suprmum property of $Hf(x)$, there exists a $r_x > 0$ such that

$$\frac{1}{m(B_{r_x}(x))} \int_{B_{r_x}(x)} |f(y)| dy = A_{r_x}|f|(x) > \alpha \quad \Leftrightarrow \quad m(B_{r_x}(x)) < \frac{1}{\alpha} \int_{B_{r_x}(x)} |f(y)| dy \tag{3.1}$$

The balls $B_{r_x}(x)$ cover E_α. So for any $c < m(E_\alpha)$, by lemma 3.4.2, there exists a finite collection $r_{x_1}, ..., r_{x_k} \in E_\alpha$ such that

$$\sum_{i=1}^{n} m(B_{r_{x_j}}(x_j)) > \frac{c}{3^n}$$

and so, combining the inequalities we have and moving values around, we get:

$$c < 3^n \sum_{i=1}^{n} m(B_{r_{x_j}}(x_j)) \overset{\text{eq. 3.1}}{\leq} \frac{3^n}{\alpha} \int_{B_{r_{x_j}}} |f(y)| dy \leq \frac{3^n}{\alpha} \int_{\mathbb{R}^n} |f(y) dy|$$

Since this is true for all c, letting $c \to m(E_\alpha)$, we get the desired result, as we sought to show.

With all this build-up, we are ready to show the result we stated at the beginning, that $A_r f(x) \to f(x)$ a.e.

Before stating the theorem, it is useful to define the notion of limit superior for real-valued functions:

Definition 3.4.8: Limit Superior

Let $\varphi : \mathbb{R} \to \mathbb{R}$. Then:

$$\limsup_{r \to R} \varphi(r) := \lim_{\epsilon \to 0} \sup_{0 < |r - R| < \epsilon} \varphi(r) = \inf_{\epsilon > 0} \sup_{0 |r - R| < \epsilon} \varphi(r)$$

As an exercise, show that:

$$\lim_{r \to R} \varphi(r) = c \quad \text{iff} \quad \limsup_{r \to R} |\varphi(r) - c| = 0$$

Theorem 3.4.9: Almost Lebesgue Differentiation Thoerem

If $f \in L^1_{\text{loc}}$. Then

$$\lim_{r \to 0} A_r f(x) = f(x)$$

for a.e. $x \in \mathbb{R}^n$. Phrased another way:

$$\lim_{r \to 0} \frac{1}{mB_r(x)} \int_{B_r(x)} f(x) - f(y) dy = 0$$

for a.e. $x \in \mathbb{R}^n$.

***Proof*:**
First, let's show we can limit our attention to function in L^1. In particular, If $A_r f(x) \to f(x)$ for a.e. x, for any $N \in \mathbb{N}$, we can bound what x's we choose by $|x| \leq N$. Since for any $r \leq 1$, the values $A_r f(x)$ depend only on $f(y)$ for $|y| \leq N+1$, we can replace f with $f\chi_{B_{N+1}(0)}$, so $f \in L^1$

By theorem 2.7.2, for every $\epsilon > 0$, there exists a continuous integrable function g such that $\int |g(y) - f(y)| dy < \epsilon$. Since g is continuous, for every $x \in \mathbb{R}^n$, for all $\delta > 0$, there exists an $r > 0$ such that $|y - x| < r$ implies $|g(y) - g(x)| < \delta$. From this, we get the result to work for g:

$$\begin{aligned}
|A_r g(x) - g(x)| &= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y) dy - g(x) \right| \\
&= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y) dy - \frac{m(B_r(x))}{m(B_r(x))} g(x) \right| \\
&= \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} g(y) - g(x) dy \right| \\
&< \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} \delta dy \right| \\
&= \frac{1}{m(B_r(x))} |m(B_r(x))\delta| \\
&= \delta
\end{aligned}$$

and so, since δ can be anything, we hve that $A_r g(x) \to g(x)$ for a.e. x.

Now we use this result to prove it for f. We will use the Maximal theorem to bound the result. Let:

$$\begin{aligned}
\limsup_{r \to 0} |A_r f(x) - f(x)| &= \limsup_{r \to 0} |A_r f(x) - A_r g(x) + A_r g(x) - g(x) + g(x) - f(x)| \\
&= \limsup_{r \to 0} |A_r (f - g)(x) + (A_r g - g)(x) + (g - f)(x)| \\
&\leq H(f - g)(x) + 0 + |f - g|(x)
\end{aligned}$$

Or, in terms of sets, if

$$E_\alpha = \left\{ x \;\middle|\; \limsup_{r \to 0} |A_r f(x) - f(x)| > \alpha \right\} \qquad F_\alpha = \{ x \mid |f - g| > \alpha \}$$

then

$$E_\alpha \subseteq F_{\alpha/2} \cup \{x \mid H(f-g)(x) > \alpha/2\}$$

Furthermore, we have that

$$(\alpha/2)m(F_{\alpha/2}) \leq \int_{F_{\alpha/2}} |f(x) - g(x)|dx < \epsilon$$

Thus, by the maximal theroem:

$$m(E_\alpha) \leq \frac{2\epsilon}{\alpha} + \frac{2C\epsilon}{\alpha}$$

and since ϵ is arbitrary, $m(E_\alpha) = 0$ for all $\alpha > 0$. However, $\lim_{r\to 0} A_r f(x)$ for all $x \notin \bigcup_{i=1}^{\infty} E_{1/n}$, finishing the proof.

In the theorem, we proved it for $f(x)$; it can in fact be shown that it can be proven for $|f|$. To show this, we define the following concept for reference:

Definition 3.4.10: Lebesgue Set

Let f be a measurable function. Then the *Lebesgue set* is :

$$L_f := \left\{ x \;\middle|\; \lim_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(x) - f(y)|dy = 0 \right\}$$

The following theorem shows that the points on which the Lebesgue set fails on is measure 0, and so the result is true for L_f as well

Theorem 3.4.11: Upgrade To Lebesgue Set

Let $f \in L^1_{\text{loc}}$. Then $m((L_f)^c) = 0$

***Proof*:**

This proof will take advantage of the zero measure set in theorem 3.4.9, and that we can form a dense set of $\mathbb{C}$ which will correspond the zero-measure set from this previous theorem, meaning we can make a limit argument that will work for the proof.

First, for each $c \in \mathbb{C}$, we can apply theorem 3.4.9 to $g_c = |f(x) - c|$ to get that:

$$\lim_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - c|dy = |f(x) - c|$$

up to some Lebesgue zero measure set, say N_c. Now, Let D be any countable dense subset of $\mathbb{C}$ and let $N = \bigcup_{c\in D} N_c$ (so $m(N) = 0$). If $x \notin N$, then for any $\epsilon > 0$, we can choose some $c \in D$ such that $|f(x) - c| < \epsilon$. Thus:

$$|f(y) - f(x)| < |f(y) - c| + \epsilon$$

Thus, we have that:

$$\limsup_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - f(x)|dy \leq |f(x) - c| + \epsilon \leq 2\epsilon$$

and since ϵ can be arbitrary small, we have that $\limsup_{r\to 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - f(x)|dy = 0$, and so if $x \notin N$, the theorem applies. But then $0 = m(N) = m(L_f)^c)$, as we sought to show.

We have essentially proven the main results of this section; the last thing we want to do is generalize the types of sets that shrink to x. So far, we have been using balls, however that is not necessary. Thus, we will generalize the types of sets and state the most general version of this "averaging" theorem

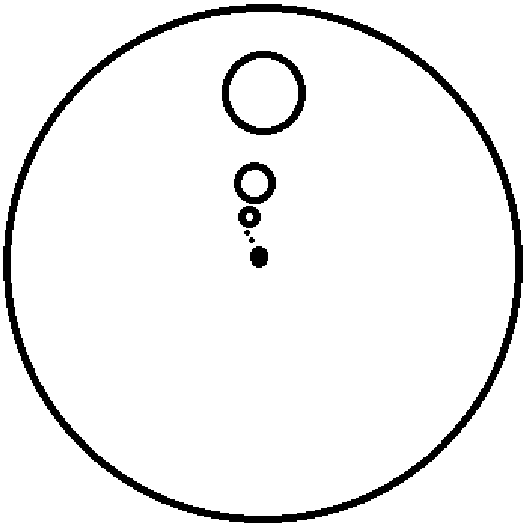
Definition 3.4.12: Sets That Shrink Nicely

A family of set $\{E_r\}_{r>0}$ of Borel subsets of $\mathbb{R}^n$ is said to *shrink nicely* if :

1. $E_r \subseteq B_r(x)$ for each r (so $m(E_r) \leq m(B_r(x))$)
2. There is a constant $\alpha > 0$ that is indepenent of r such that $m(E_r) > \alpha m(B_r(x))$

Example 3.4.13: Nicely Shrinking Sets

1. Let $U \subseteq B_1(0)$ be any borel subset. such that $m(U) > 0$, and consider $E_r = \{x + ry \mid y \in U\}$. You can imagine that U is some circle of $B_1(0)$ and see that, if $x = 0$:



   Then we see that E_r shrinks to to x (even if $x \notin B_1(0)$)

2. in $\mathbb{R}$, the set $E_r = (x, x + r]$ also shrink nicely.

With this, we can finally state the result we've building up to in full generality:

Theorem 3.4.14: The Lebesgue Differentiation Theorem

Let $f \in L^1_{\text{loc}}$. Then for every x in the Lebesgue set, that is almost every x:

$$\lim_{r\to 0}\frac{1}{m(E_r)}\int_{E_r}|f(y)-f(x)|dy = 0$$

and

$$\lim_{r\to 0}\frac{1}{m(E_r)}\int_{E_r}f(y)dy = f(x)$$

for every family of nicely shrinking sets $\{E_r\}_{r>0}$ that shrink nicely to x

Written in terms of measure's this shows that if $d\nu = fd\mu$, then:

$$\lim_{r\to 0}\frac{\nu(E_r)}{m(E_r)} = f(x)$$

***Proof*:**
To prove the first equality, we simply use the definition of nicely shrinking sets:

$$\begin{aligned}\frac{1}{m(E_r)}\int_{E_r}|f(y)-f(x)|dy &\le \frac{1}{m(E_r)}\int_{B_r(x)}|f(y)-f(x)|dy\\ &\le \frac{1}{\alpha m(B_r(x))}\int_{E_r}|f(y)-f(x)|dy\end{aligned}$$

so we can use theorem 3.4.11.

For the second equality, intead of putting absolute value, can start without them, in which case we would do the same think and then use theorem 3.4.9, and move the $f(x)$ to the other side.

3.4.15 Points As Average Notes that this also means that points can be interpreted as the *limit of averages*; in a sense they are inherently "singular". In a sense, it is more natural to work not by evaluating at points but by integrating against function in a small neighbourhood of a point (so that it capture local information). This is exactly what will be done in [13, Distribution Theory] and the Lebesgue Differentiation theorem will insure functions in the traditional sense will be a special case of distributions.

3.4.1 Regular Measures

We will apply these results to a special class of measures which, as is common with such definitions, we take some aspects that we like from the Lebesgue measure on $\mathbb{R}^n$, and want to study measure's more generally (in this case, borel measures on $\mathbb{R}^n$) that have the same property:

Definition 3.4.16: Regular Measure

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, \nu)$ be a borel measure space. Then ν is called *regular* if:

1. $\nu(K) < \infty$ for all compact subset $K \subseteq \mathbb{R}^n$
2. $\nu(E) = \inf\{\nu(U) \mid U \text{ open}, E \subseteq U\}$ for every $E \in \mathcal{B}_{\mathbb{R}^n}$

It is in fact provable that the first condition implies the second. We have shown this for $\mathbb{R}$ (i.e. $n = 1$) in theorem 1.4.2 and 1.4.4. In the study of Radon measures, this proof can be done for arbitrary n (Folland chapter 7.2).

Notice that a regular measure is automatically σ-finite, since $\mathbb{R}^n$ is σ-compact. As usual, a signed or complex measure ν is regular if $|\nu|$ is regular.

If $f \in L^+(\mathbb{R}^n)$, then the measure fdm is regular if and only if $f \in L^1_{\text{loc}}$. This means that we can generalize the Lebesgue differentiation theorem to regular measures in general instead of measures defined on densities which are locally integrable. Let's first check that that assertion is indeed the case. First, it should be clear that $f \in L^1_{\text{loc}}$ is equivalent to $\nu(K) < \infty$ for all compact sets. To show 2, we'll first show it works for bounded borel sets E. By theorem 2.7.1, for any $\delta > 0$, there exists an open set $U \supseteq E$ such that $\mu(U) < \mu(E) + \delta$, thus $\mu(U \setminus E) < \delta$. By theorem 3.2.3, for all $\epsilon > 0$, there exists a $U \supseteq E$ such that $\int_{U \setminus E} fdm < \epsilon$, so $\int_U fdm < \int_E fdm + \epsilon$, thus $\nu(U) < \nu(E) + \epsilon$. This completes the requirement of the infimum. For the unbounded case, take $E = \cup_i E_i$ where each E_i is finite, finding it in each case while choosing $\int_{U_i \setminus E_i} fdm < \epsilon 2^{-i}$.

With this equivalence, it becomes worthwile exploring Lebesgue Differentiation Theorem in the case of regular measures:

Theorem 3.4.17: Lebesgue Differentiation With Regular Measure

Let ν be a regular signed or complex Borel measure on $\mathbb{R}^n$, and let $d\nu = d\lambda + fdm$ be its Lebesgue decomposition. Then for m-almost every $x \in \mathbb{R}^n$:

$$\lim_{r \to 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

for every nicely shrinking family $\{E_r\}_{r>0}$ that nicely shrinks to x

***Proof*:**
We have essentially already done this proof, except now we have to consider we are adding $d\lambda$, $d\nu = d\lambda + fdm$, and so we would like to show that $\frac{\lambda(E_r)}{m(E_r)} \to 0$.

First, notice that $d|\nu| = d|\lambda| + |f|d\mu$ (or $|f|dm$?). Thus, regularity in ν implies regularity of λ and fdm (this should be proven, though it is a bit tedious[a]). Since $f \in L^1_{\text{loc}}$, it is enough to show that if $\lambda \perp m$, then for all x, $\lambda(E_r)/m(E_r) \to 0$ m-a.e as $r \to 0$. Without loss of generality, we can let $E_r = B_r(x)$ and assume λ is positive since:

$$\left|\frac{\lambda(E_r)}{m(E_r)}\right| \leq \frac{|\lambda|(E_r)}{m(E_r)} \leq \frac{|\lambda|(B_r(x))}{m(E_r)} \leq \frac{|\lambda|(B_r(x))}{\alpha m(B_r(x))}$$

for some $\alpha > 0$. Now, let's say $\lambda \geq 0$ (if $\lambda = 0$, then we'd be done), and take $A \in \mathcal{B}_{\mathbb{R}^n}$ such that $\lambda(A) = m(A^c) = 0$ (considering that $\lambda \perp m$). We will do the usual trick when it comes to showing that something has measure zero by assuming it doesn't, havign some sets define define in terms of a " $1/k$ sequence", so to speak, and show we reach a contradiction. Consider:

$$F_k = \left\{ x \in A \;\middle|\; \limsup_{r \to 0} \frac{\lambda(B_r(x))}{m(B_r(x))} > \frac{1}{k} \right\}$$

We will show that $m(F_k) = 0$ for all k, meaning the set of points for which $\lambda(E_r)/m(E_r) \to 0$ is 0 m-a.e. First, since λ is regular, for all $\epsilon > 0$, there exists a $U_\epsilon \supseteq A$ such that $\lambda(U_\epsilon) < \epsilon$. Since U_ϵ is open, for each point $x \in U_\epsilon$, there exists a ball such that $B(x) \subseteq U_\epsilon$. By definition of F_k, $\lambda(B(x)) > \frac{m(B(x))}{k}$. or $k\lambda(B(x)) > m(B(x))$. Next, take $V_\delta = \cup_x B(x)$. Then by construction $F_k \subseteq V_\epsilon$. We'll show that $m(V_\epsilon) \leq 3^n k\epsilon$ for all ϵ, and so $m(F_k) = 0$ or all k. To do this, notice that by assumption there is a constant c such that $c < m(V_\epsilon)$. Thus, by lemma 3.4.2, there exist disjoint balls around points $x_1, ..., x_n$ such that

$$c < 3^n \sum_1^n m(B_{x_i}) \leq 3^n k \sum_1^n \lambda(B_{x_i}) \leq 3^n k\lambda(V_\epsilon) \leq 3^n k\lambda(U_\epsilon) \leq 3^n k\epsilon$$

showing that $m(F_k) \leq m(V_\epsilon) < 3^n k\epsilon$, and since ϵ is arbitrary, we have that $m(F_k) = 0$ for all k, as we sought to show.

[a] It was homework 8, q26

3.5 Functions of Bounded Variation

We will now work towards the Fundamental Theorem of Calculus in its generalized form. This generalization is not in the direction of higher dimensions, as is the case with Stokes Theorem which is a generalization into higher dimensions, but requires much nicer conditions like working over smooth manifolds and differential forms. Rather, we want to stay in $\mathbb{R}$ (or $\mathbb{C}$) and see if we can weaken the conditions (i.e. Weaker than differentiable) and see when we can still apply FTC. Essentially, we want to see if we can define a *density* from probability using the closest notion of a derivative of a function.

In the case of $\mathbb{R}$, we have alraedy explored in section 1.4 that Borel measures defined on $\mathbb{R}$ are defined through an associated distribution F, which is a right-continuous increasing function. In particular, we defined μ_F based on a increasing right-continuous function F (i.e. a distribution) to be $\mu([a, b]) = F(b) - F(a)$. We will use the tools we build up in the last section to finally prove the Fundamental theorem of Calculus in full generality. We will then explore when we can define distributions when we have a complex function, and substitute the notion of increasing function with a function of *bounded variation.*

The first thing we will do is show how increasing functions are a.e. differentiable (which to me is sort of crazy)!!

Proposition 3.5.1: Increasing Then a.e. Differentiable

Let $F : \mathbb{R} \to \mathbb{R}$ be an increasing function. Then the set of points where F is discontinuous is countable.
Furthermore, if $G(x) = \lim_{y \to x^+} F(y) = F(x+)$, then F and G are differentiable a.e. and $F' = G'$ a.e.

***Proof*:**
First, if x is continuous at x, then $F(x+) = F(x-)$, so the interval $(F(x-), F(x+))$ is empty. If it is not empty, then since F is increasing, each $(F(x-), F(x+))$ must be disjoint. So, for any $N \in \mathbb{N}$, for all $|x| \leq N$, we have that $(F(x-), F(x+)) \subseteq (F(-N), F(N))$. Now, here is the trick of this proof: notice that

$$\sum_{|x|<N} [F(x+) - F(x-)] \leq F(N) - F(-N) < \infty$$

since the $F(x-)$ is equal to $F(x_+)$ at appropriate locations, and so the sum cancels out (note that this sum is absolutely continuous since $F(x+) \geq F(x-)$). To see more clearly what's going on ,if we label all the points in $|x| < N$ through an indexing set A, and let $x_\alpha = (F(x_\alpha -) - F(x_\alpha -))$, then:

$$\sum_{\alpha < A} x_\alpha < \infty$$

From here, we use a the fact that this the sum of uncountably many positive x_α, and use the fact that the sum of uncountably many positive elements must be infinite if more than countably elements are non-zero, but since the sum *must* be finite $\{x \in (-N, N) \mid f(x+) \neq F(x-)\}$ is countable. Since this is true for all N, the set of discontinuities of F must be countable, proving the first assertion.

For the second assertion, first note that G is increasing and right continuous and $G = F$ at all points excepts where F is discontinuous. To show G is differentaible a.e, by proposition 1.4.2

$$G(x+h) - G(x) = \begin{cases} \mu_G((x, x+h]) & \text{if } h > 0 \\ \mu_G((x+h, x]) & \text{if } h < 0 \end{cases}$$

Notice that $\{(x - r, x]\}$ and $\{(x, x + r]\}$ shrink nicely to x as $r = |h| \to 0$. Furhtermore μ_G is regual by theorem 1.4.2. Thus, by theorem 3.4.17, we have that:

$$G'(x) = \lim_{r \to 0} \frac{G(x+h) - G(x)}{h} = \lim_{r \to 0} \frac{\mu_G(E_r)}{m(E_r)}$$

exists for a.e. x. The final part of this proof is to show that if $H = G - F$, then H' exists and is equal to zero a.e.

Let $\{x_i\}$ be an enumeration of the points where $H \neq 0$. Since $H = G - F$, $H(x_i) > 0$. Let's say $\delta_i : \mathbb{R} \to \mathbb{R}$ is a function where $\delta_i(x_i) = 1$ and zero eveywehre else. Then for any $N \in \mathbb{N}$, since H is increasing we have $\sum_{\{i \mid |x_i| < N\}} H(x_i) < \infty$. Thus, we can define a new measure:

$$\mu = \sum_i H(x_i) \delta_i$$

By what we've shown, μ is finite on all compact sets by the previous lines reasoning, and so by the theorem's we've been using μ is regular. Furthermore, $\mu \perp m$ since $m(E) = \mu(E^c) = 0$ where $E = \{x_i\}_{i=1}^{\infty}$. Thus:

$$\left| \frac{H(x+h) - H(x)}{h} \right| \leq \frac{H(x+h) + H(x)}{|h|} \overset{!}{\leq} \frac{\mu\left(x - 2|h|, x + 2|h|\right)}{|h|}$$

where $\overset{!}{\leq}$ comes from trying to ingulf $H(x+h)$ and $H(x)$ into an interval.

Since all measure are regular, by theorem 3.4.17 this limit exists and $H' = 0$ a.e., completing the proof.

With this, we now have the tools to try and explore how to think about "increasing function" of the form $F : \mathbb{R} \to \mathbb{C}$. Since $\mathbb{C}$ has no inheret order, we must be more precise on how we define "increasing" and how do we make sure it doesn't explode (just like how when defining the borel measures through a distribution)

To understand the motivation behind the following definitions, imagine the following scenario: let F be the function representing the movement of a particle as it travels accross in time with respect to t (the graph being the position of the particle relative to, say, where it started). The *total variation* from time a to time b of the particle would be the total distance the particle has travelled in that time frame. if F were differentiable, this would simply be:

$$\int_a^b |F'(x)| dx$$

However, what if F is not differentiable? In that case, we will take the following approach: partition $[a, b]$ into $[a, t_1], [t_1, t_2], ..., [t_{n-1}, b]$. Then, linearly approximate F by a straight line from $(t_{i-1}, F(t_{i-1}))$ to $(t_i, F(t_i))$. Then, take the limit over all partitions (in particular, as partitions get finer). If you are worried that such a limit is not well-defined, remember that polynomials can arbitrarily well approximate continuous functions on $\mathbb{R}$, and in most cases F is right-continuous. This heuristic might be more rigorous, but it at least make me feel comfortable with doing this idea

We will do a similar idea for the complex case, except we will take the norm (which we'll also call absolute value) of the distances:

Definition 3.5.2: Total Variation

Let $f : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$. Then define the *total variation of* F to be:

$$T_F(x) = \sup \left\{ \sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| \,\middle|\, n \in \mathbb{N},\ -\infty < x_0 < x_1 < \cdots < x_n = x \right\}$$

I should be clear by construction that T_F is an increasing function. Since we put absolute values inside the sum, then as we refine the partition, the sum will get bigger. Therefore, If we have $a < b$ and consider $T_F(b)$, it is harmless to assume a is one of the subdivision points (since we can refine and get a partition with a being part of it). It follows that we have:

$$T_F(b) - T_F(a) = \sup \left\{ \sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| \,\middle|\, n \in \mathbb{N},\ a = x_0 < x_1 < \cdots < x_n = x \right\}$$

Using this fact, we can define the following:

Definition 3.5.3: Bounded Variation

Let T_F be the total variation of the function F. Then if:

$$\lim_{t\to\infty} T_F(t) = T_F(\infty) < \infty$$

Then F is said to be of *bounded variation* (on $\mathbb{R}$). We denote the set of of all functions of bounded variation as BV. If we are concerned with all function of bounded variation on an interval $[a,b]$, we will denote this as $BV([a,b])$.

Note that if $F \in BV$, then $F \in BV([a,b])$, since it's total variation on $[a,b]$ is $T_F(b) - T_F(a)$ which is less thatn the total vriation of T_F. If $F \in BV([a,b])$, then we can extend F to be in BV by letting $F(x) = F(a)$ for all $x \leq a$ and $F(b) = F(x)$ for all $b \leq x$. Further note that we rarely actually compute $T_F(b)$, but we use the fact that a function is in BV to get many theoretical results:

Example 3.5.4: Bounded Variation

1. If $F : \mathbb{R} \to \mathbb{R}$ is bounded and increasing, then $F \in BV$ $(T_f(x) = F(x) - F(-\infty))$
2. If $F, G \in BV$ and $a, b \in \mathbb{C}$ then $aF + bG \in BV$
3. If F is differentiable on $\mathbb{R}$ and F' is bounded, then $F \in BV([a,b])$ for $-\infty < a < b < \infty$ (by the mean value theorem)
4. If $F(x) = \sin(x)$, then $F \in BV([a,b])$ for all $-\infty < a < b < \infty$, but $F \in BV$
5. if $F = x\sin\left(\frac{1}{x}\right)$ for $x \neq 0$ and $F(0) = 0$, then $F \notin BV([a,b])$ for $a \leq 0 < b$ and $a < 0 \leq b$

Before we prove stuff about BV functions, let's first prove a lemma:

Lemma 3.5.5: Sum Is Increasing

Let $F \in BV$ be a real valued function. Then $T_F + F$ and $T_F - F$ are increasing functions.

***Proof*:**
Take $x < y$ and $\epsilon > 0$. Since T_F is a supremum, take appropriat $x_0 < \cdots x_n = x$ such that

$$\sum_{i}^{n} |F(x_i) - F(x_{i-1})| \geq T_F(x) - \epsilon$$

Next, notice that we can make this into an element of the supremum set of $T_f(y)$ (also known as an approximating sum) by doing

$$\sum_{i}^{n} |F(x_i) - F(x_{i-1})| + |f(y) - f(x)|$$

With this sum, take $F(y) = [F(y) - F(x)] + F(x)$. With some manipulation, we get:

$$T_f(y) \pm F(y) \geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x)$$

Since F is a real valued function, we cna simply take cases to notice that:

+	F(x) - F(y)	+	F(x) - F(y)	$2\lvert F(y) - F(x)\rvert$
+	F(x) - F(y)	-	F(x) - F(y)	0
-	F(y) - F(x)	+	F(x) - F(y)	0
-	F(y) - F(x)	-	F(x) - F(y)	$2\lvert F(y) - F(x)\rvert$

$$\begin{aligned} T_f(y) \pm F(y) &\geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x) \\ &\geq T_f(x) - \epsilon \pm F(x) \end{aligned}$$

letting $\epsilon \to 0$, we see that:

$$T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$$

and since x and y was arbitrary, we have establish that $T_F + F$ and $T_F - F$ are both increasing functions, we as sought to show

Theorem 3.5.6: Properties Of BV

1. $T_F \in BV$ is an increasing function
2. $F \in BV$ if and only if Re$F \in BV$ and $ImF \in BV$
3. If $F : \mathbb{R} \to \mathbb{R}$ then $F \in BV$ if and only if F is the difference of two bounded increasing functions; for $F \in BV$ these functions may be taken to be $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$
4. If $F \in BV$, then $F(x+) = \lim_{y \to x^-} F(y)$ and $F(x-) = \lim_{y \to x^+} F(y)$ exist for all $x \in \mathbb{R}$, as do $F(\pm\infty) = \lim_{y \to \pm\infty} F(y)$
5. If $F \in BV$, the set of points at which F is discontinuous is countable
6. If $F \in BV$ and $G(x) = F(x+)$, then F' and G' are exist and are equal a.e.

***Proof*:**

1. This essentially comes from the definition
2. This is essentially immediate, and so is left as exam review
3. Look at example 3.5.4 to see how the $\Rightarrow$ direction goes. For the $\Leftarrow$ direction, by lemma 3.5.5, the equation $F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$ express F as the difference of two increasing

functions. To show boundedness, we showed that $T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$ for $y > x$. Re-arranging and taking advantage of the fact that $F \in BV$, we get:

$$|F(y) - F(x)| \leq T_F(y) - T_F(x) \leq T_F(\infty) - T_F(-\infty) < \infty$$

showing that F (and hence T_F) is bounded

4,5,6 follow from 1,2,3 along with proposition 3.5.1.

Note that (6) is particularly powerful since we now have that a function simply needs to be of bounded variation to have a derivative a.e. (not necessarily increasing anymore). The converse is false: being differentiable a.e. (even everywhere) does not imply BV (think of $\sin(1/x)$ on $(0, 1]$).

The equation in part 2 are important enough to be given a name:

Definition 3.5.7: Jordan Decomposition

Let $F \in BV$ be a real-valued function. Then the *jordan decomposition* of F is:

$$F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$$

where $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$ are called the *positive variation of* F and *negative variation of* F respectively.

Since $x^+ = \max(x, 0) = \frac{1}{2}(|x| + x)$ and $x^- = \max(-x, 0) = \frac{1}{2}(|x| - x)$ for $x \in \mathbb{R}$, we get that:

$$\frac{1}{2}(T_F \pm F)(x) = \sum \left\{ \sum_i^n [F(x_i)F(x_{i-1})]^{\pm} \;\middle|\; x_0 < \cdots < x_n = x \right\} \pm \frac{1}{2}F(-\infty)$$

We are almost at a stage to connect borel measures to BV functions. There is one more peice of the puzzle that we need:

Definition 3.5.8: Normalized Bounded Variation (NBV)

Let $F \in BV$ be a function of bounded variation. Then if F is right-continuous and $F(-\infty) = 0$, then F is said to be a function of *normalized bounded variation*. We will denote the collection of such functions as:

$$NBV := \{F \in BV \mid F \text{ is right continuous and } F(-\infty) = 0\}$$

Example 3.5.9: NBV

Let $F \in BV$. Define $G = F(x+) - F(-\infty)$. Then $G \in NBV$, and $F' = G'$ a.e.

First, to show that $G \in BV$, by theorem 3.5.6[2,3], if F is real and $F = F_1 - F_2$ where F_1, F_2 are increasing, then $G(x) = F_1(x+) - [F_2(x+) - F(-\infty)]$, which is the difference of two increasing functions. That $G(-\infty) = 0$ comes by definition. Hence $G \in NBV$.

In fact, every $F \in BV$ almost defines an NBV function through T_F:

Lemma 3.5.10: BV, then T_F almost NBV

If $F \in BV$, then $T_F(-\infty) = 0$. If F is also right continuous, then T_F is also right-continuous (making $T_F \in NBV$)

Proof:
Let $\epsilon > 0$ so that for any x and $x_0 < \cdots < x$ we ahve:

$$\sum F(x_{i+1}) - F(x_i) = T_f(x) - T_f(x_0) > T_F(x) - \epsilon$$

So $T_f(x_0) < \epsilon$ or more generally for any $y \leq x_0$ $T_f(y) < \epsilon$. Since epislon is arbitray, as we let $y \to 0$ we get $T_f(-\infty) = 0$

For the next assertion, suppose F is right-continuous. I'll finish this proof later:

Now suppose that F is right continuous. Given $x \in \mathbb{R}$ and $\epsilon > 0$, let $\alpha = T_F(x+) - T_F(x)$, and choose $\delta > 0$ so that $|F(x+h) - F(x)| < \epsilon$ and $T_F(x+h) - T_F(x+) < \epsilon$ whenever $0 < h < \delta$. For any such h, by (3.24) there exist $x_0 < \cdots < x_n = x + h$ such that

$$\sum_1^n |F(x_j) - F(x_{j-1})| \geq \tfrac{3}{4}[T_F(x+h) - T_F(x)] \geq \tfrac{3}{4}\alpha,$$

and hence

$$\sum_2^n |F(x_j) - F(x_{j-1})| \geq \tfrac{3}{4}\alpha - |F(x_1) - F(x_0)| \geq \tfrac{3}{4}\alpha - \epsilon.$$

Likewise, there exist $x = t_0 < \cdots < t_m = x_1$ such that $\sum_1^n |F(t_j) - F(t_{j-1})| \geq \frac{3}{4}\alpha$, and hence

$$\begin{aligned} \alpha + \epsilon &> T_F(x+h) - T_F(x) \\ &\geq \sum_1^m |F(t_j) - F(t_{j-1})| + \sum_2^n |F(x_j) - F(x_{j-1})| \\ &\geq \tfrac{3}{2}\alpha - \epsilon. \end{aligned}$$

Thus $\alpha < 4\epsilon$, and since ϵ is arbitrary, $\alpha = 0$. ∎

We finally get to establish the connection between complex meaures and functions in BV:

Theorem 3.5.11: Complex Meaures And BV

If μ is a complex Borel meaure on $\mathbb{R}$ (i.e. a finite signed measure) and $F(x) = \mu((-\infty, x])$, then $F \in NBV$. Conversely, if $F \in NBV$, then there is a unique complex Borel measure μ_F such that $F(x) = \mu_F((-\infty, x])$

***Proof*:**
Let μ a complex measure. Then we can decompose it into $\mu = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$ where $\mu_i^\pm$ are positive finite measures. If $F_i^\pm(x) = \mu_i^\pm((-\infty, x])$, then $F_i^\pm$ is increasing, right continuous, $F_i^\pm(-\infty) = 0$, and $F_i^\pm(\infty) = \mu_i^\pm(\mathbb{R}) < \infty$. Thus, by theorem 3.5.6[2,3], the function

$$F = F_1 + -F_1^- + i(F_2^+ - F_2^-) \in NBV \tag{3.2}$$

Conversely, by theorem 3.5.6 and lemma 3.5.10, any $F \in NBV$ can be written in the from of equation (3.2) with the $F_i^\pm$ increasing and in NBV. By Theorem 1.4.2, each $F_i^\pm$ gives a measure $\mu_i^\pm$, so $F(x) = \mu_F((-\infty, x])$ where $\mu_F = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$.

Finally, to show $|\mu_F| = \mu_{T_F}$ (this was left as an outlined exercise 28)

Corollary 3.5.12: Total Variation Of μ_F

Let $F \in NBV$, μ_F be the associated complex measure on $\mathbb{R}$, and $G(x) = |\mu_F|((-\infty, x])$. Then:

$$|\mu_F| = \mu_{T_F}$$

***Proof*:**
We break down the proof in the following steps:

1. From the definition of T_F, $T_F \leq G$

   Proof. We need to show $T_F(x) \leq G(x)$ In the previous homework, we showed that:

   $$|\mu_F|((-\infty, x]) = \sup\left\{\sum_i^\infty |\mu_F(E_i)| \,\middle|\, \coprod_i E_i = (-\infty, x]\right\}$$

   In particular, we can choose are disjoint sets to be h-intervals and get:

   $$\begin{aligned}|\mu_F|((-\infty, x]) &= \sup\left\{\sum_i^\infty |\mu_F((x_i, x_{i+1}])| \,\middle|\, \coprod_i (x_i, x_{i+1}] = (-\infty, x]\right\} \\ &= \sup\left\{\sum_i^\infty |F(x_{i+1}) - F(x_i)| \,\middle|\, \coprod_i (x_i, x_{i+1}] = (-\infty, x]\right\}\end{aligned}$$

   and by definition of T_F

   $$T_F(x) = \sup\left\{\sum_i^N |F(x_{i+1}) - F(x_i)| \,\middle|\, N \in \mathbb{N},\ x_0 < \cdots < x_N = x\right\}$$

Notice that the set for $|\mu_F|$ contains that of $T_F(x)$ since we can set the rest of the infinite value for T_F to 0, and so $T_F \leq G = |\mu_F|$ □

2. $|\mu_F(E)| \leq \mu_{T_F}(E)$ when E is an interval, and hence E is a borel set.

Proof. Without loss of generality, we will work with intervals of the form $(a, b]$. We only require these type of intervals for the σ-algebra that will be generatedi n part (c).

First, since $F \in NBV$, then it is right-continuous, and so by lemma 3.28 T_F is right-continuous, and so by lemma 3.28 again $T_F \in NBV$. Thus, by theorem 3.29, there is an assocaited complex borel meaure such that $T_F(x) = \mu_{T_F}((-\infty, x])$. So $\mu_{T_F}((a, b]) = T_F(b) - T_F(a)$. With that, we proceed with the proof.

First, the emptyset (the trivial interval) is immediate since $|\mu_F(\emptyset)| = 0 = \mu_{T_F}(\emptyset)$. Next, to show the inequality is true for bounded intervals, notice that:

$$|\mu_F((a, b])| = |\mu_F((-\infty, b]) + \mu_F((-\infty, a]) = |F(b) - F(a)|$$

Next, recall that:

$$\mu_{T_F}((a, b]) = T_F(b) - T_F(a) = \sup \left\{ \sum_i^n |F(x_{i+1}) - F(x_i)| \;\middle|\; n \in \mathbb{N}\ a = x_0 < \cdots < x_n = b \right\}$$

So, $|F(b) - F(a)|$ is inside the supremum set, and so:

$$= |F(b) - F(a)| \leq T_F(b) - T_F(a) = \mu_{T_F}((a, b])$$

Next, for rays, notice we can split a ray into the countable union of h-intervals:

$$|\mu_F((-\infty, b])| = \left| \sum_{i=0}^{\infty} \mu_F((b - i - 1, i - k]) \right| \leq \sum_{i=0}^{\infty} |\mu_F((b - i - 1, i - k])|$$

where we can now apply the results for bounded intervals to get that:

$$\leq \sum_{i=0}^{\infty} |\mu_F((b - i - 1, i - k])| \leq \sum_{i=0}^{\infty} \mu_{T_F}((a, b]) = \mu_{T_F}((-\infty, b])$$

Thus $|\mu_F((-\infty, b])| \leq \mu_{T_F}((-\infty, b])$, and similarly for rays of the form (a, ∞).

To show it's true for all Borel sets, first recall that h-intervals form an algebra (their disjoint union forms an elementary family, and hence by proposition 1.7). Next, let $\mathcal{C} = \{E \in \mathcal{B}_{\mathbb{R}} \mid |\mu_F(E)| \leq \mu_{T_F}(E)\}$. This set clearly contains all h-intervals by what we've just shown. We'll show that this set is in fact a monotone class, and hence by the monotone class lemma contains all Borel sets.

This is simply a matter of using the appropraite definitions. Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{C}$ be an increasing sequence. Then:

$$\begin{aligned}
\left|\mu_F\left(\bigcup_i^\infty E_i\right)\right| &= \left|\lim_{i\to\infty}\mu_F\left(\bigcup_i^\infty E_i\right)\right| && \text{definition}\\
&= \lim_{i\to\infty}\left|\mu_F\left(\bigcup_i^\infty E_i\right)\right| && \text{limit property}\\
&\leq \lim_{i\to\infty}\mu_{T_F}(E_i) && \text{all } E_i\in\mathcal{C}\\
&= \mu_{T_F}\left(\bigcup_i^\infty E_i\right) && \text{definition}
\end{aligned}$$

and hence $\mathcal{C}$ is closed under increasing sequences. A similar argument works for decreasing sequences (in particular since μ_{T_F} is finite, it will hold true for any decreasing sequence). Hence, by the monotone class lemma, $\mathcal{C}$ contains the Borel set, showing that $|\mu_F(E)| \leq \mu_{T_F}(E)$ for all $E \in \mathcal{B}_{\mathbb{R}}$, completing the proof.

□

3. $|\mu_F| \leq \mu_{T_F}$, and hence $G \leq T_F$ (use exercise 21.)

Proof. Recall that

$$|\mu_F|(E) = \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}$$

where I modified what's inside the parenthesis to make it more explicit that the sets are inside $\mathcal{B}_{\mathbb{R}}$. Using what we have just proven:

$$\begin{aligned}
|\mu_F|(E) &= \sup\left\{\sum_i^\infty |\mu_F(E_i)| \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}\\
&\leq \sup\left\{\sum_i^\infty \mu_{T_F}(E_i) \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}\\
&\leq \sup\left\{\mu_{T_F}(\coprod_i E_i) \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\} && \text{disjoint}\\
&\leq \sup\left\{\sum_i^\infty \mu_{T_F}(E) \;\middle|\; E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E\right\}\\
&= \mu_{T_F}(E)
\end{aligned}$$

Thus, $|\mu_F|(E) \leq \mu_{T_F}(E)$. If we take $E = (-\infty, x]$, then we get $|\mu_F|(E) = G(E) \leq \mu_{T_F}(E)F$, completing the proof.

In particular, since we've shown the other direction, we have that $G = \mu_{T_F}$. □

With the connection established for between NBV functions and complex measures, it is natural to ask which function in NBV correspond to measure μ such that $\mu \perp m$ or $\mu \ll m$? The following proposition answers that:

Proposition 3.5.13: NBV And Radon-Nikodym with respect to m

If $F \in NBV$, then $F' \in L^1(m)$. Moreover:

1. $\mu_F \perp m$ if and only if $F' = 0$ a.e.
2. $\mu_F \ll m$ if and only if $F(x) = \int_{-\infty}^{x} F'(t)dt$

***Proof*:**
First, note that μ_F is regular by theorem 1.4.4. Next, observe that:

$$F'(x) = \lim_{r \to 0} \frac{\mu_F(E_r)}{m(E_r)}$$

where $E_r = (x, x+r]$ or $(x-r, x]$ by applying theorem 3.4.17

Since we have now characterized μ_F in terms of F, we can characterize $\mu_F \ll m$ in terms of F:

Definition 3.5.14: Absolute Continuity w.r.t. F

A function $F : \mathbb{R} \to \mathbb{C}$ is said to be *absolutely continuous* if

$$\forall \epsilon > 0, \ \exists \delta > 0 \text{ such that}$$

$$\text{for any set of disjoint intervals } (a_1, b_1), ..., (a_N, b_N),$$

$$\sum_{1}^{N} (b_i - a_i) < \delta \Rightarrow \sum_{i}^{N} |F(b_i) - F(a_i)| < \epsilon$$

More generally, F is absolutely continuous on $[a, b]$ if all the open intervals lie in $[a, b]$.

Notice that if F is absolutely continuous, then F is automatically uniformly continuous (pick $N = 1$). To understand why this is, and how it relates to other forms of continuity, it is helpful to use a driving analogy. Imagine driving a car, where the function $F(x)$ represents your distance traveled at time x:

- *Continuous:* You cannot teleport. You must drive through every point.
- *Uniformly Continuous:* You cannot have a vertical asymptote (you can't travel an infinite distance in finite time). It places a "global" rule: for a given time window δ, your distance will never exceed ϵ. However, it only looks at *one continuous interval at a time.*
- *Lipschitz Continuous:* You have a *strict, hard speed limit.* Your speed (the derivative $F'(x)$) can never exceed M.
- *Absolutely Continuous:* You *do not have a strict speed limit*, but your total fuel (or total movement) is restricted by the time allowed. You are allowed to momentarily hit "infinite" speed, but only for such a fleeting instant that the total distance gained during that spike is minuscule.

Why do we need absolute continuity if we have uniform continuity? Uniform continuity has a blind spot: it allows a function to concentrate all of its growth into a scattered collection of tiny, microscopic intervals whose total length is practically zero.

Example 3.5.15: The Devil's Staircase

A prime example is the *Cantor Function* (the "Devil's Staircase"). It is uniformly continuous on $[0, 1]$. It starts at 0 and ends at 1. However, it does 100% of its climbing on the Cantor set—a fractal set that has a total length (measure) of *exactly zero*.

Absolute continuity was invented to prevent this "Devil's Staircase" loophole. By checking a *collection* of N disjoint intervals whose *combined* total width is $< \delta$, absolute continuity ensures you cannot "sneak" a large amount of vertical growth into a small amount of horizontal space, even if you scatter that space into a million tiny pieces.

Proof. Recall the definition of absolute continuity: for every $\epsilon > 0$, there exists a $\delta > 0$ such that for any finite collection of disjoint open intervals (a_i, b_i) in $[0, 1]$,

$$\sum_{i=1}^{N}(b_i - a_i) < \delta \Rightarrow \sum_{i=1}^{N}|F(b_i) - F(a_i)| < \epsilon$$

We will negate this statement. Let $\epsilon = 1/2$. We must show that for any $\delta > 0$, we can find a collection of disjoint intervals whose total length is less than δ, but over which the sum of the variations of F is at least $1/2$.

Recall the construction of the Cantor set C. At the n-th stage of the construction, we have a set C_n consisting of 2^n disjoint closed intervals, each of length 3^{-n}. The total length of the intervals making up C_n is $(2/3)^n$.

Because $\lim_{n\to\infty}(2/3)^n = 0$, given any $\delta > 0$, we can choose an n large enough such that $(2/3)^n < \delta$. Let these 2^n intervals be denoted by $[a_k, b_k]$ for $k = 1, \ldots, 2^n$. Their interiors (a_k, b_k) form a disjoint collection of open intervals whose total length is:

$$\sum_{k=1}^{2^n}(b_k - a_k) = \left(\frac{2}{3}\right)^n < \delta$$

By the definition of the Cantor function, F is constant on the intervals removed during the construction of the Cantor set. Therefore, all of the increase of F from $F(0) = 0$ to $F(1) = 1$ happens precisely over the intervals of C_n. Across each interval $[a_k, b_k]$ in C_n, F increases by exactly 2^{-n}.

Summing the variation of F over these open intervals gives:

$$\sum_{k=1}^{2^n}|F(b_k) - F(a_k)| = \sum_{k=1}^{2^n}2^{-n} = 2^n \cdot 2^{-n} = 1$$

Since $1 \geq \epsilon = 1/2$, the condition for absolute continuity fails. Thus, F is not absolutely continuous. $\square$

To see why absolute continuity is weaker than Lipschitz, consider $F(x) = \sqrt{x}$ on $[0, 1]$. It is not Lipschitz because as $x \to 0$, the slope approaches infinity (no maximum speed limit). However, it *is* absolutely continuous because the "infinite speed" lasts for such a short duration that the accumulated distance remains bounded.

If F is everywhere differentiable and F' is bounded, then F is absolutely continuous, since $|F(b_i) - F(a_i)| \leq (\max |F'|)(b_i - a_i)$ by the Mean Value Theorem. More generally, on a compact set, we have that:

$$\text{absolutely continuous} \subseteq \text{uniformly continuous} \subseteq \text{continuous}$$

The rules change on a compact interval because a closed and bounded domain forces "nice" behavior. The function cannot escape to infinity, preventing unexpected asymptotic stretching. For example, $F(x) = x^2$ is continuous but not uniformly continuous on $\mathbb{R}$. But on a compact interval like $[0, 5]$, it is uniformly continuous (by the Heine-Cantor Theorem) because the domain is capped, and there is a "steepest point."

Because of this bounded nature, on a compact interval, the following hierarchy holds beautifully:

$$\begin{gathered}\text{continuously differentiable}\\ \subseteq\\ \text{Lipschitz continuous}\\ \subseteq\\ \text{absolutely continuous}\\ \subseteq\\ \text{Bounded Variation}\\ \subseteq\\ \text{differentiable a.e.}\end{gathered}$$

where Lipschitz continuous means for all $x, y \in \mathbb{R}$, there exists an M such that $|F(x)-F(y)| \leq M|x-y|$. After going over the FTC for Lebesgue integrals, you can prove as an exercise that Lipschitz continuous implies absolutely continuous. Usually, Lipschitz continuity is used as a simple way of finding absolutely continuous functions.

In modern mathematics, absolute continuity is the right condition for the question: *"When does $F(x) = F(a) + \int_a^x F'(t)dt$ hold true?"* For the Cantor function, the derivative is 0 almost everywhere, so integrating it gives 0, even though the function climbed from 0 to 1. Absolute continuity strictly outlaws this unmeasurable growth, ensuring the derivative accounts for 100% of the function's growth.

To make sure this new notion links with the absolute continuity of the measure:

Proposition 3.5.16: Linking Absolute Continuity Of F And Measures

Let $F \in NBV$. Then F is absolutely continuous if and only if $\mu_F \ll m$

***Proof*:**

($\Leftarrow$) Let $\mu_F \ll m$. Then since μ_F is finite, we can apply theorem 3.2.3 to the sets $E = \bigcup_i^N (a_i, b_i]$, and get that F is absolutely continuous

($\Rightarrow$) Let E be a borel set such that $m(E) = 0$. Let $\epsilon > 0$ so that we can choose a δ that satisfies the definition of absolute continuity of F. By theorem 1.4.4, we can find open sets $U \supseteq U_1 \cdots \supseteq E$ such that $m(U_1) < \delta$ (which also implies $m(U_i) < \delta$ by monotonicity) and

$\mu_F(U_i) \to \mu_F(E)$. Since each U_i is the countable union of disjoint intervals $(a_{i,j}, b_{i,j})$, for all $N \in \mathbb{N}$:

$$\sum_k^N |\mu_F((a_{i,j}, b_{i,j}))| \le \sum_k^N |F(b_{i,j}) - F(a_{i,j})| < \epsilon$$

by absolute continuity. Letting $N \to \infty$, we get that $|\mu_F(U_i)| < \epsilon$, and so $|\mu_F(E)| < \epsilon$. Since this is true for all epsilon, it must be that $|\mu_F(E)| = 0$, showing that $\mu_F \ll m$, as we sought to show

With this, we can relate L^1 functions to NBV functions:

Corollary 3.5.17: Relating L^1 And NBV Functions

Let $f \in L^1$. Then the function $F(x) = \int_{-\infty}^x f(t)dt$ is in NBV, is absolutely continuous, and $f = F'$ a.e. Conversely, if $F \in NBV$, is absolutely continuous, then $F' \in L^1(m)$ and $F(x) = \int_{-\infty}^x F'(t)dt$

***Proof*:**
This follows from proposition 3.5.13 and 3.5.16.

If the function is bounded, then we get a slightly stronger result:

Lemma 3.5.18

If F is absolutely continuous on $[a, b]$, then $F \in BV([a, b])$

***Proof*:**
We will find an upper bound for the total variation of F, that is, we'll find an N such that $T_f(b) < N$. Choose $\epsilon = 1$ and let δ be the appropriate choice to satisfy absolute continuity: for any finite set of disjoint intervals (a_i, b_i):

$$\sum(b_i - a_i) < \delta \Rightarrow \sum |F(b_i) - F(a_i)| < \epsilon$$

Choose N to be the greatest integer that is smaller than $\delta^{-1}(b-a)+1$. Then, for any subdivision $a = x_0 < x_1 < \cdots < x_n = b$, we can collect the subintervals (x_i, x_{i+1}) into at most N groups such that the sum of each group is less than δ (adding subdivision's if necessary). Then the sum $\sum |F(x_{i+1}) - F(x_i)|$ over each group is 1, and so the total variation of F on $[a, b]$ is at most N

Is the converse true: If $F \in BV$, then does it imply that F is aboslutely continuous? The answer is no. Even better, F can even be NBV, uniformally continuous, and increasing (so differentiable a.e.), but still not absolutely continuous:

Example 3.5.19: NBV, Continuous, But Not Absolutely Continuous
Let F be the cantor function. Note that F is increasing, $F(-\infty) = 0$, and $T_F(\infty) = 1/2$ (since it's continuous, it is integrable, and integarting give $1/2$). Thus μ_F is defined. However, it is not absolutely continuous with respect to m. However, F is not absolutely continuous (you should prove this).

Theorem 3.5.20: Fundamental Theorem Of Calculus For Lebesgue Integrals

Let $-\infty < a < b < \infty$ and $f : [a,b] \to \mathbb{C}$. Then the following are equivalent:

(a) F is absolutely continuous

(b) $F(x) - F(a) = \int_a^x f(t)dt$ for some $f \in L^1([a,b],m)$

(c) F is differentiable a.e. on $[a,b]$, $F' \in L^1([a,b],m)$ and $F(x) - F(a) = \int_a^x F'(t)dt$

***Proof*:**
First, to show $(a) \Rightarrow (c)$, we can subtract some constant from F so that $F(a) = 0$. Doing the usual extension of setting $F(x) = 0$ for $x < a$ and $F(x) = F(b)$ for $x > b$, then $F \in NBV$. Thus, by lemma 3.5.18 using corrollary 3.5.17. Then, that $(c) \Rightarrow (b)$ comes immetidaly since (c) is a stronger condition than (b). For $(b) \Rightarrow (a)$, if we set $f(t) = 0$ for $t \notin [a,b]$ then $f \in L^1(m)$ and so by corrollary 3.5.17 we get the final result.

If $F \in NBV$, then we usually write an integral of a function g with respect to the associated μ_F as $\int gdF$ or $\int g(x)dF(x)$. These types of integrals are called *Lebesgue-Stieltjes integrals.* I'm not sure why they were introduced, but this theorem was presented for generalizing integration by parts to these types of integrals

Theorem 3.5.21: Integration By Parts For Lebesgue-Stieltjes Integrals

Let $F, G \in NBV$ where at least one of them is continuous. Then for $-\infty < a < b < \infty$

$$\int_{(a,b]} FdG + \int_{(a,b]} GdF = F(b)G(b) - F(a)G(a)$$

***Proof*:**
F and G are linear combinations of increasing functions in NBV (by Theorem 3.27(a,b) in Folland), so a simple calculation shows that it suffices to assume F and G increasing. Suppose for the sake of definiteness that G is continuous, and let $\Omega = \{(x,y) : a < x \leq y \leq b\}$. We use Fubini's theorem to compute $\mu_F \times \mu_G(\Omega)$ in two ways:

$$\begin{aligned}\mu_F \times \mu_G(\Omega) &= \int_{(a,b]} \int_{(a,y]} dF(x)\, dG(x) = \int_{(a,b]} [F(y) - F(a)]\, dG(y) \\ &= \int_{(a,b]} F\, dG - F(a)[G(b) - G(a)],\end{aligned}$$

and since $G(x) = G(x-)$,

$$\begin{aligned}\mu_F \times \mu_G(\Omega) &= \int_{(a,b]} \int_{[x,b]} dG(y)\, dF(x) = \int_{(a,b]} [G(b) - G(x)]\, dF(x) \\ &= G(b)[F(b) - F(a)] - \int_{(a,b]} G\, dF.\end{aligned}$$

Subtracting these two equations, we get the desired result.

Finally, we end this chapter with a quick discussion about different terminology when it comes down to decomposing measures.

Definition 3.5.22: Discrete And Continuous Measures

Let μ be a complex Borel measure on $\mathbb{R}^n$

1. μ is called *discrete* if if there is a coutnable subset $\{x_i\}_i \subseteq \mathbb{R}^n$ such that $\sum_i c_i < \infty$ and $\mu = \sum c_i \delta_{x_i}$ where $\delta_{x_i}(x_i) = 1$ and is zero everywhere else (i.e. the pointmass at x).

2. μ is called *continuous* if for all $x \in \mathbb{R}^n$, $\mu(\{x\}) = 0$

Any measure can be broken down into $\mu = \mu_d + \mu_c$ where μ_d is discrete and μ_c is continuous. The main thing is to insure that μ_d is over a countable set: let $E = \{x \mid \mu(\{x\}) \neq 0\}$. Taking any countable subset $F \subseteq E$, then $\sum_{x\in F} \mu(\{x\})$ converges absolutely to $\mu(F)$ so $\left\{x \in E \;\middle|\; \mu(\{x\}) > k^{-1}\right\}$ is finite, and so it must be that E is countable, so $\mu_d(A) = \mu(A \cap E)$ is discrete and $\mu_c(A) = \mu(A \setminus E)$ is continuous.

If μ is discrete, then $\mu \perp m$, and if μ is continuous then $\mu \ll m$. Thus, we can further break down the decomposition into:

$$\mu = \mu_d + \mu_{ac} + \mu_{sc}$$

where μ_d is the discrete part, μ_{ac} is the absolutely continuous part, and μ_{sc} is the *singularly continuous* part. So far, we have mainly worked with measures with zero singularly continuous parts. There do exist measures with nonzero singularly continuous parts (Folland says the surface measure of the sphere for $n > 1$ is an example). An example for $n = 1$ is rarer, but they do exist. By proposition 3.5.13, we know that we are looking for a corresponding $F \in NBV$ such that $F' = 0$ a.e. In fact, the cantor function is such an example (when extended in the way we keep doing to make sure functions are in NBV). In fact, using the cantor function, we can construct an increasing function F that is continuous, $F' = 0$ a.e., and that is *strictly increasing*:

Example 3.5.23: Interesting Singularly Continuous Function

1. If $\{F_i\}_{i=1}^{\infty}$ is a sequence on nonegative increasing functions on $[a, b]$ such that $F(x) = \sum_1^{\infty} F_i(x) < \infty$ for all $x \in [a, b]$, then $F'(x) = \sum_1^{\infty} F_i'(x)$ for a.e. $x \in [a, b]$ (It suffices to assume $F_i \in NBV$. Consider the measure μ_{F_i})

***Proof*:**
First, notice that since the F_i are nonnegative, by letting $F(x) = 0$ for $x < a$ and $F(b) = F(y)$ for $b < y$, we have that $F_i(-\infty) = 0$. Furthermore, the the function $G(x) = F(x+)$ and $G_i(x) = F_i(x+)$ is equal a.e. with the same derivative wherever it is defined (which it is a.e.), and so if G and G_i has a particular derivative, so do F and F_i. Thus, Without loss of generality, we'll assume $F, F_i \in NBV$.

Since they are in NBV, consider the induced complex borel measure μ_F and μ_{F_i} given by theorem 3.29. Since these measures are finite, they are σ-finite, and so admit a Radon-Nykodym derivative:

$$d\mu_F = f dm + d\lambda \qquad d\mu_{F_i} = f_i dm + d\lambda_i$$

where $f dm \ll m$, $f_i dm \ll m$ and $d\lambda \perp m$, $d\lambda_i \perp m$. We will show that $f dm = \sum f_i dm$. First, we need to show that $\mu_f = \sum \mu_{F_i}$, and then use the previous homework exercise (question 8) to conclude the equality.

By definition of F, $\mu_F((-\infty, b]) = \sum_i^\infty \mu_{F_i}((-\infty, b])$. Since the two measures agree on a generating set (closed rays form a generated set), they agree everywehre, meaning $\mu_F = \sum_i^\infty \mu_{F_i}$. Thus, we have that (using absolute convergence to re-arrange the sum)

$$f dm + d\lambda = d\mu_F = \sum_{i=1}^{\infty} d\mu_{F_i} = \sum f_i dm + \sum \lambda_i$$

Since $(\sum d\lambda_i) \perp m$ and $(\sum f_i dm) \ll m$, by question 8:

$$d\lambda = \sum d\lambda_i \qquad f dm = \sum f_i dm$$

Finally, we can invoke theorem 3.22 to connect the information about f and f_i to derivative information. By letting $E_r = (x, x+r]$ (since the F and F_i's are always positive), we get:

$$\begin{aligned}
F'(x) &= \lim_{r\to 0} \frac{F(x+r) - F(x)}{r} \\
&= \lim_{r\to 0} \frac{\mu_F(E_r)}{m(E_r)} \\
&= f(x) && \text{thm. 3.22} \\
&= \sum f_i(x) && m\text{-a.e.} \\
&= \sum \left(\lim_{r\to 0} \frac{\mu_{F_i}(E_r)}{m(E_r)} \right) && \text{thm 3.22} \\
&= \sum \left(\lim_{r\to 0} \frac{F_i(x+r) - F_i(x)}{r} \right) \\
&= \sum F'(x)
\end{aligned}$$

as we sought to show.

2. Let F denote the cantor function on $[0, 1]$ (see 1.5), and set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Let $\{[a_n, b_n]\}$ be an enumeration of the closed subintervals of $[0, 1]$ with rational endpoints, and let $F_n(x) = F\left(\frac{x-a_n}{b_n-a_n}\right)$. Then $G = \sum_1^\infty 2^{-n} F_n$ is continuous and strictly

increasing on $[0,1]$ and $G' = 0$ a.e. (use exercise 39)

***Proof*:**
We break this proof down in many steps.

First, F is differentiable a.e. Notice that F is increasing and bounded, and so by theorem 3.27(e) F is a.e. differentiable. Notice that we can extend F as $F(x) = F(0)$ $x \leq 0$ and $F(1) = F(y)$ $1 \leq y$. In this way, $F \in NBV$.

Next, $F' = 0$ a.e. To see this, notice that by definition of F, the function is constant on C^c where C is the cantor set. Since the Cantor set is closed, C^c is open. On any open interval of C^c, F is constant, and so $F' = 0$ on C^c.

As a consequence, every F_n has zero derivative almost everywhere. At any point where F is differentiable:

$$F_n' = F'\left(\frac{x-a_n}{b_n-a_n}\right)\left(\frac{x-a_n}{b_n-a_n}\right)' = 0\left(\frac{x-a_n}{b_n-a_n}\right)' = 0$$

Showing that F_n' is a.e. 0.

Using this result, we can apply exercise 39 to get $G' = \sum_n 2^{-n}F_n$. Since each $2^{-n}F_n$ is a non-negative increasing function has derivative 0 a.e.:

$$G' = \sum_{n=1}^{\infty} 2^{-n}F_n' = 0 \qquad \text{a.e.}$$

Which gives us our first result.

Next, let's prove G is continuous. To see this, let $G_n = \sum_{k=1}^{n} 2^{-k}F_k$. Notice that $2^{-k}F_k$ is continuous being the product of two continuous functions (F_k is continuous being the composition of two continuous functions). By definition of F_k, the largest value if 1, and so $\left|\sum_1^n 2^{-k}F_n\right| \leq \sum_1^n 2^{-k} = 1 - 2^{-n}$. Then we can combine these two pieces of information to show that G_n uniformally converges to G. In particular, for all $\epsilon > 0$, chosoe $n > \frac{1}{2^\epsilon}$ so that $|G - G_n| < \epsilon$ for all $x \in [0,1]$. Thus, G is continuous.

Finally, to show G is strictly increasing, choose $x, y \in [a,b]$ such that $x < y$. Since x is strictly less than y, there exists an ϵ such that $x \notin [y-\epsilon, y]$. Since the rational's are dense, we can choose $[a_n, b_n]$ from our enumeration such that $x < a_n < b_n < y$ so that $x, y \notin [a_n, b_n]$. As a consequence, notice that $G(x)$ must be at least 2^{-n} away from $G(y)$, since:

$$\sum_n^{\infty} 2^{-n}F_n(x) < \sum_n^{\infty} 2^{-n}F_n(y)$$

and so G is strictly increasing.

I also want to give a quick word on convexity. I'd like to write more about convex functions, like their definition and some properties, but for now I'll just write down in bullet form what I found interesting:

1. Convex functions on compact intervals are absolutely continuous
2. As a consequence, FTC applies. In fact convex functions can be characterized like so : f is convex if and only if f' is strictly increasing. If f is twice differentiable, this leads to the

common notion of $f''(x) \geq 0$

3. In a sense, linear functions are a sepcial type of convex function. Convex functions are closed under addition and sclara multiplication. Thus, linear function being both convex and concave mean they are closd under subraction, which is why they form a vector-space, but convex functions do not. The inequalities of linear function are also all “trivial”, preciesly because of linearity, while those of convex functions are more complex.

4

Radon Measures and the Riesz Representation Theorem

Measures on the real line were built in chapter 2 from below, out of their values on intervals: a distribution function determines a premeasure, and theorem 1.4.2 extends it to a Borel measure. That route is tied to the order structure of $\mathbb{R}$ and does not extend to a general space. On a locally compact Hausdorff space there is no distribution function to start from, and yet the measures one cares about there - the ones that integrate continuous functions of compact support - are pinned down by a single piece of data: the linear functional $f \mapsto \int f \, d\mu$ they induce on $C_c(X)$. This chapter shows that this functional loses nothing. Every positive linear functional on $C_c(X)$ is integration against a measure, and against essentially only one; this is the Riesz-Markov representation theorem, the second of the two bridges between set functions and functionals that this book builds (the first being the Lebesgue-Radon-Nikodym theorem).

The theorem turns the construction of a measure into the construction of an invariant, or otherwise constrained, functional - often a far easier task, since a functional can be built by averaging where a measure cannot. That is exactly the leverage the next chapter needs: on a locally compact group one writes down a left-invariant functional on $C_c(G)$ almost by hand, and Riesz-Markov converts it into the left-invariant measure - Haar measure (chapter 6) - on which all of harmonic analysis rests. The first section assembles the topological tools the representation theorem needs on a locally compact Hausdorff space; the second proves the theorem.

4.1 Radon Measures and $C_c(X)$ on Locally Compact Hausdorff Spaces

This section assembles the three ingredients the representation theorem needs: the spaces it lives on, the test functions that carry the functional, and the measures it produces. The spaces are the locally compact Hausdorff ones, general enough to include $\mathbb{R}^n$, every compact space, and every discrete space, yet rigid enough to give the two technical tools it needs: bump functions and partitions of unity. The test functions are the continuous functions of compact support; the target measures are the Radon measures, singled out by a regularity package that couples a measure to the topology tightly enough for it to be recovered from its values on continuous functions.

The setting is not $\mathbb{R}^n$ but any space in which points can be separated and every point sits inside a compact neighbourhood. Both conditions are needed, and each does distinct work: the Hausdorff axiom makes compact sets closed and lets Urysohn's lemma run, while local compactness gives the compact neighbourhoods that shrink test functions down to compact support.

Definition 4.1.1: Locally Compact Hausdorff Space

A topological space X is *locally compact* if every point has a compact neighbourhood, that is, for each $x \in X$ there is a compact set whose interior contains x. It is *Hausdorff* if any two distinct points have disjoint open neighbourhoods. A *locally compact Hausdorff space* (LCH space) is a space that is both.

The single structural fact drawn from these axioms, used at every turn below, is that compact sets can be surrounded by open sets with compact closure. An open set V is *precompact* if its closure $\overline{V}$ is compact. In a general space precompact opens need not exist in any useful number; in an LCH space they are abundant, and abundant precisely where they are needed.

Proposition 4.1.2: Separating a Compact Set from the Boundary

Let X be an LCH space, $K \subseteq X$ compact, and $U \subseteq X$ open with $K \subseteq U$. Then there is a precompact open set V with

$$K \subseteq V \subseteq \overline{V} \subseteq U$$

***Proof*:**
Every point of K has a compact neighbourhood, so it has an open neighbourhood with compact closure; in a Hausdorff space, shrinking within that closure, one can arrange this precompact neighbourhood to sit inside U. Concretely, fix $x \in K$. Local compactness gives an open $W_x \ni x$ with $\overline{W_x}$ compact. The set $\overline{W_x}$ is a compact Hausdorff space, hence normal, and $\overline{W_x} \cap (X \setminus U)$ is a closed subset of it not containing x; normality separates x from it inside $\overline{W_x}$, producing an open $V_x \ni x$ with $\overline{V_x}$ compact (a closed subset of $\overline{W_x}$) and $\overline{V_x} \subseteq U$. As x ranges over K the sets V_x form an open cover; extract a finite subcover $V_{x_1}, \dots, V_{x_m}$ and set $V = V_{x_1} \cup \cdots \cup V_{x_m}$. Then V is open, $K \subseteq V \subseteq U$, and $\overline{V} = \overline{V_{x_1}} \cup \cdots \cup \overline{V_{x_m}}$ is a finite union of compact sets, hence compact and contained in U, completing the proof.

The point-set groundwork here is exactly the local-compactness layer developed for topological groups in [9]. What that discussion gives, and what proposition 4.1.2 repackages for the present purpose, is the capacity to insert a compact buffer $\overline{V}$ between an arbitrary compact set and any open set containing it. Every construction below threads a function through that buffer.

With the spaces fixed, the test functions come next. The functional in the representation theorem acts on continuous functions, but not all of them: a continuous function on $\mathbb{R}$ can fail to be integrable, and worse, the sup defining the eventual measure would diverge. Restricting to compact support removes both problems at once and keeps the space closed under the operations the proof needs.

Definition 4.1.3: $C_c(X)$ and Support

Let X be a topological space and $f\colon X \to \mathbb{R}$ continuous. The *support* of f is the closure of the set where f is nonzero,

$$\operatorname{supp} f = \overline{\{x \in X : f(x) \neq 0\}}$$

The space $C_c(X)$ consists of all continuous $f\colon X \to \mathbb{R}$ whose support is compact.

When complex scalars are needed, as in Lusin's theorem and the density theorem below, $C_c(X)$ denotes the corresponding space of continuous compactly supported $f\colon X \to \mathbb{C}$. Which space is meant is always determined by the codomain of the functions under discussion, and every statement about the complex space follows from the real case applied to real and imaginary parts.

Two pieces of notation compress the constraints that recur in every argument below. They record, respectively, that a function caps a compact set from above and that a function is squeezed inside an open set.

Definition 4.1.4: The Relations $K \prec f$ and $f \prec U$

Let X be an LCH space, $K \subseteq X$ compact, and $U \subseteq X$ open. For $f \in C_c(X)$:

- write $K \prec f$ if $0 \le f \le 1$ on X and $f \equiv 1$ on K;
- write $f \prec U$ if $0 \le f \le 1$ on X and $\operatorname{supp} f \subseteq U$.

The combined assertion $K \prec f \prec U$ means both hold: $f \in C_c(X)$ takes values in $[0,1]$, equals 1 on K, and is supported in U.

The relation $K \prec f \prec U$ is the exact profile of a *bump function*: a continuous plateau of height 1 over K, tapering to 0 before leaving U, and vanishing outside a compact set. The next lemma asserts that such bumps always exist. On $\mathbb{R}^n$ one can write a bump down by hand, mollifying the indicator of a slightly enlarged K; the content of the lemma is that the topology of an LCH space alone, with no smooth structure and no metric, already forces enough continuous functions into existence.

Lemma 4.1.5: Locally Compact Urysohn Lemma

Let X be an LCH space, $K \subseteq X$ compact, and $U \subseteq X$ open with $K \subseteq U$. Then there exists $f \in C_c(X)$ with

$$K \prec f \prec U$$

***Proof*:**
The strategy is to trap the whole construction inside a single compact Hausdorff space, where the classical Urysohn lemma applies, and then to check that the resulting function extends by zero to a genuine element of $C_c(X)$.

Step 1: Trap K in a compact buffer. By proposition 4.1.2 choose a precompact open V with

$$K \subseteq V \subseteq \overline{V} \subseteq U$$

The closure $\overline{V}$ is compact, and a compact Hausdorff space is normal; this normality is what the next step uses.

Step 2: Apply the normal-space Urysohn lemma on $\overline{V}$. Inside $\overline{V}$ consider the two sets

$$A = K, \qquad B = \overline{V} \setminus V$$

Both are closed in $\overline{V}$: the set K is compact, hence closed in the Hausdorff space $\overline{V}$, and B is the intersection of the closed set $X \setminus V$ with $\overline{V}$. They are disjoint, since $K \subseteq V$ misses $B = \overline{V} \setminus V$. By the normal-space Urysohn lemma ([9, Urysohn, Compact Hausdorff Spaces Are Normal]), there is a continuous $g \colon \overline{V} \to [0,1]$ with $g \equiv 1$ on $A = K$ and $g \equiv 0$ on $B = \overline{V} \setminus V$.

Step 3: Extend by zero. Define $f \colon X \to \mathbb{R}$ by

$$f(x) = \begin{cases} g(x) & x \in \overline{V} \\ 0 & x \in X \setminus \overline{V} \end{cases}$$

This is well-defined: the two clauses are prescribed on the disjoint sets $\overline{V}$ and $X \setminus \overline{V}$, which partition X, so exactly one clause applies at each point. It remains to check that f is continuous, and that $\operatorname{supp} f$ is compact and contained in U.

Continuity. Because g vanishes on $\overline{V} \setminus V$, the function f vanishes on the open set $X \setminus \overline{V}$ as well as on $\overline{V} \setminus V$, hence on all of $X \setminus V$. Thus X is covered by two *closed* sets on each of which f is continuous: on $\overline{V}$, where $f = g$ is continuous, and on $X \setminus V$, where $f \equiv 0$. These two closed sets cover X, since $\overline{V} \cup (X \setminus V) = X$, and f is continuous on each; by the closed pasting lemma (a finite closed cover on each member of which a function is continuous glues to a continuous function) f is continuous on X.

Support. The set $\{f \neq 0\}$ is contained in V, so

$$\operatorname{supp} f = \overline{\{f \neq 0\}} \subseteq \overline{V} \subseteq U$$

As $\operatorname{supp} f$ is a closed subset of the compact set $\overline{V}$, it is compact, so $f \in C_c(X)$. By construction $0 \le f \le 1$ and $f \equiv 1$ on K, so $K \prec f$, and $\operatorname{supp} f \subseteq U$ gives $f \prec U$. Hence $K \prec f \prec U$, as we sought to show.

The lemma is Urysohn's separation theorem transported from normal spaces to LCH spaces, with compact support thrown in for free. The mechanism deserves a second look: local compactness is used exactly once, in proposition 4.1.2, to manufacture the compact buffer $\overline{V}$; once inside that buffer, the argument is purely the theory of normal spaces. This is the reason LCH is the natural home for the representation theorem. An LCH space need not be normal globally, but every compact set sits inside a compact, hence normal, buffer, and that local normality is all the separation the theory ever calls on.

Example 4.1.6: A Bump on the Real Line

Take $X = \mathbb{R}$, $K = [-1, 1]$, and $U = (-2, 2)$. The recipe of lemma 4.1.5 produces a continuous $f \in C_c(\mathbb{R})$ with $[-1, 1] \prec f \prec (-2, 2)$; an explicit representative is the piecewise-linear tent that ramps down to 0 over $[1, \frac{3}{2}]$, well before the endpoints of U,

$$f(x) = \begin{cases} 1 & |x| \leq 1 \\ 3 - 2|x| & 1 \leq |x| \leq \frac{3}{2} \\ 0 & |x| \geq \frac{3}{2} \end{cases}$$

Direct inspection confirms the profile: f is continuous, $0 \leq f \leq 1$, $f \equiv 1$ on $[-1, 1]$, and $\mathrm{supp} f = [-\frac{3}{2}, \frac{3}{2}] \subseteq (-2, 2)$, so indeed $\mathrm{supp} f$ is compact and contained in U. Ramping to 0 strictly inside U is what makes $\mathrm{supp} f \subseteq U$, hence $f \prec U$, hold. Here the precompact buffer of proposition 4.1.2 may be taken as $V = (-\frac{3}{2}, \frac{3}{2})$, with $\overline{V} = [-\frac{3}{2}, \frac{3}{2}]$ the compact interval in which the classical Urysohn lemma operates.

A single bump handles one compact set inside one open set. To integrate a function against a functional, or to compare a functional's values across an open cover, the bump must be replicated across several opens at once and the pieces made to sum to 1. This is the finite partition of unity, and it is the second and last topological tool the chapter needs. Its proof is a bootstrap: shrink the cover so that the closures stay inside the original opens, bump each shrunken piece, and normalise.

Lemma 4.1.7: Finite Partition of Unity

Let X be an LCH space and $K \subseteq X$ compact, covered by finitely many open sets $U_1, \ldots, U_n$. Then there exist $h_1, \ldots, h_n \in C_c(X)$ with $h_i \prec U_i$ for each i, such that

$$\sum_{i=1}^{n} h_i \equiv 1 \quad \text{on } K, \qquad 0 \leq \sum_{i=1}^{n} h_i \leq 1 \quad \text{on } X$$

The collection $\{h_i\}$ is called a *partition of unity on K subordinate to the cover $\{U_i\}$*.

***Proof*:**

Step 1: Shrink the cover. Each point $x \in K$ lies in some U_i, and by proposition 4.1.2 applied to the compact set $\{x\}$ and the open set U_i, it has a precompact open neighbourhood W_x with $\overline{W_x}$ compact and $\overline{W_x} \subseteq U_i$. The sets $\{W_x\}_{x \in K}$ cover the compact set K, so finitely many $W_{x_1}, \ldots, W_{x_m}$ suffice. For each index i let

$$K_i = \bigcup \{ \overline{W_{x_j}} : \overline{W_{x_j}} \subseteq U_i \}$$

a finite union of compact sets, hence compact, with $K_i \subseteq U_i$. Every $\overline{W_{x_j}}$ lands in at least one U_i, so $K \subseteq \bigcup_j \overline{W_{x_j}} \subseteq \bigcup_i K_i$; thus $\{K_i\}$ is a compact refinement with $K_i \subseteq U_i$ covering K.

Step 2: Bump each piece. By the locally compact Urysohn lemma lemma 4.1.5, for each i choose $g_i \in C_c(X)$ with

$$K_i \prec g_i \prec U_i$$

Then $g_i \equiv 1$ on K_i, $0 \leq g_i \leq 1$, and $\mathrm{supp} g_i \subseteq U_i$. Since $K \subseteq \bigcup_i K_i$, at every point of K some g_i equals 1, so the sum $\sum_i g_i \geq 1$ on K.

Step 3: Normalise, carefully at the boundary. The naive normalisation $g_i / \sum_j g_j$ fails where the denominator vanishes, so first separate K from that danger. Set $g = \sum_{i=1}^n g_i$; it is continuous, and $g \geq 1$ on K, so K is contained in the open set $\{g > 0\}$. Apply lemma 4.1.5 once more to obtain $g_0 \in C_c(X)$ with

$$K \prec g_0 \prec \{g > 0\}$$

Now $g + (1 - g_0)$ is strictly positive everywhere. Fix $x \in X$. If $g(x) > 0$ the sum is positive because $1 - g_0 \geq 0$. If instead $g(x) = 0$, then $x \notin \{g > 0\}$, and since $\operatorname{supp} g_0 \subseteq \{g > 0\}$ this forces $g_0(x) = 0$, so $1 - g_0(x) = 1 > 0$. Either way $g(x) + (1 - g_0(x)) > 0$, so the denominator below never vanishes. Define

$$h_i = \frac{g_i}{g + (1 - g_0)} \qquad (i = 1, \ldots, n)$$

Each h_i is continuous, since the denominator never vanishes, and $\operatorname{supp} h_i \subseteq \operatorname{supp} g_i \subseteq U_i$ with $\operatorname{supp} h_i$ compact, so $h_i \prec U_i$. On K one has $g_0 = 1$, whence the denominator equals g, and therefore

$$\sum_{i=1}^n h_i = \frac{\sum_i g_i}{g} = \frac{g}{g} = 1 \quad \text{on } K$$

On all of X, $\sum_i h_i = g/(g + (1 - g_0)) \leq 1$ because $1 - g_0 \geq 0$, and $\sum_i h_i \geq 0$ since each $h_i \geq 0$. Thus $0 \leq \sum_i h_i \leq 1$ throughout and $\sum_i h_i \equiv 1$ on K, completing the proof.

The partition of unity converts a global statement about K into a sum of local statements, one per open U_i. Writing $\sum_i h_i \equiv 1$ on K decomposes the constant function 1 (restricted to K) into pieces h_i each confined to a single U_i, so any object that behaves well on each U_i separately can be reassembled over K by summing the h_i against it. In the representation theorem this is precisely how a functional's local behaviour, controlled by a single bump at a time, is stitched into a global integral; the partition is the instrument that turns lemma 4.1.5 from a pointwise into a global tool.

Example 4.1.8: Splitting the Interval $[0, 1]$

Cover the compact set $K = [0, 1] \subseteq \mathbb{R}$ by the two opens $U_1 = (-1, \frac{2}{3})$ and $U_2 = (\frac{1}{3}, 2)$. Confining the ramps to the transition window $[\frac{5}{12}, \frac{7}{12}]$, which sits strictly inside the overlap $U_1 \cap U_2 = (\frac{1}{3}, \frac{2}{3})$, keeps each support clear of the boundaries of its U_i. An explicit subordinate partition of unity is

$$h_1(x) = \begin{cases} 1 & 0 \leq x \leq \frac{5}{12} \\ 6(\frac{7}{12} - x) & \frac{5}{12} \leq x \leq \frac{7}{12} \\ 0 & \text{otherwise} \end{cases} \qquad h_2(x) = \begin{cases} 6(x - \frac{5}{12}) & \frac{5}{12} \leq x \leq \frac{7}{12} \\ 1 & \frac{7}{12} \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Both are continuous with $\operatorname{supp} h_1 = [0, \frac{7}{12}] \subseteq U_1$ and $\operatorname{supp} h_2 = [\frac{5}{12}, 1] \subseteq U_2$ (each support lands strictly inside its open, since $\frac{7}{12} < \frac{2}{3}$ and $\frac{5}{12} > \frac{1}{3}$), so $h_1 \prec U_1$ and $h_2 \prec U_2$, and both lie in $C_c(\mathbb{R})$. On the transition window $[\frac{5}{12}, \frac{7}{12}]$ one computes $h_1(x) + h_2(x) = 6(\frac{7}{12} - x) + 6(x - \frac{5}{12}) = 1$, while on $[0, \frac{5}{12}]$ the sum is $1 + 0$ and on $[\frac{7}{12}, 1]$ it is $0 + 1$; thus $h_1 + h_2 \equiv 1$ on all of $K = [0, 1]$. Off $[0, 1]$ neither bump exceeds 1 and both taper to 0, so $0 \leq h_1 + h_2 \leq 1$ everywhere.

The last ingredient is the class of measures the representation theorem produces. A positive functional on $C_c(X)$ carries only as much information as continuous functions can see, so the measure it represents cannot be an arbitrary Borel measure; it must be one that its values on open and compact sets already determine. The regularity conditions below encode exactly that determinacy. Outer regularity computes any Borel set's measure from the open sets above it, inner regularity computes an open set's measure from the compact sets below it, and finiteness on compacts keeps the functional's values finite. Together they define a Radon measure.

Definition 4.1.9: Radon Measure

Let X be an LCH space and μ a Borel measure on X, that is, a measure defined on the Borel σ-algebra of X. Call μ a *Radon measure* if it satisfies:

1. *Finite on compacts:* $\mu(K) < \infty$ for every compact $K \subseteq X$;
2. *Outer regular on Borel sets:* for every Borel set E,
$$\mu(E) = \inf\{\mu(U) \mid E \subseteq U,\ U \text{ open}\}$$
3. *Inner regular on open sets:* for every open set U,
$$\mu(U) = \sup\{\mu(K) \mid K \subseteq U,\ K \text{ compact}\}$$

The asymmetry between conditions (2) and (3) is deliberate and worth dwelling on. Outer regularity is demanded on all Borel sets, but inner regularity only on open sets; requiring inner regularity everywhere would exclude some legitimate measures on pathological non-σ-finite spaces, whereas the open-set version is exactly what the representation theorem delivers and exactly what its applications use. On well-behaved spaces the gap closes: when X is σ-compact, a Radon measure is automatically inner regular on all Borel sets of finite measure, so the distinction is invisible on $\mathbb{R}^n$ and every space encountered in practice. The definition is stated at its most general so that the representation theorem can assert uniqueness without a side hypothesis.

The measures built earlier in this book from distribution functions are the prototype, and they are already Radon; nothing new need be checked, since the two regularity axioms were established for them in the chapters preceding chapter 4.

Example 4.1.10: Lebesgue-Stieltjes Measures Are Radon

Let $F\colon \mathbb{R} \to \mathbb{R}$ be increasing and right-continuous, and let μ_F be the Borel measure it defines by theorem 1.4.2, characterised by $\mu_F((a,b]) = F(b) - F(a)$. Then μ_F is a Radon measure on $\mathbb{R}$.

Each Radon axiom is a result already in hand. *Finiteness on compacts:* a compact $K \subseteq \mathbb{R}$ lies in some bounded interval $(a,b]$, so $\mu_F(K) \le \mu_F((a,b]) = F(b) - F(a) < \infty$. *Outer and inner regularity:* these are precisely the two displayed identities of theorem 1.4.4, which computes $\mu_F(E) = \inf\{\mu_F(U) : U \supseteq E \text{ open}\}$ for every measurable E and $\mu_F(E) = \sup\{\mu_F(K) : K \subseteq E \text{ compact}\}$; the first is outer regularity on Borel sets, and the second gives inner regularity in particular on open sets. Thus every Lebesgue-Stieltjes measure is Radon, and taking $F(x) = x$ recovers Lebesgue measure on $\mathbb{R}$ as the archetypal Radon measure.

The example closes the loop with which chapter 4 opened. The question posed there, which set functions on a space deserve to be called measures, was answered on $\mathbb{R}$ by the distribution-function construction; the Radon axioms lift that answer to any LCH space by isolating the regularity that the $\mathbb{R}$ construction enjoyed automatically. What remains is to show that these axioms are not merely a convenient bookkeeping class but the exact image of the positive functionals on $C_c(X)$, which is the content of section 4.2.

A Radon measure was required to be inner regular only on open sets. On the σ-finite part of the space that extends to every Borel set, and it is this stronger regularity that the applications of the representation theorem draw on.

Proposition 4.1.11: Inner Regularity on σ-Finite Sets

Every Radon measure μ on an LCH space X is inner regular on each of its σ-finite Borel sets: if E is a Borel set that is σ-finite for μ, then

$$\mu(E) = \sup\{\mu(K) \mid K \subseteq E,\ K \text{ compact}\}$$

***Proof*:**
Consider first the case $\mu(E) < \infty$. Fix $\varepsilon > 0$. Outer regularity gives an open $U \supseteq E$ with $\mu(U) < \mu(E) + \varepsilon$, and inner regularity on the open set U gives a compact $F \subseteq U$ with $\mu(F) > \mu(U) - \varepsilon$. The set F need not lie in E, but its overspill sits in $U \setminus E$, a Borel set of measure $\mu(U \setminus E) = \mu(U) - \mu(E) < \varepsilon$. Outer regularity applied to $U \setminus E$ produces an open $V \supseteq U \setminus E$ with $\mu(V) < \varepsilon$. Put $K = F \setminus V$: it is a closed subset of the compact F, hence compact, and $K \subseteq E$ because everything of F outside E was covered by V. Finally, using $\mu(F) > \mu(U) - \varepsilon \geq \mu(E) - \varepsilon$,

$$\mu(K) = \mu(F) - \mu(F \cap V) \geq \mu(F) - \mu(V) > (\mu(E) - \varepsilon) - \varepsilon = \mu(E) - 2\varepsilon$$

As $K \subseteq E$ is compact and ε is arbitrary, the supremum of $\mu(K)$ over compact $K \subseteq E$ equals $\mu(E)$.

If instead $\mu(E) = \infty$, write $E = \bigcup_n E_n$ as an increasing union of Borel sets with $\mu(E_n) < \infty$; then $\mu(E_n) \to \infty$ by continuity from below. Given any N, choose n with $\mu(E_n) > N$; the finite case gives a compact $K \subseteq E_n \subseteq E$ with $\mu(K) > N$. Hence the supremum is infinite, matching $\mu(E)$.

Corollary 4.1.12: Regularity of σ-Finite Radon Measures

Every σ-finite Radon measure is regular, that is, outer and inner regular on all Borel sets. In particular, if X is σ-compact then every Radon measure on X is regular.

***Proof*:**
Outer regularity on all Borel sets is part of definition 4.1.9, and when μ is σ-finite every Borel set is σ-finite, so proposition 4.1.11 gives inner regularity on all Borel sets. If $X = \bigcup_n K_n$ is σ-compact with each K_n compact, then $\mu(K_n) < \infty$ by finiteness on compacts, so μ is σ-finite and the first statement applies.

4.2 The Riesz-Markov Representation Theorem

The bump functions and partitions of unity of section 4.1 are the tools; the measure they build is the payoff. A positive linear functional I on $C_c(X)$ assigns a number to each compactly supported continuous function, monotonically in the sense that $f \leq g$ forces $I(f) \leq I(g)$. Integration against a Radon measure is the obvious example: $f \mapsto \int_X f\,d\mu$ is linear, and it is positive because $f \geq 0$ makes the integrand nonnegative. The content of this section is that there are no other examples. Every positive functional on $C_c(X)$ integrates against one Radon measure, and against exactly one. This converts the analytic datum I, a rule for averaging functions, into the geometric datum μ, a rule for sizing sets, and it is the mechanism by which the construction of Haar measure in chapter 6 will produce a measure out of the invariance of an averaging operation.

The functional records only what the measure does to continuous functions, so the measure must be reconstructed from that restriction. The reconstruction runs through an outer measure: define the size of an open set as the largest mass I can place inside it, extend to all sets by outer approximation, and appeal to the Carathéodory machine of theorem 1.3.4 to cut down to a σ-algebra on which the result is a genuine measure. The work is then to check that this measure is Radon, that it represents I, and that it is forced.

Theorem 4.2.1: Riesz-Markov Representation

Let X be a locally compact Hausdorff space and let $I\colon C_c(X) \to \mathbb{R}$ be a positive linear functional, so that $I(f) \geq 0$ whenever $f \geq 0$. Then there is a unique Radon measure μ on X such that

$$I(f) = \int_X f\,d\mu$$

for every $f \in C_c(X)$.

The proof occupies the remainder of the section. Existence is a construction in five stages: build a candidate set function, promote it to an outer measure, verify Carathéodory measurability of the open sets, extract the regularity that makes it Radon, and finally match I against integration. Uniqueness is a short argument once the regularity is in hand.

***Proof*:**
Write $C_c^+(X)$ for the nonnegative functions in $C_c(X)$. Throughout, $f \prec U$ means $f \in C_c^+(X)$ with $0 \leq f \leq 1$ and $\operatorname{supp} f \subseteq U$, and $K \prec f$ means $f \in C_c^+(X)$ with $0 \leq f \leq 1$ and $f \equiv 1$ on K, as fixed in section 4.1.

Step 1: The candidate set function. For $U \subseteq X$ open define

$$\mu(U) = \sup\{I(f) \mid f \prec U\} \tag{4.1}$$

and for an arbitrary $E \subseteq X$ define

$$\mu^*(E) = \inf\{\mu(U) \mid E \subseteq U,\ U \text{ open}\} \tag{4.2}$$

Two consistency points must be settled before μ^* can be used. First, the supremum in (4.1) is over a nonempty set: the zero function satisfies $0 \prec U$, so $\mu(U) \geq I(0) = 0$, and $\mu(\emptyset) = 0$ because $f \prec \emptyset$ forces $f = 0$. Second, if U is itself open then the two definitions agree. Indeed $U \subseteq U$ is an admissible cover in (4.2), so $\mu^*(U) \leq \mu(U)$; and if $U \subseteq V$ with V open, then any $f \prec U$ also satisfies $f \prec V$, whence $\mu(U) \leq \mu(V)$, so $\mu(U) \leq \inf_V \mu(V) = \mu^*(U)$. Thus $\mu^*(U) = \mu(U)$ for open U, and no ambiguity arises from writing μ for both. The same monotonicity of (4.1) in the open set will be used repeatedly: $U \subseteq V$ open implies $\mu(U) \leq \mu(V)$.

Step 2: μ^* is an outer measure. The three axioms of definition 1.3.1 must be verified. That $\mu^*(\emptyset) = 0$ is immediate from $\mu(\emptyset) = 0$. Monotonicity is equally direct: if $A \subseteq B$ then every open $U \supseteq B$ also contains A, so the infimum in (4.2) defining $\mu^*(A)$ is over a larger collection and hence no larger, giving $\mu^*(A) \le \mu^*(B)$.

The one substantial axiom is countable subadditivity. The heart of it is the open case, where the partition of unity enters. Let $U = \bigcup_{j=1}^{\infty} U_j$ with each U_j open. The claim is

$$\mu(U) \le \sum_{j=1}^{\infty} \mu(U_j) \tag{4.3}$$

To see this, take any $f \prec U$. Its support $K = \operatorname{supp} f$ is compact and contained in $\bigcup_j U_j$, so by compactness finitely many of the U_j already cover K, say $K \subseteq U_1 \cup \cdots \cup U_n$ after relabelling. By lemma 4.1.7 there are functions $h_i \prec U_i$ for $i = 1, \ldots, n$ with $\sum_{i=1}^{n} h_i \equiv 1$ on K. Since f is supported in K, on all of X one has $f = \sum_{i=1}^{n} f h_i$: at a point of K this is $f \cdot 1$, and off K every term vanishes because f does. Each product fh_i lies in $C_c^+(X)$, is bounded by $h_i \le 1$, and is supported in $\operatorname{supp} h_i \subseteq U_i$, so $fh_i \prec U_i$. Linearity of I then gives

$$I(f) = \sum_{i=1}^{n} I(fh_i) \le \sum_{i=1}^{n} \mu(U_i) \le \sum_{j=1}^{\infty} \mu(U_j)$$

where the first inequality is (4.1) applied to each $fh_i \prec U_i$. Taking the supremum over $f \prec U$ yields (4.3).

The passage to arbitrary sets is now formal. Let $\{E_j\}_{j=1}^{\infty}$ be any sets and put $E = \bigcup_j E_j$. If some $\mu^*(E_j) = \infty$ the inequality $\mu^*(E) \le \sum_j \mu^*(E_j)$ is trivial, so assume every $\mu^*(E_j)$ is finite. Fix $\varepsilon > 0$. By (4.2) choose for each j an open $U_j \supseteq E_j$ with

$$\mu(U_j) \le \mu^*(E_j) + \varepsilon 2^{-j}$$

Then $U = \bigcup_j U_j$ is open and contains E, so by (4.2), (4.3), and the choice of the U_j,

$$\mu^*(E) \le \mu(U) \le \sum_{j=1}^{\infty} \mu(U_j) \le \sum_{j=1}^{\infty} \mu^*(E_j) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, $\mu^*(E) \le \sum_j \mu^*(E_j)$. Thus μ^* is an outer measure.

Step 3: Open sets are Carathéodory-measurable. By definition 1.3.3 a set E is μ^*-measurable exactly when

$$\mu^*(S) \ge \mu^*(S \cap E) + \mu^*(S \setminus E) \quad \text{for every } S \subseteq X \tag{4.4}$$

the reverse inequality being automatic from subadditivity. It suffices to establish (4.4) for every open E, and then only for test sets S with $\mu^*(S) < \infty$, since (4.4) is trivial when $\mu^*(S) = \infty$.

Fix an open E and a test set S with $\mu^*(S) < \infty$. Let $\varepsilon > 0$. By (4.2) choose an open $U \supseteq S$ with $\mu(U) \le \mu^*(S) + \varepsilon$. It is enough to bound $\mu^*(U \cap E) + \mu^*(U \setminus E)$, because $S \cap E \subseteq U \cap E$

and $S \setminus E \subseteq U \setminus E$ and μ^* is monotone. The set $U \cap E$ is open, so by (4.1) there is $f \prec U \cap E$ with

$$I(f) > \mu(U \cap E) - \varepsilon \tag{4.5}$$

Now $K = \operatorname{supp} f$ is a compact subset of $U \cap E$, so $U \setminus K$ is open and contains $U \setminus E$. Apply (4.1) again on this open set: there is $g \prec U \setminus K$ with

$$I(g) > \mu(U \setminus K) - \varepsilon \geq \mu^*(U \setminus E) - \varepsilon \tag{4.6}$$

the last inequality by monotonicity, since $U \setminus E \subseteq U \setminus K$ (which holds because $K \subseteq U \cap E \subseteq E$). The two bump functions have disjoint supports: $\operatorname{supp} f \subseteq K$ while $\operatorname{supp} g \subseteq U \setminus K$. Consequently $f + g$ takes values in $[0, 1]$, because at any point at most one summand is nonzero, and $\operatorname{supp}(f + g) \subseteq U$, so $f + g \prec U$. Therefore $I(f) + I(g) = I(f + g) \leq \mu(U)$ by (4.1). Using $\mu^*(U \cap E) = \mu(U \cap E)$ from Step 1 and then (4.5), (4.6),

$$\mu^*(U \cap E) + \mu^*(U \setminus E) = \mu(U \cap E) + \mu^*(U \setminus E) < \big(I(f) + \varepsilon\big) + \big(I(g) + \varepsilon\big) \leq \mu(U) + 2\varepsilon$$

Since $\mu(U) \leq \mu^*(S) + \varepsilon$ and $S \cap E \subseteq U \cap E$, $S \setminus E \subseteq U \setminus E$, monotonicity gives

$$\mu^*(S \cap E) + \mu^*(S \setminus E) \leq \mu(U) + 2\varepsilon \leq \mu^*(S) + 3\varepsilon$$

Letting $\varepsilon \to 0$ yields (4.4). Hence every open set is μ^*-measurable.

The μ^*-measurable sets form a σ-algebra containing every open set, so they contain the σ-algebra generated by the open sets, that is, the Borel sets. By theorem 1.3.4 the collection $\mathcal{M}$ of μ^*-measurable sets is a σ-algebra and $\mu = \mu^*|_{\mathcal{M}}$ is a complete measure. Restricting attention to the Borel σ-algebra $\mathcal{B}_X \subseteq \mathcal{M}$, the resulting μ is a Borel measure. From now on μ denotes this measure; (4.1) continues to compute it on open sets by Step 1.

Step 4: μ is a Radon measure. Three properties from definition 4.1.9 require checking: finiteness on compact sets, outer regularity on Borel sets, and inner regularity on open sets.

For finiteness on compacts, the following bound does the work: for any compact K and any $\varphi \in C_c^+(X)$ with $\varphi \geq 1$ on K,

$$\mu(K) \leq I(\varphi) \tag{4.7}$$

To prove (4.7), put $U_\alpha = \{x \in X \mid \varphi(x) > \alpha\}$ for $0 < \alpha < 1$, an open set containing K because $\varphi \geq 1 > \alpha$ there. If $g \prec U_\alpha$, then $g \leq \alpha^{-1}\varphi$ everywhere: on U_α one has $\alpha^{-1}\varphi > 1 \geq g$, while off U_α the function g vanishes because $\operatorname{supp} g \subseteq U_\alpha$. Positivity of I then gives $I(g) \leq \alpha^{-1} I(\varphi)$. Taking the supremum over $g \prec U_\alpha$ and using monotonicity of μ,

$$\mu(K) \leq \mu(U_\alpha) \leq \alpha^{-1} I(\varphi)$$

Letting $\alpha \to 1$ yields (4.7). In particular, for a given compact K, lemma 4.1.5 gives some f with $K \prec f$, and then $f \geq 1$ on K gives $\mu(K) \leq I(f) < \infty$. Thus μ is finite on every compact set.

Outer regularity is built into the construction. For a Borel set E, definition (4.2) is exactly $\mu(E) = \inf\{\mu(U) \mid E \subseteq U,\ U \text{ open}\}$. Thus μ is outer regular on all Borel sets.

Inner regularity on open sets is the content of (4.1). Let U be open and let $c < \mu(U)$. By (4.1) there is $f \prec U$ with $I(f) > c$. Set $K = \operatorname{supp} f$, a compact subset of U. For any open V with $K \subseteq V$ one has $f \prec V$, so $I(f) \leq \mu(V)$; taking the infimum over such V and using $\mu(K) = \inf\{\mu(V) \mid K \subseteq V,\ V \text{ open}\}$ from (4.2) gives $I(f) \leq \mu(K)$. Hence $\mu(K) \geq I(f) > c$ with $K \subseteq U$ compact. Since $c < \mu(U)$ was arbitrary,

$$\mu(U) = \sup\{\mu(K) \mid K \subseteq U,\ K \text{ compact}\}$$

so μ is inner regular on open sets. Therefore μ is a Radon measure.

Step 5: $I(f) = \int_X f\,d\mu$ for all $f \in C_c(X)$. It suffices to prove

$$I(f) \leq \int_X f\,d\mu \quad \text{for every real } f \in C_c(X) \tag{4.8}$$

for then applying (4.8) to $-f$ gives $-I(f) \leq -\int_X f\,d\mu$, hence the reverse inequality, and the two together give equality. The complex case follows by taking real and imaginary parts.

Fix a real $f \in C_c(X)$ and let $K = \operatorname{supp} f$. Since f is continuous with compact support it is bounded, say $f(X) \subseteq [a, b]$ with $a < b$. Fix $\varepsilon > 0$ and choose real numbers

$$y_0 < a < y_1 < y_2 < \cdots < y_n = b$$

with $y_i - y_{i-1} < \varepsilon$ for each i and with $y_0 < a$ so that $f(x) > y_0$ for all x. Define the Borel sets

$$E_i = \{x \in K \mid y_{i-1} < f(x) \leq y_i\}, \qquad i = 1, \ldots, n$$

These are disjoint and their union is K, since every $x \in K$ has $a < f(x) \leq b$ and so falls in exactly one band. Each E_i is the intersection of K with the preimage of a half-open interval under the continuous f, hence Borel.

The plan is to trap f between step functions on the E_i and to compare I and the integral band by band. By outer regularity choose open sets $U_i \supseteq E_i$ with

$$\mu(U_i) < \mu(E_i) + \frac{\varepsilon}{n} \tag{4.9}$$

and, by shrinking, arrange in addition that $f(x) < y_i + \varepsilon$ for all $x \in U_i$: this is possible because $f < y_i + \varepsilon$ on the open set $\{f < y_i + \varepsilon\} \supseteq E_i$, and intersecting the chosen U_i with this open set preserves both $E_i \subseteq U_i$ and (4.9). The sets $U_1, \ldots, U_n$ cover the compact set K, so by lemma 4.1.7 there are $h_i \prec U_i$ with $\sum_{i=1}^n h_i \equiv 1$ on K. Since $\operatorname{supp} f = K$, exactly as in Step 2 one has $f = \sum_{i=1}^n f h_i$ on all of X.

Now estimate $I(f)$ from above. The function $\varphi = \sum_{i=1}^n h_i$ lies in $C_c^+(X)$ and equals 1 on K, so (4.7) applies to it. By linearity $I(\varphi) = \sum_i I(h_i)$, hence

$$\mu(K) \leq \sum_{i=1}^{n} I(h_i) \tag{4.10}$$

On U_i the function f is bounded above by $y_i + \varepsilon$, and h_i is supported in U_i with $0 \leq h_i \leq 1$, so $fh_i \leq (y_i + \varepsilon)h_i$ pointwise. Monotonicity and linearity of I then give $I(fh_i) \leq (y_i + \varepsilon)I(h_i)$, for either sign of $y_i + \varepsilon$. Summing over i,

$$I(f) = \sum_{i=1}^{n} I(fh_i) \leq \sum_{i=1}^{n} (y_i + \varepsilon) I(h_i)$$

The bound $I(h_i) \leq \mu(U_i)$ from $h_i \prec U_i$ can only be inserted where the coefficient of $I(h_i)$ is nonnegative, so shift by the constant $|a| + \varepsilon$: write $y_i + \varepsilon = (y_i + \varepsilon + |a| + \varepsilon) - (|a| + \varepsilon)$, where the first factor is nonnegative because $y_i > a \geq -|a|$. The subtracted constant contributes $-(|a| + \varepsilon)\sum_i I(h_i) \leq -(|a| + \varepsilon)\mu(K)$ by (4.10), since $|a| + \varepsilon \geq 0$. The nonnegative first factor multiplies $I(h_i) \leq \mu(U_i)$ termwise. Carrying this out,

$$\begin{aligned} I(f) &\leq \sum_{i=1}^{n} (y_i + \varepsilon + |a| + \varepsilon)\, \mu(U_i) - (|a| + \varepsilon)\, \mu(K) \\ &< \sum_{i=1}^{n} (y_i + \varepsilon + |a| + \varepsilon) \left(\mu(E_i) + \frac{\varepsilon}{n} \right) - (|a| + \varepsilon)\, \mu(K) \end{aligned}$$

Since $\bigsqcup_i E_i = K$, the terms carrying the constant $|a| + \varepsilon$ telescope: $\sum_i (|a| + \varepsilon)\mu(E_i) = (|a| + \varepsilon)\mu(K)$, cancelling the final term up to the ε/n remainders. Expanding and regrouping,

$$I(f) \leq \sum_{i=1}^{n} (y_i + \varepsilon)\, \mu(E_i) + \frac{\varepsilon}{n} \sum_{i=1}^{n} (y_i + \varepsilon + |a| + \varepsilon)$$

The second sum is a remainder. Assuming $\varepsilon \leq 1$, each term satisfies $y_i + \varepsilon + |a| + \varepsilon \leq b + |a| + 2$, so summing n terms and multiplying by ε/n bounds the remainder by $C\varepsilon$ with $C = b + |a| + 2$, a constant depending only on f through the fixed interval $[a, b]$. For the main term, $x \in E_i$ gives $y_i < y_{i-1} + \varepsilon < f(x) + \varepsilon$, hence $y_i + \varepsilon < f(x) + 2\varepsilon$ on E_i. Since $y_i + \varepsilon$ is constant on E_i, integrating this pointwise bound gives

$$\sum_{i=1}^{n} (y_i + \varepsilon)\, \mu(E_i) \leq \sum_{i=1}^{n} \int_{E_i} (f + 2\varepsilon)\, d\mu = \int_K f\, d\mu + 2\varepsilon\, \mu(K) = \int_X f\, d\mu + 2\varepsilon\, \mu(K)$$

where $\int_K f\, d\mu = \int_X f\, d\mu$ because f vanishes off K. Collecting the two bounds,

$$I(f) \leq \int_X f\, d\mu + 2\varepsilon\, \mu(K) + C\varepsilon$$

Both $\mu(K) < \infty$ and C are finite and independent of ε, so letting $\varepsilon \to 0$ gives (4.8). This establishes $I(f) = \int_X f\, d\mu$ for all $f \in C_c(X)$, completing the existence half.

Step 6: Uniqueness. Suppose μ and ν are Radon measures with $\int_X f\,d\mu = I(f) = \int_X f\,d\nu$ for all $f \in C_c(X)$. It suffices to show $\mu(K) = \nu(K)$ for every compact K. Indeed, each open U is inner regular, $\mu(U) = \sup\{\mu(K) \mid K \subseteq U,\ K \text{ compact}\}$, so agreement on compacts forces $\mu(U) = \nu(U)$ on open sets; and each Borel E is outer regular, $\mu(E) = \inf\{\mu(U) \mid E \subseteq U,\ U \text{ open}\}$, so agreement on open sets forces $\mu(E) = \nu(E)$ on all Borel sets.

Fix a compact K and $\varepsilon > 0$. By outer regularity of ν choose an open $U \supseteq K$ with $\nu(U) < \nu(K)+\varepsilon$; shrinking U if necessary, arrange $\overline{U}$ compact using local compactness. By lemma 4.1.5 take $f \in C_c(X)$ with $K \prec f \prec U$, so $\chi_K \leq f \leq \chi_U$ pointwise. Then

$$\mu(K) = \int_X \chi_K\,d\mu \leq \int_X f\,d\mu = \int_X f\,d\nu \leq \int_X \chi_U\,d\nu = \nu(U) < \nu(K) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, $\mu(K) \leq \nu(K)$. Interchanging the roles of μ and ν gives $\nu(K) \leq \mu(K)$, hence $\mu(K) = \nu(K)$. By the reduction above, μ and ν agree on every open set and therefore on every Borel set. The Radon measure representing I is unique, completing the proof.

The theorem translates between two descriptions of the same data. A positive functional is an averaging rule with no set-theoretic content; a Radon measure is an assignment of size to sets. The construction shows that the functional already encodes the measure: the value $\mu(U) = \sup\{I(f) \mid f \prec U\}$ reads off the largest averaged mass I can concentrate inside U, and outer approximation then sizes every set. Regularity is not an extra hypothesis imposed on the output but the exact structure the construction produces, which is why the representing measure comes out Radon rather than merely Borel. Uniqueness records the converse, that a Radon measure is determined by its action on $C_c(X)$: two Radon measures agreeing on all continuous compactly supported functions agree on compact sets by squeezing χ_K between bump functions, and regularity carries that agreement to the whole Borel σ-algebra.

The identification is only as sharp as the class of measures it ranges over. Positivity of I is what forces the candidate to be a measure at all, since it is what makes (4.1) monotone in the open set. Restricting to Radon measures is what makes the correspondence a bijection: without regularity, distinct Borel measures can integrate every $f \in C_c(X)$ to the same value, so the measure is recovered only up to its Radon regularization. On a space where the topology already tames measures, this subtlety collapses. When X is second countable and locally compact, every Borel measure finite on compacts is automatically outer and inner regular, so the Radon qualifier is redundant and $I \mapsto \mu$ is a bijection onto all locally finite Borel measures.

Example 4.2.2: Lebesgue Measure from the Riemann Integral

Take $X = \mathbb{R}$ with its usual topology, a locally compact Hausdorff space, and let I be the Riemann integral,

$$I(f) = \int_{-\infty}^{\infty} f(x)\,dx$$

which is defined on every $f \in C_c(\mathbb{R})$ because such an f is continuous and supported in a compact interval, hence Riemann integrable there. Linearity is standard, and positivity holds because a nonnegative continuous function has nonnegative Riemann integral. Theorem 4.2.1 produces a unique Radon measure μ on $\mathbb{R}$ with $\int_{\mathbb{R}} f\,d\mu = \int_{-\infty}^{\infty} f(x)\,dx$ for all $f \in C_c(\mathbb{R})$.

This μ is Lebesgue measure. To read off its value on an interval $[a,b]$, sandwich the indicator $\chi_{[a,b]}$ between two continuous trapezoids. For small $\delta > 0$ with $2\delta < b-a$, let $f_\delta \in C_c(\mathbb{R})$ equal 1 on $[a,b]$, equal 0 outside $(a-\delta, b+\delta)$, and interpolate linearly on the two flanking intervals of width δ, and let $g_\delta \in C_c(\mathbb{R})$ equal 1 on $[a+\delta, b-\delta]$, equal 0 outside (a,b), and interpolate linearly on the two intervals of width δ. Then $g_\delta \leq \chi_{[a,b]} \leq f_\delta$ pointwise, so monotonicity of the integral gives

$$\int_{\mathbb{R}} g_\delta \, d\mu \leq \mu([a,b]) \leq \int_{\mathbb{R}} f_\delta \, d\mu$$

The two integrals are Riemann integrals by the representation. Computing each trapezoid's area as a central rectangle plus two triangles,

$$\int_{-\infty}^{\infty} g_\delta(x)\, dx = (b-a) - \delta, \qquad \int_{-\infty}^{\infty} f_\delta(x)\, dx = (b-a) + \delta$$

since f_δ adds two triangles of combined area δ to the rectangle $b-a$, while g_δ removes the same. Hence $(b-a) - \delta \leq \mu([a,b]) \leq (b-a) + \delta$, and letting $\delta \to 0$ gives $\mu([a,b]) = b-a$. Thus μ agrees with the Lebesgue measure built in theorem 1.4.2 from the distribution function $F(x) = x$. The Riesz-Markov route recovers Lebesgue measure from the bare positivity of the Riemann integral, with no explicit outer-measure construction on intervals: the general theorem does that work once and for all.

The same regularity that pins down the correspondence also lets an arbitrary measurable function be edited, off a set of small measure, into a continuous one. This is Lusin's theorem, the precise form of the slogan that "measurable functions are nearly continuous."

Theorem 4.2.3: Lusin's Theorem

Let X be a locally compact Hausdorff space, μ a Radon measure on X, and $f\colon X \to \mathbb{C}$ a measurable function vanishing outside a set of finite measure. For every $\varepsilon > 0$ there is $g \in C_c(X)$ with

$$\mu\big(\{x \in X \mid f(x) \neq g(x)\}\big) < \varepsilon,$$

and g may be taken with $\|g\|_\infty \leq \|f\|_\infty$.

***Proof*:**
Let $A = \{x \mid f(x) \neq 0\}$, so $\mu(A) < \infty$. By inner regularity (corollary 4.1.12) fix a compact K_0 with $\mu(A \setminus K_0) < \varepsilon/2$ and a relatively compact open $U \supseteq K_0$; it suffices to produce $g \in C_c(X)$ supported in $\overline{U}$ agreeing with f off a subset of K_0 of measure $< \varepsilon/2$. Decomposing f into real and imaginary, positive and negative parts and rescaling, we may assume $0 \leq f < 1$ on K_0 and write the binary expansion

$$f = \sum_{n=1}^{\infty} 2^{-n} \mathbf{1}_{T_n}, \qquad T_n \subseteq K_0 \text{ measurable.}$$

For each n, regularity (corollary 4.1.12) gives a compact K_n and open V_n with $K_n \subseteq T_n \subseteq V_n \subseteq U$ and $\mu(V_n \setminus K_n) < \varepsilon\, 2^{-n-1}$, and the locally compact Urysohn lemma (lemma 4.1.5) gives $h_n \in C_c(X)$ with $\mathbf{1}_{K_n} \leq h_n \leq \mathbf{1}_{V_n}$. The series $g = \sum_n 2^{-n} h_n$ converges uniformly, so $g \in C_c(X)$ with support in $\overline{U}$, and on $K_0 \setminus \bigcup_n (V_n \setminus K_n)$ each h_n equals $\mathbf{1}_{T_n}$, so $g = f$ there. Hence $f \neq g$

only on $(A \setminus K_0) \cup \bigcup_n (V_n \setminus K_n)$, of measure $< \varepsilon/2 + \sum_n \varepsilon\, 2^{-n-1} = \varepsilon$. Finally, composing g with the radial retraction $z \mapsto z\, \min(1, \|f\|_\infty/|z|)$ of $\mathbb{C}$ onto the closed disc of radius $\|f\|_\infty$ leaves g in $C_c(X)$, does not enlarge the disagreement set, and enforces $\|g\|_\infty \le \|f\|_\infty$.

Lusin's theorem edits one function at a time. Fed the simple approximants of theorem 2.3.10, it yields a genuine density statement: the continuous functions of compact support, the very test class from which this chapter built its measures, come arbitrarily close to every function whose p-th power is integrable. The notation $\|f\|_p = (\int |f|^p\, d\mu)^{1/p}$ is carried over from chapter 2, as pure shorthand for the integral it abbreviates.

Theorem 4.2.4: $C_c(X)$ Dense in L^p

Let X be an LCH space, μ a Radon measure on X, and $1 \le p < \infty$. For every measurable $f\colon X \to \mathbb{C}$ with $\int_X |f|^p\, d\mu < \infty$ and every $\varepsilon > 0$ there is $g \in C_c(X)$ with

$$\|f - g\|_p < \varepsilon$$

that is, $C_c(X)$ is dense in $L^p(\mu)$.

***Proof*:**
Step 1: reduce to a simple function. By theorem 2.3.10 there is a simple function φ, which we write in canonical form as $\varphi = \sum_{i=1}^m a_i \chi_{E_i}$ with distinct nonzero values a_i and disjoint sets E_i, such that

$$\int_X |f - \varphi|^p\, d\mu < \left(\frac{\varepsilon}{4}\right)^p$$

The finite-measure clause is not a courtesy of the choice but automatic: a simple function with $\int |\varphi|^p\, d\mu < \infty$ assigns finite measure to each nonzero level set, since $|a_i|^p\, \mu(E_i) \le \int |\varphi|^p\, d\mu$. This is exactly the hypothesis Lusin's theorem requires, for φ vanishes outside $E_1 \cup \cdots \cup E_m$, a set of finite measure. If φ vanishes identically, then $g = 0$ already satisfies $\|f - g\|_p < \varepsilon/4$ and the proof is complete. Otherwise set $M = \max_i |a_i| = \sup_X |\varphi| > 0$.

Step 2: apply Lusin's theorem. Theorem 4.2.3 applied to φ with parameter $\delta = (\varepsilon/(8M))^p$ produces $g \in C_c(X)$ with $\mu(\{x \in X \mid \varphi(x) \ne g(x)\}) < \delta$ and $\|g\|_\infty \le \|\varphi\|_\infty = M$, so $|g| \le M$ everywhere. Off the disagreement set the integrand $|\varphi - g|$ vanishes, and on it $|\varphi - g| \le |\varphi| + |g| \le 2M$, so

$$\int_X |\varphi - g|^p\, d\mu \le (2M)^p\, \mu(\{x \in X \mid \varphi(x) \ne g(x)\}) < (2M)^p \left(\frac{\varepsilon}{8M}\right)^p = \left(\frac{\varepsilon}{4}\right)^p$$

Step 3: combine. The triangle inequality for $\|\cdot\|_p$ (Minkowski's inequality) belongs to the theory of L^p as a normed space and is proved in [13, Minkowski's Inequality], so a cruder bound is used instead. For $s, t \ge 0$ one has $s + t \le 2\max(s, t)$, and raising both sides to the p-th power gives $(s+t)^p \le 2^p \max(s,t)^p \le 2^p(s^p + t^p)$. Applying this pointwise to $|f - g| \le |f - \varphi| + |\varphi - g|$ and integrating (monotonicity of the integral and proposition 2.2.7),

$$\int_X |f-g|^p \, d\mu \leq 2^p \left(\int_X |f-\varphi|^p \, d\mu + \int_X |\varphi - g|^p \, d\mu \right) < 2^{p+1} \left(\frac{\varepsilon}{4}\right)^p \leq 4^p \left(\frac{\varepsilon}{4}\right)^p = \varepsilon^p$$

where $2^{p+1} \leq 4^p$ because $p \geq 1$. Taking p-th roots gives $\|f-g\|_p < \varepsilon$, completing the proof.

The proof is a triangle whose two sides were already drawn. Theorem 2.3.10 trades an arbitrary p-th-power integrable function for one with finitely many values, and Lusin's theorem trades those measurable values for continuity at the price of a small exceptional set. The $\|g\|_\infty \leq \|\varphi\|_\infty$ refinement is what keeps that exceptional set cheap: the error is confined to a set of measure δ on which the integrand is bounded by $2\|\varphi\|_\infty$, giving the estimate $\|\varphi - g\|_p \leq 2\|\varphi\|_\infty \, \delta^{1/p}$ that drives Step 2. Regularity enters only through Lusin's theorem, but there it is essential. On a Radon measure space the topology is coupled to the measure tightly enough that the test functions of the representation theorem see everything that integration sees.

The density is also the standard bridge from computations on test functions to statements about all integrable functions. On a locally compact group with Haar measure (chapter 6) it is this theorem, for $p = 1$ and $p = 2$, that transports identities verified on $C_c(G)$, where invariance is a pointwise computation, to all of L^1 and L^2, where convolution and the theory of unitary representations [10] live.

5

Hausdorff Dimension and Kakeya Sets

The Lebesgue measure m assigns a notion of "volume" to measurable subsets of $\mathbb{R}^n$, and throughout the preceding chapters this has been the principal tool for quantifying the size of sets. But m is a blunt instrument for the many sets it assigns measure zero: a single point, a line segment in $\mathbb{R}^2$, and the classical Cantor set C all satisfy $m = 0$, yet they differ in every geometric sense. A single point is 0-dimensional, a line segment is 1-dimensional, and the Cantor set—uncountable, perfect, totally disconnected (proposition 1.4.9)—seems to occupy some intermediate regime. To make this distinction precise requires a family of measures that probes sets at every "dimensional scale," and the critical scale at which the measure transitions from infinity to zero defines a notion of dimension that applies to arbitrary subsets of $\mathbb{R}^n$.

This chapter develops this notion—the *Hausdorff dimension*—and then applies it to a geometric problem whose simplicity belies its difficulty: if a subset of $\mathbb{R}^n$ contains a unit line segment in every direction, how small can it be? Such sets, called *Besicovitch sets* or *Kakeya sets*, can have Lebesgue measure zero. The Kakeya conjecture asserts that despite having zero volume, they must still have full Hausdorff dimension n. The conjecture is resolved for $n = 2$ (Davies, 1971) and $n = 3$ (Wang–Zahl, 2025), but remains open in all higher dimensions. The tools developed in this chapter—Hausdorff measure, Minkowski dimension, and the geometry of directional sets—form the foundation for this problem and its connections to harmonic analysis.

5.1 Hausdorff Measure and Hausdorff Dimension

The construction of Hausdorff measure follows the same outer-measure paradigm used to build the Lebesgue measure in proposition 1.3.2: cover the set by countably many pieces, assign a cost to each piece, and take the infimum over all such covers. For Lebesgue measure the cost of a piece is its volume; for Hausdorff measure the cost is the s-th power of its diameter, where $s \geq 0$ is a free parameter. By varying s, one obtains a one-parameter family of measures: when s is large, coverings are cheap and the measure collapses to zero; when s is small, no covering is efficient and the measure blows up. The transition point between these two regimes is the Hausdorff dimension.

Definition 5.1.1: Hausdorff Measure

Let $E \subseteq \mathbb{R}^n$ and $s \geq 0$. For each $\delta > 0$, define the *δ-approximate s-dimensional Hausdorff measure* of E by

$$\mathcal{H}^s_\delta(E) = \inf \left\{ \sum_{j=1}^{\infty} (\operatorname{diam} U_j)^s \;\middle|\; E \subseteq \bigcup_{j=1}^{\infty} U_j, \quad \operatorname{diam} U_j < \delta \right\}$$

where the infimum is taken over all countable covers of E by sets $U_j \subseteq \mathbb{R}^n$ with $\operatorname{diam} U_j < \delta$. The convention $0^0 = 1$ is adopted.
As δ decreases, fewer covers are admissible, so $\mathcal{H}^s_\delta(E)$ is non-decreasing in $1/\delta$. Define the *s-dimensional Hausdorff measure* of E by

$$\mathcal{H}^s(E) = \lim_{\delta \to 0} \mathcal{H}^s_\delta(E) = \sup_{\delta > 0} \mathcal{H}^s_\delta(E) \in [0, \infty]$$

The parameter δ forces the cover to use "fine" pieces: one cannot cheat by covering E with a single large set and paying a cost proportional only to the diameter of that set. As $\delta \to 0$, only the local geometry of E contributes. The parameter s controls the "dimensional weight" assigned to each covering set: a set of diameter r is penalized as r^s, so higher s makes small-diameter covers cheaper. This interplay between δ (resolution) and s (dimensional weight) is what allows $\mathcal{H}^s$ to detect dimension.

Proposition 5.1.2: $\mathcal{H}^s$ Is a Metric Outer Measure

For each $s \geq 0$, the function $\mathcal{H}^s : P(\mathbb{R}^n) \to [0, \infty]$ is an outer measure. Furthermore, $\mathcal{H}^s$ is a *metric outer measure*: if $A, B \subseteq \mathbb{R}^n$ satisfy $\operatorname{dist}(A, B) > 0$, then

$$\mathcal{H}^s(A \cup B) = \mathcal{H}^s(A) + \mathcal{H}^s(B)$$

As a consequence, by theorem 1.3.4, all Borel sets in $\mathbb{R}^n$ are $\mathcal{H}^s$-measurable.

***Proof*:**
Step 1: $\mathcal{H}^s$ is an outer measure. The argument follows the same template as proposition 1.3.2. First, $\mathcal{H}^s(\emptyset) = 0$: the empty cover (no sets) has cost 0. Monotonicity is immediate: if $A \subseteq B$, every cover of B is a cover of A, so $\mathcal{H}^s_\delta(A) \leq \mathcal{H}^s_\delta(B)$ for each δ, and the inequality persists in the limit.

For countable subadditivity, let $\{E_i\}_{i=1}^{\infty}$ be a countable collection and fix $\delta > 0$ and $\epsilon > 0$. For each i, choose a cover $\{U_j^i\}_{j=1}^\infty$ of E_i with diam $U_j^i < \delta$ and

$$\sum_{j=1}^{\infty} (\text{diam } U_j^i)^s \leq \mathcal{H}^s_\delta(E_i) + \frac{\epsilon}{2^i}$$

Then $\{U_j^i\}_{i,j}$ covers $\bigcup_i E_i$, so

$$\mathcal{H}^s_\delta\left(\bigcup_i E_i\right) \leq \sum_{i=1}^{\infty}\sum_{j=1}^{\infty} (\text{diam } U_j^i)^s \leq \sum_{i=1}^{\infty} \mathcal{H}^s_\delta(E_i) + \epsilon$$

Since ϵ was arbitrary, $\mathcal{H}^s_\delta(\bigcup_i E_i) \leq \sum_i \mathcal{H}^s_\delta(E_i)$. Taking $\delta \to 0$, countable subadditivity of $\mathcal{H}^s$ follows.

Step 2: $\mathcal{H}^s$ is a metric outer measure. Let $d = \text{dist}(A, B) > 0$ and choose any $\delta < d$. If $\{U_j\}$ is a cover of $A \cup B$ with diam $U_j < \delta < d$, then each U_j satisfies $U_j \cap A = \emptyset$ or $U_j \cap B = \emptyset$: if some $x \in U_j \cap A$ and $y \in U_j \cap B$ existed, then $|x - y| \leq \text{diam } U_j < d = \text{dist}(A, B)$, a contradiction. Partitioning the cover into $\mathcal{A} = \{U_j \mid U_j \cap A \neq \emptyset\}$ and $\mathcal{B} = \{U_j \mid U_j \cap B \neq \emptyset\}$ gives

$$\sum_j (\text{diam } U_j)^s = \sum_{U_j \in \mathcal{A}} (\text{diam } U_j)^s + \sum_{U_j \in \mathcal{B}} (\text{diam } U_j)^s \geq \mathcal{H}^s_\delta(A) + \mathcal{H}^s_\delta(B)$$

Taking the infimum over all such covers: $\mathcal{H}^s_\delta(A \cup B) \geq \mathcal{H}^s_\delta(A) + \mathcal{H}^s_\delta(B)$. The reverse inequality holds by subadditivity, so equality holds for each $\delta < d$, and hence in the limit $\delta \to 0$.

Step 3: Borel measurability. The fact that every metric outer measure on $\mathbb{R}^n$ makes all Borel sets measurable (in the sense of definition 1.3.3) follows from a standard argument: the metric property implies that every open set G satisfies the Carathéodory condition $\mathcal{H}^s(S) = \mathcal{H}^s(S \cap G) + \mathcal{H}^s(S \cap G^c)$ for all test sets S, and since open sets generate the Borel σ-algebra $\mathcal{B}_{\mathbb{R}^n}$, the claim follows from theorem 1.3.4. The verification that open sets satisfy the Carathéodory condition uses a telescoping decomposition of $S \cap G$ into annular shells $A_k = \left\{x \in S \cap G \;\middle|\; \frac{1}{k+1} \leq d(x, G^c) < \frac{1}{k}\right\}$ with pairwise positive separation, and is left as an exercise.

This gives a second major application of theorem 1.3.4: the Carathéodory machinery, applied to the Hausdorff outer measure, produces a Borel measure for each dimensional parameter s. The resulting measure space $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, \mathcal{H}^s)$ is the arena in which Hausdorff dimension is defined. Note that the σ-algebra of $\mathcal{H}^s$-measurable sets may be strictly larger than $\mathcal{B}_{\mathbb{R}^n}$ (just as the Lebesgue σ-algebra $\mathcal{L}$ is strictly larger than $\mathcal{B}_{\mathbb{R}}$), but for the purposes of dimension theory the Borel sets suffice.

A point of contrast with the Lebesgue outer measure from proposition 1.3.2 is worth emphasizing. The Lebesgue outer measure is constructed from a premeasure on intervals (or rectangles), and its restriction to Borel sets is the unique extension of that premeasure (theorem 1.3.7). The Hausdorff outer measure has no such premeasure origin—it is defined directly on all subsets of $\mathbb{R}^n$ using coverings by *arbitrary* sets, not just intervals. The cost functional $(\text{diam})^s$ plays the role that the volume functional played for the Lebesgue construction. This generality is what allows $\mathcal{H}^s$ to detect fractal behavior: by optimizing over all possible coverings (not just grids), the measure can adapt to the local geometry of irregular sets.

Before defining dimension, it is useful to ground the new construction in the familiar. The case $s = n$ recovers the Lebesgue measure up to a normalizing constant:

Proposition 5.1.3: Hausdorff Measure and Lebesgue Measure

Let m denote the Lebesgue measure on $\mathbb{R}^n$. Then for every Lebesgue measurable set $E \subseteq \mathbb{R}^n$:

$$\mathcal{H}^n(E) = \frac{2^n}{\omega_n}\, m(E)$$

where $\omega_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)}$ is the volume of the unit ball in $\mathbb{R}^n$. The constant $c_n = 2^n/\omega_n$ arises because $\mathcal{H}^n$ measures sets using $(\mathrm{diam})^n$ while m uses side-lengths (or ball volumes).

***Proof Sketch*:**

For the inequality $\mathcal{H}^n(E) \le c_n\, m(E)$: fix $\delta > 0$. By the definition of Lebesgue measure (or by theorem 2.7.1), E can be covered by countably many cubes $\{Q_j\}$ of side length $\delta/\sqrt{n}$, each having diameter $\mathrm{diam}(Q_j) = \delta/\sqrt{n} \cdot \sqrt{n} = \delta$. The s-cost of each cube is $(\mathrm{diam}\ Q_j)^n = \delta^n$, while its Lebesgue measure is $(\delta/\sqrt{n})^n$. Since $\sum_j m(Q_j) \ge m(E)$, the number of cubes needed is at least $m(E)/(\delta/\sqrt{n})^n = m(E)n^{n/2}/\delta^n$. Then

$$\mathcal{H}^n_\delta(E) \le \sum_j \delta^n = N \cdot \delta^n$$

where N is the number of cubes. By choosing the cover to have $\sum_j m(Q_j) \le m(E) + \epsilon$, the bound becomes $\mathcal{H}^n_\delta(E) \le (m(E) + \epsilon)n^{n/2}$. Taking $\delta \to 0$ and $\epsilon \to 0$: $\mathcal{H}^n(E) \le n^{n/2} m(E)$. (The sharper constant $c_n = 2^n/\omega_n$ requires a more careful argument using ball covers instead of cube covers.)

The reverse inequality $\mathcal{H}^n(E) \ge c_n\, m(E)$ uses the *isodiametric inequality*: among all sets of diameter d in $\mathbb{R}^n$, the ball of diameter d has the largest volume, namely $\omega_n(d/2)^n$. Hence for any cover $\{U_j\}$ of E with $\mathrm{diam}\ U_j < \delta$:

$$m(E) \le \sum_j m(U_j) \le \sum_j \omega_n \left(\frac{\mathrm{diam}\ U_j}{2}\right)^n = \frac{\omega_n}{2^n}\sum_j (\mathrm{diam}\ U_j)^n$$

Taking the infimum over all such covers: $m(E) \le \frac{\omega_n}{2^n}\mathcal{H}^n_\delta(E)$. Taking $\delta \to 0$: $m(E) \le \frac{\omega_n}{2^n}\mathcal{H}^n(E)$, which rearranges to $\mathcal{H}^n(E) \ge \frac{2^n}{\omega_n} m(E) = c_n\, m(E)$.

5.1.4 Remark: The Isodiametric Inequality The isodiametric inequality states that among all subsets of $\mathbb{R}^n$ with a given diameter, the ball achieves the maximal volume. A proof using Steiner symmetrization can be found in Evans–Gariepy *Measure Theory and Fine Properties of Functions*. The inequality is logically independent of the rest of this chapter and is used only in the proof of proposition 5.1.3.

The normalization means $\mathcal{H}^n$ and m carry the same information on $\mathbb{R}^n$. Some authors define $\mathcal{H}^n$ with the normalizing constant built in; the convention used here (no constant in the definition, constant appearing in the comparison) is more common in fractal geometry.

Example 5.1.5: Hausdorff Measure at Integer Dimensions

1. $\mathcal{H}^0$ is the counting measure. If $|E| = k$ is finite, then $\mathcal{H}^0(E) = k$: cover each point by a set of diameter $< \delta$, and each contributes $(\text{diam})^0 = 1$. If E is infinite, $\mathcal{H}^0(E) = \infty$. This is consistent with the convention $0^0 = 1$.

2. If $\gamma : [a, b] \to \mathbb{R}^2$ is a C^1 curve with $|\gamma'(t)| > 0$, then $\mathcal{H}^1(\gamma([a, b]))$ equals the arc length $\int_a^b |\gamma'(t)|\, dt$. The Hausdorff 1-measure thus recovers the classical notion of length for smooth curves. More generally, $\mathcal{H}^k$ restricted to a smooth k-dimensional submanifold of $\mathbb{R}^n$ agrees (up to a normalizing constant) with the k-dimensional volume form induced by the ambient metric.

3. For non-integer values of s, the measure $\mathcal{H}^s$ does not correspond to any classical geometric quantity. The set E at which $\mathcal{H}^s(E)$ transitions from infinity to zero has "effective dimension" s—it is "too large" to be measured by any $\mathcal{H}^t$ with $t < s$ (giving infinity) and "too small" for $\mathcal{H}^t$ with $t > s$ (giving zero). The next proposition makes this transition precise.

The central observation about $\mathcal{H}^s$ is that for a given set E, the value $\mathcal{H}^s(E)$ jumps from ∞ to 0 at a single critical exponent. This jump is a consequence of the power-law relationship between the cost functions at different exponents:

Proposition 5.1.6: Critical Exponent

Let $E \subseteq \mathbb{R}^n$. If $\mathcal{H}^s(E) < \infty$ for some $s \geq 0$, then $\mathcal{H}^t(E) = 0$ for all $t > s$. Equivalently, if $\mathcal{H}^t(E) > 0$ for some t, then $\mathcal{H}^s(E) = \infty$ for all $s < t$.

***Proof*:**
Fix $\delta > 0$ and let $\{U_j\}$ be a cover of E with diam $U_j < \delta$. For $t > s$:

$$\sum_j (\text{diam } U_j)^t = \sum_j (\text{diam } U_j)^{t-s} (\text{diam } U_j)^s \leq \delta^{t-s} \sum_j (\text{diam } U_j)^s$$

Taking the infimum: $\mathcal{H}^t_\delta(E) \leq \delta^{t-s} \mathcal{H}^s_\delta(E)$. If $\mathcal{H}^s(E) < \infty$, then $\mathcal{H}^s_\delta(E) \leq \mathcal{H}^s(E) + 1$ for small δ, so $\mathcal{H}^t_\delta(E) \leq \delta^{t-s}(\mathcal{H}^s(E) + 1) \to 0$ as $\delta \to 0$. Thus $\mathcal{H}^t(E) = 0$.

This jump behavior defines a unique threshold:

Definition 5.1.7: Hausdorff Dimension

Let $E \subseteq \mathbb{R}^n$. The *Hausdorff dimension* of E is

$$\dim_H(E) = \inf\{s \geq 0 \mid \mathcal{H}^s(E) = 0\} = \sup\{s \geq 0 \mid \mathcal{H}^s(E) = \infty\}$$

where the two expressions are equal by proposition 5.1.6. At the critical value $s_0 = \dim_H(E)$, the measure $\mathcal{H}^{s_0}(E)$ can be 0, ∞, or any value in between.

By proposition 5.1.3, $\dim_H(E) = n$ whenever $m(E) > 0$, and $\dim_H(E) \leq n$ for all $E \subseteq \mathbb{R}^n$. The utility of $\dim_H$ is precisely for sets where $m(E) = 0$: it provides a finer stratification of null sets.

A countable set has dimension 0; a smooth curve in $\mathbb{R}^2$ has dimension 1; a "thick dust" like the Cantor set has some intermediate fractional dimension. The value $\mathcal{H}^{s_0}(E)$ at the critical exponent $s_0 = \dim_H(E)$ itself is often difficult to compute and can be 0, positive and finite, or ∞. For many problems in geometric measure theory (including the Kakeya conjecture), only the dimension matters, and the precise value of $\mathcal{H}^{s_0}(E)$ is secondary.

Proposition 5.1.8: Properties of Hausdorff Dimension

Let $E, F \subseteq \mathbb{R}^n$ and let $\{E_i\}_{i=1}^{\infty} \subseteq \mathbb{R}^n$ be a countable collection.

1. **(Monotonicity)** If $E \subseteq F$, then $\dim_H(E) \le \dim_H(F)$.
2. **(Countable stability)** $\dim_H\left(\bigcup_{i=1}^{\infty} E_i\right) = \sup_i \dim_H(E_i)$.
3. **(Countable sets)** If E is countable, then $\dim_H(E) = 0$.
4. **(Lipschitz maps)** If $f : E \to \mathbb{R}^m$ satisfies $|f(x) - f(y)| \le L|x - y|$ for all $x, y \in E$, then $\dim_H(f(E)) \le \dim_H(E)$. In particular, $\dim_H$ is invariant under bi-Lipschitz maps.

***Proof*:**

1. If $E \subseteq F$, then $\mathcal{H}^s(E) \le \mathcal{H}^s(F)$ for all s, so $\mathcal{H}^s(F) = 0$ implies $\mathcal{H}^s(E) = 0$, giving $\dim_H(E) \le \dim_H(F)$.
2. Let $d = \sup_i \dim_H(E_i)$. By monotonicity, $\dim_H(\bigcup_i E_i) \ge d$. For the reverse, if $s > d$, then $\mathcal{H}^s(E_i) = 0$ for all i, so by countable subadditivity $\mathcal{H}^s(\bigcup_i E_i) \le \sum_i \mathcal{H}^s(E_i) = 0$, hence $\dim_H(\bigcup_i E_i) \le s$. Since $s > d$ was arbitrary, $\dim_H(\bigcup_i E_i) \le d$.
3. A single point $\{x\}$ has $\dim_H(\{x\}) = 0$ since $\mathcal{H}^s(\{x\}) = 0$ for all $s > 0$ (cover by a ball of diameter δ: cost $= \delta^s \to 0$). By countable stability, any countable set has dimension 0.
4. If $\{U_j\}$ covers E with diam $U_j < \delta$, then $\{f(U_j \cap E)\}$ covers $f(E)$ with diam $f(U_j \cap E) \le L\,\mathrm{diam}\,U_j < L\delta$. Thus

$$\mathcal{H}^s_{L\delta}(f(E)) \le \sum_j (L\,\mathrm{diam}\,U_j)^s = L^s \sum_j (\mathrm{diam}\,U_j)^s$$

   Taking the infimum and the limit $\delta \to 0$: $\mathcal{H}^s(f(E)) \le L^s\mathcal{H}^s(E)$. If $\mathcal{H}^s(E) = 0$ then $\mathcal{H}^s(f(E)) = 0$, giving $\dim_H(f(E)) \le \dim_H(E)$. If f is bi-Lipschitz, apply the same argument to f^{-1} to get the reverse inequality.

Property (4) implies that $\dim_H$ is a bi-Lipschitz invariant but *not* a topological invariant. The fat Cantor set C_α of measure $\alpha > 0$ from lemma 1.4.10 satisfies $\dim_H(C_\alpha) = 1$ (since $m(C_\alpha) > 0$ forces $\mathcal{H}^1(C_\alpha) > 0$), yet C_α is homeomorphic to the standard Cantor set C by corollary 1.4.16. Since the homeomorphism need not be Lipschitz, the dimensions are free to differ. This distinction matters: the Kakeya conjecture (which asserts that certain sets have full Hausdorff dimension) would be trivially false if dimension were a topological invariant, since any compact set is homeomorphic to a set of dimension 0 in a sufficiently large ambient space.

Property (2)—countable stability—is the feature that most sharply distinguishes Hausdorff dimension from the Minkowski dimension studied in the next section. It ensures that the dimension of a countable union is controlled by the dimensions of the individual pieces, a property that is indispensable for measure-theoretic arguments but that the Minkowski dimension fails to satisfy.

The upper bound on dimension is typically obtained by exhibiting an explicit covering and computing its cost, as in the standard representation of the Cantor set by stage-k intervals. Lower bounds are harder: one must show that *no* covering can be too cheap. The most effective general technique is to "distribute mass" on the set and use the following duality:

Lemma 5.1.9: Mass Distribution Principle

Let $E \subseteq \mathbb{R}^n$ and suppose there exists a Borel measure μ with $\mu(E) > 0$ and a constant $C > 0$ such that

$$\mu(B(x,r)) \leq Cr^s \qquad \text{for all } x \in \mathbb{R}^n \text{ and all } r > 0$$

Then $\dim_H(E) \geq s$.

***Proof*:**
Let $\{U_j\}$ be any cover of E with diam $U_j < \delta$. Each U_j is contained in a ball of radius diam U_j, so $\mu(U_j) \leq C(\text{diam } U_j)^s$. Then

$$0 < \mu(E) \leq \mu\left(\bigcup_j U_j\right) \leq \sum_j \mu(U_j) \leq C \sum_j (\text{diam } U_j)^s$$

Hence $\sum_j (\text{diam } U_j)^s \geq \mu(E)/C > 0$ for every admissible cover, giving $\mathcal{H}^s_\delta(E) \geq \mu(E)/C > 0$ and thus $\mathcal{H}^s(E) > 0$, so $\dim_H(E) \geq s$.

The lemma reduces dimension lower bounds to the construction of a measure with controlled local growth. The idea is a duality: to show $\dim_H(E) \geq s$, one produces a measure supported on E that "sees" the set as s-dimensional. The condition $\mu(B(x,r)) \leq Cr^s$ says that μ distributes its mass at rate r^s per ball—matching the growth rate of an s-dimensional object. If such a measure exists, no covering can be too efficient, and the Hausdorff measure at exponent s must be positive.

For self-similar sets, the natural "equidistributed" measure typically satisfies the required ball condition. For the Cantor set, this is the Cantor measure, which assigns equal mass to each interval at every stage of the construction.

Example 5.1.10: Hausdorff Dimension of the Middle-Thirds Cantor Set

Let C denote the standard middle-thirds Cantor set (definition 1.4.8). Then $\dim_H(C) = \frac{\log 2}{\log 3} \approx 0.631$.

Upper bound. At stage k of the construction, C is covered by 2^k closed intervals, each of length 3^{-k}. These intervals have diameter 3^{-k}, so for $s = \frac{\log 2}{\log 3}$:

$$\mathcal{H}^s_{3^{-k}}(C) \leq 2^k \cdot (3^{-k})^s = 2^k \cdot 3^{-k \log 2/\log 3} = 2^k \cdot 2^{-k} = 1$$

The computation uses $3^{-\log 2/\log 3} = 2^{-1}$, which follows from exponentiating $-\frac{\log 2}{\log 3} \cdot \log 3 = -\log 2$. Thus $\mathcal{H}^s(C) \leq 1 < \infty$, giving $\dim_H(C) \leq s$. Note that for any $s' < s$, the same covering gives $\mathcal{H}^{s'}_{3^{-k}}(C) \leq 2^k \cdot 3^{-ks'} = (2 \cdot 3^{-s'})^k$, and since $s' < s$ implies $2 \cdot 3^{-s'} > 2 \cdot 3^{-s} = 1$, this diverges as $k \to \infty$. The covering at exponent s is thus "perfectly balanced"—the cost neither diverges nor converges to zero.

Lower bound. Define the Cantor measure μ on C as follows: at stage k, assign mass 2^{-k} to each of the 2^k intervals I_k^j. Concretely, if E is a union of stage-k intervals, then $\mu(E) =$ (number of stage-k intervals in E) $\cdot 2^{-k}$. This definition is consistent across stages (each stage-k interval contains exactly two stage-$(k+1)$ intervals, and $2 \cdot 2^{-(k+1)} = 2^{-k}$), so it defines a premeasure on the algebra of finite unions of stage-k intervals, and extends to a Borel measure μ on C with $\mu(C) = 1$ by theorem 1.3.7. The claim is that $\mu(B(x,r)) \leq Cr^s$ for a constant C and $s = \frac{\log 2}{\log 3}$.

For any $x \in C$ and $r > 0$, choose k so that $3^{-(k+1)} \leq r < 3^{-k}$. The ball $B(x,r)$ intersects at most 2 stage-k intervals (since consecutive stage-k intervals are separated by a gap of length $\geq 3^{-k}/2$). Therefore

$$\mu(B(x,r)) \leq 2 \cdot 2^{-k} = 2^{-(k-1)}$$

Since $r \geq 3^{-(k+1)}$, raising to the power s:

$$r^s \geq 3^{-(k+1)s} = 3^{-s} \cdot 3^{-ks} = 3^{-s} \cdot 2^{-k}$$

Combining: $\mu(B(x,r)) \leq 2^{-(k-1)} = 2 \cdot 2^{-k} \leq 2 \cdot 3^s \cdot r^s$. By the mass distribution principle, $\dim_H(C) \geq s = \frac{\log 2}{\log 3}$.

The same technique applies to any self-similar Cantor set where a fixed fraction is removed at each step. The formula $\dim_H = \log(\text{pieces})/\log(\text{scaling factor})$ is a general phenomenon for self-similar sets satisfying an "open set condition," though the proof of this general statement requires additional machinery beyond the mass distribution principle.

Example 5.1.11: Further Dimension Computations

1. **Fat Cantor sets.** The fat Cantor set C_α with $m(C_\alpha) = \alpha > 0$ satisfies $\dim_H(C_\alpha) = 1$. The lower bound follows from $\mathcal{H}^1(C_\alpha) \geq c_1\, m(C_\alpha) > 0$ by proposition 5.1.3, and $\dim_H(C_\alpha) \leq 1$ since $C_\alpha \subseteq \mathbb{R}$. Despite being homeomorphic to the standard Cantor set by corollary 1.4.16, the two sets have different Hausdorff dimensions ($\log 2/\log 3 < 1$). Hausdorff dimension is thus a metric—not topological—invariant.

2. **The rationals and irrationals.** $\dim_H(\mathbb{Q}) = 0$ since $\mathbb{Q}$ is countable. By proposition 5.1.8(2), $1 = \dim_H(\mathbb{R}) = \max\{\dim_H(\mathbb{Q}), \dim_H(\mathbb{R} \setminus \mathbb{Q})\}$, so $\dim_H(\mathbb{R} \setminus \mathbb{Q}) = 1$.

3. **Smooth submanifolds.** If $M \subseteq \mathbb{R}^n$ is a smooth k-dimensional submanifold (for instance, a smooth curve in $\mathbb{R}^2$ or a smooth surface in $\mathbb{R}^3$), then $\dim_H(M) = k$. The upper bound follows by covering M with coordinate patches, each of which is bi-Lipschitz equivalent to an open subset of $\mathbb{R}^k$. The lower bound follows from proposition 5.1.3 applied within each coordinate chart: the k-dimensional Lebesgue measure of the chart is positive, so $\mathcal{H}^k(M) > 0$. This confirms that Hausdorff dimension extends the classical notion of dimension for smooth objects.

The computations above illustrate a general principle: for "tame" sets (smooth manifolds, self-similar fractals satisfying the open set condition), Hausdorff dimension is computable and agrees with geometric intuition. The subtlety arises for sets constructed through limiting processes without self-similar structure—such as the Besicovitch sets studied in the next section, where even establishing a lower bound on $\dim_H$ requires fundamentally new ideas.

Exercise 5.1.1

1. Let C_r be the Cantor set obtained by removing a central interval of length $r \in (0,1)$ at each stage (the middle-thirds Cantor set corresponds to $r = 1/3$). Show that $\dim_H(C_r) = \frac{\log 2}{\log(2/(1-r))}$.
2. Let $f : [0,1] \to \mathbb{R}$ be α-Hölder continuous, that is, $|f(x)-f(y)| \leq C|x-y|^\alpha$ for some $C > 0$ and $\alpha \in (0,1]$. Show that $\dim_H(\mathrm{graph}(f)) \leq 2-\alpha$, where $\mathrm{graph}(f) = \{(x, f(x)) \mid x \in [0,1]\} \subseteq \mathbb{R}^2$. (*Hint*: cover $[0,1]$ by N intervals of length $1/N$; each piece of the graph is contained in a rectangle of width $1/N$ and height C/N^α.)
3. Show that every metric outer measure on $\mathbb{R}^n$ makes all Borel sets measurable. (*Hint*: it suffices to show that every open set G satisfies the Carathéodory condition. For a test set S, write $S \cap G = \bigcup_{k=1}^\infty A_k$ where $A_k = \left\{ x \in S \cap G \;\middle|\; \frac{1}{k+1} \leq d(x, G^c) < \frac{1}{k} \right\}$, and use the metric property on A_k's with indices differing by ≥ 2.)
4. Show that $\dim_H(rE) = \dim_H(E)$ for all $r \neq 0$ and $\dim_H(E + a) = \dim_H(E)$ for all $a \in \mathbb{R}^n$ (compare with proposition 1.4.7).
5. Construct an uncountable set $E \subseteq \mathbb{R}$ with $\dim_H(E) = 0$. (*Hint*: consider a Cantor-type construction where the pieces shrink faster than geometrically.)

5.2 Minkowski Dimension

The Hausdorff dimension has excellent theoretical properties—countable stability, Lipschitz invariance, and a clean relationship to Hausdorff measure—but computing it can be demanding, since the definition requires optimizing over *all* countable covers. In practice, a coarser notion that counts δ-balls rather than arbitrary covers is often more tractable. This leads to the *Minkowski dimension* (also called the box-counting dimension), which uses only one type of covering set (balls or cubes of a fixed radius) and extracts the dimension from the growth rate of the covering number.

Definition 5.2.1: Minkowski Dimension

Let $E \subseteq \mathbb{R}^n$ be a bounded set. For each $\delta > 0$, let $N(E, \delta)$ denote the smallest number of sets of diameter at most δ needed to cover E. Define the *upper Minkowski dimension* and *lower Minkowski dimension* of E by

$$\overline{\dim}_M(E) = \limsup_{\delta \to 0} \frac{\log N(E,\delta)}{-\log \delta} \qquad \underline{\dim}_M(E) = \liminf_{\delta \to 0} \frac{\log N(E,\delta)}{-\log \delta}$$

If the two coincide, their common value is the *Minkowski dimension* $\dim_M(E)$.

The ratio $\log N(E, \delta)/(-\log \delta)$ measures the exponent in the scaling $N(E, \delta) \sim \delta^{-d}$: if covering E at resolution δ requires $\sim \delta^{-d}$ balls, then E "looks d-dimensional" at that scale. The $\limsup$ and $\liminf$ capture the worst-case and best-case scaling behavior as $\delta \to 0$.

There is an equivalent volume-based formulation that is sometimes more convenient. Let $E_\delta = \{x \in \mathbb{R}^n \mid d(x, E) < \delta\}$ denote the open δ-neighborhood of E. The volume $m(E_\delta)$ grows as E is "thickened" by δ, and the rate of growth encodes the dimension:

$$\overline{\dim}_M(E) = n - \liminf_{\delta \to 0} \frac{\log m(E_\delta)}{\log \delta}$$

To see the connection: E_δ is approximately the union of the $N(E, \delta)$ balls of radius δ in an optimal cover, so $m(E_\delta) \approx N(E, \delta) \cdot \omega_n \delta^n$. Taking logarithms and dividing by $\log \delta$ recovers the covering-number formulation. This perspective is particularly useful for relating Minkowski dimension to Lebesgue measure estimates, which is the context in which it appears in the Kakeya conjecture.

The covering number $N(E, \delta)$ admits several equivalent formulations, and these equivalences are useful in different computational contexts:

Proposition 5.2.2: Equivalent Formulations

Let $E \subseteq \mathbb{R}^n$ be bounded. The following quantities all have the same $\limsup$ and $\liminf$ as $\delta \to 0$ when divided by $-\log \delta$:

1. $N(E, \delta)$: smallest number of sets of diameter $\leq \delta$ covering E.
2. $N'(E, \delta)$: smallest number of closed balls of radius δ covering E.
3. $M(E, \delta)$: largest number of disjoint balls of radius δ centered at points of E.
4. $N_{\text{cube}}(E, \delta)$: number of cubes in a δ-grid of $\mathbb{R}^n$ that intersect E.

***Proof*:**
The equivalences follow from volume comparison. By definition, $N(E, \delta) \leq N'(E, \delta)$ since balls of radius δ have diameter 2δ, but a cover by sets of diameter $\leq \delta$ can be replaced by enclosing balls, giving $N'(E, \delta) \leq N(E, \delta)$. For $M(E, \delta)$ and $N'(E, \delta)$: any ball of radius δ covers at most one center of a disjoint packing, so $M(E, \delta) \leq N'(E, \delta)$. Conversely, the balls of radius 2δ centered at packing points cover E, so $N'(E, 2\delta) \leq M(E, \delta)$.

For N_{cube}: any cube of side δ has diameter $\delta\sqrt{n}$, and any ball of radius δ is contained in at most 3^n cubes of side δ, giving $N_{\text{cube}}(E, \delta) \leq 3^n N'(E, \delta)$ and $N'(E, \delta\sqrt{n}) \leq N_{\text{cube}}(E, \delta)$.

In each case, passing from δ to $c\delta$ for a fixed constant c shifts $\log N$ by at most $\log(c^n)$, which is $O(1)$ and hence negligible compared to $-\log \delta$ as $\delta \to 0$. Therefore all formulations yield the same $\limsup$ and $\liminf$.

The relationship between Minkowski and Hausdorff dimensions is governed by a one-sided inequality:

Proposition 5.2.3: Hausdorff Versus Minkowski

For any bounded $E \subseteq \mathbb{R}^n$:

$$\dim_H(E) \leq \underline{\dim}_M(E) \leq \overline{\dim}_M(E)$$

***Proof*:**
Fix $s > \overline{\dim}_M(E)$ and $\delta > 0$. By definition of the upper Minkowski dimension, for all sufficiently

small δ there holds $N(E,\delta) \leq \delta^{-s}$. Cover E by $N(E,\delta)$ sets of diameter $\leq \delta$:

$$\mathcal{H}^s_\delta(E) \leq N(E,\delta) \cdot \delta^s \leq \delta^{-s} \cdot \delta^s = 1$$

Hence $\mathcal{H}^s(E) = \lim_{\delta\to 0} \mathcal{H}^s_\delta(E) \leq 1 < \infty$, so $\dim_H(E) \leq s$. Since $s > \overline{\dim}_M(E)$ was arbitrary, $\dim_H(E) \leq \overline{\dim}_M(E)$. The inequality $\underline{\dim}_M(E) \leq \overline{\dim}_M(E)$ is immediate from $\liminf \leq \limsup$.

The inequality $\dim_H \leq \underline{\dim}_M$ can be strict: the Hausdorff dimension optimizes over all covers (including covers with variable-diameter sets), while the Minkowski dimension uses only covers at a single scale δ. This constraint prevents Minkowski dimension from "adapting" to sets with non-uniform local structure. The proof of proposition 5.2.3 makes the source of the inequality transparent: the uniform cover at scale δ used to bound $\mathcal{H}^s_\delta(E)$ is only one of the many possible covers over which $\mathcal{H}^s_\delta$ takes an infimum. The Hausdorff measure can find cheaper covers, which is why $\dim_H$ can be strictly smaller.

Despite this deficiency, Minkowski dimension has a significant practical advantage: the covering number $N(E,\delta)$ can often be computed or estimated directly from the geometry of E, without the need to optimize over all covers. For this reason, many results in geometric measure theory (including the Kakeya conjecture) are first formulated in terms of Minkowski dimension, and the stronger Hausdorff statement is established separately. The volume-based formulation of Minkowski dimension also connects it directly to the "δ-tube" estimates that appear in the discretized Kakeya problem.

Example 5.2.4: The Set $\{1/n\}$: Separation of Dimensions

Let $E = \{1/n : n \geq 1\} \subseteq \mathbb{R}$. Since E is countable, $\dim_H(E) = 0$ by proposition 5.1.8(3). The Minkowski dimension is positive: the clustering of points near the origin inflates the box count.

To compute $N(E,\delta)$, partition E into a "sparse" part and a "dense" part. Fix $\delta > 0$ small and let $n_0 = \lfloor 1/\sqrt{\delta} \rfloor$.

Sparse part. For $n \leq n_0$, consecutive points $1/n$ and $1/(n+1)$ have spacing $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \geq \frac{1}{n_0(n_0+1)} \geq \frac{\delta}{2}$ for δ small. Each such point requires its own δ-interval, contributing $n_0 \sim \delta^{-1/2}$ intervals.

Dense part. For $n > n_0$, all points lie in $[0, 1/n_0] \subseteq [0, 2\sqrt{\delta}]$. This interval is covered by at most $\lceil 2\sqrt{\delta}/\delta \rceil = \lceil 2/\sqrt{\delta} \rceil \sim \delta^{-1/2}$ intervals of length δ.

Combining both parts: $N(E,\delta) \sim \delta^{-1/2}$. Therefore

$$\dim_M(E) = \lim_{\delta\to 0} \frac{\log(\delta^{-1/2})}{-\log\delta} = \frac{1}{2}$$

The gap $\dim_H(E) = 0 < 1/2 = \dim_M(E)$ arises because the points cluster near 0, inflating the box count at small scales. The Hausdorff dimension ignores this clustering by using a cover with variable-diameter sets (large balls for the sparse region, small balls near 0). The Minkowski dimension is constrained to a single scale δ and pays a penalty for the dense cluster.

For well-behaved sets such as self-similar fractals, the two dimensions coincide:

Example 5.2.5: Minkowski Dimension of the Cantor Set

The middle-thirds Cantor set C satisfies $\dim_M(C) = \log 2 / \log 3$.

At stage k, C is covered by 2^k intervals of length 3^{-k}, so $N(C, 3^{-k}) \leq 2^k$. Conversely, each of the 2^k stage-k intervals contains a point of C at distance $\geq 3^{-k}/2$ from the others, so $N(C, 3^{-k}) \geq 2^k$. Thus $N(C, 3^{-k}) = 2^k$ for each k, and

$$\dim_M(C) = \lim_{k\to\infty} \frac{\log 2^k}{\log 3^k} = \frac{\log 2}{\log 3}$$

This confirms $\dim_M(C) = \dim_H(C)$.

Products of sets interact predictably with Minkowski dimension, which is useful for constructing higher-dimensional examples:

Proposition 5.2.6: Minkowski Dimension of Products

Let $A \subseteq \mathbb{R}^m$ and $B \subseteq \mathbb{R}^n$ be bounded. Then

$$\overline{\dim}_M(A \times B) \leq \overline{\dim}_M(A) + \overline{\dim}_M(B)$$

If $\dim_M(A)$ and $\dim_M(B)$ both exist (upper equals lower), then $\dim_M(A \times B) = \dim_M(A) + \dim_M(B)$.

***Proof*:**

Any cover of A by N_A sets of diameter $\leq \delta$ and cover of B by N_B sets of diameter $\leq \delta$ induces a cover of $A \times B$ by $N_A \cdot N_B$ sets of diameter $\leq 2\delta$ (using the product metric). Therefore $N(A \times B, 2\delta) \leq N(A, \delta) \cdot N(B, \delta)$, giving

$$\frac{\log N(A \times B, 2\delta)}{-\log(2\delta)} \leq \frac{\log N(A, \delta) + \log N(B, \delta)}{-\log(2\delta)}$$

Taking $\limsup$ and using $-\log(2\delta) \sim -\log \delta$ establishes the upper Minkowski bound. The lower bound $\underline{\dim}_M(A \times B) \geq \underline{\dim}_M(A) + \underline{\dim}_M(B)$ follows from a packing argument (a maximal δ-separated set in $A \times B$ projects to δ-separated sets in A and B), and when the limits exist, the two bounds combine to give equality.

The product formula confirms the intuition that taking a Cartesian product "adds dimensions": the product of a d_1-dimensional set and a d_2-dimensional set is $(d_1 + d_2)$-dimensional. For Hausdorff dimension, the analogous inequality $\dim_H(A \times B) \geq \dim_H(A) + \dim_H(B)$ holds, but equality can fail without additional regularity assumptions (though equality does hold when one of the factors has equal upper and lower Minkowski dimensions).

5.2.7 Remark: Countable Stability Fails for Minkowski Dimension Unlike Hausdorff dimension, Minkowski dimension is *not* countably stable. The set $E = \{1/n\}$ is a countable union of points, each of Minkowski dimension 0, yet $\dim_M(E) = 1/2$. This failure is the principal theoretical deficiency of Minkowski dimension. For compact sets arising from geometric constructions—including the Besicovitch sets studied in the next section—the two dimensions often coincide, and the Kakeya conjecture asserts that both equal n.

The tools developed in this section and the previous one—Hausdorff dimension for sharp theoretical bounds, Minkowski dimension for computationally tractable estimates, and the inequality $\dim_H \leq \dim_M$ relating the two—form the dimensional framework for studying the Kakeya conjecture. The next section constructs the sets to which this framework will be applied: the Besicovitch sets, which contain a line segment in every direction yet have zero Lebesgue measure.

Exercise 5.2.1

1. Let $E_\alpha = \{n^{-\alpha} : n \geq 1\}$ for $\alpha > 0$. Show that $\dim_M(E_\alpha) = \frac{1}{1+\alpha}$.
2. Show that $\dim_M(\mathbb{Q} \cap [0,1]) = 1$ while $\dim_H(\mathbb{Q} \cap [0,1]) = 0$. This gives a maximal gap between the two dimensions.
3. Verify proposition 5.2.6 explicitly for $A = B = C$ (the middle-thirds Cantor set). Compute $\dim_M(C \times C)$ and confirm it equals $2\log 2/\log 3$.
4. Show that if $E \subseteq \mathbb{R}^n$ is bounded and $\dim_M(E)$ exists, then $\dim_M(\overline{E}) = \dim_M(E)$.
5. Show that for any bounded $E \subseteq \mathbb{R}^n$ and $\lambda > 0$, $\overline{\dim}_M(\lambda E) = \overline{\dim}_M(E)$ where $\lambda E = \{\lambda x \mid x \in E\}$. Contrast this with the behavior of Lebesgue measure under dilation (proposition 1.4.7).

5.3 Besicovitch Sets

In 1917, Sōichi Kakeya posed the following question: what is the smallest area of a region in the plane within which a unit needle can be continuously rotated through 180 degrees? If the needle is rotated about its center, the region is a circle of area $\pi/4$. A more efficient motion traces a *deltoid* (a three-cusped hypocycloid) of area $\pi/8$. In 1928, Pál proved that among *convex* regions, the equilateral triangle of unit height has the smallest area. One might expect that the answer for general (non-convex) regions is some small but positive number.

Two distinct questions emerge from Kakeya's original problem. The first, called the *Kakeya needle problem*, asks for the infimum of areas of regions in which a needle can be *continuously* rotated. Besicovitch showed in 1928 that this infimum is zero: the needle can be rotated continuously through all directions within a set of arbitrarily small area, using a "shifting" motion that translates the needle to avoid sweeping large regions. The second question, which is the focus of this section, drops the continuity requirement entirely and asks: what is the smallest measure of a set that *contains* a unit segment in every direction (without any requirement that the segments be related by a continuous motion)? Besicovitch answered this in 1919: such sets can have Lebesgue measure *zero*. This result predates his solution to the needle problem and is the more fundamental of the two—the set problem concerns the static geometry of directional containment, stripped of all topological or kinematic constraints.

The existence of measure-zero sets containing a segment in every direction is one of the most counterintuitive facts in geometric measure theory. Such a set carries full directional information—it "sees" every direction in S^{n-1}—yet is negligible from the perspective of Lebesgue measure. The connection to the Cantor set constructions in proposition 1.4.9 is real: both phenomena exploit the gap between geometric richness and measure-theoretic size.

Definition 5.3.1: Besicovitch Set

A set $K \subseteq \mathbb{R}^n$ is called a *Besicovitch set* (or *Kakeya set*) if for every direction $e \in S^{n-1}$, there exists a point $a \in \mathbb{R}^n$ such that the unit segment $\{a + te \mid 0 \leq t \leq 1\}$ is contained in K.

A Besicovitch set need not be measurable *a priori*, but all known constructions produce Borel (in fact, compact) sets. Any closed disk of radius $1/2$ or larger is a trivial Besicovitch set—every unit segment can be translated to fit inside it—and has positive measure. The content of Besicovitch's theorem is that the measure can be driven to zero while retaining all directions. The existence question thus splits into two parts: first, construct sets of arbitrarily small measure containing all directions; then, pass to a set of measure exactly zero.

The first part uses the *Perron tree* construction (named after Oskar Perron, who gave an early version of this argument in 1928). The key geometric mechanism is that thin triangles pointing in nearly parallel directions can be translated to overlap heavily: the overlap region is large near the apex (where the triangles are narrow and close together) and small near the base (where they spread apart). This produces a net reduction in area while preserving all directional content.

Lemma 5.3.2: Perron Tree

Let $T \subseteq \mathbb{R}^2$ be a triangle with base of length b on a horizontal line and apex at height h above the base. For each point P on the base, the segment from P to the apex defines a direction $\theta(P) \in (0, \pi)$, and as P ranges over the base, $\theta(P)$ sweeps an interval $[\theta_{\min}, \theta_{\max}]$. Then for any $\epsilon > 0$, there exists a compact set $\mathcal{P} \subseteq \mathbb{R}^2$ with $m(\mathcal{P}) < \epsilon$ containing, for each $\theta \in [\theta_{\min}, \theta_{\max}]$, a line segment of length h in direction θ.

***Proof*:**
The construction proceeds by iterated splitting and sliding.

Step 1: One splitting. Divide the base of T into 2 equal segments, producing two sub-triangles T_L and T_R sharing the same apex. Each sub-triangle contains segments in half of the directions: T_L handles $[\theta_{\min}, \theta_{\text{mid}}]$ and T_R handles $[\theta_{\text{mid}}, \theta_{\max}]$. Translate T_R horizontally so that its apex coincides with that of T_L. In the resulting figure $\mathcal{P}_1 = T_L \cup T_R'$, all original directions are present, but the two sub-triangles overlap near the apex, reducing the total area.

To compute the overlap, observe that each sub-triangle narrows linearly from base to apex. At height y above the base ($0 \leq y \leq h$), each sub-triangle occupies a horizontal interval of width $\frac{b}{2}(1 - y/h)$: the factor $(1 - y/h)$ reflects the linear taper from base width $b/2$ to apex width 0. Before the translation, T_L and T_R sit side-by-side with no overlap. After translating T_R so that the two apexes coincide, the horizontal offset between the sub-triangles at height y is $\frac{b}{2}(1 - y/h)$ (the same as the width, since the apex shift equals the base half-width). The overlapping region at height y therefore has width

$$2 \cdot \frac{b}{2}(1 - y/h) - \frac{b}{2} = \frac{b}{2}\left(1 - 2\frac{y}{h} + \frac{y}{h}\right) = \frac{b}{2} \cdot \frac{y}{h}$$

(valid for y where the sub-triangles genuinely overlap). Integrating the area of the overlap:

$$m(T_L \cap T_R') = \int_0^h \frac{b}{2} \cdot \frac{y}{h}\, dy = \frac{bh}{4}$$

Since $m(T_L) + m(T_R) = m(T) = bh/2$, the area of the union is $m(\mathcal{P}_1) = bh/2 - bh/4 = bh/4 = m(T)/2$. The overlap saved exactly half the original area.

Step 2: Iteration. Apply the same splitting-and-sliding procedure to each of the two sub-triangles in $\mathcal{P}_1$, obtaining 4 sub-sub-triangles with total area $m(T)/4$. After k iterations, the construction produces 2^k thin triangles with total area $\leq m(T)/2^k$. Choosing k large enough that $m(T)/2^k < \epsilon$ completes the construction.

Step 3: Directional preservation. At each stage, every sub-triangle retains segments in its assigned sub-interval of directions, and the translation preserves these segments (it shifts their starting points but does not rotate them). The union $\mathcal{P}_k$ therefore contains a segment of length $h(1-2^{-k}) \geq h/2$ in each direction $\theta \in [\theta_{\min}, \theta_{\max}]$. A final rescaling by a factor of $h/(h \cdot (1-2^{-k}))$ adjusts the segment lengths to exactly h and increases the area by a bounded factor. Since k can be chosen arbitrarily large, the total area can still be made $< \epsilon$.

The area estimate $m(\mathcal{P}_k) = m(T)/2^k$ computed above is for a specific splitting scheme. In fact, with a more aggressive sliding strategy (translating the sub-triangles further before each iteration), the area can be reduced even faster. The precise optimum is not needed for the existence result; what matters is that the area can be made arbitrarily small.

The construction should be visualized as follows. At stage 0, the set is a single triangle—a solid wedge containing segments in all directions in a fixed angular interval. At stage 1, the triangle is split into two thinner wedges that are slid together so their tips overlap, producing a figure shaped like an elongated "V" or a narrow tree with two branches. At stage 2, each branch is split and slid again, producing four branches. The process continues: at stage k, the set consists of 2^k very thin triangular "branches" that overlap extensively near their tips. The total area shrinks exponentially with k, yet every original direction is still represented by some branch. The visual resemblance to a tree with an increasingly dense canopy and decreasing total leaf area explains the name "Perron tree."

Example 5.3.3: Perron Tree: First Two Stages

Start with an isosceles triangle T with base $[-1, 1] \times \{0\}$ and apex $(0, 1)$. This triangle has area 1 and contains segments from the base to the apex sweeping directions in the interval $[\pi/4, 3\pi/4]$ (approximately, for this aspect ratio). A segment from the base point $(-1, 0)$ to the apex $(0, 1)$ points in direction $\arctan(1/1) = \pi/4$, and a segment from $(1, 0)$ to $(0, 1)$ points in direction $\pi - \pi/4 = 3\pi/4$. As the base point sweeps from -1 to 1, the direction sweeps from $\pi/4$ to $3\pi/4$.

Stage 1. Split the base at 0. The left sub-triangle T_L has vertices $(-1, 0)$, $(0, 0)$, $(0, 1)$; the right sub-triangle T_R has vertices $(0, 0)$, $(1, 0)$, $(0, 1)$. Translate T_R left by 1 unit so its apex coincides with T_L's. The translated T_R' has vertices $(-1, 0)$, $(0, 0)$, $(-1, 1)$. The union $T_L \cup T_R'$ is a parallelogram-like figure of area 1/2: the overlap occurs in the upper portion where both triangles taper to width zero. T_L retains segments in directions $[\pi/4, \pi/2]$ and T_R' retains directions in

$[\pi/2, 3\pi/4]$, so all original directions are present.

Stage 2. Split each of T_L and T_R' at their base midpoints, obtaining 4 thin triangles. Slide each pair together as before. The resulting figure has area $1/4$ and still contains segments in all directions in $[\pi/4, 3\pi/4]$. The figure now consists of 4 thin triangles that share significant overlap in their upper portions, creating a branching pattern.

After k stages, the area is 2^{-k}, the set consists of 2^k thin triangles of base width 2^{1-k}, and the figure visually resembles a fractal "tree"—hence the name Perron tree. The tree structure arises because at each stage, the splitting doubles the number of branches while the sliding compresses them together.

With the Perron tree in hand, the passage to a measure-zero Besicovitch set proceeds in two steps: first cover all directions, then iterate the compression to reduce the area to zero.

Theorem 5.3.4: Existence of Measure-Zero Besicovitch Sets

There exists a compact Besicovitch set $K \subseteq \mathbb{R}^2$ with $m(K) = 0$.

***Proof*:**
Step 1: Covering all directions. The Perron tree from lemma 5.3.2 produces a set containing segments in all directions within a single angular interval. Since $[0, \pi)$ is compact, finitely many such intervals cover it: choose N triangles $T^{(1)}, \dots, T^{(N)}$, suitably rotated, so that their angular intervals cover $[0, \pi)$. For any $\epsilon > 0$, applying lemma 5.3.2 to each $T^{(j)}$ produces compact sets $\mathcal{P}^{(j)}$ with $m(\mathcal{P}^{(j)}) < \epsilon/N$. Their union $K_\epsilon = \bigcup_{j=1}^N \mathcal{P}^{(j)}$ is compact, satisfies $m(K_\epsilon) < \epsilon$, and contains a segment (of some fixed length $\ell > 0$, independent of ϵ) in every direction. By rescaling, a segment of length ℓ in every direction can be promoted to a *unit* segment in every direction, and multiplies the area by $(1/\ell)^2$; since ϵ was arbitrary, the area of the rescaled set can still be made $< \epsilon$.

Step 2: Passage to measure zero. The sets K_ϵ from Step 1 are not nested (different values of ϵ may produce geometrically unrelated configurations), so taking $K = \bigcap_n K_{1/n}$ does not yield a set of measure zero—the intersection may even be empty. Instead, the construction must be refined to produce a nested sequence.

Choose a compact set K_0 containing unit segments in all directions (for instance, a sufficiently large closed disk). The strategy is to apply the Perron tree compression *within* the existing set K_0, producing a subset $K_1 \subseteq K_0$ of smaller measure that still contains all directions, and then iterate.

Concretely, partition the directions $[0, \pi)$ into N sub-intervals $I_1, \dots, I_N$ and identify the corresponding "sectors" of K_0: for each I_j, the set of points in K_0 lying on unit segments with directions in I_j forms a roughly triangular region Σ_j. Within each sector Σ_j, apply the splitting-and-sliding procedure of lemma 5.3.2, translating sub-triangles *inward* (toward the interior of Σ_j). The resulting set $K_1 = \bigcup_{j=1}^N \mathcal{P}_j$ consists of thin sub-triangles that are contained in K_0 (the inward sliding ensures this), and satisfies $m(K_1) \le m(K_0)/2$. Each direction in $[0, \pi)$ is still represented, since the Perron tree within each sector preserves all directions in I_j.

Iterating this procedure produces a nested sequence $K_0 \supseteq K_1 \supseteq K_2 \supseteq \cdots$ with $m(K_k) \le m(K_0)/2^k$.

Define $K = \bigcap_{k=0}^\infty K_k$. Since $m(K_0) < \infty$ and the sequence is nested, continuity from above (proposition 1.2.5) gives $m(K) = \lim_{k\to\infty} m(K_k) = 0$.

Step 3: K contains all directions. For each direction $\theta \in [0, \pi)$ and each k, the set of starting points a such that $\{a + te_\theta : 0 \le t \le 1\} \subseteq K_k$ is a nonempty compact set $S_k(\theta) \subseteq K_k$ (nonempty by construction, compact as a closed subset of K_k). The sequence $S_0(\theta) \supseteq S_1(\theta) \supseteq \cdots$ is nested and nonempty, so $\bigcap_k S_k(\theta) \neq \emptyset$ by the finite intersection property. Any point $a \in \bigcap_k S_k(\theta)$ satisfies $\{a + te_\theta : 0 \le t \le 1\} \subseteq K_k$ for all k, hence $\{a + te_\theta : 0 \le t \le 1\} \subseteq K$. Therefore K is a Besicovitch set.

The passage from "small area" to "zero area" in Step 2 is the most delicate part of the construction. The argument that the Perron tree compression can be made to produce nested sets requires geometric care—in particular, the sub-triangles must be slid toward the interior of the parent triangle, not away from it. A complete treatment of this nesting, with explicit coordinates at each stage, can be found in Falconer's *The Geometry of Fractal Sets* or Mattila's *Geometry of Sets and Measures in Euclidean Spaces*.

The comparison to the Vitali set construction (theorem 1.0.1) is instructive. Vitali sets are *unmeasurable*—they cannot be assigned any consistent measure. Besicovitch sets are measurable (in fact, compact and Borel) with measure *exactly zero*. Both constructions reveal fundamental limitations: Vitali sets show that not all subsets of $\mathbb{R}$ can be measured; Besicovitch sets show that measure zero does not preclude rich geometric structure. However, Vitali sets require the axiom of choice, while Besicovitch sets are constructed explicitly (within ZF). This constructivity makes the Besicovitch set a more "concrete" pathology—it is not an artifact of set-theoretic axioms but a genuine feature of Euclidean geometry.

The existence of measure-zero Besicovitch sets shows that the Lebesgue measure is too coarse to capture the directional complexity of subsets of $\mathbb{R}^n$. A finer invariant is needed, and Hausdorff dimension from section 5.1 provides exactly this: while $m(K) = 0$ for all Besicovitch sets K, the Hausdorff dimension $\dim_H(K)$ can still be positive and potentially as large as n. The question of how large $\dim_H(K)$ must be is the Kakeya conjecture.

The construction generalizes to higher dimensions via a product argument:

Proposition 5.3.5: Besicovitch Sets in $\mathbb{R}^n$

For every $n \ge 2$, there exists a compact Besicovitch set $K \subseteq \mathbb{R}^n$ with $m(K) = 0$.

***Proof*:**
Let $K_2 \subseteq \mathbb{R}^2$ be a compact Besicovitch set with $m_2(K_2) = 0$ (where m_2 denotes the 2-dimensional Lebesgue measure). The product $K_2 \times [0,1]^{n-2} \subseteq \mathbb{R}^n$ has n-dimensional measure $m_n(K_2 \times [0,1]^{n-2}) = m_2(K_2) \cdot 1 = 0$ by theorem 2.6.10. However, this product set only contains segments whose directions lie in the 2-plane $\mathbb{R}^2 \times \{0\}^{n-2}$: a unit segment in $K_2 \times [0,1]^{n-2}$ has the form $(a + te_1, a_3, \ldots, a_n)$ where $e_1 \in S^1 \subseteq \mathbb{R}^2$ and $a_3, \ldots, a_n$ are fixed. Any direction $e \in S^{n-1}$ with a nonzero component outside the first two coordinates is missing.

To capture all directions in S^{n-1}, take a finite collection of orthogonal transformations $R_1, \ldots, R_M \in O(n)$ such that

$$\bigcup_{j=1}^{M} R_j(\mathbb{R}^2 \times \{0\}^{n-2})$$

covers a neighborhood of every direction in S^{n-1}. Such a finite collection exists by a compactness argument: for each direction $e \in S^{n-1}$, there is a rotation R mapping e into $\mathbb{R}^2 \times \{0\}^{n-2}$ (and in fact into a neighborhood of directions), and the resulting open cover of S^{n-1} has a finite subcover. For each R_j, the set $R_j(K_2 \times [0,1]^{n-2})$ is compact and has $m_n = 0$ (orthogonal transformations preserve Lebesgue measure by theorem 2.7.4 since $|\det R_j| = 1$). The union

$$K = \bigcup_{j=1}^{M} R_j(K_2 \times [0,1]^{n-2})$$

is a finite union of measure-zero compact sets, hence compact and measure zero. For any direction $e \in S^{n-1}$, some R_j maps e into $\mathbb{R}^2 \times \{0\}^{n-2}$, and the corresponding rotated copy $R_j(K_2 \times [0,1]^{n-2})$ contains a unit segment in direction e. Therefore K is a Besicovitch set.

This product-and-rotate construction is crude—it gives a Besicovitch set but reveals nothing about its dimension. To understand why, note that $K_2 \times [0,1]^{n-2}$ has Hausdorff dimension at most $\dim_H(K_2) + (n-2) \leq 2 + (n-2) = n$ by proposition 5.2.6 (applied to Hausdorff dimension, where the analogous inequality holds). Rotating and taking a finite union does not change the dimension, so the construction gives $\dim_H(K) \leq n$—which is already known from $K \subseteq \mathbb{R}^n$. The construction provides no *lower* bound on $\dim_H(K)$ beyond the trivial $\dim_H(K) \geq 1$ (from the presence of line segments).

The dimension of a Besicovitch set is the central open question. By proposition 5.1.8(1), every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) \leq n$ (since $K \subseteq \mathbb{R}^n$). The question is whether $\dim_H(K) = n$ always holds—can the dimension be strictly less than n? In $\mathbb{R}^2$, the answer is no:

Theorem 5.3.6: Davies' Theorem

Every Besicovitch set in $\mathbb{R}^2$ has Hausdorff dimension 2.

Davies' proof (1971) uses a projection argument that is specific to $\mathbb{R}^2$. The key idea is as follows. If $K \subseteq \mathbb{R}^2$ is a Besicovitch set, consider the projection $\pi_\theta : \mathbb{R}^2 \to \mathbb{R}$ onto the line perpendicular to direction θ. Since K contains a unit segment in direction θ, the projection $\pi_\theta(K)$ contains an interval of length $\sin(\alpha)$ for some angle α depending on θ. By a result of Marstrand (1954) relating the Hausdorff dimension of a set to the dimensions of its projections, the fact that $\pi_\theta(K)$ has positive length for *every* direction θ forces $\dim_H(K) \geq 2$. (In precise terms: if $\dim_H(K) < 2$, then $\pi_\theta(K)$ would have zero length for *almost every* θ by Marstrand's projection theorem, contradicting the Besicovitch property.)

This argument fails in $\mathbb{R}^n$ for $n \geq 3$ because the relationship between a set and its projections becomes weaker in higher dimensions. A set of Hausdorff dimension $d < n$ in $\mathbb{R}^n$ projects to a set of positive measure in *some* directions but not necessarily in enough directions to force $d = n$. The combinatorial structure of how lines in $\mathbb{R}^n$ interact with lower-dimensional projections is far richer and less constrained than in $\mathbb{R}^2$, and exploiting the Besicovitch property (a segment in *every* direction) requires fundamentally different techniques. The natural generalization of Davies' theorem—that every Besicovitch set in $\mathbb{R}^n$ has Hausdorff dimension n—is the Kakeya conjecture, which forms the subject of the remaining sections.

To summarize the situation: Besicovitch sets of measure zero exist in every $\mathbb{R}^n$ with $n \geq 2$, so the Lebesgue measure cannot distinguish between a "large" Besicovitch set (like a disk) and a "small" one (measure zero). The Hausdorff dimension provides the finer distinction. In $\mathbb{R}^2$, Davies' theorem settles the matter: $\dim_H(K) = 2$ for every Besicovitch set. In $\mathbb{R}^3$, the Kakeya conjecture was open for over 50 years until Wang and Zahl resolved it in 2025. In $\mathbb{R}^n$ for $n \geq 4$, it remains one of the central open problems in geometric measure theory.

Exercise 5.3.1

1. Verify that at stage k of the Perron tree construction in lemma 5.3.2, the total area of the 2^k sub-triangles (before accounting for overlaps) is $m(T)$, and that the overlap estimate $m(\mathcal{P}_k) \leq m(T)/2^k$ holds by induction.

2. Show that if $K \subseteq \mathbb{R}^n$ is a Besicovitch set and $T \in GL_n(\mathbb{R})$ is an invertible linear map, then $T(K)$ is also a Besicovitch set. Does this remain true if T is injective but not surjective?

3. Show that a Besicovitch set $K \subseteq \mathbb{R}^n$ cannot be contained in any affine hyperplane. Conclude that $\dim_H(K) \geq 1$ for all $n \geq 2$. (This lower bound is far from optimal.)

4. Explain why the product construction $K_2 \times [0,1]^{n-2}$ does not directly contain unit segments in all directions of S^{n-1} (identify the missing directions), and verify that the finite rotation argument in proposition 5.3.5 resolves this.

5. (Nikodym set) A *Nikodym set* $N \subseteq \mathbb{R}^2$ is a set such that for every point $x \in \mathbb{R}^2$, there exists a line ℓ_x through x with $\ell_x \setminus \{x\} \subseteq N$. Show that Nikodym sets of measure zero exist. (*Hint*: the construction is closely related to the Besicovitch construction.)

6. Show that the Besicovitch set K constructed in theorem 5.3.4 is nowhere dense (definition 1.4.12). Conclude that K is topologically "small" (meager, being a compact set with empty interior) yet geometrically "rich" (containing a unit segment in every direction). Compare this with the discussion of the Cantor set in proposition 1.4.13.

5.4 The Kakeya Conjecture and Maximal Functions

The preceding section established that Besicovitch sets—compact sets of Lebesgue measure zero containing a unit segment in every direction—exist in every $\mathbb{R}^n$ with $n \geq 2$. Davies' theorem (theorem 5.3.6) showed that in $\mathbb{R}^2$, such sets must have full Hausdorff dimension despite having zero measure. The central open problem is whether this remains true in all dimensions:

Conjecture 5.4.1: Kakeya Set Conjecture

Every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) = n$. Equivalently (since $\dim_H \leq \dim_M$ by proposition 5.2.3), every Besicovitch set satisfies $\dim_M(K) = n$.

The equivalence stated above is one-directional: the Hausdorff statement is strictly stronger than the Minkowski statement (since $\dim_H \leq \dim_M$, full Hausdorff dimension implies full Minkowski dimension, but not conversely). Both versions are commonly referred to as "the Kakeya conjecture," and both are open for $n \geq 4$. In $n = 2$, both are settled by theorem 5.3.6. In $n = 3$, both were resolved by Wang and Zahl in 2025, as discussed at the end of this chapter.

The conjecture admits a natural discretization in terms of "δ-tubes" that makes it more amenable to combinatorial and analytic methods. The passage from continuous sets to discrete collections of tubes is a standard technique in geometric measure theory: it replaces the measure-zero set K with a family of small-but-positive-volume objects, converting a dimension problem into a volume problem.

Definition 5.4.2: δ-Tube

Let $e \in S^{n-1}$ be a direction, $a \in \mathbb{R}^n$ a center point, and $\delta > 0$. The *δ-tube* centered at a in direction e is the set

$$T_e^\delta(a) = \{x \in \mathbb{R}^n \mid |(x-a) - ((x-a)\cdot e)e| \le \delta, \quad 0 \le (x-a)\cdot e \le 1\}$$

This is a cylinder of length 1, radius δ, with its long axis in direction e and centered at the point $a + \frac{1}{2}e$. Its volume is $|T_e^\delta(a)| = \omega_{n-1}\delta^{n-1}$, where ω_{n-1} is the volume of the unit ball in $\mathbb{R}^{n-1}$.

A δ-tube should be thought of as a "thickened" line segment: as $\delta \to 0$, it shrinks to the unit segment $\{a + te : 0 \le t \le 1\}$. A Besicovitch set K containing a segment in every direction can be "thickened" by replacing each segment with a δ-tube, and the Kakeya conjecture can be rephrased in terms of the volume of the resulting union of tubes.

A family of directions is called *δ-separated* if any two directions $e, e' \in S^{n-1}$ satisfy $|e - e'| \ge \delta$. A maximal δ-separated subset of S^{n-1} has cardinality $\sim \delta^{-(n-1)}$ (since S^{n-1} has $(n-1)$-dimensional surface area $n\omega_n$ and each δ-cap has area $\sim \delta^{n-1}$). The discretized Kakeya conjecture asserts that any such family of tubes has nearly maximal union volume:

Proposition 5.4.3: Discretized Kakeya Conjecture

The Kakeya set conjecture (**??** 5.4.1) is equivalent to the following statement: for every $\epsilon > 0$, there exists a constant $C_\epsilon > 0$ such that if $\mathbb{T}$ is a collection of δ-tubes $\{T_{e_j}^\delta(a_j)\}_{j=1}^N$ with δ-separated directions and $N \ge c\delta^{-(n-1)}$, then

$$\left| \bigcup_{j=1}^N T_{e_j}^\delta(a_j) \right| \ge C_\epsilon\, \delta^\epsilon$$

The bound $|\bigcup T_j| \ge C_\epsilon \delta^\epsilon$ is a weak lower bound: it says the volume cannot decay faster than any power of δ. For comparison, the total volume of all N tubes (if disjoint) would be $N \cdot \omega_{n-1}\delta^{n-1} \sim 1$, while the trivial lower bound from a single tube is $\omega_{n-1}\delta^{n-1}$. The conjecture sits between these extremes, asserting that the union cannot be much smaller than a fixed positive constant (independent of δ).

***Proof*:**
Step 1: Kakeya conjecture implies the discretized statement. Assume the Kakeya set conjecture holds, and let $\mathbb{T} = \{T_{e_j}^\delta(a_j)\}_{j=1}^N$ be a collection of δ-tubes with δ-separated directions and $N \ge c\delta^{-(n-1)}$. Define $K = \bigcap_{k=1}^\infty \overline{\bigcup_{j=1}^N T_{e_j}^{1/k}(a_j)}$, which is a compact set containing a unit segment in each direction e_j. By extending to a full Besicovitch set (adding segments in the finitely many missing directions), the Kakeya conjecture gives $\dim_M(K) = n$.

The δ-neighborhood $K_\delta = \{x \mid d(x, K) < \delta\}$ contains all the tubes $T_{e_j}^\delta(a_j)$. By the volume formulation of Minkowski dimension (section 5.2), $\dim_M(K) = n$ implies that for any $\epsilon > 0$ and all sufficiently small δ:

$$|K_\delta| \ge c_\epsilon\, \delta^{n-n+\epsilon} = c_\epsilon\, \delta^\epsilon$$

More precisely, the definition $\dim_M(K) = n$ means $\lim_{\delta\to 0} \frac{\log|K_\delta|}{\log\delta} = n - n = 0$ (using the n-dimensional volume formulation), so $|K_\delta|$ decays slower than any power of δ. Since $|\bigcup T_j| \le |K_\delta|$, the discretized bound follows.

Step 2: Discretized statement implies the Kakeya conjecture. Conversely, let K be a Besicovitch set and fix $\epsilon > 0$. For each $\delta > 0$, choose a maximal δ-separated set of directions $\{e_j\}_{j=1}^N$ on S^{n-1} (so $N \sim \delta^{-(n-1)}$). For each e_j, the set K contains a unit segment in direction e_j, so there exist center points a_j such that the δ-tube $T_{e_j}^\delta(a_j)$ is contained in the δ-neighborhood K_δ. In particular, $|\bigcup_j T_{e_j}^\delta(a_j)| \le |K_\delta|$.

By the discretized statement, $|K_\delta| \ge |\bigcup T_j| \ge C_\epsilon \delta^\epsilon$. Taking logarithms and dividing by $\log\delta$ (which is negative for $\delta < 1$):

$$\frac{\log|K_\delta|}{\log\delta} \le \frac{\log C_\epsilon + \epsilon\log\delta}{\log\delta} = \epsilon + \frac{\log C_\epsilon}{\log\delta} \to \epsilon \quad \text{as } \delta \to 0$$

By the volume formulation of Minkowski dimension, $n - \dim_M(K) = \lim_{\delta\to 0} \frac{\log|K_\delta|}{\log\delta} \le \epsilon$, giving $\dim_M(K) \ge n - \epsilon$. Since $\epsilon > 0$ was arbitrary, $\dim_M(K) = n$.

The passage from $\dim_M(K) = n$ to $\dim_H(K) = n$ requires an additional argument: the discretized statement must be applied at multiple scales simultaneously (not just a single δ) to produce efficient covers for the Hausdorff measure. The details of this "multi-scale" refinement are standard and can be found in Wolff's lecture notes or Mattila's textbook.

The discretized formulation transforms the Kakeya conjecture from a question about abstract Hausdorff dimension into a concrete volume estimate for unions of tubes. This reformulation is the form in which the conjecture is typically attacked, since tubes are rigid geometric objects whose intersections can be analyzed explicitly. The ϵ-loss in the bound $|\bigcup T_j| \ge C_\epsilon\delta^\epsilon$ may seem artificial, but it is essential: a bound of the form $|\bigcup T_j| \ge c$ (with no δ-dependence) is equivalent to the stronger statement that K has positive Lebesgue measure, which is false for Besicovitch sets. The δ^ϵ factor allows the volume to decay, but only sub-polynomially.

The intersection structure of tubes is governed by a basic geometric estimate: two tubes at angle α intersect in a region of controlled volume. This estimate is the foundation for all results on the Kakeya problem:

Lemma 5.4.4: Tube Intersection Estimate

Let $T_e^\delta(a)$ and $T_{e'}^\delta(a')$ be two δ-tubes in $\mathbb{R}^n$ with $|e - e'| = \alpha \ge \delta$. Then

$$|T_e^\delta(a) \cap T_{e'}^\delta(a')| \lesssim \frac{\delta^{n-1}}{\alpha}$$

where the implicit constant depends only on n. In particular, when $\alpha \sim 1$ (nearly perpendicular tubes), the intersection has volume $\sim \delta^{n-1}$, while when $\alpha \sim \delta$ (nearly parallel tubes), the intersection can be as large as $\sim \delta^{n-2}$.

***Proof*:**
The intersection of two cylinders of radius δ and length 1 whose axes meet at angle α is contained in a "slab" of thickness $\delta/\sin\alpha$ in the direction bisecting the two axes, and of cross-section $\sim \delta^{n-2}$ in the remaining directions. Since $\sin\alpha \geq c\alpha$ for $\alpha \in [0, \pi/2]$, the volume of the intersection is bounded by

$$|T_e^\delta(a) \cap T_{e'}^\delta(a')| \lesssim \delta^{n-2} \cdot \frac{\delta}{\alpha} = \frac{\delta^{n-1}}{\alpha}$$

completing the estimate.

The tube intersection estimate controls the degree to which two tubes can overlap as a function of their angular separation. When $\alpha \sim 1$, the tubes cross nearly perpendicularly, and their intersection is a small "dot" of volume $\sim \delta^{n-1}$. When $\alpha \sim \delta$, the tubes are nearly parallel and can overlap along their entire length, giving an intersection of volume $\sim \delta^{n-2}$—much larger. The worst case for the Kakeya problem is when many tubes are nearly parallel, since their overlaps waste volume. The Kakeya conjecture asserts that no arrangement of tubes can exploit this overlap efficiently enough to make the total union volume small.

The total overlap of a collection of N tubes with δ-separated directions can be quantified. Denoting $F = \sum_{j=1}^N \chi_{T_j}$, the L^2 norm squared of F counts pairwise intersections:

$$\|F\|_{L^2}^2 = \sum_{i,j} |T_i \cap T_j|$$

The diagonal terms contribute $N \cdot |T_j| \sim N\delta^{n-1}$. The off-diagonal terms are controlled by the tube intersection estimate and depend on the distribution of angles between pairs of tubes. Bounding this L^2 norm is the key to Córdoba's argument and the starting point for all higher-dimensional approaches.

With the tube intersection estimate in hand, the Kakeya problem can be approached through a maximal function that encodes the "worst-case" averaging behavior of a function along tubes. The connection to the Hardy–Littlewood maximal function (definition 3.4.6) is direct: the Hardy–Littlewood function takes suprema over balls centered at a point, while the Kakeya maximal function takes suprema over tubes in a given direction. The directional constraint is what makes the Kakeya problem harder: the Hardy–Littlewood maximal theorem (theorem 3.4.7) gives L^p bounds for $p > 1$ with constants independent of the ball radius, while the Kakeya maximal function incurs a loss of $\delta^{-\epsilon}$ (and the conjecture asserts this loss is the worst possible).

Definition 5.4.5: Kakeya Maximal Function

Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $\delta > 0$. The *Kakeya maximal function of f* is the function $f_\delta^* : S^{n-1} \to [0, \infty]$ defined by

$$f_\delta^*(e) = \sup_{a \in \mathbb{R}^n} \frac{1}{|T_e^\delta(a)|} \int_{T_e^\delta(a)} |f(x)|\, dx$$

where the supremum is over all positions $a \in \mathbb{R}^n$ of the δ-tube in direction e.

For each direction e, $f_\delta^*(e)$ captures the maximum average value of $|f|$ over any tube pointing in direction e. If $f = \chi_K$ for a Besicovitch set K, then $f_\delta^*(e) \geq c > 0$ for every direction (since K contains a segment, hence a substantial fraction of a tube, in every direction), and bounding f_δ^* from above constrains how "concentrated" the set K can be. The Kakeya maximal conjecture asserts a quantitative L^p bound:

Conjecture 5.4.6: Kakeya Maximal Conjecture

For every $\epsilon > 0$, there exists $C_\epsilon > 0$ such that for all $\delta > 0$ and $f \in L^n(\mathbb{R}^n)$:

$$\|f_\delta^*\|_{L^n(S^{n-1})} \leq C_\epsilon \, \delta^{-\epsilon} \|f\|_{L^n(\mathbb{R}^n)}$$

The exponent $p = n$ is critical: for $p < n$, the bound fails (as can be shown using a bush-like configuration), and for $p > n$, the bound follows from simpler covering arguments. Testing the maximal conjecture on $f = \chi_K$ recovers the set conjecture: if $\|f_\delta^*\|_{L^n} \leq C_\epsilon \delta^{-\epsilon}$ and $f_\delta^*(e) \geq c$ for every e, then the surface area of S^{n-1} gives $c^n \cdot n\omega_n \leq C_\epsilon^n \delta^{-n\epsilon} \|f\|_{L^n}^n$, forcing $\|f\|_{L^n}^n \geq c' \delta^{n\epsilon}$, hence $|K_\delta| \geq c' \delta^{n\epsilon}$ for any ϵ, establishing $\dim_M(K) = n$. The maximal conjecture is thus strictly stronger than the set conjecture—it provides quantitative control, not just a dimension bound.

Example 5.4.7: The Maximal Function for Simple Configurations

Consider $n = 2$ and $f = \chi_R$ where R is a rectangle of width δ and length 1 aligned with the x-axis. Then $f_\delta^*(e_1) = 1$ (the tube in direction $e_1 = (1, 0)$ can be chosen to coincide with R), while $f_\delta^*(e) \lesssim \delta/|e - e_1|$ for $|e - e_1| \gg \delta$ (a tube at angle α to R intersects R in a set of area $\sim \delta^2/\alpha$, and dividing by the tube area δ gives the average $\sim \delta/\alpha$). Integrating over S^1: $\|f_\delta^*\|_{L^2}^2 \lesssim 1 + \delta^2 \sum_{k=1}^{1/\delta} 1/(k\delta)^2 \cdot \delta \sim \log(1/\delta)$. This shows the $\log(1/\delta)$ factor in Córdoba's estimate is achieved (up to constants) by a single rectangle, confirming the bound is sharp.

In $\mathbb{R}^2$, the maximal conjecture is a theorem, proved by Córdoba in 1977 using a technique known as the "bush argument." The proof is a model for higher-dimensional approaches and uses only the tube intersection estimate and the Cauchy–Schwarz inequality:

Theorem 5.4.8: Córdoba's Estimate

In $\mathbb{R}^2$, for every $\delta > 0$ and $f \in L^2(\mathbb{R}^2)$:

$$\|f_\delta^*\|_{L^2(S^1)} \leq C \left(\log \frac{1}{\delta} \right)^{1/2} \|f\|_{L^2(\mathbb{R}^2)}$$

In particular, the Kakeya maximal conjecture holds in $\mathbb{R}^2$ (since $(\log 1/\delta)^{1/2} \leq C_\epsilon \delta^{-\epsilon}$ for any $\epsilon > 0$).

***Proof*:**

By a standard linearization argument, it suffices to prove the dual estimate: if $\{T_j\}_{j=1}^N$ is a collection of δ-tubes with δ-separated directions (so $N \sim 1/\delta$), and $\{a_j\}_{j=1}^N$ are non-negative coefficients, then

$$\left\| \sum_{j=1}^N a_j \chi_{T_j} \right\|_{L^2(\mathbb{R}^2)}^2 \leq C \log \frac{1}{\delta} \sum_{j=1}^N a_j^2 |T_j|$$

Taking $a_j = 1$ for all j gives a bound on the L^2 norm of the sum of characteristic functions, which by duality controls the maximal function.

Step 1: Expanding the L^2 norm. Compute

$$\left\|\sum_j a_j \chi_{T_j}\right\|_{L^2}^2 = \sum_{i,j} a_i a_j |T_i \cap T_j|$$

Split this into diagonal terms ($i = j$) and off-diagonal terms ($i \neq j$).

Step 2: Diagonal terms. The diagonal contributes $\sum_j a_j^2 |T_j| = \omega_1 \delta \sum_j a_j^2$, where $|T_j| = \omega_1 \delta$ is the area of a δ-tube in $\mathbb{R}^2$ (a rectangle of width 2δ and length 1).

Step 3: Off-diagonal terms. For $i \neq j$, the tube intersection estimate (lemma 5.4.4) gives $|T_i \cap T_j| \lesssim \delta^2/|e_i - e_j|$ in $\mathbb{R}^2$ (since $n = 2$, the bound is $\delta^{n-1}/\alpha = \delta/\alpha$; including the width of the tubes gives δ^2/α after a more careful accounting). Order the directions so that $e_1, e_2, \ldots, e_N$ are arranged consecutively around S^1 with $|e_j - e_{j+1}| \sim \delta$. Then $|e_i - e_j| \gtrsim |i - j|\delta$, and by Cauchy–Schwarz ($a_i a_j \leq \frac{1}{2}(a_i^2 + a_j^2)$):

$$\begin{aligned}
\sum_{i \neq j} a_i a_j |T_i \cap T_j| &\lesssim \sum_{i \neq j} a_i a_j \frac{\delta^2}{|i - j|\delta} \\
&= \delta \sum_{i \neq j} \frac{a_i a_j}{|i - j|} \\
&\leq \delta \sum_j a_j^2 \sum_{k=1}^{N} \frac{1}{k} \\
&\lesssim \delta \log N \sum_j a_j^2 \\
&\sim \delta \log \frac{1}{\delta} \sum_j a_j^2
\end{aligned}$$

where the second inequality uses the Cauchy–Schwarz step and the symmetry of the sum. The harmonic sum $\sum_{k=1}^N 1/k \sim \log N \sim \log(1/\delta)$ produces the logarithmic factor.

Step 4: Combining. Adding the diagonal and off-diagonal contributions:

$$\left\|\sum_j a_j \chi_{T_j}\right\|_{L^2}^2 \lesssim \delta \log \frac{1}{\delta} \sum_j a_j^2 = \log \frac{1}{\delta} \sum_j a_j^2 |T_j|$$

The duality between $\sum a_j \chi_{T_j}$ and the maximal function f_δ^* (via testing against f) then yields the claimed bound $\|f_\delta^*\|_{L^2} \leq C(\log 1/\delta)^{1/2} \|f\|_{L^2}$.

The factor $(\log 1/\delta)^{1/2}$ in Córdoba's bound is an artifact of the harmonic sum over tube intersections. It is sub-polynomial in $1/\delta$ (growing slower than any power $\delta^{-\epsilon}$), so the Kakeya maximal conjecture follows immediately. As a corollary, the Kakeya set conjecture in $\mathbb{R}^2$ follows from the maximal conjecture, giving a second proof of theorem 5.3.6.

The structure of the proof reveals why $n = 2$ is tractable. In $\mathbb{R}^2$, the directions live on S^1 and can be linearly ordered. The harmonic sum $\sum_{k=1}^N 1/k \sim \log N$ captures the total pairwise overlap, and this sum grows slowly enough that the Cauchy–Schwarz step produces a useful bound. In $\mathbb{R}^n$ for $n \geq 3$, the directions live on S^{n-1} and cannot be linearly ordered. The pairwise intersection sum becomes a double sum over an $(n-1)$-dimensional index set, and the geometric structure of the intersections depends not just on the angle between two tubes but also on their relative positions in space. A direct repetition of Córdoba's approach in $\mathbb{R}^n$ yields the bound $\dim_H(K) \geq (n+1)/2$, which for $n = 3$ gives $\dim_H \geq 2$—weaker than Wolff's $5/2$. The additional gain in Wolff's argument comes from exploiting the *linear* structure of tubes (not just their pointwise intersections):

Theorem 5.4.9: Wolff's Bound

Every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) \geq (n+2)/2$.

***Proof Key ideas*:**

The proof introduces the "hairbrush" argument, which improves on Córdoba's "bush" argument by exploiting the linear structure of tubes (rather than just their pointwise intersections).

Step 1: Bush versus hairbrush. In Córdoba's argument, the worst case is a "bush": many tubes passing through a common δ-ball. In a bush, the tubes spread out radially, and their union has volume $\sim N\delta^{n-1}$ (each tube contributes its full volume beyond the common ball). This gives $|\bigcup T_j| \geq c$ when $N \sim \delta^{-(n-1)}$, which is the full conjecture—but only if a bush actually occurs. When tubes are spread out and no bush exists, Córdoba's argument loses efficiency.

Step 2: Wolff's observation. Instead of looking for a point through which many tubes pass, look for a *tube* T_0 that is intersected by many other tubes. The tubes intersecting T_0 form a "hairbrush" rooted at T_0. The intersections are spread *along* T_0 (not concentrated at a point), and the tubes emanating from these intersection points spread into the ambient space.

Step 3: Volume estimate. The hairbrush structure produces a volume gain over the naive estimate. By a pigeonhole argument, there exists a tube T_0 intersected by $\gtrsim N^{1/2}$ other tubes (this uses the total intersection count). The $N^{1/2}$ tubes of the hairbrush have δ-separated intersection points along T_0 and δ-separated directions, so their union has volume $\gtrsim N^{1/2} \cdot \delta^{n-1} \sim \delta^{(n-2)/2}$. This leads to the bound $\dim_M(K) \geq (n+2)/2$ (and with more work, $\dim_H(K) \geq (n+2)/2$).

The pigeonhole step is the heart of the argument. The total number of "incidences" (pairs (T_i, T_j) with $T_i \cap T_j \neq \emptyset$ and $i \neq j$) is at least $cN^2\delta$ (from a counting argument using the volume of the union). Distributing these incidences among N tubes, at least one tube T_0 must have $\geq cN\delta \sim N^{1/2} \cdot N^{1/2}\delta$ incidences. The $N^{1/2}\delta$ factor, combined with the geometry of tube intersections along T_0, gives the improved volume bound.

For $n = 3$, Wolff's bound gives $\dim_H(K) \geq 5/2$. The gap between $5/2$ and the conjectured value of 3 resisted all attempts for nearly three decades, until Wang and Zahl closed it in 2025 (see the end of this chapter). For $n \geq 4$, Wolff's bound of $(n+2)/2$ remains the best known general result, though incremental improvements have been obtained using arithmetic combinatorics (Bourgain, Katz–Tao) and polynomial methods (Guth).

The following table summarizes the state of knowledge for the Kakeya set conjecture in low dimensions, illustrating the progression from Wolff's bound to the resolution in $\mathbb{R}^3$:

n	Conjectured $\dim_H$	Best lower bound	Reference
2	2	2 (resolved)	Davies (1971)
3	3	3 (resolved)	Wang–Zahl (2025)
4	4	3	Wolff (1995)
5	5	7/2	Wolff (1995)
n	n	$(n+2)/2$	Wolff (1995)

Exercise 5.4.1

1. Verify the tube intersection estimate (lemma 5.4.4) explicitly in $\mathbb{R}^2$: show that two rectangles of width 2δ and length 1 whose long axes meet at angle α have intersection area $\lesssim \delta^2/\alpha$.
2. Show that the Kakeya maximal conjecture implies the Kakeya set conjecture. Specifically, if K is a Besicovitch set and $f = \chi_{K_\delta}$ (the δ-neighborhood of K), show that the maximal conjecture gives $|K_\delta| \geq c_\epsilon \delta^{n\epsilon}$, and conclude $\dim_M(K) = n$.
3. In $\mathbb{R}^2$, use Córdoba's estimate directly (not through the maximal function) to show that a union of $N = 1/\delta$ tubes with δ-separated directions has area $\gtrsim 1/\log(1/\delta)$. (*Hint*: let $F = \sum_j \chi_{T_j}$. Compute $\|F\|_{L^1}$ and $\|F\|_{L^2}$, then apply Cauchy–Schwarz: $\|F\|_{L^1}^2 \leq |\operatorname{supp}(F)| \cdot \|F\|_{L^2}^2$.)
4. Compute the volume of the intersection of two δ-tubes in $\mathbb{R}^3$ at angle α, and verify the estimate $\lesssim \delta^2/\alpha$.
5. Show that the "bush" configuration—N tubes all passing through a common δ-ball—achieves the volume lower bound $|\bigcup T_j| \gtrsim 1$ when the directions are δ-separated. This shows the Kakeya conjecture would be trivially true if all Besicovitch sets had a "bush" structure.
6. Explain why the Kakeya maximal conjecture remains open in $\mathbb{R}^3$ even after the Wang–Zahl resolution of the set conjecture. What additional information would the maximal conjecture provide?

5.5 The Finite Field Kakeya Conjecture

In 1999, Wolff proposed studying the Kakeya problem over finite fields as a model for the Euclidean case. The finite field setting strips away all analytic structure—there are no limits, no measures, no covering arguments—and reduces the problem to pure combinatorics. A "line" in $\mathbb{F}_q^n$ is a set of q points $\{x + ty : t \in \mathbb{F}_q\}$ for $x, y \in \mathbb{F}_q^n$ with $y \neq 0$, and a "Kakeya set" is a set containing a line in every direction. The question is: how large must such a set be?

Wolff hoped that techniques for the finite field problem would transfer to $\mathbb{R}^n$. The finite field analogue was resolved in 2008 by Dvir, using an algebraic method—the "polynomial method"—that is entirely different from the combinatorial and analytic approaches used in the Euclidean setting. The proof is short and self-contained, making it a compelling application of algebraic techniques to a combinatorial problem.

Definition 5.5.1: Finite Field Kakeya Set

Let $\mathbb{F}_q$ be a finite field with q elements and $\mathbb{F}_q^n$ the n-dimensional vector space over $\mathbb{F}_q$. A set $K \subseteq \mathbb{F}_q^n$ is called a *Kakeya set* if for every direction $y \in \mathbb{F}_q^n \setminus \{0\}$, there exists $x \in \mathbb{F}_q^n$ such that $\{x + ty : t \in \mathbb{F}_q\} \subseteq K$.

Since $\mathbb{F}_q^n$ is a finite set of cardinality q^n, the question is quantitative: what is the minimum cardinality of a Kakeya set? There are $(q^n - 1)/(q - 1)$ directions (points of projective space $\mathbb{P}^{n-1}(\mathbb{F}_q)$), and each line has q points, so a Kakeya set contains at least $(q^n - 1)/(q - 1)$ lines. If these lines were disjoint, the minimum size would be $q \cdot (q^n - 1)/(q - 1) \sim q^n$. The finite field Kakeya conjecture asserts that even with maximal overlap, the size is still $\sim q^n$.

The analogy with the Euclidean case is as follows. A "Besicovitch set" in $\mathbb{F}_q^n$ would be a Kakeya set of size $o(q^n)$—much smaller than the ambient space. The finite field conjecture states that no such set exists: every Kakeya set has size $\geq cq^n$, meaning it must occupy a positive fraction of $\mathbb{F}_q^n$. In the Euclidean setting, Besicovitch sets of Lebesgue measure zero *do* exist, so the analogy is not perfect. The finite field conjecture is closer in spirit to the Minkowski dimension statement: the "dimension" of a Kakeya set (measured by the exponent in $|K| \sim q^d$) must equal n.

Before Dvir's work, partial results on the finite field conjecture were obtained by Wolff (1999), Bourgain–Katz–Tao (2004), and others, using combinatorial methods that mirrored the Euclidean techniques (bush and hairbrush arguments). These gave bounds of the form $|K| \geq cq^{\alpha n}$ for various $\alpha < 1$. Dvir's proof, by contrast, gives the optimal exponent $\alpha = 1$ in a single stroke, using an entirely algebraic approach.

The key to Dvir's proof is a vanishing lemma that connects the geometry of lines to the algebra of polynomials:

Lemma 5.5.2: Vanishing on Kakeya Sets

Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set and let $P \in \mathbb{F}_q[x_1, \ldots, x_n]$ be a polynomial of degree $\deg P < q$. If P vanishes on K (i.e., $P(x) = 0$ for all $x \in K$), then P is the zero polynomial.

***Proof*:**
Suppose $P \neq 0$ and $\deg P = d < q$. Write $P = P_d + P_{d-1} + \cdots + P_0$ where P_j is the homogeneous component of degree j and $P_d \neq 0$.

Let $y \in \mathbb{F}_q^n \setminus \{0\}$ be any direction. Since K is a Kakeya set, there exists $x \in \mathbb{F}_q^n$ such that $\{x + ty : t \in \mathbb{F}_q\} \subseteq K$. Consider the univariate polynomial $Q(t) = P(x + ty)$. Since P vanishes on K and $x + ty \in K$ for all $t \in \mathbb{F}_q$, the polynomial Q vanishes at all q elements of $\mathbb{F}_q$. The degree of Q as a polynomial in t is at most $d < q$. A univariate polynomial of degree $< q$ over $\mathbb{F}_q$ with q roots must be the zero polynomial. Therefore $Q \equiv 0$.

The leading coefficient of $Q(t) = P(x + ty)$ (as a polynomial in t) is $P_d(y)$: expanding $P(x + ty)$, the terms of degree d in t come exclusively from the homogeneous part P_d, giving the leading term $t^d P_d(y)$. Since $Q \equiv 0$, all coefficients vanish, and in particular $P_d(y) = 0$.

This holds for every $y \in \mathbb{F}_q^n \setminus \{0\}$. Since P_d is homogeneous, $P_d(0) = 0$ as well (every monomial in P_d has positive degree). Thus P_d vanishes on all of $\mathbb{F}_q^n$. A nonzero polynomial of degree d in n variables over $\mathbb{F}_q$ can vanish on at most $d \cdot q^{n-1}$ points of $\mathbb{F}_q^n$ (by the Schwartz–Zippel lemma). Since $d < q$, this bound is $< q^n = |\mathbb{F}_q^n|$, so P_d cannot vanish on all of $\mathbb{F}_q^n$ unless $P_d = 0$. This contradicts $\deg P = d$, completing the proof.

The vanishing lemma has a clean algebraic content: a Kakeya set is "algebraically large" in the sense that no low-degree polynomial can vanish on it. The theorem then follows by a dimension count:

Theorem 5.5.3: Dvir's Theorem

Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set. Then $|K| \geq \binom{q+n-1}{n} \geq \frac{q^n}{n!}$.

***Proof*:**
The vector space $V = \{P \in \mathbb{F}_q[x_1, \dots, x_n] \mid \deg P < q\}$ of polynomials of degree $< q$ in n variables has dimension

$$\dim V = \binom{q+n-1}{n}$$

over $\mathbb{F}_q$ (the number of monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 + \cdots + a_n < q$). Consider the evaluation map $\mathrm{ev} : V \to \mathbb{F}_q^K$ defined by $\mathrm{ev}(P) = (P(x))_{x \in K}$. By lemma 5.5.2, $\ker(\mathrm{ev}) = \{0\}$, so ev is injective. Since the codomain has dimension $|K|$ over $\mathbb{F}_q$:

$$|K| \geq \dim V = \binom{q+n-1}{n} \geq \frac{q^n}{n!}$$

The last inequality follows from $\binom{q+n-1}{n} = \frac{(q+n-1)(q+n-2)\cdots q}{n!} \geq \frac{q^n}{n!}$.

The proof is three lines once the vanishing lemma is established: the polynomial space is too large to be annihilated by evaluation on a small set. The logic is: (1) a Kakeya set cannot support the vanishing of any nonzero low-degree polynomial (by the vanishing lemma), so (2) the evaluation map from the space of low-degree polynomials to $\mathbb{F}_q^K$ must be injective, and (3) injectivity forces $|K| \geq \dim V$. The proof thus converts a geometric property (containing a line in every direction) into an algebraic property (no low-degree polynomial vanishes on K), and then uses linear algebra to extract a size bound.

The bound $|K| \geq q^n/n!$ is sharp up to the constant $1/n!$. The factor $n!$ comes from the binomial coefficient and can be improved: Dvir–Kopparty–Saraf–Sudan (2013) showed $|K| \geq q^n/2^n$ by a refinement involving the multiplicity of vanishing rather than just vanishing itself. The optimal constant remains open, though for large q the bound $|K| \geq (1 + o(1))q^n/2^n$ is expected to be close to sharp.

A natural question is whether the polynomial method gives any information about the *structure* of Kakeya sets, not just their size. Over finite fields, the answer is partially yes: the method shows that Kakeya sets must be "algebraically spread" (they cannot be contained in a low-degree variety), and subsequent work has used this to study the geometric distribution of points in Kakeya sets. Over $\mathbb{R}^n$, as the following remark explains, the method requires substantial modification.

Example 5.5.4: Dvir's Bound in Small Cases

1. Let $q = 3$ and $n = 2$, so $\mathbb{F}_3^2$ has 9 points. The theorem gives $|K| \geq \binom{4}{2} = 6$. A Kakeya set in $\mathbb{F}_3^2$ must contain a line (of 3 points) in each of the 4 directions of $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0), (0,1), (1,1), (1,2)\}$. The 4 lines have 12 points in total, but with overlaps. One can verify that the minimum Kakeya set has $|K| = 7$ (achievable by arranging the 4 lines to

share a common point and one additional overlap), consistent with the lower bound of 6.

2. Let $q = 2$ and $n = 3$, so $\mathbb{F}_2^3$ has 8 points. The theorem gives $|K| \geq \binom{4}{3} = 4$. There are $(2^3 - 1)/(2 - 1) = 7$ directions in $\mathbb{P}^2(\mathbb{F}_2)$. Each "line" has only 2 points ($\{x, x + y\}$ for $t \in \{0, 1\} = \mathbb{F}_2$). With 7 lines of 2 points each, a total of 14 point-memberships must be distributed among at most 8 points. The complement of any single point in $\mathbb{F}_2^3$ has 7 points and can be checked to be a Kakeya set (every direction is represented), so $|K| = 7$ is achievable. The lower bound of 4 is far from tight in this case, reflecting the looseness of the $1/n!$ constant for small q.

5.5.5 Remark: Why the Polynomial Method Does Not Transfer to $\mathbb{R}^n$ The polynomial method is effective over $\mathbb{F}_q$ because a nonzero polynomial of degree d can vanish on at most $d \cdot q^{n-1}$ points of $\mathbb{F}_q^n$ (the Schwartz–Zippel lemma)—a quantitative bound that makes the dimension count work. Over $\mathbb{R}$, no such bound exists: a nonzero polynomial of degree d can vanish on an uncountable set (e.g., the zero set of $x_1^2 + x_2^2 - 1$ is the unit circle, which has the cardinality of $\mathbb{R}$). The discrete counting argument that powers Dvir's proof has no continuous analogue.

The conceptual barrier is that over finite fields, "small" sets are sets with few elements, and "large" sets have many elements—a distinction captured by cardinality. Over $\mathbb{R}^n$, "small" sets are sets of low Hausdorff dimension, and "large" sets have high dimension—a distinction that cardinality cannot detect ($\mathbb{R}$ and $\mathbb{R}^2$ have the same cardinality but different dimensions). The polynomial method over finite fields exploits cardinality, not dimension, which is why it does not transfer.

A partial descendant of the polynomial method in the Euclidean setting is the *polynomial partitioning* technique introduced by Guth and Katz (2010), which uses a polynomial to partition $\mathbb{R}^n$ into cells of controlled geometry rather than to vanish on a set. The idea is to choose a polynomial P of degree d such that $\mathbb{R}^n \setminus Z(P)$ (the complement of the zero set) splits into $\sim d^n$ cells, each containing a controlled number of tubes. The zero set $Z(P)$ itself captures the tubes that lie on an algebraic variety, which can be handled separately. This technique has led to progress on related problems (such as the Erdős distinct distances problem) but has not yet resolved the Euclidean Kakeya conjecture.

The contrast between the finite field and Euclidean settings encapsulates a broader theme in modern combinatorial geometry: problems that are "purely combinatorial" over finite fields often require analytic and multi-scale techniques over $\mathbb{R}$. The Kakeya problem is a paradigmatic example of this phenomenon, and the interplay between algebraic, combinatorial, and analytic methods remains an active area of research.

Exercise 5.5.1

1. Verify Dvir's bound for $q = 2$, $n = 3$: compute the number of directions in $\mathbb{P}^2(\mathbb{F}_2)$, construct a Kakeya set in $\mathbb{F}_2^3$ explicitly, and check $|K| \geq \binom{4}{3} = 4$.

2. Show that lemma 5.5.2 fails if the degree condition is relaxed to $\deg P \leq q$ (rather than $\deg P < q$). (*Hint*: consider the polynomial $P(x) = \prod_{a \in \mathbb{F}_q}(x_1 - a)$, which has degree q and vanishes on $\mathbb{F}_q^n$ regardless of whether K is a Kakeya set.)

3. (Schwartz–Zippel lemma) Prove that a nonzero polynomial $P \in \mathbb{F}_q[x_1, \ldots, x_n]$ of total degree d has at most $d \cdot q^{n-1}$ zeros in $\mathbb{F}_q^n$. (*Hint*: induct on n. For $n = 1$ this is the fundamental

theorem of algebra over $\mathbb{F}_q$. For $n > 1$, write $P = \sum_{j=0}^{d} P_j(x_1, \ldots, x_{n-1})x_n^j$ and consider $a \in \mathbb{F}_q$ such that $P(\cdot, a)$ is nonzero.)

4. Show that a random subset $K \subseteq \mathbb{F}_q^n$ of size $|K| = q^{n-1}/2$ is, with high probability, *not* a Kakeya set when q is large. (*Hint*: for a fixed direction, the probability that a specific line lies in K is $(|K|/q^n)^q$. Apply a union bound over all directions.)

5. The *joints problem* asks: given L lines in $\mathbb{R}^3$, how many "joints" (points where three non-coplanar lines meet) can there be? Guth and Katz (2010) proved the sharp bound $O(L^{3/2})$ using the polynomial method. Formulate and prove the analogous result over $\mathbb{F}_q^3$: if $\mathcal{L}$ is a set of lines in $\mathbb{F}_q^3$ and J is the set of joints, show $|J| \leq C|\mathcal{L}|^{3/2}$ using a vanishing argument analogous to lemma 5.5.2. (*Hint*: if $|J| > c|\mathcal{L}|^{3/2}$, find a polynomial of degree $< (|\mathcal{L}|/c)^{1/2}$ vanishing on J, then show its gradient vanishes at every joint, contradicting non-coplanarity.)

5.6 The Wang–Zahl Theorem and Open Frontiers

The results presented so far establish the Kakeya set conjecture in $\mathbb{R}^2$ (Davies, 1971) and provide the lower bound $\dim_H(K) \geq (n+2)/2$ in $\mathbb{R}^n$ (Wolff, 1995). In $\mathbb{R}^3$, Wolff's bound gives $\dim_H(K) \geq 5/2$, leaving a gap of 1/2 below the conjectured value of 3. This gap was narrowed incrementally over three decades through the introduction of techniques from arithmetic combinatorics, additive number theory, and polynomial methods. The following is a summary of the main advances:

1. Wolff (1995): $\dim_H(K) \geq (n+2)/2$. (Hairbrush argument.)
2. Katz–Łaba–Tao (2000): $\dim_M(K) > 5/2 + \epsilon_0$ in $\mathbb{R}^3$ for an explicit small $\epsilon_0 > 0$. (Additive combinatorics, sticky Kakeya configurations.)
3. Katz–Tao (2002): $\dim_H(K) \geq (2 - \sqrt{2})(n-4) + 3$ for $n \geq 4$. (Sum-product estimates from arithmetic combinatorics.)
4. Katz–Zahl (2019): $\dim_H(K) \geq 5/2 + \epsilon_1$ in $\mathbb{R}^3$ for a small $\epsilon_1 > 0$. (Polynomial partitioning and degree reduction.)
5. Wang–Zahl (2022): Structural classification of near-extremal Kakeya sets in $\mathbb{R}^3$ into "sticky" and "non-sticky" regimes.
6. Wang–Zahl (2025): $\dim_H(K) = 3$ and $\dim_M(K) = 3$ for all Besicovitch sets in $\mathbb{R}^3$.

Each step introduced new ideas that reshaped the problem.

The Katz–Łaba–Tao result (2000) was the first to break Wolff's 5/2 barrier in $\mathbb{R}^3$. Their approach introduced the notion of "sticky" Kakeya sets: configurations where, at every intermediate scale $\delta < r < 1$, the tubes cluster into thin slabs. They showed that sticky configurations are in some sense the "hardest" case, and handled non-sticky configurations by a refined hairbrush argument. The improvement $5/2 + \epsilon_0$ was small but conceptually significant: it demonstrated that the barrier at 5/2 was not fundamental.

The Katz–Tao papers (2002) brought arithmetic combinatorics—specifically, sum-product estimates over $\mathbb{R}$—into the Kakeya problem. The connection is through the "arithmetic Kakeya conjecture":

if the set of slopes in a Kakeya configuration has additive structure (e.g., is close to an arithmetic progression), then sum-product estimates force the set to be large. This approach established the first bounds that improve over $(n+2)/2$ in high dimensions.

The Katz–Zahl paper (2019) adapted the polynomial partitioning technique from incidence geometry to the Kakeya setting. By partitioning $\mathbb{R}^3$ using the zero set of a polynomial, they decomposed the Kakeya problem into "cellular" and "algebraic" components, handling each with different methods. This gave a second independent proof that the $5/2$ barrier is not sharp.

The Wang–Zahl papers (2022, 2025) unified and extended these ideas into a comprehensive structural theory of Kakeya sets at multiple scales, culminating in the resolution of the three-dimensional case.

Theorem 5.6.1: Wang–Zahl Theorem

Every Besicovitch set $K \subseteq \mathbb{R}^3$ has Hausdorff dimension 3 and Minkowski dimension 3.

The proof, contained in the preprint "Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions" (arXiv:2502.17655, February 2025), is approximately 130 pages. The core argument is an induction on scales, and the key innovation is a reformulation of the Kakeya problem in terms of a more general "super-Kakeya" condition involving convex sets (called "grains") rather than tubes. The term "super-Kakeya" refers to the fact that the reformulated problem is *stronger* than the original Kakeya conjecture: it asserts a volume bound for unions of convex sets satisfying a condition that includes the Kakeya condition as a special case. The stronger formulation is paradoxically easier to prove, because it is preserved under the induction on scales (the original Kakeya condition is not, since passing from scale δ to scale $\sqrt{\delta}$ distorts tubes into more general convex sets).

5.6.2 Remark: Key Ideas of the Proof The proof of theorem 5.6.1 has three main components.
1. Induction on scales. The problem at scale δ (tubes of radius δ) is reduced to a problem at scale $\sqrt{\delta}$ (tubes of radius $\sqrt{\delta}$). At the coarser scale, the tubes are replaced by "grains"—convex sets of controlled dimensions that capture the large-scale geometry. The inductive step shows that if the volume bound fails at scale δ, it must also fail at scale $\sqrt{\delta}$ with a quantitatively worse failure, which eventually contradicts the trivial bound at scale ~ 1.
2. Sticky versus non-sticky configurations. In their 2022 paper, Wang and Zahl classified near-extremal Kakeya configurations into two regimes. In the "sticky" regime, tubes cluster into thin slabs at every intermediate scale, creating a self-similar structure. In the "non-sticky" regime, tubes spread out at intermediate scales. The two regimes require different arguments: the non-sticky case can be handled by a refined version of Wolff's hairbrush argument (combined with polynomial methods), while the sticky case requires a new inductive structure using the grain decomposition.
3. The role of dimension 3. When a grain in the inductive step degenerates (becomes flat or thin), the problem reduces to a lower-dimensional Kakeya problem: a 2-dimensional problem for flat grains, or a 1-dimensional problem for thin grains. In $\mathbb{R}^3$, these lower-dimensional problems are already solved (by Davies' theorem and trivial estimates). This "dimension reduction" is the mechanism that closes the induction. In $\mathbb{R}^n$ for $n \geq 4$, the degenerate cases reduce to Kakeya problems in $\mathbb{R}^3, \mathbb{R}^4, \ldots$, which remain open, blocking the induction.

The Wang–Zahl theorem settles the Kakeya set conjecture in $\mathbb{R}^3$, but several related problems remain open. The state of the art and the main open directions are summarized below.

5.6.3 Remark: Open Problems and Directions *The conjecture in higher dimensions.* The Kakeya set conjecture remains open for all $n \geq 4$. The barrier is the dimension reduction step in the Wang–Zahl argument: degenerate configurations in $\mathbb{R}^n$ reduce to Kakeya problems in $\mathbb{R}^{n-1}$ (and lower), which are not yet solved for $n-1 \geq 4$. Resolving the conjecture in $\mathbb{R}^4$ would likely require either a different inductive scheme or new ideas to handle the degenerate cases without reducing to lower dimensions.
The Kakeya maximal conjecture. Even in $\mathbb{R}^3$, the Kakeya maximal conjecture (**??** 5.4.6) remains open. The Wang–Zahl proof establishes the set conjecture but does not produce the quantitative L^p bounds needed for the maximal statement. Strengthening the result to a maximal function estimate would have immediate consequences for the restriction and Bochner–Riesz conjectures in harmonic analysis.
Furstenberg sets. A *Furstenberg set* is a variant of a Besicovitch set where, instead of requiring a full unit segment in every direction, one requires a subset of Hausdorff dimension $\geq s$ inside a unit segment in every direction. The Furstenberg set conjecture predicts the minimum Hausdorff dimension of such sets as a function of s and n. Wang and Zahl's techniques yield partial results on Furstenberg sets, but the full conjecture remains open.
The (n, k)-Besicovitch conjecture. An (n, k)-Besicovitch set is a set of measure zero in $\mathbb{R}^n$ containing a translate of every k-dimensional unit disk. For $k > 1$, it is conjectured that no such set exists (in contrast to $k = 1$, where they do exist). The case $k = 1$ is the classical Kakeya problem; the higher-k cases are open for all n.
Connections to number theory. The Kakeya conjecture has implications for problems in analytic number theory, including estimates for exponential sums and bounds on the distribution of primes in arithmetic progressions. Montgomery's conjecture on Dirichlet polynomials implies the Kakeya conjecture in all dimensions; the resolution of the three-dimensional case does not immediately yield progress on Montgomery's conjecture, but it strengthens the evidence and provides new techniques that may be applicable. For the number-theoretic connections, see [1].

The resolution of the three-dimensional Kakeya conjecture represents a major advance in geometric measure theory, and the techniques introduced by Wang and Zahl—induction on scales with grain decomposition, the sticky/non-sticky dichotomy, and the interaction with polynomial methods—are expected to have applications beyond the Kakeya problem itself.

The chapter has traced a progression from abstract measure theory (Hausdorff dimension as a refinement of Lebesgue measure) through geometric construction (Besicovitch sets as measure-zero objects with full directional content) to combinatorial analysis (tube intersection estimates and the Kakeya maximal function) and algebra (the polynomial method over finite fields). This progression reflects the interdisciplinary nature of the Kakeya problem: it sits at the intersection of geometric measure theory, combinatorics, algebra, and harmonic analysis. The harmonic-analytic connections—in particular, the relationship between the Kakeya conjecture and the restriction, Bochner–Riesz, and local smoothing conjectures for the Fourier transform—are treated later in this book.

The remaining challenges (higher dimensions, maximal function estimates, Furstenberg sets) provide concrete targets for future work, and the tools developed in this chapter—Hausdorff dimension, Minkowski dimension, tube intersection estimates, and the relationship between directional geometry and volume bounds—are the foundation on which that progress will be built. The reader who wishes to go further is directed to the following references:

1. Wolff, *Lectures on Harmonic Analysis* (AMS, 2003): lecture notes covering the Kakeya problem, restriction estimates, and related topics.
2. Mattila, *Geometry of Sets and Measures in Euclidean Spaces* (Cambridge, 1995): the standard reference for Hausdorff dimension and geometric measure theory.
3. Falconer, *The Geometry of Fractal Sets* (Cambridge, 1985): a more elementary treatment of dimension theory and Besicovitch sets.
4. Wang and Zahl, "Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions" (arXiv:2502.17655, 2025): the proof of the three-dimensional case.

Exercise 5.6.1

1. Verify that Wolff's bound (theorem 5.4.9) gives $\dim_H(K) \geq 5/2$ for Besicovitch sets in $\mathbb{R}^3$, and compare this to the Wang–Zahl result $\dim_H(K) = 3$.
2. Show that if the Kakeya set conjecture holds in $\mathbb{R}^n$, it automatically holds in $\mathbb{R}^m$ for all $m \leq n$. (*Hint*: embed $\mathbb{R}^m$ into $\mathbb{R}^n$ and use proposition 5.1.8(4).)
3. Explain in your own words why the Wang–Zahl argument does not immediately extend to $\mathbb{R}^4$. Identify the specific step that fails and what additional input would be needed.
4. The Besicovitch set K constructed in theorem 5.3.4 has $m(K) = 0$ and $\dim_H(K) = 2$ (by theorem 5.3.6). Show that K must have $\dim_H(K) \geq 2$ directly from the Kakeya maximal function estimate in $\mathbb{R}^2$ (theorem 5.4.8), without appealing to Davies' theorem.
5. Show that a Besicovitch set $K \subseteq \mathbb{R}^3$ with $m(K) = 0$ is nowhere dense (has empty interior). Relate this to definition 1.4.12: K is topologically small (meager) yet has full Hausdorff dimension by the Wang–Zahl theorem. Compare with proposition 1.4.13.

6

Haar Measure

The Riesz-Markov theorem of chapter 4 turned the construction of a measure into the construction of a functional. This chapter cashes that in on the one class of spaces where a canonical functional presents itself: a locally compact group. On such a group the operation of left translation acts on functions, and one asks for a measure that translation cannot see, a *left-invariant* Radon measure. That there is such a measure, unique up to scale, is the theorem of Haar, and it is the single fact on which all of harmonic analysis on groups rests: without it there is no $L^2(G)$ to decompose, no convolution algebra $L^1(G)$, no unitary regular representation. The point-set theory of locally compact groups is developed in [9]; here they enter only as the spaces on which the invariant measure is built.

The construction is Weil's, and it is worth seeing why a construction is needed at all rather than a formula. On $\mathbb{R}^n$ or a Lie group an invariant measure can be written down from a volume form, but a general locally compact group carries no such differentiable structure. Weil's idea replaces the missing formula with a comparison: the *covering number* $(f : g)$ counts how many left-translates of a fixed small bump g are needed to dominate f, a discrete stand-in for the ratio of the integrals of f and g. As the bump g concentrates at the identity this ratio stabilises, and the limit is the invariant functional; Riesz-Markov then converts it into Haar measure. The limit is extracted by a compactness argument, an application of Tychonoff's theorem, and this is one of its most striking uses. The first section carries out the construction and the second proves uniqueness and introduces the modular function that measures the failure of left-invariance to be two-sided.

6.1 Existence of Haar Measure

On the line, Lebesgue measure is singled out by translation invariance: the length of a set does not change when the set is slid along $\mathbb{R}$. A locally compact group carries the same operation of sliding, now written as multiplication by a fixed element, and the question this section answers is whether an invariant measure still exists in the passage from $\mathbb{R}$ to an arbitrary such group. The affirmative answer is Weil's, and its engine is the Riesz-Markov theorem of chapter 4: rather than build the measure set by set, the argument constructs an invariant way of averaging continuous functions and lets theorem 4.2.1 convert that averaging operation into a measure. The whole labour is therefore in manufacturing a positive, left-invariant linear functional on $C_c(G)$ out of nothing but the group operation and compactness.

Throughout, G is a locally compact Hausdorff group in the sense of [9, Topological Group]: a group carrying a locally compact Hausdorff topology for which multiplication and inversion are continuous. The identity is e. The group acts on functions by *left translation*: for $s \in G$ and $f\colon G \to \mathbb{R}$,

$$L_s f(x) = f(s^{-1}x)$$

the inverse in $s^{-1}x$ being what makes $s \mapsto L_s$ a left action, $L_{st} = L_s L_t$. Left translation carries $C_c(G)$ into itself, since $L_s f$ is continuous and $\operatorname{supp}(L_s f) = s \operatorname{supp} f$ is compact. Write $C_c^+(G)$ for the nonnegative functions in $C_c(G)$. The construction below produces a measure invariant under left translation, a *left* Haar measure; the right-invariant theory is its mirror image, obtained by replacing L_s with right translation throughout, and the comparison of the two is the subject of section 6.2.

The one analytic input, needed only at the final additivity step, is that a compactly supported continuous function cannot oscillate too fast under small translations. On $\mathbb{R}$ this is uniform continuity; on a group the correct formulation is uniform continuity with respect to the group's own notion of smallness, a neighbourhood of the identity.

Lemma 6.1.1: Left-Uniform Continuity

Let G be a locally compact Hausdorff group and $f \in C_c(G)$. For every $\varepsilon > 0$ there is a neighbourhood V of the identity e such that

$$|f(sx) - f(x)| < \varepsilon \quad \text{for all } x \in G \text{ and all } s \in V$$

Equivalently, $\|L_{s^{-1}} f - f\|_\infty < \varepsilon$ for every $s \in V$. The same argument produces a neighbourhood V' of e with the right-translate estimate

$$|f(xt) - f(x)| < \varepsilon \quad \text{for all } x \in G \text{ and all } t \in V'$$

***Proof*:**
The equivalence of the two formulations is a change of variable: $L_{s^{-1}} f(x) = f(sx)$, so $\|L_{s^{-1}} f - f\|_\infty = \sup_x |f(sx) - f(x)|$, and the displayed bound over all x is exactly $\|L_{s^{-1}} f - f\|_\infty \le \varepsilon$. It suffices to prove the pointwise bound.

Step 1: Localize using continuity at each point. Let $K = \operatorname{supp} f$, a compact set. Fix $\varepsilon > 0$. Because f is continuous, for each point $y \in G$ there is an open neighbourhood of y on which f varies by less than $\varepsilon/2$. Continuity of multiplication lets this neighbourhood be taken in translated form: there is an open neighbourhood W_y of the identity e with

$$|f(wy) - f(y)| < \frac{\varepsilon}{2} \quad \text{for all } w \in W_y$$

since $w \mapsto wy$ is continuous and equals y at $w = e$. By the symmetric-neighbourhood halving lemma of [9, Halving a Neighbourhood], choose for each y a symmetric open neighbourhood V_y of e with $V_y V_y \subseteq W_y$.

Step 2: Cover the support and extract a finite subcover. As y ranges over K, the open sets $V_y y$ cover K, each containing its own y. Compactness of K yields finitely many points $y_1, \ldots, y_n$ with

$$K \subseteq \bigcup_{i=1}^{n} V_{y_i} y_i$$

Set $V = \bigcap_{i=1}^{n} V_{y_i}$, an open neighbourhood of e as a finite intersection; shrinking if necessary, take V symmetric. This V is the neighbourhood claimed.

Step 3: The uniform estimate. Fix $s \in V$ and $x \in G$; the goal is $|f(sx) - f(x)| < \varepsilon$. Consider two cases according to whether x or sx meets the support.

Suppose first that $x \in K$. Then $x \in V_{y_i} y_i$ for some i, so $x = v y_i$ with $v \in V_{y_i}$; since V_{y_i} is symmetric this gives $x y_i^{-1} = v \in V_{y_i} \subseteq W_{y_i}$ and hence, applying the Step 1 bound with $w = x y_i^{-1}$,

$$|f(x) - f(y_i)| = |f\big((x y_i^{-1}) y_i\big) - f(y_i)| < \frac{\varepsilon}{2}$$

For the translated point, $sx = (sv) y_i$ with $s \in V \subseteq V_{y_i}$ and $v \in V_{y_i}$, so $sv \in V_{y_i} V_{y_i} \subseteq W_{y_i}$, and the same bound gives

$$|f(sx) - f(y_i)| = |f\big((sv) y_i\big) - f(y_i)| < \frac{\varepsilon}{2}$$

The triangle inequality now yields $|f(sx) - f(x)| < \varepsilon$.

If instead $x \notin K$ but $sx \in K$, apply the previous paragraph with x replaced by sx and s by $s^{-1} \in V$ (using V symmetric): from $sx \in K$ one gets $|f(s^{-1}(sx)) - f(sx)| < \varepsilon$, which is $|f(x) - f(sx)| < \varepsilon$. Finally, if neither x nor sx lies in K, then $f(x) = f(sx) = 0$ and the bound is trivial. In every case $|f(sx) - f(x)| < \varepsilon$, which is the left-translate estimate.

Step 4: The right-translate estimate. The argument for V' is identical with the two sides of every product interchanged. Continuity of f at y and continuity of the map $w \mapsto yw$ furnish an open neighbourhood W'_y of e with $|f(yw) - f(y)| < \varepsilon/2$ for $w \in W'_y$; halving gives a symmetric V'_y with $V'_y V'_y \subseteq W'_y$; the sets $y_i V'_{y_i}$ cover K; and $V' = \bigcap_i V'_{y_i}$, taken symmetric, satisfies $|f(xt) - f(x)| < \varepsilon$ for all x and all $t \in V'$, by the same three-case analysis on whether x or xt meets K. This proves both estimates, completing the proof.

Left-uniform continuity says that translating f by an element close to e perturbs it uniformly little in supremum norm. The role reserved for it is delicate: the covering numbers built next are only *sub*additive in general, and the reason they become exactly additive in the limit is that, over a small enough neighbourhood, a function and its translates are indistinguishable to within ε, so the overcounting in subadditivity is squeezed to zero. Nothing else in the construction uses continuity of f beyond boundedness; this lemma is where the topology of G enters the analysis.

Example 6.1.2: Left-Uniform Continuity on $\mathbb{R}$ and on the Circle

On the additive group $\mathbb{R}$, left translation by s is the shift $L_s f(x) = f(x-s)$, and lemma 6.1.1 reduces to the classical statement that a continuous function of compact support is uniformly continuous:

given ε, a symmetric interval $V = (-\delta, \delta)$ works, since $|x - y| < \delta$ forces $|f(x) - f(y)| < \varepsilon$. The neighbourhood V of 0 is exactly the modulus of continuity δ.

On the circle group $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, which is compact, every continuous function has compact support, so $C_c(\mathbb{T}) = C(\mathbb{T})$ and the lemma applies to all continuous functions. A rotation by a small angle moves each point a small arc-length, and a continuous function on the compact circle is uniformly continuous, so again a small symmetric arc V about the identity controls $|f(sx) - f(x)|$ uniformly. The compactness of $\mathbb{T}$ is what removes the compact-support hypothesis; on $\mathbb{R}$ the hypothesis cannot be dropped, as the non-uniformly-continuous $\sin(x^2)$ shows.

With the analytic tool in hand, the combinatorial core of the construction can begin. The idea is to measure the size of one function relative to another by asking how many translated copies of the second are needed to cover the first. If g is thought of as a fixed unit of mass, concentrated near a small region, then covering f by weighted left-translates of g and adding the weights estimates the total mass of f in units of g.

Definition 6.1.3: Haar Covering Number

Let G be a locally compact Hausdorff group and let $f, g \in C_c^+(G)$ with g not identically zero. The *Haar covering number* of f with respect to g is

$$(f : g) = \inf \left\{ \sum_{i=1}^{n} c_i \;\middle|\; n \in \mathbb{N},\ c_i \geq 0,\ s_i \in G,\ f \leq \sum_{i=1}^{n} c_i \, L_{s_i} g \right\}$$

the infimum of the total weight over all finite dominations of f by nonnegative combinations of left-translates of g (including the empty sum, $n = 0$, which dominates $f \equiv 0$ with total weight 0).

The covering number is a discrete stand-in for the ratio of integrals $\left(\int f\right)/\left(\int g\right)$. If a left-invariant integral $\int$ existed, then $f \leq \sum_i c_i L_{s_i} g$ would integrate to $\int f \leq \left(\sum_i c_i\right) \int g$ by invariance, so $\left(\int f\right)/\left(\int g\right) \leq (f : g)$; the covering number is an a priori upper bound for a ratio that does not yet exist, computed purely from the order structure and the group action. The construction extracts the integral by making g shrink toward a point mass at e, at which stage the upper bound becomes exact.

Example 6.1.4: Covering Numbers of Intervals on $\mathbb{R}$

Take $G = \mathbb{R}$. Let g be the tent function of height 1 supported on $[-1, 1]$, equal to $1 - |x|$ there, and let f be the trapezoid of height 2 supported on $[-3, 3]$, equal to 2 on $[-2, 2]$ and tapering linearly to 0 at ± 3. To dominate f by translates $c_i L_{s_i} g(x) = c_i \, g(x - s_i)$, place tents along the interval: translate g to sit centred at each half-integer $-\frac{5}{2}, -2, -\frac{3}{2}, \ldots, \frac{5}{2}$ and give each weight 2. Adjacent unit tents of weight 2 overlap so that their sum stays at least 2 across the plateau $[-2, 2]$ and dominates the tapering flanks, so this domination by eleven translates of weight 2 gives $(f : g) \leq 22$. In the other direction, proposition 6.1.5(f) below gives $(f : g) \geq (\sup f)/(\sup g) = 2/1 = 2$. The true value lies between; its precise size is unimportant, because only the ratio $(f : g)/(f_0 : g)$ against a fixed reference f_0 survives into the limit. What matters is that g concentrated near 0 makes the covering count proportional to the length of the base of f, which is the geometric content of Lebesgue measure.

The covering number obeys a package of order and invariance properties, each read directly off the definition. They are the axioms an integral must satisfy, verified here at the discrete level; the limit will inherit every one of them that is preserved by a passage to a cluster point.

Proposition 6.1.5: Properties of the Covering Number

Let $f, f_1, f_2, g, h \in C_c^+(G)$ with g, h not identically zero. Then:

1. *Finiteness:* $(f : g) < \infty$.
2. *Subadditivity and homogeneity:* $(f_1 + f_2 : g) \leq (f_1 : g) + (f_2 : g)$ and $(cf : g) = c\,(f : g)$ for every $c \geq 0$.
3. *Left invariance:* $(L_s f : g) = (f : g)$ for every $s \in G$.
4. *Submultiplicativity:* $(f : h) \leq (f : g)\,(g : h)$.
5. *Monotonicity:* $f_1 \leq f_2$ implies $(f_1 : g) \leq (f_2 : g)$.
6. *Lower bound:* $\dfrac{\sup f}{\sup g} \leq (f : g)$.

***Proof*:**
(a) Finiteness. The support $K = \operatorname{supp} f$ is compact. Since g is continuous and not identically zero, it is positive somewhere; pick a point x_0 with $g(x_0) = a > 0$ and an open neighbourhood N of x_0 with $g > a/2$ on N. The left-translates sN, as s ranges over G, cover G, so finitely many $s_1 N, \ldots, s_n N$ cover K. For each i the translate $L_{s_i} g$ satisfies $L_{s_i} g(x) = g(s_i^{-1} x) > a/2$ whenever $x \in s_i N$, because then $s_i^{-1} x \in N$, where $g > a/2$. Set $c = 2 \sup f / a$ and consider the domination candidate $\sum_{i=1}^n c\, L_{s_i} g$. At any point $x \in K$, some index i has $x \in s_i N$, so that term alone gives

$$\sum_{i=1}^{n} c\, L_{s_i} g(x) \geq c\, L_{s_i} g(x) > c \cdot \frac{a}{2} = \sup f \geq f(x)$$

using that all terms are nonnegative; off K the function f vanishes so the inequality $f \leq \sum_i c\, L_{s_i} g$ is automatic there. This exhibits a finite domination of total weight $nc = 2n \sup f / a$, so $(f : g) \leq 2n \sup f / a < \infty$.

(b) Subadditivity and homogeneity. If $f_1 \leq \sum_i c_i L_{s_i} g$ and $f_2 \leq \sum_j d_j L_{t_j} g$ are dominations, then adding them gives $f_1 + f_2 \leq \sum_i c_i L_{s_i} g + \sum_j d_j L_{t_j} g$, a domination of $f_1 + f_2$ with total weight $\sum_i c_i + \sum_j d_j$. Taking infima over the two families separately yields $(f_1 + f_2 : g) \leq (f_1 : g) + (f_2 : g)$. For homogeneity with $c > 0$, scaling a domination of f by c gives a domination of cf and multiplies every admissible total weight by c, so $(cf : g) = c(f : g)$; the case $c = 0$ is $(0 : g) = 0$, taking the empty sum.

(c) Left invariance. Left translation is an order isomorphism commuting with the action: if $f \leq \sum_i c_i L_{s_i} g$ then applying L_s to both sides, and using $L_s L_{s_i} = L_{s s_i}$, gives $L_s f \leq \sum_i c_i L_{s s_i} g$, a domination of $L_s f$ with the same weights. Thus $(L_s f : g) \leq (f : g)$, and applying the same to

$L_{s^{-1}}$ gives the reverse inequality, so $(L_s f : g) = (f : g)$.

(d) Submultiplicativity. Suppose $f \leq \sum_i c_i L_{s_i} g$ and $g \leq \sum_j d_j L_{t_j} h$. Substituting the second domination into the first and using $L_{s_i}\big(\sum_j d_j L_{t_j} h\big) = \sum_j d_j L_{s_i t_j} h$ (again $L_{s_i} L_{t_j} = L_{s_i t_j}$, and L_{s_i} preserves the pointwise inequality $g \leq \sum_j d_j L_{t_j} h$ since it only relabels the argument),

$$f \leq \sum_i c_i L_{s_i} g \leq \sum_i c_i \sum_j d_j L_{s_i t_j} h = \sum_{i,j} c_i d_j \, L_{s_i t_j} h$$

a domination of f by translates of h with total weight $\big(\sum_i c_i\big)\big(\sum_j d_j\big)$. Taking infima over the two families gives $(f : h) \leq (f : g)(g : h)$.

(e) Monotonicity. If $f_1 \leq f_2$ then any domination $f_2 \leq \sum_i c_i L_{s_i} g$ also dominates f_1, so the admissible weights for f_1 include those for f_2, and the infimum can only decrease: $(f_1 : g) \leq (f_2 : g)$.

(f) Lower bound. Let $f \leq \sum_i c_i L_{s_i} g$ be any domination. Evaluating at a point x^* where $f(x^*) = \sup f$,

$$\sup f = f(x^*) \leq \sum_i c_i \, g(s_i^{-1} x^*) \leq \Big(\sum_i c_i\Big) \sup g$$

since each $g(s_i^{-1} x^*) \leq \sup g$. Hence $\sum_i c_i \geq (\sup f)/(\sup g)$ for every domination, and taking the infimum gives $(\sup f)/(\sup g) \leq (f : g)$, completing the proof.

The six properties are precisely what an invariant integral should do, minus one: the covering number is subadditive, but genuine additivity, $(f_1 + f_2 : g) = (f_1 : g) + (f_2 : g)$, fails, because a single translate of g can straddle the supports of f_1 and f_2 and be counted once toward covering their sum while it would have to be counted twice to cover them separately. Recovering additivity is the whole difficulty, and it is resolved only in the limit as g shrinks. Before taking that limit, the covering number is normalized so that its values live in a fixed compact range independent of the shrinking parameter g, which is what will make a limit available.

Fix once and for all a reference function $f_0 \in C_c^+(G)$ with f_0 not identically zero, and for $f \in C_c^+(G)$ and $g \in C_c^+(G)$ (both nonzero) define the normalized ratio

$$I_g(f) = \frac{(f : g)}{(f_0 : g)}$$

The point of dividing by $(f_0 : g)$ is that the arbitrary unit g cancels to first order, leaving a quantity pinned between two bounds that do not mention g at all. Applying submultiplicativity, proposition 6.1.5(d), twice, once as $(f : g) \leq (f : f_0)(f_0 : g)$ and once as $(f_0 : g) \leq (f_0 : f)(f : g)$, gives after dividing by $(f_0 : g)$ the two-sided estimate

$$\frac{1}{(f_0 : f)} \leq I_g(f) \leq (f : f_0) \tag{6.1}$$

valid for every admissible g. The lower bound is nonzero because $(f_0 : f) < \infty$ by proposition 6.1.5(a). Thus each $I_g(f)$ is confined to the compact interval $[1/(f_0 : f),\ (f : f_0)]$, whose endpoints depend on f but not on g.

This uniform confinement is exactly what a compactness argument feeds on. As g shrinks, the family $\{I_g\}$ has no reason to converge, but it cannot escape to infinity, and Tychonoff's theorem converts that boundedness into the existence of a limiting functional along a subnet.

The functionals I_g, for g ranging over $C_c^+(G) \setminus \{0\}$, are points of the product space

$$X = \prod_{f \in C_c^+(G)\setminus\{0\}} \left[\tfrac{1}{(f_0 : f)},\ (f : f_0) \right]$$

one coordinate for each nonzero $f \in C_c^+(G)$, the f-coordinate of the point I_g being the number $I_g(f)$. That this is a legitimate point of X is precisely the bound (6.1). Each factor is a compact interval in $\mathbb{R}$, so by *Tychonoff's theorem* [9, Tychonoff's Theorem], which states that an arbitrary product of compact spaces is compact in the product topology, the space X is compact.

To take the limit as g concentrates at e, index the family not by the functions g themselves but by the neighbourhoods of e they can be squeezed into. Let $\mathcal{N}$ be the collection of open neighbourhoods of e, directed by reverse inclusion: declare $V' \succeq V$ when $V' \subseteq V$. This is a genuine directed set, since any two neighbourhoods V_1, V_2 have the common refinement $V_1 \cap V_2 \succeq V_1, V_2$, itself a neighbourhood of e. For each $V \in \mathcal{N}$ choose, once and for all, a nonzero $g_V \in C_c^+(G)$ with $\operatorname{supp} g_V \subseteq V$; such a g_V exists by lemma 4.1.5, applied to a compact neighbourhood of e sitting inside V. The assignment $V \mapsto I_{g_V}$ is then a net in the compact space X. A net in a compact space has a cluster point, so there is a point $I \in X$ that the net $\{I_{g_V}\}$ frequents: for every neighbourhood of I in X and every $V_0 \in \mathcal{N}$, some $V \subseteq V_0$ has I_{g_V} in that neighbourhood. Concretely, since the topology on X is the product topology and the g_V can be squeezed into arbitrarily small neighbourhoods of e, I being a cluster point means that for every finite set of functions $f_1, \ldots, f_k$, every $\varepsilon > 0$, and every neighbourhood U of e, there is a nonzero $g \in C_c^+(G)$ with $\operatorname{supp} g \subseteq U$ and

$$|I_g(f_j) - I(f_j)| < \varepsilon \quad \text{for } j = 1, \ldots, k \tag{6.2}$$

This $I\colon C_c^+(G) \setminus \{0\} \to \mathbb{R}$ is the candidate functional. It remains to show that it has the algebraic properties of an integral, at which point the Riesz-Markov theorem finishes the job. The properties fall into two groups: those that hold for every I_g and are inherited by the cluster point because they are closed conditions, and additivity, which holds for I_g only approximately and becomes exact precisely in the limit.

Closed properties inherited by I. Each of the following holds for every I_g, hence for the cluster point I, because it is a conjunction of equalities and non-strict inequalities among finitely many coordinates, and such conditions are closed in the product topology and preserved under passing to a cluster point. First, normalization: $I_g(f_0) = (f_0 : g)/(f_0 : g) = 1$ for every g, so $I(f_0) = 1$; in particular I is not identically zero. Second, positive homogeneity: $I_g(cf) = cI_g(f)$ for $c \geq 0$ by proposition 6.1.5(b), so $I(cf) = cI(f)$. Third, monotonicity: $f_1 \leq f_2$ gives $I_g(f_1) \leq I_g(f_2)$ by proposition 6.1.5(e), so $I(f_1) \leq I(f_2)$; in particular $I(f) \geq 0$. Fourth, left invariance: $I_g(L_s f) = I_g(f)$ by proposition 6.1.5(c), so $I(L_s f) = I(f)$. Fifth, subadditivity: $I_g(f_1 + f_2) \leq I_g(f_1) + I_g(f_2)$ by proposition 6.1.5(b), so $I(f_1 + f_2) \leq I(f_1) + I(f_2)$.

The one property still missing is the reverse inequality, superadditivity, which together with subadditivity forces additivity. This is the only place the shrinking of g does real work, and the only place left-uniform continuity is used.

Lemma 6.1.6: Approximate Additivity of the Covering Number

Let $f_1, f_2 \in C_c^+(G)$ and $\varepsilon > 0$. There is a neighbourhood U of e such that for every nonzero $g \in C_c^+(G)$ with $\mathrm{supp} g \subseteq U$,

$$(f_1 : g) + (f_2 : g) \leq (1 + \varepsilon)\,(f_1 + f_2 : g)$$

***Proof*:**
Step 1: A master function separating the supports. Let $K = \mathrm{supp}(f_1 + f_2)$, a compact set, and choose by lemma 4.1.5 a function $\varphi \in C_c^+(G)$ with $\varphi \equiv 1$ on K. Fix a parameter $\delta > 0$ to be chosen at the end, and set

$$F = f_1 + f_2 + \delta\varphi$$

which is strictly positive on K (indeed $F \geq \delta$ there) and lies in $C_c^+(G)$. Define, for $k = 1, 2$, the functions

$$h_k(x) = \begin{cases} \dfrac{f_k(x)}{F(x)} & F(x) > 0 \\ 0 & F(x) = 0 \end{cases}$$

Each h_k is continuous: on the open set $\{F > 0\}$ it is a quotient of continuous functions, so continuous there. At a point x with $F(x) = 0$ the function h_k vanishes together with f_k throughout a whole neighbourhood, so it is continuous at x as well. Indeed $\mathrm{supp} f_k \subseteq \mathrm{supp}(f_1 + f_2) = K$ and $\varphi \equiv 1$ on K, so $F = f_1 + f_2 + \delta\varphi \geq \delta > 0$ on K; hence $F(x) = 0$ forces $x \notin K$, and since K is closed, x has a neighbourhood disjoint from K. On that neighbourhood $f_k \equiv 0$ (as $\mathrm{supp} f_k \subseteq K$) and therefore $h_k \equiv 0$, so h_k is locally constant and in particular continuous at x. Thus $h_k \in C_c^+(G)$, $\mathrm{supp} h_k \subseteq \mathrm{supp} f_k$, and $h_1 + h_2 \leq 1$ everywhere, with $h_1 + h_2 = (f_1 + f_2)/F \leq 1$ where $F > 0$. Crucially $f_k = h_k F$ pointwise.

Step 2: Uniform continuity of the h_k. By the right-translate estimate of lemma 6.1.1 applied to h_1 and h_2, there is a neighbourhood U of e such that

$$|h_k(xt) - h_k(x)| < \delta \quad \text{for all } x \in G,\ t \in U,\ k = 1, 2 \tag{6.3}$$

Shrinking U, take it symmetric. This U is the neighbourhood claimed; the estimate below holds for any g with $\mathrm{supp} g \subseteq U$.

Step 3: Transport a domination of F into a joint domination of f_1, f_2. Fix a nonzero $g \in C_c^+(G)$ with $\mathrm{supp} g \subseteq U$, and let $F \leq \sum_i c_i L_{s_i} g$ be any domination of F by translates of g. The claim is that this single domination controls f_1 and f_2 simultaneously with only an $O(\delta)$ loss. Consider a point x and an index i contributing to the sum, meaning $L_{s_i} g(x) = g(s_i^{-1}x) \neq 0$; then $z := s_i^{-1}x \in \mathrm{supp} g \subseteq U$, so $x = s_i z$ with $z \in U$. Applying (6.3) with argument s_i and $t = z$ gives $|h_k(s_i z) - h_k(s_i)| < \delta$, that is,

$$h_k(x) \leq h_k(s_i) + \delta \quad \text{whenever } g(s_i^{-1}x) \neq 0 \tag{6.4}$$

Now estimate $f_k = h_k F$. At any point x, using $f_k = h_k F \leq h_k \sum_i c_i L_{s_i} g$ and inserting (6.4) term by term (each term with $L_{s_i} g(x) \neq 0$ satisfies $h_k(x) \leq h_k(s_i) + \delta$, and terms with $L_{s_i} g(x) = 0$

contribute nothing),

$$f_k(x) = h_k(x)F(x) \le h_k(x) \sum_i c_i\, L_{s_i} g(x) \le \sum_i c_i \big(h_k(s_i) + \delta\big) L_{s_i} g(x)$$

Hence $f_k \le \sum_i c_i \big(h_k(s_i) + \delta\big) L_{s_i} g$ is a domination of f_k, so by definition of the covering number

$$(f_k : g) \le \sum_i c_i \big(h_k(s_i) + \delta\big)$$

Summing over $k = 1, 2$ and using $h_1(s_i) + h_2(s_i) \le 1$,

$$(f_1 : g) + (f_2 : g) \le \sum_i c_i \big(h_1(s_i) + h_2(s_i) + 2\delta\big) \le (1 + 2\delta) \sum_i c_i$$

This holds for every domination of F, so taking the infimum over such dominations gives

$$(f_1 : g) + (f_2 : g) \le (1 + 2\delta)\,(F : g) \tag{6.5}$$

Step 4: Absorb the master term and choose δ. It remains to compare $(F : g)$ with $(f_1 + f_2 : g)$. By subadditivity and homogeneity, proposition 6.1.5(b),

$$(F : g) = (f_1 + f_2 + \delta\varphi : g) \le (f_1 + f_2 : g) + \delta\,(\varphi : g)$$

The term $(\varphi : g)$ is bounded by $(f_1 + f_2 : g)$ up to a g-independent constant. By submultiplicativity, proposition 6.1.5(d),

$$(\varphi : g) \le (\varphi : f_1 + f_2)\,(f_1 + f_2 : g)$$

and $A = (\varphi : f_1 + f_2)$ is finite by proposition 6.1.5(a) and depends only on f_1, f_2, φ, not on g. Substituting,

$$(F : g) \le \big(1 + \delta A\big)\,(f_1 + f_2 : g)$$

Combining with (6.5),

$$(f_1 : g) + (f_2 : g) \le (1 + 2\delta)(1 + \delta A)\,(f_1 + f_2 : g)$$

Given $\varepsilon > 0$, choose δ small enough that $(1 + 2\delta)(1 + \delta A) \le 1 + \varepsilon$; since A depends only on f_1, f_2, this fixes δ, and with it the master function F and the neighbourhood U from Step 2, before any g is chosen. The resulting inequality is the claim, completing the proof.

Dividing lemma 6.1.6 through by $(f_0 : g)$ turns it into a statement about the normalized functionals: for g supported in the neighbourhood $U = U(\varepsilon)$,

$$I_g(f_1) + I_g(f_2) \le (1 + \varepsilon)\, I_g(f_1 + f_2) \tag{6.6}$$

This is the near-superadditivity that the cluster point converts into exact superadditivity.

Proposition 6.1.7: The Limit Functional Is Additive

The cluster point I of the net $\{I_g\}$ satisfies $I(f_1 + f_2) = I(f_1) + I(f_2)$ for all $f_1, f_2 \in C_c^+(G)$.

***Proof*:**
Subadditivity, $I(f_1 + f_2) \le I(f_1) + I(f_2)$, was already inherited from proposition 6.1.5(b) as a closed condition. Only superadditivity remains.

Fix $\varepsilon > 0$ and let $U = U(\varepsilon)$ be the neighbourhood provided by lemma 6.1.6, so that (6.6) holds for every g with $\operatorname{supp} g \subseteq U$. Apply the cluster-point property (6.2) to the three functions $f_1, f_2, f_1 + f_2$ with tolerance ε and the neighbourhood U: there is a single g with $\operatorname{supp} g \subseteq U$ such that simultaneously

$$|I_g(f_1) - I(f_1)| < \varepsilon, \qquad |I_g(f_2) - I(f_2)| < \varepsilon, \qquad |I_g(f_1 + f_2) - I(f_1 + f_2)| < \varepsilon$$

For this g, both (6.6) and these three approximations hold. Combining them,

$$I(f_1) + I(f_2) < I_g(f_1) + I_g(f_2) + 2\varepsilon \le (1+\varepsilon)I_g(f_1 + f_2) + 2\varepsilon < (1+\varepsilon)\big(I(f_1 + f_2) + \varepsilon\big) + 2\varepsilon$$

The right-hand side is $I(f_1 + f_2) + \varepsilon\, I(f_1 + f_2) + \varepsilon(1+\varepsilon) + 2\varepsilon$. As $\varepsilon \to 0$ the correction terms vanish, since $I(f_1 + f_2)$ is a fixed finite number, and the inequality passes to the limit as $I(f_1) + I(f_2) \le I(f_1 + f_2)$. Together with subadditivity this gives $I(f_1) + I(f_2) = I(f_1 + f_2)$, as we sought to show.

Additivity is the last missing axiom, and its proof is the reason the whole construction runs through a shrinking net rather than a single covering number: only in the limit do the translated copies of g stop straddling the supports of f_1 and f_2, and lemma 6.1.1, through lemma 6.1.6, is what quantifies the shrinking overcounting. The functional I is now additive and positively homogeneous on $C_c^+(G)$, hence extends by $I(f) = I(f^+) - I(f^-)$ to a positive linear functional on all of $C_c(G)$, real-valued and, by the closed properties above, left-invariant and nonzero.

The extension deserves one line of care, since additivity and homogeneity were established only on the nonnegative cone. Every $f \in C_c(G)$ decomposes as $f = f^+ - f^-$ with $f^\pm \in C_c^+(G)$, and setting $I(f) = I(f^+) - I(f^-)$ is well-defined and linear by the standard argument: if $f = u - v$ with $u, v \in C_c^+(G)$ then $u + f^- = v + f^+$, so additivity on the cone gives $I(u) + I(f^-) = I(v) + I(f^+)$, whence $I(u) - I(v) = I(f^+) - I(f^-)$, and linearity over $\mathbb{R}$ follows from additivity and homogeneity on the cone. Positivity is immediate: $f \ge 0$ means $f = f^+$, so $I(f) = I(f^+) \ge 0$. Left invariance and $I(f_0) = 1 \ne 0$ persist.

Everything is now in place. A positive, left-invariant, nonzero linear functional on $C_c(G)$ has been built from the group structure alone, and theorem 4.2.1 turns any such functional into a measure.

Theorem 6.1.8: Existence of Haar Measure

Every locally compact Hausdorff group G admits a nonzero left-invariant Radon measure μ: a Radon measure, not identically zero, with $\mu(sE) = \mu(E)$ for every $s \in G$ and every Borel set $E \subseteq G$.

***Proof*:**
Let $I: C_c(G) \to \mathbb{R}$ be the positive, left-invariant, nonzero linear functional constructed above, with $I(f_0) = 1$. By the Riesz-Markov representation theorem, theorem 4.2.1, applied to the locally compact Hausdorff space G, there is a unique Radon measure μ on G with

$$I(f) = \int_G f \, d\mu$$

for all $f \in C_c(G)$. The measure is not identically zero, since $\int_G f_0 \, d\mu = I(f_0) = 1$ forces μ to give positive mass to the support of f_0.

It remains to transfer left invariance of I to left invariance of μ. Fix $s \in G$ and define the translated measure $\mu_s(E) = \mu(s^{-1}E)$, again a Radon measure, since $E \mapsto s^{-1}E$ is a homeomorphism preserving the Radon axioms. For an indicator, $\int_G \chi_E \, d\mu_s = \mu(s^{-1}E) = \int_G \chi_{s^{-1}E} \, d\mu$, and $\chi_{s^{-1}E}(x) = \chi_E(sx) = L_{s^{-1}}\chi_E(x)$; extending by linearity and monotone limits, $\int_G f \, d\mu_s = \int_G L_{s^{-1}} f \, d\mu$ for every $f \in C_c(G)$. Therefore

$$\int_G f \, d\mu_s = \int_G L_{s^{-1}} f \, d\mu = I(L_{s^{-1}} f) = I(f) = \int_G f \, d\mu$$

where the middle equality is left invariance of I, proposition 6.1.5(c) carried to the limit. Both μ and μ_s are Radon measures representing the same functional I, so by the uniqueness clause of theorem 4.2.1 they coincide: $\mu(s^{-1}E) = \mu(E)$ for every Borel E. Replacing s by s^{-1} gives $\mu(sE) = \mu(E)$, which is left invariance. Thus μ is a nonzero left-invariant Radon measure, completing the proof.

The theorem is the group-theoretic counterpart of the fact that Lebesgue measure exists on $\mathbb{R}$, and the proof exhibits the same mechanism in a form stripped of the line's special features. Nothing was used but the group operation, the topology, and compactness; no coordinates, no metric, and no explicit formula for the measure appear. The measure is defined only implicitly, as the one whose integral averages functions the way the invariant functional I does, which is why the Riesz-Markov theorem is indispensable: it is the passage from the analytic object, an invariant integral, to the geometric object, an invariant measure. The invariant measure so produced is named for its discoverer.

Definition 6.1.9: Haar Measure

Let G be a locally compact Hausdorff group. A *left Haar measure* on G is a nonzero left-invariant Radon measure on G, that is, a Radon measure $\mu \neq 0$ with $\mu(sE) = \mu(E)$ for every $s \in G$ and every Borel set E. A *right Haar measure* is defined identically with right translation, $\mu(Es) = \mu(E)$.

Theorem 6.1.8 is exactly the assertion that a left Haar measure exists on every locally compact Hausdorff group. Existence is only half of what makes Haar measure useful; the measure would be of little value if it depended on the arbitrary choices made in its construction, the reference f_0 and the cluster point I. That it does not, that any two left Haar measures differ only by a positive scalar, is the uniqueness theorem of section 6.2, and the relation between left and right Haar measures is the modular function studied there.

Example 6.1.10: Haar Measure on the Classical Groups
The construction specializes to familiar invariant integrals. On $G = \mathbb{R}^n$ under addition, left translation is $L_s f(x) = f(x - s)$, and Lebesgue measure vol is left-invariant and Radon, so it is a Haar measure; the construction reproduces it up to the scalar fixed by the choice of f_0. On the multiplicative group $G = \mathbb{R}_{>0}$, left translation is $L_s f(x) = f(x/s)$, and the measure $d\mu(x) = dx/x$ is invariant, since substituting $x = su$ gives $d(su)/(su) = du/u$; this is the Haar measure of $\mathbb{R}_{>0}$, and it is the reason the logarithm converts multiplication to translation. On the general linear group $G = \mathrm{GL}_n(\mathbb{R})$, an open subset of $\mathbb{R}^{n^2}$ and hence locally compact, the measure

$$d\mu(A) = \frac{dA}{|\det A|^n}$$

where dA is Lebesgue measure on the n^2 matrix entries, is left-invariant: left multiplication by S is a linear map on the entries with Jacobian $|\det S|^n$, which the denominator exactly cancels. In each case the abstract theorem guarantees existence, while the explicit formula, available here because the group sits concretely inside a Euclidean space, identifies the measure the theorem produces.

6.1.11 Remark: Cartan's Choice-Free Route The construction above extracts I as a cluster point of the net $\{I_g\}$, an appeal to compactness of the product X and hence to Tychonoff's theorem, which rests on the axiom of choice. A second route, due to Cartan, avoids this. Instead of taking a cluster point, one shows directly that the net $\{I_g(f)\}$ is a *Cauchy* net as $\mathrm{supp} g \to \{e\}$: left-uniform continuity, lemma 6.1.1, forces $I_g(f)$ and $I_{g'}(f)$ to agree to within any prescribed tolerance once both g, g' have small enough support, so completeness of $\mathbb{R}$ gives a genuine limit $I(f) = \lim_g I_g(f)$ with no subnet and no compactness. The Cauchy estimate simultaneously yields additivity and, with a little more work, uniqueness of the limit, so the choice-free argument delivers existence and uniqueness together. The two routes prove the same theorem; the covering-number combinatorics and the role of lemma 6.1.1 are identical, and only the final extraction of the limit differs. See Deitmar and Echterhoff for the Cauchy-net development in full.

6.2 Uniqueness and the Modular Function

The construction of section 6.1 produces a left Haar measure but says nothing about how many there are. A measure is a choice of scale, and an abstract group carries no distinguished unit of volume, so the most one can ask is uniqueness up to a positive multiple. That is what holds: two left Haar measures on the same group differ only by a constant factor. The proof applies the Fubini-Tonelli theorem (theorem 2.6.10) to a product of the two measures, the first place where an invariant measure is integrated against another.

Once uniqueness is available, the failure of a left-invariant measure to be right-invariant can be measured. Right translation carries a left Haar measure to another left Haar measure, so by uniqueness the two differ by a scalar depending on the translating element. The resulting positive function on the group, the modular function, records the discrepancy between left and right; it is identically 1 exactly when the two notions of invariance agree.

Throughout, G is a locally compact Hausdorff group ([9, Topological Group]) with left translation

$L_s f(x) = f(s^{-1}x)$, and a left Haar measure is a nonzero left-invariant Radon measure (definition 6.1.9). Two such measures are compared on $C_c^+(G)$, the nonzero nonnegative members of $C_c(G)$, since a Radon measure is determined by its integrals against $C_c(G)$ (the uniqueness clause of Riesz-Markov, theorem 4.2.1) and every $f \in C_c(G)$ is a difference of two members of $C_c^+(G) \cup \{0\}$.

The mechanism of the proof is to show that the ratio $\int f\,d\nu / \int f\,d\mu$ does not depend on f. Once this common value is a single constant c, the two measures satisfy $\int f\,d\nu = c \int f\,d\mu$ on $C_c^+(G)$, hence on $C_c(G)$, and Riesz-Markov upgrades the equality of functionals to $\nu = c\mu$. The independence of f comes from a double integral $\iint f(yx)g(x)\,d\mu(x)\,d\nu(y)$ evaluated two ways: once by left-invariance, which produces $\int f\,d\mu$ times a ν-integral of g, and once by letting g concentrate at the identity, which extracts $\int f\,d\nu$. Reconciling the two limits forces the ratio. Left-uniform continuity (lemma 6.1.1) is what makes the limit converge, and the Fubini-Tonelli theorem is what licenses the two orders.

Theorem 6.2.1: Uniqueness of Haar Measure

Let G be a locally compact Hausdorff group and let μ and ν be left Haar measures on G. Then there is a constant $c > 0$ with $\nu = c\mu$.

***Proof*:**
Integrals over G are written without their domain. Every integrand below is a product of members of $C_c^+(G)$ with translated arguments, hence continuous with compact support; since a Radon measure is finite on compact sets (definition 4.1.9), each integral is a finite real number, and each double integral has a nonnegative continuous integrand of compact support in $G \times G$, so the Tonelli half of theorem 2.6.10 permits evaluating it in either order. Both μ and ν are left-invariant, so $\int \varphi(sx)\,d\mu(x) = \int \varphi\,d\mu$ and likewise for ν, for all $s \in G$ and $\varphi \in C_c(G)$.

Fix $f \in C_c^+(G)$; the claim to be proved is that $\int f\,d\nu / \int f\,d\mu$ is a positive constant independent of f.

Step 1: A double integral evaluated by left-invariance. Let $g \in C_c^+(G)$ be symmetric under inversion, $g(x) = g(x^{-1})$; such g exist, since for any nonzero $\varphi \in C_c^+(G)$ the sum $\varphi(x) + \varphi(x^{-1})$ is symmetric and again in $C_c^+(G)$, inversion being a homeomorphism of G. Consider

$$J = \iint f(yx)\,g(x)\,d\mu(x)\,d\nu(y)$$

Integrating over x first, substitute $x \mapsto y^{-1}x$ in the inner μ-integral, valid by left-invariance of μ. Then $f(yx)g(x)$ becomes $f(x)g(y^{-1}x)$, so

$$\int f(yx)\,g(x)\,d\mu(x) = \int f(x)\,g(y^{-1}x)\,d\mu(x)$$

and hence, by Tonelli,

$$J = \iint f(x)\,g(y^{-1}x)\,d\mu(x)\,d\nu(y) = \int f(x)\left[\int g(y^{-1}x)\,d\nu(y)\right]d\mu(x)$$

The inner integral $S(x) = \int g(y^{-1}x)\,d\nu(y)$ is constant on G: for $u \in G$, substituting $y \mapsto uy$ by left-invariance of ν,

$$S(ux) = \int g(y^{-1}ux)\,d\nu(y) = \int g\big((uy)^{-1}ux\big)\,d\nu(y) = \int g(y^{-1}x)\,d\nu(y) = S(x)$$

As ux ranges over G with u, the value $S(x) = S(e) = \int g(y^{-1})\,d\nu(y) = \int g\,d\nu$ is the same everywhere, the last equality by symmetry of g. Therefore

$$J = \left(\int f\,d\mu\right)\left(\int g\,d\nu\right)$$

Step 2: The same integral evaluated by concentration. Now integrate J in the other order, over y first:

$$J = \int g(x)\left[\int f(yx)\,d\nu(y)\right]d\mu(x) = \int g(x)\,F(x)\,d\mu(x), \qquad F(x) = \int f(yx)\,d\nu(y)$$

The function F is continuous, and $F(e) = \int f\,d\nu$. To bring J close to $F(e)$, specialize g to concentrate near the identity. Fix a compact symmetric neighborhood V_0 of e, available since a locally compact group has a neighborhood base of e consisting of compact symmetric sets ([9, Halving a Neighbourhood, Locally Compact in Hausdorff Space], combined). For a symmetric neighborhood $V \subseteq V_0$ of e, choose a symmetric $g_V \in C_c^+(G)$ with $\operatorname{supp} g_V \subseteq V$ and $\int g_V\,d\mu = 1$; such g_V exists by taking a bump function supported in V (lemma 4.1.5), symmetrizing it, and normalizing by its positive μ-integral. With $g = g_V$, Step 1 reads

$$\left(\int f\,d\mu\right)\left(\int g_V\,d\nu\right) = \int g_V(x)\,F(x)\,d\mu(x)$$

Since $g_V \geq 0$, $\int g_V\,d\mu = 1$, and $\operatorname{supp} g_V \subseteq V$,

$$\left|\int g_V(x)\,F(x)\,d\mu(x) - F(e)\right| = \left|\int g_V(x)\big(F(x) - F(e)\big)\,d\mu(x)\right| \leq \sup_{x\in V}|F(x) - F(e)|$$

Step 3: Continuity of F at the identity. By the same argument as lemma 6.1.1 applied through inversion, f is right-uniformly continuous: given $\varepsilon > 0$ there is a neighborhood V of e, which may be taken inside V_0, with $|f(yx) - f(y)| < \varepsilon$ for all $y \in G$ whenever $x \in V$. For such x,

$$|F(x) - F(e)| = \left|\int \big(f(yx) - f(y)\big)\,d\nu(y)\right| \leq \int |f(yx) - f(y)|\,d\nu(y) \leq \varepsilon\,\nu(K)$$

where $K = \operatorname{supp} f \cdot V_0^{-1}$ is a compact set containing the support of every $y \mapsto f(yx) - f(y)$ with $x \in V_0$, and $\nu(K) < \infty$. Hence $\sup_{x\in V}|F(x) - F(e)| \leq \varepsilon\,\nu(K)$, which tends to 0 as V shrinks. Combining with Step 2,

$$\left(\int f\,d\mu\right)\left(\int g_V\,d\nu\right) \longrightarrow F(e) = \int f\,d\nu \qquad \text{as } V \to \{e\}$$

Step 4: The ratio is constant. Dividing the limit of Step 3 by $\int f\,d\mu > 0$ shows that the net $\left(\int g_V\,d\nu\right)$, indexed by the shrinking symmetric neighborhoods V of e, converges to

$$\frac{\int f\,d\nu}{\int f\,d\mu}$$

This net involves only the functions g_V and the measure ν, not f. A second choice $f' \in C_c^+(G)$ runs against the same net, so it converges to $\int f'\,d\nu / \int f'\,d\mu$ as well; since a net has at most one limit, the two ratios agree. Writing c for the common value, $\int f\,d\nu = c\int f\,d\mu$ for every $f \in C_c^+(G)$, and $c > 0$ because $\int f\,d\nu$ and $\int f\,d\mu$ are both positive.

Step 5: Passage to measures. From $\int f\,d\nu = c\int f\,d\mu$ for all $f \in C_c^+(G)$, and by linearity for all $f \in C_c(G)$, the two positive linear functionals $f \mapsto \int f\,d\nu$ and $f \mapsto c\int f\,d\mu$ agree on $C_c(G)$. By the uniqueness clause of Riesz-Markov (theorem 4.2.1), the Radon measures representing them coincide, giving $\nu = c\mu$, as we sought to show.

The uniqueness theorem is what makes "the" Haar measure a meaningful phrase: once a single normalization is fixed, every left-invariant Radon integral on G is determined. The remaining freedom is the choice of scale, and for two classes of groups there is a canonical way to spend it. On a compact group one normalizes so that $\mu(G) = 1$, making Haar measure a probability measure; on a discrete group one normalizes so that each point has mass 1, making Haar measure the counting measure. Between them, on $\mathbb{R}^n$ the normalization $\mu([0,1]^n) = 1$ recovers Lebesgue measure, which is why Lebesgue measure is invisible to translation.

The proof rests on evaluating one double integral two ways. Left-invariance alone gives the clean value $\int f\,d\mu \cdot \int g\,d\nu$, but only when g is symmetric, which is what lets the constant $S(x) = \int g\,d\nu$ emerge from the inner integral. The second evaluation replaces g by an approximate identity concentrated at e, and it is left-uniform continuity (lemma 6.1.1), applied to f through inversion, that forces the smoothed integral $\int g_V(x)F(x)\,d\mu(x)$ to converge to the point value $F(e) = \int f\,d\nu$. The role of the modular function, developed next, is exactly to measure what right-invariance would have given for free had it been available; its absence is why the argument routes through a limit rather than a single substitution.

Example 6.2.2: Uniqueness on the Circle

Take $G = \mathbb{T} = \mathbb{R}/\mathbb{Z}$, the circle group, which is compact and abelian. Arc length $d\theta$, normalized to total mass 1, is a left Haar measure, since translation $\theta \mapsto \theta + t$ preserves arc length. Let ν be any other left Haar measure on $\mathbb{T}$. By theorem 6.2.1, $\nu = c\,d\theta$ for some $c > 0$. If ν is also normalized to $\nu(\mathbb{T}) = 1$, then

$$1 = \nu(\mathbb{T}) = c\int_{\mathbb{T}} d\theta = c$$

forces $c = 1$, so $\nu = d\theta$. The normalized Haar measure on $\mathbb{T}$ is thus the unique invariant probability measure, and it assigns to an arc of length ℓ the mass ℓ. This is the fact underlying Fourier series: the characters $\theta \mapsto e^{2\pi i n\theta}$ are orthonormal in $L^2(\mathbb{T}, d\theta)$ precisely because $d\theta$ is the invariant measure.

With uniqueness in hand, right translation can be analyzed. Left translation preserves a left Haar measure by definition, but right translation need not: replacing $f(x)$ by $f(xs)$ transports the measure by a right shift, and there is no reason the outcome should again be μ. What can be said is that the transported measure is still a left Haar measure, so uniqueness pins it down to a scalar multiple of μ. That scalar is the object to be named.

Definition 6.2.3: Right Translation of a Left Haar Measure

Let μ be a left Haar measure on a locally compact Hausdorff group G, and let $s \in G$. The *right translate* $R_s\mu$ is the Borel measure defined on Borel sets E by

$$(R_s\mu)(E) = \mu(Es)$$

equivalently, by the integration formula $\int f \, d(R_s\mu) = \int f(xs^{-1}) \, d\mu(x)$ for $f \in C_c(G)$.

The two descriptions agree: $(R_s\mu)(E) = \mu(Es)$ makes $R_s\mu$ the pushforward of μ under $x \mapsto xs^{-1}$, whose integral of f is $\int f(xs^{-1}) \, d\mu(x)$. The right translate $R_s\mu$ is again a Radon measure, and it is left-invariant: for $t \in G$ and a Borel set E, left-invariance of μ gives $(R_s\mu)(tE) = \mu(tEs) = \mu(Es) = (R_s\mu)(E)$, since tEs is the left translate of Es by t. It is nonzero because μ is. Thus $R_s\mu$ is a left Haar measure, and theorem 6.2.1 applies.

Definition 6.2.4: Modular Function

Let μ be a left Haar measure on a locally compact Hausdorff group G. For each $s \in G$, the right translate $R_s\mu$ is a left Haar measure, so by theorem 6.2.1 there is a unique positive constant $\Delta(s)$ with

$$R_s\mu = \Delta(s) \, \mu$$

The function $\Delta \colon G \to \mathbb{R}_{>0}$ so defined is the *modular function* of G.

The definition appears to depend on the chosen left Haar measure μ. The first thing to record is that it does not, so that Δ deserves the name *the* modular function *of* G.

Lemma 6.2.5: The Modular Function is Independent of μ

Let G be a locally compact Hausdorff group and let μ and μ' be left Haar measures on G. Then for every $s \in G$, the constant $\Delta(s)$ that definition 6.2.4 attaches to μ equals the constant it attaches to μ'.

***Proof*:**
By theorem 6.2.1 there is a constant $a > 0$ with $\mu' = a\mu$. Fix $s \in G$ and write $\Delta(s)$ for the constant defined from μ, so $R_s\mu = \Delta(s)\mu$. For a Borel set E, definition 6.2.3 gives

$$(R_s\mu')(E) = \mu'(Es) = a \, \mu(Es) = a \, (R_s\mu)(E)$$

so $R_s\mu' = a \, R_s\mu = a \, \Delta(s)\mu = \Delta(s) \, (a\mu) = \Delta(s) \, \mu'$. Since the constant in definition 6.2.4 is unique, the constant attached to μ' is $\Delta(s)$ as well.

The modular function is thus an invariant of G alone, not of the chosen normalization. Its value $\Delta(s)$ measures how a right shift by s scales left-invariant volume, a reading made precise in the proposition below.

The next proposition records the two structural facts about Δ: it is a homomorphism to the multiplicative group of positive reals, and it is continuous. Both follow from uniqueness and the group law, and together they make Δ a continuous character of G valued in $\mathbb{R}_{>0}$.

Proposition 6.2.6: Properties of the Modular Function

Let G be a locally compact Hausdorff group with left Haar measure μ and modular function Δ. Then

1. $\Delta\colon G \to (\mathbb{R}_{>0}, \times)$ is a group homomorphism: $\Delta(st) = \Delta(s)\Delta(t)$ for all $s, t \in G$, and $\Delta(e) = 1$.
2. For every $f \in C_c(G)$ and $s \in G$,
$$\int f(xs)\, d\mu(x) = \Delta(s)^{-1} \int f(x)\, d\mu(x)$$
3. Δ is continuous.

***Proof*:**

1. Right translation is an action of G on measures: for a Borel set E,
$$(R_s R_t \mu)(E) = (R_t\mu)(Es) = \mu(Est) = \mu\big(E(st)\big) = (R_{st}\mu)(E)$$
so $R_s R_t \mu = R_{st}\mu$. Applying definition 6.2.4 twice, $R_s R_t \mu = R_s(\Delta(t)\mu) = \Delta(t)\Delta(s)\mu$, while $R_{st}\mu = \Delta(st)\mu$. Equating the two scalars gives $\Delta(st) = \Delta(s)\Delta(t)$, the homomorphism property. Taking $s = t = e$ gives $\Delta(e) = \Delta(e)^2$, hence $\Delta(e) = 1$.
2. By definition 6.2.3, $\int f\, d(R_s\mu) = \int f(xs^{-1})\, d\mu(x)$, and by definition 6.2.4, $R_s\mu = \Delta(s)\mu$, so $\int f(xs^{-1})\, d\mu(x) = \Delta(s) \int f\, d\mu$. Replacing s by s^{-1} and using $\Delta(s^{-1}) = \Delta(s)^{-1}$ from part (1),
$$\int f(xs)\, d\mu(x) = \Delta(s^{-1}) \int f\, d\mu = \Delta(s)^{-1} \int f\, d\mu$$
3. Normalizing, fix $f \in C_c^+(G)$ with $\int f\, d\mu = 1$; such f exists because μ is nonzero. By part (2), $\Delta(s)^{-1} = \int f(xs)\, d\mu(x)$, so it suffices to show $s \mapsto \int f(xs)\, d\mu(x)$ is continuous, since $\Delta(s)$ is then continuous as the reciprocal of a continuous positive function. Let $s_0 \in G$ and $\varepsilon > 0$. Right-uniform continuity of f, the analogue of lemma 6.1.1 proved by the same argument applied through inversion, gives a neighborhood V of e with $\sup_x |f(xu) - f(x)| < \varepsilon$ for $u \in V$; relabeling the supremum variable $x \mapsto xs_0^{-1}$ gives $\sup_x |f(xs) - f(xs_0)| < \varepsilon$ whenever $s_0^{-1}s \in V$. Moreover, for s ranging over the compact neighborhood $s_0\overline{V_0} = \{s : s_0^{-1}s \in \overline{V_0}\}$ of s_0, with $\overline{V_0} \subseteq V$ compact, all the functions $x \mapsto f(xs)$ are supported in the common compact set $K = \operatorname{supp} f \cdot \overline{V_0}^{-1} s_0^{-1}$. Hence, for s in this neighborhood,
$$\left| \int f(xs)\, d\mu(x) - \int f(xs_0)\, d\mu(x) \right| \le \int |f(xs) - f(xs_0)|\, d\mu(x) \le \varepsilon\, \mu(K)$$
with $\mu(K) < \infty$. As $\varepsilon > 0$ was arbitrary, $s \mapsto \int f(xs)\, d\mu(x)$ is continuous at s_0, and s_0 was arbitrary, so Δ is continuous, completing the proof.

The formula $\int f(xs)\, d\mu(x) = \Delta(s)^{-1} \int f\, d\mu$ is the working form of the modular function: right-translating the argument of f by s scales its integral by $\Delta(s)^{-1}$. In set terms this reads $\mu(Es) = \Delta(s)\mu(E)$, directly from definition 6.2.3 and definition 6.2.4, so right translation of a set by s multiplies its left-invariant volume by $\Delta(s)$. The homomorphism property says these scaling factors compose correctly under the group law, and continuity says they vary continuously with s. A character valued in $\mathbb{R}_{>0}$ is exactly the data needed to correct integrals for the asymmetry between left and right.

The placement of the inverse is fixed once and for all by definition 6.2.4: $R_s\mu = \Delta(s)\mu$ with $(R_s\mu)(E) = \mu(Es)$, so $\mu(Es) = \Delta(s)\mu(E)$ on sets and $\int f(xs)\,d\mu = \Delta(s)^{-1}\int f\,d\mu$ on functions. Some texts adopt the reciprocal, defining the modular function so that $\int f(xs)\,d\mu = \Delta(s)\int f\,d\mu$; the two conventions differ by $s \mapsto s^{-1}$ and both are in use. This book is consistent with definition 6.2.4 throughout.

Example 6.2.7: The Modular Function of an Abelian Group

Let G be abelian with left Haar measure μ. For any $s \in G$ and Borel set E, commutativity gives $Es = sE$, so

$$(R_s\mu)(E) = \mu(Es) = \mu(sE) = \mu(E)$$

the last step by left-invariance. Thus $R_s\mu = \mu$, and comparison with $R_s\mu = \Delta(s)\mu$ forces $\Delta(s) = 1$ for every s: the modular function of an abelian group is identically 1. Concretely, on $G = \mathbb{R}$ with Lebesgue measure, $\int f(x+s)\,dx = \int f(x)\,dx$, so $\Delta(s)^{-1} = 1$ for all s, recovering translation invariance of the Lebesgue integral in both directions.

The vanishing of the modular discrepancy in the abelian case instances a phenomenon worth naming. Groups on which $\Delta \equiv 1$ are those whose left Haar measure is automatically right-invariant, and they are the setting in which harmonic analysis is cleanest, since a single bi-invariant measure serves.

Definition 6.2.8: Unimodular Group

A locally compact Hausdorff group G is *unimodular* if its modular function is trivial, $\Delta \equiv 1$. Equivalently, G is unimodular if and only if every left Haar measure on G is also right-invariant.

The equivalence in the definition is immediate from proposition 6.2.6(2): right-invariance of μ means $\int f(xs)\,d\mu(x) = \int f\,d\mu$ for all f and s, and this holds if and only if $\Delta(s)^{-1} = 1$ for all s, that is $\Delta \equiv 1$. Unimodularity is therefore the exact condition under which the left and right theories of invariant integration coincide, and a large class of groups satisfies it.

Proposition 6.2.9: Classes of Unimodular Groups

Each of the following locally compact Hausdorff groups is unimodular.

1. Abelian groups.
2. Compact groups.
3. Discrete groups.

***Proof*:**

1. By example 6.2.7, an abelian group has $Es = sE$, so $R_s\mu = \mu$ and $\Delta \equiv 1$.
2. Let G be compact. By proposition 6.2.6(1) and (3), $\Delta\colon G \to \mathbb{R}_{>0}$ is a continuous

homomorphism, so its image $\Delta(G)$ is a subgroup of $(\mathbb{R}_{>0}, \times)$ and, as the continuous image of a compact set, is compact. The only compact subgroup of $\mathbb{R}_{>0}$ is $\{1\}$: any subgroup containing a point $a \neq 1$ contains all powers a^n, and either $a > 1$, whence $a^n \to \infty$, or $a < 1$, whence $a^{-n} \to \infty$, so the subgroup is unbounded and not compact. Therefore $\Delta(G) = \{1\}$, that is $\Delta \equiv 1$, and G is unimodular.

3. Let G be discrete. Counting measure μ, which assigns to a finite set its cardinality and to an infinite set ∞, is a left Haar measure: on a discrete space every compact set is finite, so μ is a Radon measure, and $\mu(sE) = |sE| = |E| = \mu(E)$ since left translation is a bijection. Counting measure is equally right-invariant, $\mu(Es) = |Es| = |E| = \mu(E)$, so by definition 6.2.8 G is unimodular. Explicitly, $R_s\mu = \mu$ gives $\Delta(s) = 1$.

Each class thus has trivial modular function, as required.

These three classes cover most groups met in a first course: finite groups, which are compact and discrete at once; $\mathbb{R}^n$, $\mathbb{T}^n$, and $\mathbb{Z}^n$, which are abelian; and the compact matrix groups $\mathrm{SU}(n)$ and $\mathrm{SO}(n)$. Semisimple and reductive Lie groups are also unimodular, though by criteria beyond the elementary ones above. Unimodularity is common enough that its failure can feel exotic, which makes an explicit non-unimodular group essential for calibration.

The standard example is the group of orientation-preserving affine transformations of the line, the $ax + b$ group. It is the smallest natural group on which left and right Haar measures differ, and computing both makes the modular function concrete rather than axiomatic.

Example 6.2.10: The $ax + b$ Group

Let G be the group of affine transformations $x \mapsto ax + b$ of $\mathbb{R}$ with $a > 0$ and $b \in \mathbb{R}$. Writing an element as the pair (a, b), the composition $(a, b)(a', b')$ sends x to $a(a'x + b') + b = (aa')x + (ab' + b)$, so the group law is

$$(a, b)(a', b') = (aa',\ ab' + b)$$

with identity $(1, 0)$ and inverse $(a, b)^{-1} = (a^{-1}, -a^{-1}b)$. As a topological space G is the open half-plane $\{(a, b) : a > 0\} \subseteq \mathbb{R}^2$, a locally compact Hausdorff group.

Left Haar measure. On the half-plane take $d\mu_L = a^{-2}\, da\, db$, meaning $\int f\, d\mu_L = \iint f(a, b)\, a^{-2}\, da\, db$ against ordinary Lebesgue measure $da\, db$. To check left-invariance, fix $s = (a_0, b_0)$; left translation by s replaces (a, b) by $s(a, b) = (a_0a,\ a_0b + b_0)$. Under the substitution $(a', b') = (a_0a, a_0b + b_0)$ the Jacobian is

$$\frac{\partial(a', b')}{\partial(a, b)} = \det \begin{pmatrix} a_0 & 0 \\ 0 & a_0 \end{pmatrix} = a_0^2$$

so $da'\, db' = a_0^2\, da\, db$, while $a' = a_0a$ gives $(a')^{-2} = a_0^{-2}a^{-2}$. Hence

$$(a')^{-2}\, da'\, db' = a_0^{-2}a^{-2} \cdot a_0^2\, da\, db = a^{-2}\, da\, db$$

so $d\mu_L$ is unchanged by left translation, a left Haar measure.

Right Haar measure. By the symmetric computation, take $d\mu_R = a^{-1}\, da\, db$. Right translation by $s = (a_0, b_0)$ replaces (a, b) by $(a, b)s = (aa_0,\ ab_0 + b)$. Under $(a', b') = (aa_0, ab_0 + b)$ the Jacobian is

$$\frac{\partial(a', b')}{\partial(a, b)} = \det \begin{pmatrix} a_0 & 0 \\ b_0 & 1 \end{pmatrix} = a_0$$

so $da'\,db' = a_0\,da\,db$, while $a' = aa_0$ gives $(a')^{-1} = a_0^{-1}a^{-1}$, and

$$(a')^{-1}\,da'\,db' = a_0^{-1}a^{-1}\cdot a_0\,da\,db = a^{-1}\,da\,db$$

so $d\mu_R = a^{-1}\,da\,db$ is right-invariant, a right Haar measure. The two differ by the factor a^{-1}, so no positive constant makes them equal, and G is not unimodular.

The modular function. To read off Δ, apply proposition 6.2.6(2) to μ_L. Right translation by $s = (a_0, b_0)$ acts on $d\mu_L = a^{-2}\,da\,db$ through the Jacobian just computed: $da'\,db' = a_0\,da\,db$ and $(a')^{-2} = a_0^{-2}a^{-2}$, so

$$\int f((a,b)s)\,d\mu_L = \iint f(a',b')\,a^{-2}\,da\,db = \iint f(a',b')\,a_0\,(a')^{-2}\,da'\,db' = a_0\int f\,d\mu_L$$

Comparing with $\int f(xs)\,d\mu_L(x) = \Delta(s)^{-1}\int f\,d\mu_L$ gives $\Delta(s)^{-1} = a_0$, so

$$\Delta(a_0, b_0) = a_0^{-1}$$

The modular function depends only on the multiplicative part a_0 and ignores the translation part b_0, consistent with Δ being a homomorphism to $\mathbb{R}_{>0}$: the map $(a_0, b_0) \mapsto a_0^{-1}$ is the projection $(a, b) \mapsto a$ followed by inversion on $\mathbb{R}_{>0}$, both homomorphisms. The relation $d\mu_R = a\,d\mu_L = \Delta^{-1}d\mu_L$ between the right and left measures is the general fact that a right Haar measure is Δ^{-1} times a left Haar measure, here made explicit.

The $ax + b$ group shows that the gap between left and right invariance is real and that the modular function measures it exactly. The reason a group can be non-unimodular is visible in the computation: left translation acts on (a, b) with Jacobian a_0^2 regardless of the base point, while right translation acts with Jacobian a_0, and the discrepancy a_0 between the two is $\Delta(s)^{-1}$. Whenever conjugation on the group distorts volume, the modular function is nontrivial, and the affine group is the first place this occurs.

7

What Next?

Measure theory is infrastructure. It is learned in order to read something else, so what this book was for is best measured by what it now opens up. Five directions lead out of it, and they leave from different chapters.

The spaces the integrable functions form

Chapter 2 produced L^1 and proved that it is complete, then stopped short of asking what structure L^p carries. That question (the norm, the dual, the geometry of the unit ball, and what a bounded operator between such spaces can do) is functional analysis, and it is the immediate sequel. *Functional Analysis* [13] picks up exactly where the density theorems here leave off, with Riesz-Fischer, Hölder and Minkowski, and the duality $(L^p)^* \cong L^q$ whose proof needs the Radon-Nikodym theorem of Chapter 3. It also repays the debt of Chapter 4: the Riesz-Markov theorem proved there is the statement that the dual of $C_0(X)$ is a space of measures, and the functional-analytic reading of that fact is where it becomes a tool. Rudin's *Real and Complex Analysis* runs the two subjects together in one volume and is the classical place to see the seam from the other side.

Fourier analysis, and why it needed Lebesgue

The historical motivation of Chapter 1 was a question about Fourier series that the Riemann integral could not answer. *Harmonic Analysis* [14] is that question answered: the Fourier transform on L^1 and L^2, Plancherel, tempered distributions, singular integrals and Calderón-Zygmund theory. It is also the book in which the Haar measure of Chapter 6 stops being a construction and becomes a setting, since Pontryagin duality is Fourier analysis with $\mathbb{R}^n$ replaced by an arbitrary locally compact abelian group. Stein and Shakarchi's *Fourier Analysis* is the gentlest entry; Grafakos's *Classical*

Fourier Analysis is the reference.

The same objects, read as chance

A probability space is a measure space of total mass one, a random variable is a measurable function, and an expectation is an integral. Nothing in *Probability* [7] asks the reader to learn a new theory. It asks them to read this one in a different dialect, and then shows what the change of vocabulary makes visible: independence as a product measure, the laws of large numbers as convergence theorems, and conditional expectation as the Radon-Nikodym derivative of Chapter 3. The martingale theory at the end of that book is where uniform integrability, introduced here in Chapter 2, does its real work. From there the tower continues into *Mathematical Statistics* [5] and, once continuous time is admitted, *Stochastic Calculus* [8]. Durrett's *Probability: Theory and Examples* is the standard graduate companion.

Measuring sets that are not smooth

Chapter 5 assigns a dimension to a set and uses it to state a conjecture that is still open. The subject that grows from there is geometric measure theory: Frostman's lemma and Riesz energies, the behaviour of dimension under projection, and the structure theorems separating rectifiable sets from purely unrectifiable ones. *Geometric Measure Theory* [4] takes that up directly on this book's Hausdorff chapter. Mattila's *Geometry of Sets and Measures in Euclidean Spaces* is the right first book; Federer's *Geometric Measure Theory* is the one everyone cites and nobody reads first.

Groups that carry their own integral

Chapter 6 constructs, on any locally compact group, a measure invariant under translation, and thereby makes "average over the group" a legal operation. That single fact is the entry point to representation theory in its analytic form. *Unitary Representation Theory* [10] builds $L^1(G)$, the group C^*-algebra, and the Plancherel theorem on top of it, and *Lie Groups* [15] uses invariant integration to prove Peter-Weyl. Folland's *A Course in Abstract Harmonic Analysis* is the standard reference for the measure-theoretic side of that story.

Index

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